MICROPROSE · 1994 · MS-DOS

Sid Meier's COLONIZATION

Technical Reference

The shipped MS-DOS release, documented from its binaries and data files. Executables and overlay architecture · asset containers · terrain and the map compositor · colonies, market, and the native economy · units, movement, and combat · powers, diplomacy, and revolution · events and strings · the UI engine, screen by screen · the map editor · data structures.

RECONSTRUCTED BY DISASSEMBLY OF VICEROY.EXE AND MAPEDIT.EXE · EVERY FIGURE TRACED TO A FILE OFFSET, A DATA-FILE FIELD, OR LIVE GAME MEMORY — OR MARKED UNMAPPED

Contents

The shipped 1994 MS-DOS release, documented from its binaries and data files. Facts in this volume were established by disassembly of the shipped executables, by reads of live game memory under emulation, and by pixel-level comparison of screens reconstructed from these pages against the running game. Where a value could not be established it is marked unmapped or runtime — no figure in this volume is invented.

PART I

The machine and its files

§1 Executables and overlay architecture

The shipped game directory contains six DOS MZ executables. One of them — VICEROY.EXE, the game proper — carries essentially all of the game logic, packed into an RTLink Plus (Pocket Soft) version-2 overlay image appended to a small resident load image. The other programs are satellites: a map editor, an intro player, an endgame player, and two installer utilities. This section maps the executable set, then dissects VICEROY's overlay machinery in enough detail to resolve any far call in the binary to a file offset.

1.1 The executable inventory

All six are plain 16-bit real-mode MZ executables (no NE/PE extended header). Values measured from the shipped files:

EXE	FILE BYTES	IMAGE	HEADER	CODE+DATA	OVERLAY	ENTRY CS:IP	RELOCS
VICEROY.EXE	494,910	132,709	9,216	123,493	362,201	110D:071D	2,260
MAPEDIT.EXE	145,292	114,185	5,632	108,553	31,107	1388:001E	1,365
OPENING.EXE	89,178	67,271	3,072	64,199	21,907	0452:015C	664
CLOSING.EXE	83,246	62,769	2,560	60,209	20,477	037D:0150	595
MPSCOPY.EXE	38,620	4,986	512	4,474	33,634	0000:0031	0
INSTALL.EXE	51,285	51,285	32	51,253	0	0C35:000E	0

Five of the six carry a post-image overlay region. A byte-pattern survey (counts of `55 8B EC` prologues, `C8` ENTER prologues, `9A` far calls, `CD 21` DOS calls) shows that **only VICEROY's overlay contains loadable code** — 8,507 LCALL instructions and 1,323 ENTER prologues. The MAPEDIT/OPENING/CLOSING overlay regions are linker *debug data* (symbol tables and source-file directories), not executed code. VICEROY.EXE is 73% overlay by bytes; that is where the bulk of the game lives. The build stack is Microsoft C 6.0 medium model (confirmed by byte-pattern match of `__aFlmul` / `__aFldiv` and the canonical MSC 6.0 `rand()` LCG constants 0x000343FD / 0x00269EC3) over the MicroProse MADS asset engine and the RTLink Plus overlay runtime.

1.2 VICEROY.EXE — MZ header and file geography

FIELD	VALUE	NOTE
signature	MZ	
relocation count (+0x06)	2,260	4-byte (off,seg) entries
header paragraphs (+0x08)	576	code base = 576×16 = file 0x2400
entry IP (+0x14)	0x071D	
entry CS (+0x16)	0x110D	runtime/load-image entry segment
reloc table offset (+0x18)	0x1E	

File 0x2400 is paragraph 0 of the load image; all segment arithmetic below is relative to it. Landmark strings inside the image:

MARKER	FILE OFFSET	MEANING
`RTLink`	0x1A25D	identifies the RTLink-linked image
`Enter directory for $`	0x1A5B7	immediately precedes the thunk table
`MS Run-Time`	0x1D9A8	start of the static data segment − 8

DGROUP (the static data segment) begins at file `0x1D9A0` and spans 0x2CD0 bytes of initialised data (its overlay-directory record gives load paragraph 0x1B5A). Every static datum at DGROUP offset `D` therefore lives at file `0x1D9A0 + D` — the rule used throughout this manual to byte-read strings and initialised tables (e.g. the string `"ORDERS"` at DGROUP 0x225D = file 0x1FBFD). At run time the loaded DGROUP segment is not a static constant; in an instrumented session (1994 binary under DOSBox) it was found at **segment `0x1CFD`** (physical 0x1CFD0) by anchoring on the contiguous section-name table `UNIT\0ORDERS\0ACTIONS\0` which sits at DGROUP offset 0x2258. DGROUP offsets are preserved from the file image, so any static `DGROUP:0xNNNN` citation is readable live at `phys(DGROUP_base) + 0xNNNN`.

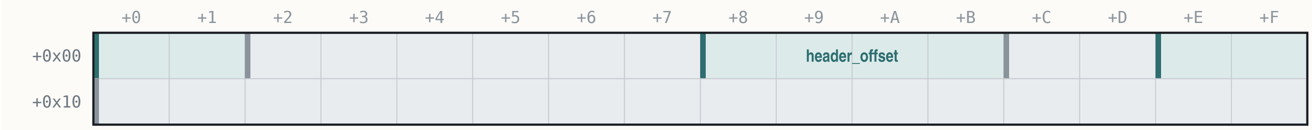
Gross file layout:

REGION	FILE RANGE	CONTENTS
MZ header + relocs	0x0..0x2400	header, 2,260 relocation entries
Load image (resident)	0x2400..0x20665	C runtime, RTLink loader, glyph/blit/message cores, resident helpers
— overlay segment list	0x192F0	31 × 32-byte records (§1.3)
— thunk table	0x1A5F0..~0x1D610	1,023 far-call stubs (§1.4)
— DGROUP image	0x1D9A0..0x20670	initialised data + strings
Overlay region	0x20670..end	31 demand-paged code segments (§1.3)

1.3 RTLink Plus version 2 — overlay pages

VICEROY uses the "embedded overlay" flavour of RTLink Plus version 2: overlay segments are appended inside the EXE itself (no separate .OVL file) and demand-loaded by a resident runtime. The **segment list** at file `0x192F0` is a table of 32-byte records, one per overlay segment, with `segmentNum` incrementing 2..32 (31 records):

RTLinkSegmentRecord one segment-list record @0x192F0 + 32*i



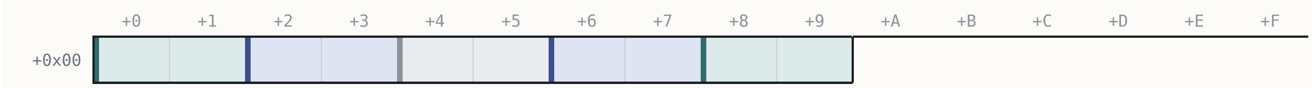
32 bytes (0x20) · mapped 8 · unmapped 24

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	uint16_t	`load_segment`	in-memory load paragraph (`0x0000` or `0x0040`)
+0x02..+0x07	6	–	–	unmapped(6 bytes; dword at +`0x04` zero in this binary)
+0x08..+0x0B	4	uint32_t	`header_offset`	file offset of the segment's own header
+0x0C..+0x0D	2	–	–	unmapped(2 bytes)
+0x0E..+0x0F	2	uint16_t	`segment_num`	incrementing id 2,3,4,... (not an index)
+0x10..+0x1F	16	–	–	unmapped(16 bytes, zero)

(Location quirk: the generic V2 layout puts this list 48 bytes after the zero-offset relocation entry; in VICEROY that relocation is at file `0x192B0` and the list starts +64, with 16 zero bytes between.)

Each `header_offset` points at a per-segment header, followed by that segment's internal relocations, followed by the code:

RTLinkSegmentHeader per-segment header, at header_offset



10 bytes (0xA) · mapped 8 · unmapped 2

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	uint16_t	`seg_paragraphs`	total paragraphs (header + code)
+0x02..+0x03	2	uint16_t	`hdr_paragraphs`	header paragraphs
+0x04..+0x05	2	–	–	unmapped(2 bytes, small value)
+0x06..+0x07	2	uint16_t	`reloc_start`	== 0 in V2
+0x08..+0x09	2	uint16_t	`num_relocs`	count of internal fixups

```
// code_offset = header_offset + hdr_paragraphs*16
// code_size   = (seg_paragraphs - hdr_paragraphs)*16
```

The **page-id model** resolves overlay calls: `page_id = segment_num - 1`, and a code target is `code_offset(page) + (ljmp_seg << 4) + offset_in_segment` (the `ljmp_seg` term is the segment word retained in the thunk — zero for 506 of the 658 type-A thunks; nonzero for 152, e.g. page `0x1A`'s second sub-segment `ljmp_seg 0x11E` giving base `0x72090` + `0x11E0` = `0x73270`). The complete page map, byte-verified against the EXE:

PAGE	HEADEROFF	CODEOFF	CODESIZE	RELOCS	LOADSEG
0x01	0x20670	0x20EE0	0x3D10	527	0x0000
0x02	0x24BF0	0x25900	0x7200	826	0x0000
0x03	0x2CB00	0x2CFD0	0x2B20	296	0x0040
0x04	0x2FAF0	0x30550	0x6440	652	0x0040
0x05	0x36990	0x37340	0x4040	608	0x0040
0x06	0x3B380	0x3B900	0x3160	340	0x0040
0x07	0x3EA60	0x3ECF0	0x1400	152	0x0040
0x08	0x400F0	0x404B0	0x2420	228	0x0040
0x09	0x428D0	0x42C50	0x17B0	214	0x0000
0x0A	0x44400	0x44540	0x16E0	70	0x0000
0x0B	0x45C20	0x45D00	0x0900	43	0x0000
0x0C	0x46600	0x46DE0	0x4C70	492	0x0000
0x0D	0x4BA50	0x4C1F0	0x7350	477	0x0000
0x0E	0x53540	0x53820	0x2A90	171	0x0000
0x0F	0x562B0	0x56A10	0x3F40	461	0x0040
0x10	0x5A950	0x5AF70	0x37D0	382	0x0000
0x11	0x5E740	0x5E9B0	0x1120	145	0x0000
0x12	0x5FAD0	0x5FE60	0x1E40	218	0x0040
0x13	0x61CA0	0x61E10	0x15D0	79	0x0000
0x14	0x633E0	0x63880	0x2E00	285	0x0040

PAGE	HEADEROFF	CODEOFF	CODESIZE	RELOCS	LOADSEG
0x15	0x66680	0x66850	0x2130	105	0x0040
0x16	0x68980	0x68EE0	0x2C20	332	0x0040
0x17	0x6BB00	0x6BE50	0x3A00	201	0x0000
0x18	0x6F850	0x6F8E0	0x0260	24	0x0000
0x19	0x6FB40	0x6FDF0	0x16A0	162	0x0040
0x1A	0x71490	0x72090	0x4340	757	0x0040
0x1B	0x763D0	0x764D0	0x08A0	51	0x0040
0x1C	0x76D70	0x76E50	0x0A30	43	0x0040
0x1D	0x77880	0x77990	0x0470	58	0x0000
0x1E	0x77E00	0x77ED0	0x06D0	42	0x0040
0x1F	0x785A0	0x78640	0x0700	30	0x0040

Total overlay code ≈ 0x52CA0 bytes across 31 segments. `load_segment` takes only two values (0x0000 / 0x0040): segments sharing a value alias the same physical memory window and are paged in on demand — which is exactly why the load image cannot far-call overlay code directly and must go through the thunk table.

1.4 The thunk table and the stub-segment windows

The resident far-call **thunk table** begins at file 0x1A5F0 (the first `0x9A` byte after the `Enter directory for $` marker) and holds **1,023 thunks: 658 type-A and 365 type-B**, at variable record sizes (10/12/14 bytes, a few longer at the tail).

Type A (overlay call — this is the record the runtime patches at load time):

```
9A AB 0D 0D 11      LCALL 0x110D:0x0DAB   ; overlay-loader entry ("with page-id")
EA <off16> <seg16>  JMPF  target          ; seg16 = 0 on disk — PATCHED at load
<page16>            trailer word = page_id (= segment_num − 1)
```

Type B (`LCALL 0x110D:0x0D91` then a fixed JMPF): the target is already resident — the JMP-FAR lands back in the load image. Most helper calls that look like overlay calls are type B and decode statically (e.g. `LCALL 0x181F:0x04D4` → file 0xC322 `random_int`, `0x181F:0x035C` → 0x48CC `clamp`, `0x181F:0x07B4` → 0xBC10 `power_attribute_bit`).

The two loader entries live at file 0x1427B (type-A entry, sets `cs:[0x39F1]=0`) and 0x14261 (type-B entry, sets `cs:[0x39F1]=0x52`); they share a body at file 0x14293 which saves registers, calls the segment-lookup helper at file 0x164A2 (which walks six storage-class sub-helpers at 0x164FE/0x164E8/0x16564/0x16516/0x16837/0x167F2), and then either returns (segment already resident), **rewrites the thunk's JMPF segment word in place** (at `[si−4]`, si = saved return IP − 5), or faults the segment in from disk first. Because the patch happens at load time, the live JMPF targets do not exist in the static file — they must be derived through the page model of §1.3.

Load-image code reaches the table through **three overlapping stub-segment windows**, whose file bases are `0x2400 + (seg << 4)`:

WINDOW SEGMENT	FILE BASE OF WINDOW
0x181F	0x1A5F0
0x191F	0x1B5F0
0x1A1F	0x1C5F0

So for any disassembled `LCALL seg:off` with seg ∈ {0x181F, 0x191F, 0x1A1F}: `thunk_file_offset = 0x2400 + (seg << 4) + off`. The windows are 0x1000 paragraphs apart, so every thunk is reachable from at least one window. Worked validation: `LCALL 0x1A1F:0x06E0` → thunk at file 0x1CCD0 = `9A AB0D 0D11 EA 5203 0000 1000` → page 0x10, offset 0x0352 → code base 0x5AF70 + 0x352 = **file 0x5B2C2** (`func_05B2C2`, which duly opens with `C8 3A 00 00` = `ENTER 0x3A`). Under the general formula 578 of the 658 type-A thunks land on clean ENTER/PUSH-BP prologues; the remainder are legitimate mid-function tail-call entries.

1.5 COLONIZE.EXE — the flat companion build

Alongside the shipped set, the project's byte record preserves **COLONIZE.EXE** (455,137 bytes, described as the launcher/front-end build; it is *not* one of the six executables in the shipped game directory, whose batch file `COLONIZE.BAT` launches VICEROY.EXE). Its structure is the mirror image of VICEROY's: image size equals file size — **no overlay at all** — with 36,864 header bytes, **9,199 relocations** and 1,244 discovered functions, all resident. Code that in VICEROY is overlay-resident (for instance the F2–F9 advisor-report painters) is directly addressable in COLONIZE.EXE, which makes it a useful decompilation cross-check; conversely, several early offsets circulated for VICEROY features (e.g. a "king-audience painter at 0x249B1") turned out to be COLONIZE.EXE offsets — and that one is merely a filename builder, not a screen painter. All offsets in this manual index VICEROY.EXE unless explicitly stated otherwise.

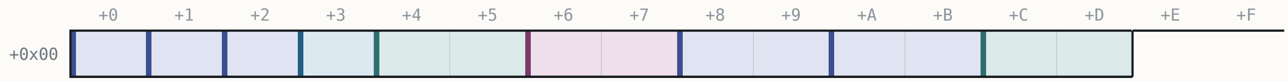
1.6 OPENING.EXE and CLOSING.EXE — separate cinematic programs

The opening credits/cinematic and the endgame cinematic are **separate programs**, not VICEROY code. Both are MZ executables built with the same MSC 6.0 runtime (four C-runtime helpers byte-matched in each) and the same MADS asset engine; both carry an appended RTLink debug-data region (21,907 / 20,477 bytes) that contains **no executable code**. Their playback is data-driven from shipped text files: OPENING.TXT (1,479 bytes; lines of `start_frame, end_frame, series, sprite`), CLOSING.TXT (762 bytes; lines of `Series, Frame, Repeats, BaseX, Delay`, driving the CLOS-FWK/BEL/HAT/LDY/MAN/MIL/BKG.SS sheets), PATH.DAT (6,459 bytes of plain-ASCII `x, y` per-frame ship trajectory for the opening voyage) and AMERICA.MOV (572 bytes of movie metadata).

1.7 MAPEDIT.EXE — standalone editor with shipped debug symbols

MAPEDIT.EXE (145,292 bytes) is a **standalone program** sharing the MADS engine and the game's assets (viceroy.pal, TERRAIN.SS, PHYS0.SS, ICONS.SS, WOODTILE.SS, FONTINTR, FONTTINY, CURSOR.SS). Uniquely among the six, it **ships with CodeView NB02 debug information**: its overlay region opens with a directory of 33 named records — 13 top-level `.obj` segment records (popup.obj, menu.obj, text.obj, stuff.obj, map_2/5/6/9/a.obj, write.obj, vicemisc.obj, terrain.obj, me_mini.obj) and 19 `.c`/`.asm` source records — in this record format:

NB02Record NB02 directory record (MAPEDIT overlay)



14 bytes (0xE) · mapped 14 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00	1	uint8_t	**sep0**	= 0x00
+0x01	1	uint8_t	**magic**	= 0x01
+0x02	1	uint8_t	**sep1**	= 0x00
+0x03	1	uint8_t	**name_len**	
+0x04..+0x05	2	uint16_t	**linker_seg_para**	linker segment paragraph (sequential 0x06D7..0x0BA3)
+0x06..+0x07	2	uint16_t	**flags**	0..0x0E
+0x08..+0x09	2	uint16_t	**size_bytes**	
+0x0A..+0x0B	2	uint16_t	**reserved**	always 0
+0x0C..+0x0D	2	uint16_t	**type_marker**	0 = top-level .obj, 1 = sub-source

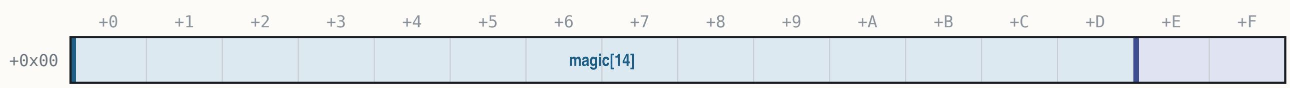
The mined symbol table yields **1,071 named symbols**, so the editor's functions are cited in this manual by their true linker names (`_write_map_file`, `_load_map_file`, `_forest_fix`, ...). A symbol at `seg:off` lives at file offset `seg·16 + off + 0x1600` (`0x1600` being unaccounted header slack below MAPEDIT's 5,632-byte MZ header region). The editor's own behaviour — startup, menus, and the .MP writer that is the ground truth for §3 — is fully symbol-resolved.

§2 Asset containers and file formats

Every graphical asset in the game — sprite sheets (.SS), full-screen backgrounds (.PIK) and bitmap fonts (.FF) — is wrapped in the same MicroProse **MADSPACK 2.0** container, with sections individually compressed by the **FAB** LZ codec. The decoders described here are ports of the in-EXE library routines (byte-verified `madspack_load` / `fab_decompress`), and their decisive correctness test is that every compressed section expands to exactly its directory-declared size across all shipped files. Sound is handled differently: the "driver" is itself a loadable DOS executable selected by a filename template.

2.1 The MADSPACK 2.0 container

MadspackHeader container header



16 bytes (0x10) · mapped 16 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x0D	14	char	**magic[14]**	"MADSPACK 2.0\x1A\x00" (loader checks first 8 bytes)
+0x0E..+0x0F	2	uint16_t	**section_count**	

MadspackDirEntry directory entry (10 bytes)

10 bytes (0xA) · mapped 10 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00	1	uint8_t	**flag**	0 = stored raw, 1 = FAB-compressed
+0x01	1	uint8_t	**mode**	observed 4 (NOT a codec selector; flag decides)
+0x02..+0x05	4	uint32_t	**unpacked_len**	
+0x06..+0x09	4	uint32_t	**packed_len**	

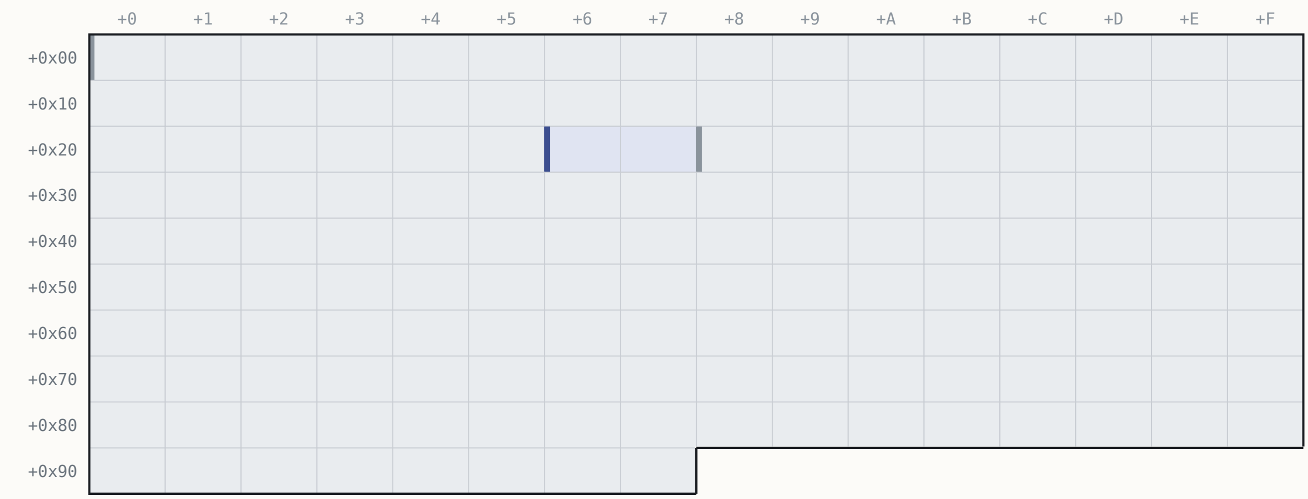
The fixed `0xB0` data start (a 160-byte reserved block follows the directory) is load-bearing: reading section data at `16 + 10·N` instead silently mis-frames every section.

FAB compression is an LZ77 bitstream with the magic `"FAB"` + a shift byte (10–13; observed 12 throughout), then a 16-bit LSB-first bit reservoir primed from bytes 4–5. Decoding: a `1` bit emits one literal byte from the byte stream; `0` then `0` is a short back-reference (2 more bits → copy length 2–5; 1-byte offset, biased −256); `0` then `1` is a long back-reference (2 bytes carrying an offset split by the shift value and a length field, where length 0 escapes to an extension byte: 0 = end of stream, 1 = bitstream realign, else length = byte+1). FAB encoding is non-deterministic at the bit level, so a re-encoded file decodes identically without byte-matching the original. The in-game loader chain is byte-verified: archive open + magic check at `0x76F26` inside `func_076E50`, buffered chunked section transfer via `func_0775EC` (chunks clamped to `0xF000`), with the per-section decode transform dispatched through a callback vtable.

2.2 .SS sprite sheets

A typical .SS has 4 sections: **(0)** a sheet header, **(1)** the frame-descriptor table, **(2)** a 768-byte palette (256 × 6-bit RGB triples), **(3)** RLE pixel data.

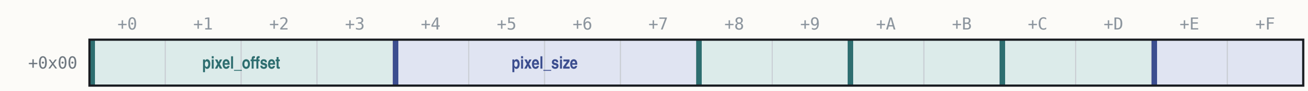
SSSheetHeader section 0 — sheet header (0x98 = 152 bytes)



152 bytes (0x98) · mapped 2 · unmapped 150

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x25	38	–	–	unmapped(38 bytes)
+0x26..+0x27	2	uint16_t	`frame_count`	number of 16-byte descriptors in section 1
+0x28..+0x97	112	–	–	unmapped(112 bytes)

SSFrameDesc section 1 — one descriptor per disk sprite (16 bytes)



16 bytes (0x10) · mapped 16 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x03	4	uint32_t	`pixel_offset`	offset of this frame's RLE stream in section 3
+0x04..+0x07	4	uint32_t	`pixel_size`	RLE byte count
+0x08..+0x09	2	int16_t	`anchor_x`	anchor = CENTRE-x (screen x = ax - floor(w/2))
+0x0A..+0x0B	2	int16_t	`anchor_y`	anchor = BOTTOM-y (screen y = ay - h + 1)
+0x0C..+0x0D	2	uint16_t	`width`	
+0x0E..+0x0F	2	uint16_t	`height`	

Anchor semantics (ruling of 2026-07-31, pixel-verified against the running game, 1994 binary under DOSBox): the descriptor's (x,y) pair is an (anchor-x = centre-x, anchor-y = bottom-y) placement anchor, evidenced twice independently — KING1.SS descriptor (94,198) for a 189×187 frame draws at (0,12), and ENGLND1.SS descriptor (118,121) draws at (32,0), both pixel-exact. Placement of such frames is asset-anchored, not code-literal.

RLE pixel encoding (per line, decoded row-major into a w×h cell pre-filled with transparent):

BYTE	MEANING
0xFC	end of sprite
0xFF	end of line (advance to next row, pixel column resets)
0xFD (as line marker)	run-mode line: repeated (count, pixel) pairs; a count byte of 0xFF ends the line
any other first byte	literal-line marker; the following bytes are literal pixels, except 0xFE which introduces a (count, pixel) run, and 0xFF which ends the line

BYTE	MEANING
`0xFD (as pixel value)`	**transparent** — the sprite colour key

Frame numbering — engine frame N = disk descriptor N−1. Engine code and all engine-derived documentation number sprite frames from 1; the descriptor table on disk is 0-based. Proof (ruling of 2026-07-31): TERRAIN.SS holds exactly **12** disk descriptors while the engine loads "frames 1..12"; WOODFRAM.SS holds **1** ("frame 1"); NAMEPLAT.SS holds **3** (frames 1–3); PHYS0.SS holds **154** descriptors (disk 0..153) so engine frame 154 is disk sprite 153. The draw verb `func_00E76A` indexes `base + frame·12 + 0x36` with no subtraction — the −1 lives in the load/record layout. A pixel render of PHYS0.SS confirms disk sprites 150–153 are the four straight-coast shorelines while disk 149 (= engine "frame 150") is a diagonal wave/hatch overlay.

2.3 .PIK full-screen images

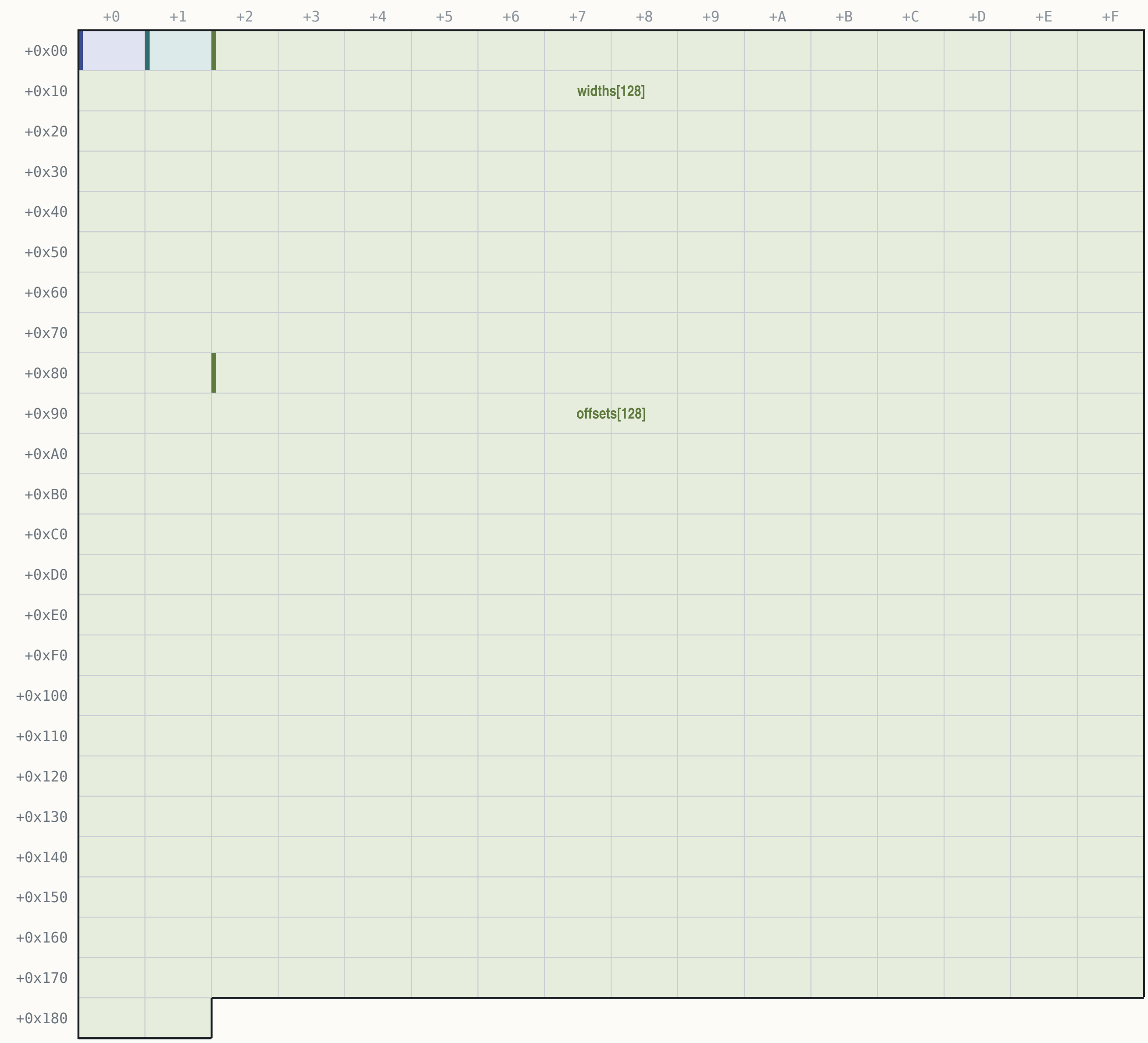
A .PIK is a single 320×200 indexed-colour background in a MADSPACK container, nominally 3 sections: image header, an optional 768-byte palette, and FAB-compressed pixels. **Palette precedence rule**: an asset with an embedded palette section renders with its own palette; only assets without one fall back to the master VICEROY.PAL. Both cases ship:

- **COLONY.PIK has no embedded palette** — it is a 2-section file, a 320×72 strip blitted at y=128 on the colony screen, whose pixel indices are authored directly against the gameplay palette (pixel-verified against the running game, 1994 binary under DOSBox).
- **EUROPE.PIK is self-contained** — 3 sections including its embedded palette, and its artwork *bakes in* screen furniture: the harbour town, the dock slots, the 16-good market grid and the red "E" button are all background art, not engine-drawn elements (pixel-verified; the Europe screen rebuild from this file was 100.00% pixel-exact outside dynamic-sprite masks).

2.4 .FF bitmap fonts

An .FF is a MADSPACK container whose single FAB section decompresses to this payload:

FFFont .FF decompressed payload



386 bytes (0x182) · mapped 386 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00	1	uint8_t	`cell_height`	glyph cell height H; engine line pitch = H+3 (@0x3AB7)
+0x01	1	uint8_t	`max_width`	
+0x02..+0x81	128	uint8_t	`widths[128]`	slot j = width of ASCII char j+1
+0x82..+0x181	256	uint16_t	`offsets[128]`	slot j = payload offset of char j+1's bitmap

The index mapping is `ch−1`: `width(ch) = widths[ch−1]`, `bitmap(ch) = payload[offsets[ch−1] .. offsets[ch]]`. This is proven both by rendering (under the naive `widths[ch]` mapping every glyph draws as its successor) and by the engine bytes: the string-blit core `func_00E51C` decrements the character before both lookups — `dec dl` at 0xE5DA, width read `mov al,[bx+si+2]` at 0xE5E9, offset read `mov si,[bx+si+0x82]` at 0xE606. Glyph bitmap size = `H · ceil(width·2/8)` (validated for 95/95 printable slots in all five shipped fonts). The four 2-bpp levels are 0 = transparent and 1/2/3 = ink shades for anti-aliasing. **Advance = width** (each glyph carries its own trailing spacing column). Decompressed payload sizes: FONTTINY 914, FONT-NP 914, FONTKING 1,219, FONTINTR 1,898, FONTSMAL 1,148 — each exactly 386 + Σ glyph sizes.

2.5 Sound: the $sound$ driver overlay and #SOUND.COL

The audio "drivers" are DOS executables. All four device files — ASOUND.COL (AdLib), GSOUND.COL (SoundBlaster/GameBlaster), PSOUND.COL (PC speaker), RSOUND.COL (Roland MT-32) — begin with the MZ magic; CONFIG.COL (20 bytes) is a small device-configuration blob (base port / IRQ words). At boot, `func_07845A` (called from 0x762E6) takes the template string **`"#SOUND.COL"`** (file 0x1FD5A), replaces the `#` with the configured device letter from `[0x2608]`, and `func_01287A` loads the resulting file via DOS int 21h AX=4B03 (load overlay) under the tag **`"$sound$ "`** (file 0x2004B). The driver's header supplies five entry vectors installed to DGROUP `0xA654`–`0xA667` by `func_012928`; command dispatch is at `0x1299A` (with an 8-deep queue when locked), and the timer ISR at `0xC6D9` clocks vector 4 every tick and vector 3 every fifth tick. Sampled audio lives in COLDIG.BIN (993,755 bytes of 8-bit unsigned PCM), indexed by the driver.

2.6 Data-file inventory

The shipped game directory holds 300 files. By type (sizes are byte totals):

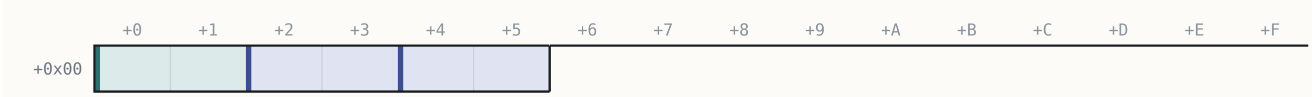
TYPE	COUNT	BYTES	ROLE
.SS	206	1,418,359	sprite sheets (§2.2): terrain/overlays (TERRAIN, PHYS0), icons, units, portraits (CC-00..24), woodcuts (WDCUT01..13), cinematic series
.PIK	35	928,431	320×200 backgrounds (§2.3): COLONY, EUROPE, REPORT1–9, KINGLSS1/2, OPENMENU, DIFFICUL, NATIONS, …
.TXT	18	186,518	game data/text: NAMES (data dictionary), GAME (510 dialog `@KEY` definitions), LABELS, MENU, MAPMENU, PEDIA, COLONY, TRIBE, WOODCUT, OPENING, CLOSING, DEBUG, MAPEDIT, README, MEMORY/2, AUTOEXEC, CONFIG
.SAV	10	288,510	save slots COLONY00–09.SAV
.EXE	6	902,531	§1.1
.COL	5	190,180	4 MZ sound drivers + CONFIG.COL (§2.5)
.FF	5	4,014	fonts FONTTINY / FONTSMAL / FONTKING / FONTINTR / FONT-NP (§4)
.DAT	3	20,646	CYCLE.DAT (34 B, palette-cycling data, format unmapped), PATH.DAT (ASCII ship path), INSTALL.DAT
.DB	2	1,619	MODULES.DB (34 engine module names), ERRORS.DB (engine constraint names, e.g. PopupTooManyLines)
.BAT	2	102	COLONIZE.BAT, COLDEMO.BAT launchers
.MP	1	12,534	AMER2.MP — the standard Americas map (§3)
.BIN	1	993,755	COLDIG.BIN PCM sample bank
.PAL	1	1,024	VICEROY.PAL master palette (§4.4)
.MOV	1	572	AMERICA.MOV cinematic metadata
.GIF / .COM / .PIF / .ION	4	129,829	installer splash, PKUNZJR.COM, install shortcuts/descriptions

§3 The .MP map format and the game's loader

The map file format is known from the best possible source: the shipped editor's own writer and loader, read under their real linker names from MAPEDIT.EXE's CodeView symbols (`_write_map_file` at MAPEDIT file `0xB840`, `_load_map_file` at `0xB700`), and cross-checked byte-for-byte against the shipped world AMER2.MP. VICEROY reads the same file — but, as live testing showed, it rewrites parts of what it reads. Terrain-id semantics follow the NAMES.TXT `$TERRAIN` sections (`@UNFORESTED`, `@FORESTED`, `@OTHER`), which the editor's own data loader corroborates.

3.1 File layout

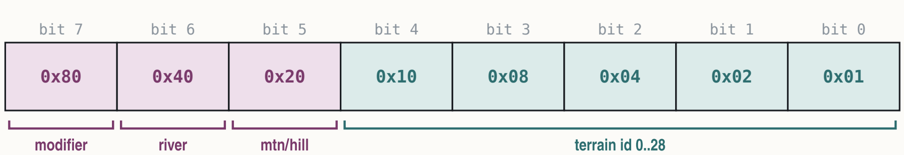
MPFile file size = 6 + 3*width*height (AMER2: 12,534 = 6 + 3*4176)



6 bytes (0x6) · mapped 6 · unmapped 0

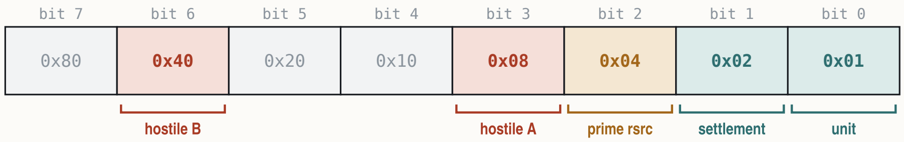
OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	uint16_t	width	AMER2 = 58 (0x3A) — full grid incl. border ring
+0x02..+0x03	2	uint16_t	height	AMER2 = 72 (0x48)
+0x04..+0x05	2	uint16_t	version	must be 4 (loader cmp @MAPEDIT 0xB76D)

Layer 1 — terrain byte per tile; ids and overlay bits (section 3)



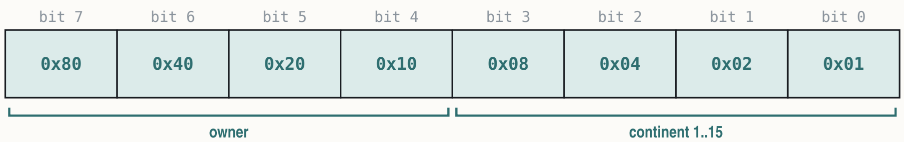
terrain id 0..28 — AND 0x1F fold at 0x620A; 8..23 = forested variants
mtn/hill — with bit 7: set = Mountains (0xA0), clear = Hills (0x20)
river — with bit 7: set = major river (0xC0), clear = minor (0x40)
modifier — qualifies bits 5/6 as above

Layer 2 — feature byte per tile; VICEROY discards this layer on load and rebuilds it



unit — unit present (MAPEDIT _is_unit @0x43C8)
settlement — colony/village/city (@0x43F6..)
prime rsrc — prime resource (_resource_at @0x45CF)
hostile A — tested with bit 6 by _is_hostile @0x44D1 (mask 0x48)

Layer 3 — continent byte per tile



continent 1..15 — 0 = border/none (_continent_at @0x428B)
owner — 0xF = none (_owner_of @0x42C5)

Nothing follows layer 3 — there are no colony/unit/settlement record arrays in .MP files (those belong to save-games).

Layer 1 — terrain byte:

BITS	MASK	MEANING
0–4	`0x1F`	terrain id 0..28
5	`0x20`	mountains/hills overlay
6	`0x40`	river overlay
7	`0x80`	modifier for bits 5/6 — with bit 5: set = Mountains (id 27), clear = Hills (28); with bit 6: set = Major River, clear = Minor River

(Paint masks at MAPEDIT `0x2744`–`0x2788`; reader `_terrain_type` at `0xB17C`.) **Forest is *not* a bit** — forested terrain is expressed in the id itself: ids 8..23 are the auto-forest range (the game's terrain-id fold, byte-verified at VICEROY file `0x6204`: read byte, mask `& 0x1F`, apply the auto-forest conversion; ids 16..23 are aliases of 8..15). The editor's `_terrain_is_forest` at `0xB222` tests exactly 8..`0xF` and `0x10`..`0x17`.

Layer 2 — feature flags (game-side; the editor passes them through): bit 0 unit present, bit 1 settlement, bit 2 prime resource, bits 3/6 tested by the editor's `_is_hostile` at `0x44D1` (semantics game-side). AMER2.MP's layer 2 is all zeros.

Layer 3 — continent/owner: low nibble = continent/region id 1..15 (0 = border/none; labels compressed by `_map_find_continents` at `0xB242`, overflow → `0xF`); high nibble = owner (`0xF` = none).

AMER2.MP layer-1 bit-combo census (byte-verified): `0x20` hills ×56 · `0x40` minor river ×178 · `0x60` hills+minor river ×1 · `0xA0` mountains ×170 · `0xC0` major river ×47; plain-id tiles ×3,724. Terrain ids used: 0..23, 25, 26.

3.2 Terrain ids 0..28

Per NAMES.TXT `$TERRAIN`, in `@UNFORESTED` / `@FORESTED` / `@OTHER` order:

ID	NAME	ID	NAME
0	Tundra	8–15	forested variants of 0–7 (Boreal Forest, …)
1	Desert	16–23	aliases of 8–15 (folded on load)
2	Plains	24	Arctic
3	Prairie	25	Ocean (blank-map fill value)
4	Grassland	26	Sea Lane
5	Savanna	27	Mountains (overlay form `0xA0`)
6	Marsh	28	Hills (overlay form `0x20`)
7	Swamp		

Mountains/Hills have a dual representation: as ids 27/28 (how `@OTHER` names them and how `_terrain_type` reports them: bit 5 set → `0x1B` + (bit7 ? 0 : 1)) and as overlay bits on a base land terrain (how tiles store them; editor paint masks: Mountains `or 0xA0 / and 0x1F`, Hills `or 0x20 / and 0x5F`, Major River `or 0xC0 / and 0x1F`, Minor River `or 0x40 / and 0x3F`, all at MAPEDIT `0x2744`–`0x2788`).

3.3 Border ring and the sea-lane column

The outermost 1-tile ring is not editable (`_change_map` bounds x,y ∈ [1..w−2]/[1..h−2] at MAPEDIT 0x31E9–0x320D) and is skipped by the continent finder; in AMER2.MP the ring is Ocean. The **sea-lane column is the right-most *playable* column x = w−2**: all 70 interior rows of AMER2's column 56 carry base id 26 (Sea Lane) — never fake this column as any other terrain. New maps are created all-Ocean (fill 0x19), hard-coded 58×72, version 4 (`_create_blank_map` at 0xB94A); the sea lane is hand-painted, not synthesised.

3.4 MAPEDIT's writer, loader and error codes

The writer emits header words, then the three layers in order (writes at MAPEDIT 0xB878–0xB8AC then 0xB8C8/0xB8E8/0xB910). The loader mirrors it, requires version == 4 (0xB76D–0xB773) and caps `w·h ≤ 0x2EE0` (12,000 tiles) at 0xB6CE. Error codes (`_map_error`):

CODE	MEANING
1	file open failed
2	header read failed
3	wrong version (≠ 4)
4 / 5 / 6	layer 1 / 2 / 3 I/O failure
9999	map too big (w·h > 0x2EE0)

Round-trips are not byte-preserving. On every load MAPEDIT runs `_forest_fix` (MAPEDIT 0x16B6), which normalises ids 0x10..0x17 → id−8 and strips forest under a mountains/hills overlay. A load→save cycle of any file containing ids 16..23 (AMER2.MP does) therefore changes bytes while preserving meaning.

3.5 VICEROY's loader behaviour

VICEROY loads AMER2.MP at the start of the standard game and accepts the same 6-byte header + three layers, but live testing with a crafted map (loaded in the running 1994 binary under DOSBox, with same-moment RAM reads of the runtime planes) established three loader-side rewrites:

1. **Layer 2 (features) is discarded.** All crafted feature bytes read back 0 live; the game rebuilds the runtime feature plane itself (bit 0 unit, bit 1 settlement). Feature-gated pixels (shore overlay, roads, resource suppression) are unreachable via .MP data.
2. **Border normalisation is enforced by the loader**: rows 0 and h−1 → Arctic; columns 0, 1 **and** w−1 → Sea Lane for all interior rows, **overwriting even land**. The sea-lane column is loader-enforced, not merely a data convention.
3. **Forest alias ids 16..23 are folded to 8..15 at load**, matching the editor's `_forest_fix`.

The runtime plane pointers live at DGROUP `[0x15C]` (terrain), `[0x160]` (feature), `[0x164]` (continent+owner) and `[0x168]` (flags; low nibble = colony-site value). In the same live campaign the full tile-render rule set derived from these bytes reproduced 100.0000% of non-overlay pixels across every captured viewport.

§4 Fonts and palette

Text in the game flows through one resident glyph engine fed by proportional 2-bpp bitmap fonts (§2.4), with colour supplied not per-glyph but through a tiny indirection — a 4-entry palette-index look-up table that maps the glyph's ink levels at draw time. All colour ultimately indexes the VGA palette: a master file palette plus per-screen palettes embedded in .PIK backgrounds, with two index bands animated by pure palette rotation.

4.1 The four loaded fonts (and one orphan)

VICEROY loads exactly **four** fonts through a single load verb, `LCALL 0x1A1F:0xA86` (target file 0x76C70), which has exactly four call sites — proving the fifth disk file, FONTSMAL.FF, is an **orphan, never loaded** (no `fontsmal` string exists in the EXE):

FONT	NAME STRING	LOAD SITE	HANDLE STORED TO	ROLE
FONTTINY	`"fonttiny"` @0x1FD32	0x760E8 (in boot loader `func_075FB6`)	latch `[0x89E]/[0x8A0]`	**boot-default body font**: HUD, inventory, popup bodies, advisor bodies
FONTINTR	`"fontintr"` @0x1FD2B	0x760C6 (same loader, loaded first)	latch `[0x268A]/[0x268C]`	titles, menus, Hall of Fame, score figures
FONTKING	`"FONTKING"` @0x1FCCB	0x754F6 (in `func_075352`)	shared active-font global `[0x1F9E]/[0x1FA0]` (no private latch; FONTTINY fallback @0x7550A)	**king-defeats screen only** — the sole load
FONT-NP	`"FONT-NP"` @0x1F8AF	0x6B7AF (in `func_06B722`)	stack local `[bp−0x3AC]` (no global)	speaker name-plate / woodcut path

Font selection is a **screen-level latch**, not a per-draw argument: paint code pushes the latched handle (`push [0x8A0]; push [0x89E]` for FONTTINY at e.g. 0x25F62 / 0x30EDE / 0x3860C; `push [0x268C]; push [0x268A]` for FONTINTR at 0x22ABE / 0x23C06 / 0x3B054). The dialog framework's `SMALLFONT` directive merely copies the latched font — it does not load a smaller one. The measure core at file 0xE6A6 accumulates `width(ch) + spacing` per character with the same `ch−1` table indexing as the blitter.

4.2 Per-font metrics

Decoded from the shipped .FF files with the corrected `ch−1` mapping (95/95 glyph-size validation per font). Line pitch = cell height + 3.

FONT	CELL H	PITCH	MAX W	SPACE	NOTABLE WIDTHS
FONTTINY	6	9	6	2	most glyphs 4; `i l` = 2, `t` = 3; `M W m w` = 6
FONTKING	7	10	8	2	`i l` = 2; digits 3–7; `M W` = 8
FONT-NP	8	11	11	5	uppercase A–Z only; `I` = 5, `S` = 5; `M W` = 11
FONTINTR	9	12	9	3	digits all 6; `I i l` = 3; `M W` = 7; `% / @` = 9
FONTSMAL (orphan)	6	9	8	3	digits 6; `M W` = 8 — present on disk, never loaded

4.3 The glyph ink LUT

The rasteriser core `func_00E51C` unpacks each glyph's 2-bpp levels (`shl ax,2` at 0xE629) and maps every ink level 0..3 through a **4-entry palette-index LUT held at far pointer `[0x269E]:[0x26A0]`** (captured into the frame at 0xE532, applied `mov ah,[bp+si−6]` at 0xE632). A LUT entry of **0xFF means transparent** — the pixel is skipped (`cmp ah,0xFF` at 0xE637). A string's colour is therefore whatever the 4-byte LUT contains at draw time; the setter family is `func_00E68A` (writes `[0x269E]=cl, [0x269F]=dl, [0x26A0]=al, [0x26A1]=al`). Anti-aliased text renders with up to three ink shades per call, and "text colour" bugs or state-dependence trace to LUT stores, not to the fonts.

4.4 VICEROY.PAL and the working palette

VICEROY.PAL is 1,024 bytes: **768 bytes of 256 × 3-byte RGB triples in VGA 6-bit precision, plus 256 trailing unused bytes** (stride 3, not 4 — a settled ruling). Conversion to 8-bit per channel is `v8 = (v6 << 2) | (v6 >> 4)`. Screens that load a .PIK with an embedded palette run on that palette instead (§2.3); indices below are resolved against the master file, decoded directly from the shipped VICEROY.PAL:

INDEX	VICEROY.PAL RGB (8-BIT)	PINNED ROLE (BYTE-CITED SITE)
12 (0x0C)	(255,0,0) red	primary red entry
14 (0x0E)	(255,255,85) yellow	selected-good cell outline (Europe screen, runtime-selected vs green 10)
15 (0x0F)	(255,255,255) white	body/number text ink across screens
0x2F	(12,12,12) near-black	Europe market **price ink** at y=194 (byte-cited, seen on screen)
0x37	(93,121,186)*	dialog **bevel-dark + selection bar** (`[0x1F40]/[0x1F42]/[0x1F48]` =0x37 @0x734E0/0x734E6)
68 (0x44)	(85,150,52) green	ink init `[0x830]` =0x44 (file 0x1E1D0+); Europe title idle green
69 (0x45)	(52,130,32) green	Europe caption ink (measured; not byte-cited)
149 (0x95)	(199,162,32) gold	ink init `[0x831]` =0x95; Europe arrival-banner gold
0xFC	(255,85,255)*	boot-menu **gold hilite** (`[0x1F4E]` =0xFC @0x734C2; live-confirmed at the boot menu); Hall-of-Fame gold resolves to (199,162,32) via its screen palette
0xFD	(255,85,255)*	dialog **bevel-light** ink (`[0x1F46]/[0x1F50]` =0xFD @0x734DA)
0xFE	(255,85,255)*	boot dialog **text** ink (`[0x1F4A]` =0xFE @0x734BC)

VICEROY.PAL — the 256-colour master palette 6-bit VGA, v8 = (v6<<2)|(v6>>4) · row = high nibble

_0	_1	_2	_3	_4	_5	_6	_7	_8	_9	_A	_B	_C	_D	_E	_F
00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F
20	21	22	23	24	25	26	27	28	29	2A	2B	2C	2D	2E	2F
30	31	32	33	34	35	36	37	38	39	3A	3B	3C	3D	3E	3F
40	41	42	43	44	45	46	47	48	49	4A	4B	4C	4D	4E	4F
50	51	52	53	54	55	56	57	58	59	5A	5B	5C	5D	5E	5F
60	61	62	63	64	65	66	67	68	69	6A	6B	6C	6D	6E	6F
70	71	72	73	74	75	76	77	78	79	7A	7B	7C	7D	7E	7F
80	81	82	83	84	85	86	87	88	89	8A	8B	8C	8D	8E	8F
90	91	92	93	94	95	96	97	98	99	9A	9B	9C	9D	9E	9F
A0	A1	A2	A3	A4	A5	A6	A7	A8	A9	AA	AB	AC	AD	AE	AF
B0	B1	B2	B3	B4	B5	B6	B7	B8	B9	BA	BB	BC	BD	BE	BF
C0	C1	C2	C3	C4	C5	C6	C7	C8	C9	CA	CB	CC	CD	CE	CF
D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	DA	DB	DC	DD	DE	DF
E0	E1	E2	E3	E4	E5	E6	E7	E8	E9	EA	EB	EC	ED	EE	EF
F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	FA	FB	FC	FD	FE	FF

Decoded from the shipped VICEROY.PAL. The water ramp, indices 54–60 (0x36–0x3C), palette-cycles at run time; 0xFD is the .SS sprite-transparent index; 0xFC–0xFE are magenta placeholders overridden by per-screen .PIK palettes (§4.4).

* Entries marked with an asterisk are placeholder values in the master file (0xFC–0xFE are magenta; 0x37 sits inside the water ramp): the screens that use these indices load their own palettes, so the on-screen colour comes from the active screen palette, not from VICEROY.PAL. The boot-mode ink block is written as a unit by the setter at 0x734BC–0x734E9 (`[0x1F44]` =0x2E frame ring, plus the entries above).

Palette animation — two rotation bands. Water is animated *entirely* by palette cycling; no pixel indices ever change:

1. **Water ramp, indices 54–60** — a descending blue ramp, (105,138,195) → (32,44,138) in the master palette. Live render-and-diff of the Europe screen is pixel-exact once the capture's cycle phase is applied to these seven entries; the pixel data is static.
2. **Sparkle band, indices 120–127** — the sea-lane/water sparkle, a second blue band rotated by the VGA palette mechanism (live capture matched at rotation phase +2; the band is shared by the map and colony screens).

The 34-byte CYCLE.DAT file is associated with the cycling mechanism; its exact format is unmapped.

PART II

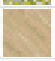
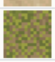
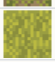
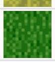
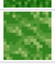
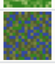
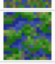
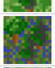
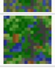
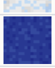
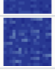
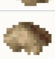
World and terrain

§5 Terrain system

Every square of the world map carries one of 29 terrain identifiers. The authoritative source for terrain identity, ordering and per-terrain statistics is the `$TERRAIN` block of the game's `NAMES.TXT` data file, which the engine parses at start-up into a 16-byte-stride record table in the data segment; the renderer, the yield calculator and the combat engine all index that one table. This section gives the complete id space, the data-file legend, the full per-terrain statistics, and the runtime record layout.

5.1 Terrain identifiers 0..28

The id is the low five bits of the stored terrain byte (`AND al,0x1F` at `0x620A`). Ids 8..23 are the forested variants of bases 0..7; ids 16..23 are a second encoding of the same eight forest types (see §5.4).

ID	TILE	HEX	NAME	NOTES
0		0x00	Tundra	base terrain
1		0x01	Desert	base terrain
2		0x02	Plains	base terrain
3		0x03	Prairie	base terrain
4		0x04	Grassland	base terrain
5		0x05	Savannah	base terrain
6		0x06	Marsh	base terrain
7		0x07	Swamp	base terrain
8		0x08	Boreal	forested Tundra
9		0x09	Scrub	forested Desert
10		0x0A	Mixed	forested Plains
11		0x0B	Broadleaf	forested Prairie
12		0x0C	Conifer	forested Grassland
13		0x0D	Tropical	forested Savannah
14		0x0E	Wetland	forested Marsh
15		0x0F	Rain	forested Swamp
16–23	—	0x10–0x17	(aliases)	second encoding of 8..15; fold `(id&7)|8` at 0x6225
24		0x18	Arctic	
25		0x19	Ocean	water
26		0x1A	Sea Lane	water; the right-edge map column is always Sea Lane (id 26, never 25)
27		0x1B	Mountains	encoded as bit flags, not a stored id (§5.4)
28		0x1C	Hills	encoded as bit flags, not a stored id (§5.4)

The map loader enforces the border convention at load time (runtime-verified against live memory): rows 0 and 71 are overwritten with Arctic (0x18), and columns 0, 1 and 57 with Sea Lane (0x1A) for y = 1..70, overwriting even land; forest alias ids 16..23 are folded to 8..15 in the live plane.

5.2 The $TERRAIN column legend

The legend is stated in the header comment of the data file itself, directly above the `@UNFORESTED` block, and the 14-column CSV rows match it exactly:

```
a) Name
b) Movement, Defensive, Improvement, Value
```

```
c) Yield (Farmer, Planter(s), Planter(t), Planter(c),
         Trapper, Lumberjack, Ore Miner, Silver Miner, Fisherman)
```

The nine yield columns map to goods as follows (in-memory column order matches the CSV order — the yield read is by good index, §5.5):

YIELD COLUMN	WORKER	GOOD PRODUCED
1	Farmer	Food
2	Planter(s)	Sugar
3	Planter(t)	Tobacco
4	Planter(c)	Cotton
5	Trapper	Furs
6	Lumberjack	Lumber
7	Ore Miner	Ore
8	Silver Miner	Silver
9	Fisherman	Fish (Food)

5.3 Per-terrain data — the full 21 distinct rows

Transcribed verbatim from the extracted `@UNFORESTED` / `@FORESTED` / `@OTHER` blocks. Columns: Mv = Movement, Df = Defensive, Im = Improvement, Vl = Value; then the nine yields in legend order (Fa Su To Co Fu Lu Or Si Fi).

Unforested bases (ids 0–7):

ID	TILE	NAME	MV	DF	IM	VL	FA	SU	TO	CO	FU	LU	OR	SI	FI
0		Tundra	1	0	4	2	2	0	0	0	0	0	2	0	0
1		Desert	1	0	3	2	1	0	0	1	0	0	2	0	0
2		Plains	1	0	3	4	4	0	0	2	0	0	1	0	0
3		Prairie	1	0	3	4	2	0	0	3	0	0	0	0	0
4		Grassland	1	0	3	4	2	0	3	0	0	0	0	0	0
5		Savannah	1	0	3	4	3	3	0	0	0	0	0	0	0
6		Marsh	2	1	5	2	2	0	2	0	0	0	2	0	0
7		Swamp	2	1	7	2	2	2	0	0	0	0	2	0	0

Forested variants (ids 8–15; ids 16–23 alias these):

ID	TILE	NAME	MV	DF	IM	VL	FA	SU	TO	CO	FU	LU	OR	SI	FI
8		Boreal	2	2	4	3	1	0	0	0	3	2	1	0	0
9		Scrub	1	2	4	1	1	0	0	1	2	1	1	0	0
10		Mixed	2	2	4	3	2	0	0	1	3	3	0	0	0
11		Broadleaf	2	2	4	3	1	0	0	1	2	2	0	0	0
12		Conifer	2	2	4	3	1	0	1	0	2	3	0	0	0
13		Tropical	2	2	6	3	2	1	0	0	2	2	0	0	0
14		Wetland	3	2	6	1	1	0	1	0	2	2	1	0	0
15		Rain	3	3	7	1	1	1	0	0	1	2	1	0	0

Other terrains (ids 24–28):

ID	TILE	NAME	MV	DF	IM	VL	FA	SU	TO	CO	FU	LU	OR	SI	FI
24		Arctic	2	0	4	0	0	0	0	0	0	0	0	0	0
25		Ocean	1	0	2	3	0	0	0	0	0	0	0	0	3
26		Sea Lane	1	0	2	0	0	0	0	0	0	0	0	0	3

ID	TILE	NAME	MV	DF	IM	VL	FA	SU	TO	CO	FU	LU	OR	SI	FI
27		Mountains	3	6	7	2	0	0	0	0	0	0	4	1	0
28		Hills	2	4	4	2	1	0	0	0	0	0	4	0	0

5.4 The auto-forest rule and the tile-byte encoding

The stored terrain byte is decoded by `get_terrain_id_from_raw` (`func_006204`, file 0x6204): read the byte, mask `AND 0x1F` (0x620A), then apply the auto-forest conversion. Ids 8..23 form the forested band; for any id in that band the decoder (mode global `[0x18E]` = 2) normalises to the eight canonical forest ids 8..15 via `(id & 7) | 8` (0x6225) — so 16→8 … 23→15: ids 16..23 are a second encoding of the same eight forest variants, not distinct terrains. Mode `[0x18E]` = 3 instead strips to the unforested base `id & 7` (0x6238); the default mode returns the raw masked id.

The remaining bits of the terrain byte (per the map-file writer and `func_0624E`): bit 5 (0x20) = relief present; bit 7 (0x80) then selects Mountains (set, id 27) versus Hills (clear, id 28) — `AND 0x80; SBB; +0x1B` in `func_0624E`. Bit 6 (0x40) = river, with bit 7 doubling as the Major (set) / Minor (clear) river selector on river tiles. Bit 7 in isolation (bit 5 clear, bit 6 clear) never occurs in shipped maps and is inert.

5.5 The runtime terrain record table (DS:0x2F74, stride 16)

At start-up a loader (`func_0745F0`, byte-stream reads via the section reader) fills one 16-byte record per terrain at `DS:0x2F74 + terrain·16`: first a name-token word, then the four `b)` columns as bytes, then the nine yield bytes.

TerrainRecord DS:0x2F74 + terrain*16, one record per terrain id

	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
+0x00												yields[9]				

16 bytes (0x10) · mapped 15 · unmapped 1

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	uint16_t	name	name token (loaded 0x74607; title read [id*16+0x2F74] at 0x69DD1)
+0x02	1	uint8_t	movement	Movement (DS:0x2F76, loaded 0x7460D)
+0x03	1	uint8_t	defensive	Defensive (DS:0x2F77, combat read at 0x7E63)
+0x04	1	uint8_t	improvement	Improvement (DS:0x2F78)
+0x05	1	uint8_t	value	Value (DS:0x2F79)
+0x07..+0x0F	9	uint8_t	yields[9]	9 yields (DS:0x2F7B+good, loop at 0x74633)
+0x06	1	–	–	unmapped(1 byte)

The yield read is byte-verified in the colony production calculator at 0x9C15 (`func_009B9C`): `SHL si,4` (terrain·16), `bx` = good index, `MOV al,[bx+si+0x2F7B]` — i.e. `yield = [terrain·16 + 0x2F7B + good]`. The Colonizopedia terrain pages title from the same table (`[id·16 + 0x2F74]` at 0x69DD1, with a "(forest)" qualifier appended for ids 8..15 at 0x69DF3), and the pedia's terrain category walks ids 0..0x1C skipping 0x10..0x18 (0x6B26E).

5.6 Terrain defence in combat: Defensive value × 25%

The defence-bonus filler `func_007D3E` reads the Defensive column at `[terrain·16 + 0x2F77]` (0x7E63) for the defender's tile and accumulates it into the defence-modifier chain. The Combat Analysis dialog (`func_05E9B0`) prints the terrain row as **+ (Defensive × 25 %)** — the row is flagged "Terrain" for the defender ("Ambush" for the attacker), draws the target tile, and is skipped when the value is 0. So the byte-verified per-terrain bonuses are: open land (Tundra/Desert/Plains/Prairie/ Grassland/Savannah) +0 %; Marsh/Swamp +25 %; forests +50 % (Rain +75 %); Hills +100 %; Mountains +150 %; Arctic/Ocean/Sea Lane +0 %.

5.7 Special resources (@RESOURCE) and overlay names (@OTHER_NAMES)

The `@RESOURCE` block ("Special resource squares & values") lists each prime-resource square with a single value byte — the production-bonus magnitude for the good implied by its name. Transcribed verbatim (the data file carries the Prime Timber row twice):

RESOURCE	VALUE	RESOURCE	VALUE
Depleted Mine	6	Beaver	6
Oasis	3	Game	6
Wheat	4	Prime Timber	6
Prime Cotton	6	Prime Timber	6
Prime Tobacco	6	Silver Deposit	12
Prime Sugar	7	Ore Deposit	6
Minerals	4	Fishery	5

Prime-resource *presence* on a tile is procedural, not stored — it is the map compositor's detail-band position hash (§6.9).

`@OTHER_NAMES` supplies the five overlay/UI names, in order: **Forest, River, Major River, Minor River, Unexplored**.

§6 The map compositor

Every visible map tile is composed at render time by a three-function chain: the visible-rectangle loop `func_0514` (`func_0685DC`, file 0x685DC..0x68897) walks the viewport; the per-tile selector `func_0513` (`func_0681A8`, 0x681A8..0x685DB) chooses every sprite frame for one tile; and the sub-cell composer `func_0512` (`func_067F50`, 0x67F50..0x681A7) dithers each tile's edges into its neighbours. The whole chain below is pixel-verified at 100.0000 % of non-overlay pixels against the running game (1994 binary under DOSBox), in three independent live tests: an all-water window (45,056 px), a coastal land window (41,540 px), and five viewport captures of a crafted test map exercising hills, rivers, river mouths, lakes and forest aliases.

Frame-numbering convention (stated once, applies throughout): all `0xNN` frame constants in this chain are *engine* frame numbers, which are 1-based over the sprite sheet's on-disk descriptors — disk sprite = engine frame − 1. (Proven by descriptor counts: TERRAIN 12, WOODFRAM 1, NAMEPLAT 3, PHYS0 154, plus pixel renders.) PHYS0.SS holds 154 disk frames (0..0x99); transparent pixels use palette index 0xFD.

6.1 O514 — the visible loop and per-tile addressing

`func_0685DC` walks the visible tile rectangle from the scroll origin `[0x8328]` (x) / `[0x832E]` (y) over the viewport span, clamped to the map extents `[0x8804]`/`[0x8806]`. Per tile it computes the linear index `(y+1)·stride[0x8548] + (x+1)` (0x6868E) — the +1s are the 1-tile border padding that keeps neighbour reads in-bounds — forms far-pointers into the four map planes (`[0x15C]` terrain, `[0x160]` feature, `[0x164]` continent/owner, `[0x168]` flags; §7.1) at that index, sets the tile's screen anchor (centre-x `[0xA5A4] = col·16 + 8` at 0x6875F, baseline-y `[0xA5A6] = (row+1)·16 − 1` at 0x68720), and calls O513. The fog mask `[0xA89E] = 1 << (player+4)` is latched at 0x685F2 (§7.2).

O513 first latches the tile bytes: `[0xA89F]` = feature byte (from `[0xA594]`), `[0xA8A1]` = terrain byte (from `[0xA598]`), `[0xA8A2]` = classified terrain id, and the tile fog byte `[0xA8A0]` (from `[0xA59C]`), at 0x681E0; world coordinates are `[0xA5A0]`/`[0xA5A2]` (engine scroll-space = plane index − 1). All drawing lands in the 16×16 per-tile composite buffer at near-pointer 0x839E, which is then blitted to screen.

6.2 Ground tiles — the 12-frame TERRAIN.SS sheet

TERRAIN.SS is the base-ground sheet (loaded at boot and on map-enter), composited *under* every PHYS0.SS overlay. `emit_ground_sprite` (`func_067E28`) blits from the sheet pointer `[0x16C:0x16E]` (plain 16-px blit thunk when zoom = 0, 0x67E3A). The sheet has exactly 12 frames; the ground-frame normaliser (`func_003436`) maps the classified id to a disk frame:

```
class 0..7            -> frames 0..7   (the eight unforested bases)
class 9 or 0x11       -> frame 8       (Scrub cactus ground)
class 0x18 Arctic     -> frame 9       (class >= 8: frame = class - 0xF)
class 0x19 Ocean      -> frame 10
class 0x1A Sea Lane   -> frame 11
```

O513's ground-id selection (0x682C0..0x68301): with `c = [0xA8A2]`, fold `c & 7` for `c < 0x18`; a forested tile grounds with its unforested base, *except* Scrub (`folded == 1`), which grounds via the fallback id 0x11 → cactus frame 8. The Ocean/Sea-Lane split (frames 10/11) is visible on screen and pixel-confirmed.

6.3 O513 draw order and the sprite-frame assignment map

The per-tile draw order on the visible path, with the PHYS0 engine-frame bands:

STEP	OVERLAY	ENGINE FRAMES	GATE / SELECTOR	SITE
1	fog tile + fog edges	`0x95; 0x69+dir`	hidden flag (§6.11)	`0x68212, 0x68244`
2	base ground	`TERRAIN 1..12`	normaliser §6.2	`0x68285 / 0x68301`
3	open-water early out	–	water, 0 land neighbours: ground + detail + O512, return	`0x68274`
4	O512 edge blend	`0x69..0x6C stencils`	every differing edge (§6.11)	`0x68315`
5	forest	`0x41 + mask4`	forested, not Scrub	`0x6833D`
6	shore hatch	`0x96`	feature byte & 0x40	`0x6834F`
7	relief	`0x21 / 0x31 + mask4`	terrain byte & 0x20 (§6.5)	`0x6835C`
8	rivers	`0x01 / 0x11 + mask4`	terrain byte & 0x40 (§6.6)	`0x6838A`
9	detail band	`0x5A + value`	position hash (§6.9)	`0x682B2 / 0x683F7 / 0x685D6`
10	surf/rumor	`0x68`	rumor hash (§6.10)	`0x68405..0x68414`
11	roads	`0x51; 0x52+d`	feature byte & 0x0A (§6.8)	`0x68417`
12	coast edges	`0x97..0x9A`	water tile, clean pattern (§6.7)	`0x6850D`
13	coast quadrants	`0x6D..0x8C`	water tile, no clean pattern (§6.7)	`0x684BC..0x684F5`
14	river mouths	`0x8D..0x90 / 0x91..0x94`	water tile, own bits & 0xC0 (§6.6)	`0x68524..0x685AC`

Rows 5–11 run on land tiles; rows 12–14 on water tiles. There is no other road or coast band: the 0x6D band is coast sub-tiles (not roads) and 0x95 is the unexplored tile (not a coast base).

6.4 Forest overlay — 0x41 + connection mask

Frame = `0x41 + mask`, where the 4-cardinal mask (`func_067C8E`) sets weights **N=8, S=4, W=2, E=1** for each neighbour that connects. A neighbour connects (`func_067C54`) iff its masked id is in the forest band 8..0x17 **and** `(id & 7) ≠ 1` — **desert Scrub never connects** (and a Scrub centre draws no forest overlay at all; its trees are the cactus ground frame). Live-confirmed: a Boreal|Scrub pair renders Boreal with the isolated mask and Scrub with no overlay.

6.5 Mountains and hills — 0x21 / 0x31 + mask

Gated by terrain-byte bit 0x20 on non-water tiles: bit 0x80 set → Mountains, base 0x21; clear → Hills, base 0x31. The 4-cardinal adjacency mask (`func_067BE4`, weights N=8/S=4/W=2/E=1) counts a neighbour iff its terrain byte satisfies `(byte & 0xA0) == (own & 0xA0)` — **hills never connect to mountains** and vice versa (live-confirmed both ways: each renders its isolated frame beside the other).

6.6 Rivers and river mouths

Rivers (draw at 0x6838A, land tiles): base = **0x01** when terrain-byte bit 0x80 is set (Major River) else **0x11** (Minor River) (0x6839E/0x683A6); add the 4-cardinal mask from `func_067B84` (terrain-plane bit 0x40, weights N=8/S=4/W=2/E=1); an isolated river (mask 0) is forced to mask **0xF** (0x683BB) and drawn `base + mask` (0x683C6). Because the mask tests only bit 0x40, **major and minor rivers interconnect** (a minor river joins an adjacent major run), and a land river beside plain ocean counts no connection there (it renders isolated) — both live-confirmed.

River mouths (water tiles, 0x68524..0x685AC): a *water* tile carrying its own river bits (`terrain & 0xC0`, latched at 0x68206 before the beach-halo substitution) draws base = **0x8D** if its bit 0x80 is set (major) else **0x91** (0x68524: `AND 0x80; CMP 1; SBB; AND 4; ADD 0x8D`), then for each cardinal d = 0..3 (N,E,S,W) draws `base + d` for every neighbour whose terrain byte has bit 0x40 **and** classifies as non-water. Negative controls confirmed live: ocean with river bits but no land-river neighbour draws nothing; plain ocean beside a land river draws no mouth.

6.7 Coast — beach halo, clean edges, quadrant fallback

`analyse_connections` (`func_067A24`) runs **only for water tiles** (gate 0x68256). It builds the 8-direction land-neighbour bitmap `[0xA8A6]` (bit d = land, order 0=N, 1=NE, 2=E, 3=SE, 4=S, 5=SW, 6=W, 7=NW; water neighbours 0x19/0x1A skipped) plus a 4-entry per-quadrant code table.

Beach-halo ground substitution (0x67AD4): while walking the cardinals, the routine *overwrites* `[0xA8A1]` with the folded class of the last cardinal land neighbour seen (visit order N,E,S,W — **W wins**), and reclassifies `[0xA8A2]` (0x67B10). O513 therefore grounds a coastal water tile with the *neighbour's land terrain*, draws the coast frames over it, and finally backfills water through the frames' 0-index holes (`func_067EEC` masked fill). The code-0 quadrant frames (disk 0x6C–0x6F) are all-zero "punch-throughs" that exist precisely to punch water through the substituted ground.

Clean edges (0x68474..0x6850D): default pattern −1, then four tests on `[0xA8A6]` assign the pattern and the draw is `0x97 + pattern`:

PATTERN	MASK TEST	FRAME	EDGE
0	`& 0xDD == 0xC1`	0x97	NW land corner
1	`& 0x77 == 0x07`	0x98	NE land corner
2	`& 0x77 == 0x70`	0x99	SW land corner
3	`& 0xDD == 0x1C`	0x9A	SE land corner

All four are drawable (engine 0x97..0x9A = disk 150..153, the 16×16 shoreline edges); a 2×2 lake exercises all four, pixel-exact.

Quadrant fallback (no clean pattern, 0x684BC..0x684F5): for q = 0..3 (TL, TR, BR, BL) draw the 8×8 frame **`0x6D + code[q]·4 + q`** at the quadrant's sub-cell offset. The quadrant code (built at 0x67ABD..0x67AEF) ORs, per quadrant: **|=4** for its own cardinal (N,E,S,W for q0..q3), **|=1** for the next-clockwise cardinal, **|=2** for its diagonal — maximum 7. All four quadrants draw unconditionally (code 0 ⇒ frame `0x6D + q`, the punch-throughs) for any water tile with ≥1 land neighbour that escapes the clean patterns. The reachable band is 0x6D..0x8C, all 8×8; a single-tile lake yields codes 7,7,7,7 → frames 0x89/0x8A/0x8B/0x8C, pixel-exact.

6.8 Roads — 0x51, then one frame per direction

Draw at 0x68417, gated: feature byte & 0x0A, mode `[0x18E]` == 0, non-water. The 8-dir mask comes from `func_067D54` over the feature plane (bits 0x0A). **Mask 0 → the single isolated frame 0x51; otherwise ONE FRAME PER SET BIT, `0x52 + d`** for each set direction d (0=N, 1=NE, … 7=NW) — *not* a combined-mask frame. The road band is 0x51..0x59 only. (Byte-decoded; road pixels are not reachable from crafted map files because the loader discards the feature plane — §7.1.)

6.9 The detail band 0x5A — the position hash IS prime resources

Sites 0x682B2 (open water), 0x683F7 (land), 0x685D6 (coast water), each gated `[0x18A] == 0` (suppressed in the colony scene panel). The hash (`func_0060A0`, with salt word `[0x190]`; salt 0 disables the band):

```
v = (x & 3)*4 + (y & 3)
h = ((y >> 2)*3 + (x >> 2) + salt - forest) & 0xF     ; forest = 1 for ids 8..0x17
draw 0x5A + DTAB[class]   iff  h == v  or  (h ^ 0xA) == v
```

`class` uses the **full id decode including the relief bits → 27/28** (`func_0624E` semantics; the plain `& 0x1F` reading was falsified by live pixels — mountains draw the ore/gold sprite 0x66 = 0x5A+DTAB[27], hills the rock 0x67 = 0x5A+DTAB[28]). The per-class table DTAB is the word array at

DS:0x192 (29 entries; runtime-read from the live game, pixel-verified): `[0,1,2,3,4,5,6,6, 9,1,8,9,10,10,6,6, <dup of 8..15 for raw 16..23>, −1 (Arctic), 7 (Ocean), −1 (Sea Lane), 12 (Mountains), 13 (Hills)]`; entry −1 = no detail, entry value 0 is replaced by 6. Feature-plane bit 0x04 suppresses the detail unless the table value is 12 (then `0x5A + 0`); tiles owned by a village (feature bit 2 with continent-plane owner ≥ 4) draw none.

This hash is the prime-resource mechanism: a hash-hit tile is what the sidebar reports as e.g. "(Prime Tobacco)" — procedural, never stored in the map (live-corroborated).

6.10 Surf / rumor circle — 0x68

Frame 0x68 is the rumor circle, drawn on the land path after the detail band (0x68405..0x68414) when the rumor hash (`func_006188`) hits:

```
((y >> 2)*0x13 + (x >> 2)*0x11 + salt + 8) & 0x1F) - ((x & 3)*4) == (y & 3)
```

Suppressed for classes 0x18/0x19/0x1A and — the live-verified owner gate (`func_005DF0` family) — whenever the continent-plane **owner nibble ≠ 0xF** (so land claimed by a village never shows a rumor circle).

6.11 Unexplored tiles and the O512 edge/fog blends

A hidden tile (hidden flag `[bp-8]` from fog mask `[0xA89E]` + tile fog byte `[0xA8A0]`, 0x681E0..0x681FE, tested at 0x6820C) draws **frame 0x95 — the fog/unexplored tile** (0x68212), then calls O512 for the fog-edge blend. 0x95 is *not* a coast sprite, and 0x69..0x6C are fog-edge/blend stencils, not coast.

O512 (`func_067F50`) loops the 4 cardinals (dx `[0,1,0,−1]` / dy `[−1,0,1,0]` = N,E,S,W, tables at DGROUP 0xA8/0xAE). Per neighbour: bounds test (engine coordinate 0 IS in bounds — plane column 1; upper bound engine width−2), read + fold its terrain, classify, read its fog state. **Water-neighbour ring-walk** (enabled only when arg2 = 0): walk the neighbour's own 8-ring counting *down* — even (cardinal) indices only, order **W → S → E → N — first non-water wins** — and use that land class as the blend class; this produces the land-side dithered beach. Skip the edge when the neighbour is still water after the walk, or when neighbour class == centre class with the neighbour visible. Otherwise draw `0x69 + dir` — a sparse index-0 dot stencil (disk 0x68..0x6B = N,E,S,W) stamped into the mask buffer 0x839E — then masked-blit the neighbour's terrain through it (`emit_terrain_sprite`, `func_067EEC`; plain thunk, or the scaled variant when `[0x184] ≠ 0`). Net effect: the neighbour's terrain bleeds into this tile's edge as a dither gradient — every biome transition, the coast's land side, and the fog boundary all come from this one composer.

The two O512 call sites in O513 (args: hidden, disable-ring, third 0): - fog path 0x68244, after the 0x95 draw: `O512(1, centre_is_water, 0)` — blends explored neighbours into a fogged tile's edge; - main path 0x68315: `O512(0, centre_is_water, 0)` — so land centres run with the ring-walk enabled (beach dither), water centres with it off (their coast is the §6.7 composition).

6.12 Zoom scaling

The zoom level `[0x184]` runs 0..3. Viewport setup (`func_06787C`): span `[0x8544] = 15 << zoom`, `[0x8546] = 12 << zoom`, tile pitch `[0x5AD4] = 16 >> zoom`; sprite scale `[0x186] = 100 >> zoom` (0x679F4) — blits go through the plain thunks at scale 100 and the scaled variants otherwise. Zoom 0..3 = 15×12 @16 px, 30×24 @8 px, 60×48 @4 px, 120×96 @2 px.

§7 Runtime map state and the map screen periphery

The playing board lives in four parallel byte planes addressed through far pointers in the data segment, drawn into a 320×200 Mode 13h screen whose left three-quarters is the tile viewport and whose right edge is a woodgrain sidebar with minimap, status lines and the selected-unit panel. This section pins the live memory layout, the fog model, the screen geometry, and the colony screen's scene panel (which reuses the §6 compositor).

7.1 The live map planes

Four far pointers hold the planes (runtime-verified against live memory):

POINTER	PLANE	CONTENTS
`[0x15C]`	terrain	terrain byte (§5.4 encoding)
`[0x160]`	feature	bit 0 unit present, bit 1 settlement; road bits 0x0A; rebuilt by the game — the map-file feature layer is DISCARDED at load
`[0x164]`	continent/owner	low nibble continent, high nibble owner (0xF = none; village-owned land 5..0xA)
`[0x168]`	flags	low nibble = colony-site value

Each plane is row-major, **stride 58**, with tile (0,0) at segment **base + 0x10**. Three coordinate frames coexist: *plane* coordinates (0..57 × 0..71, including the border ring), *engine* scroll-space `[0xA5A0]`/`[0xA5A2]` = plane − 1, and the **sidebar "Locat:" display, which shows the plane index (= engine + 1)**. O514's `(y+1)·stride + (x+1)` indexing (§6.1) converts engine to plane coordinates.

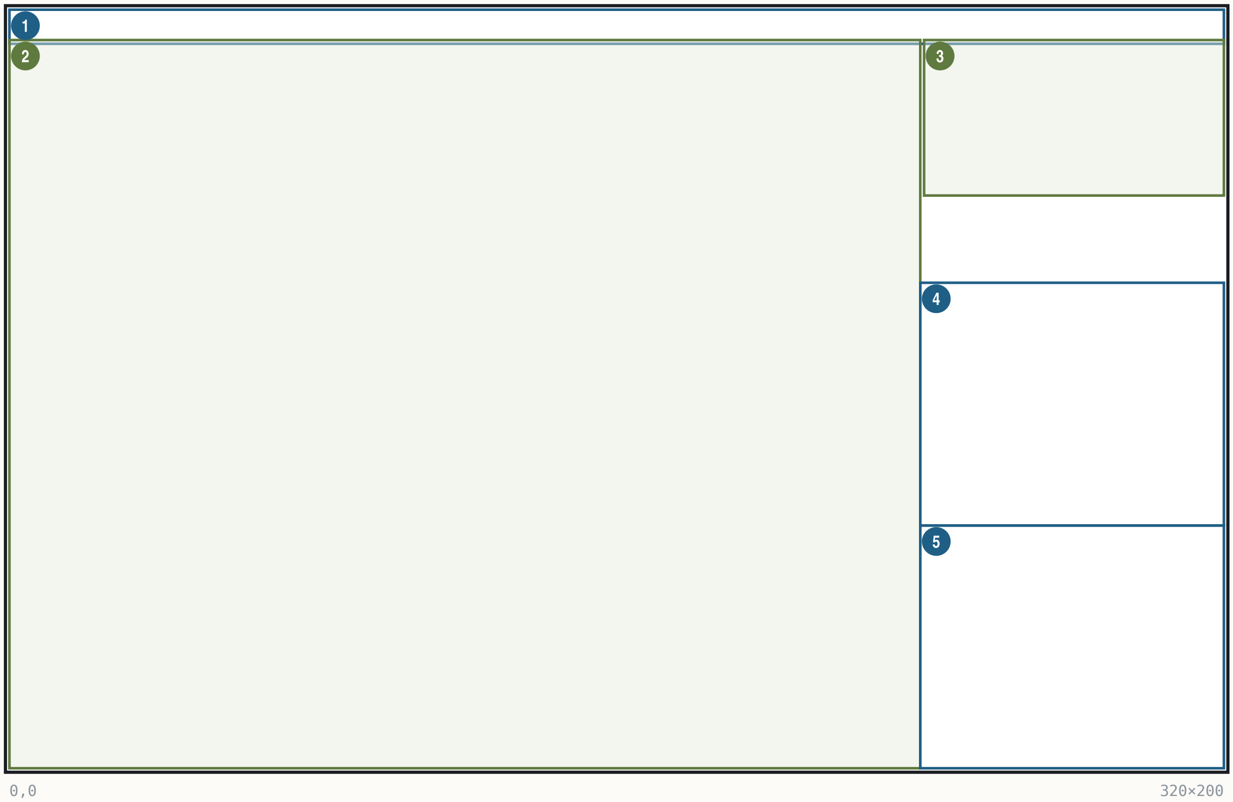
7.2 Fog and Reveal Map

The per-tile fog byte holds one bit per power in bits 4–7; the render mask is `[0xA89E] = 1 << (player + 4)` (0x685F2) — explored-by-power-3 = bit 7 = 0x80, matching runtime dumps. **Reveal Map** (cheat menu, enabled by Alt-W, I, N; the "Reveal Map → Complete Map" row) sets the full-view flag `[0x53A2]`

`= 1` and zeroes the fog mask `[0xA89E]`; per-tile fog bytes are left untouched, and O513's mask==0 path then treats every tile as visible.

7.3 The map screen — regions, fonts, keys

The map screen — regions, fonts, keys 320×200 · Mode 13h



#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×9	text	Menu strip	wood fill; pulldown titles
2	(0,8) 240×192	art	Map viewport	15<<z x 12<<z tiles at 16>>z px
3	(241,8) 79×41	art	Minimap	1 px/tile window + white viewport rect
4	(240,72) 80×64	text	Season/Gold/Tax	FONTTINY line stack
5	(240,136) 80×64	text	Selected-unit panel	unit sprite + @INFO labels

REGION	BOUNDS	KIND	CONTENT SOURCE	STATE BINDING
Menu strip	(0,0,320,9)	text	menu titles `@GAME @VIEW @ORDERS @REPORTS @TRADE @CUP @PEDIA`	pulldown state
Viewport	(0,8)–(240,200)	art	§6 compositor	scroll `[0x8328]` / `[0x832E]`, zoom `[0x184]`
Minimap	(241,8,79,41)	art	`func_066CD6` (§7.4)	scroll window, fog, owners
Status lines	(240,72,80,64)	text	season/year, Gold, Tax (§7.5)	year `[0x538A]`, power record
Unit panel	(240,136,80,64)	text	Moves/Locat/type/skill/orders/terrain	selected unit record

Fonts/inks: menu strip FONTTINY in green RGB (82,138,49); sidebar text FONTTINY in white, palette index 0x0F (set at 0x76C85); sidebar x-origin `[0x8550]` = 240 (0x71039). The UI colour slots at DS:0x830.. are loaded from the data file's `@COLORS` line — nine palette-index bytes 68,149,8,128,47,138,134,128,138 (basic, hilite, grey, enhance, shadow, select, border0..2) written to 0x830..0x839 (0x836 skipped), at 0x751A2..0x751E7.

Viewport zoom table (§6.12): **15<<z × 12<>z px**, z = `[0x184]` ∈ 0..3 = 15×12@16 / 30×24@8 / 60×48@4 / 120×96@2. In VICEROY these are the four VIEW-menu "Zoom Level" entries (plus zoom in/out); the stand-alone map editor binds the same four spans to F1..F4 (`F1` = z3 120×96 … `F4` = z0 15×12, handler `_set_zoom_level(0x29-id)` at its 0x30D8, clamp 0..3).

Navigation/keys: menu pulldowns per the seven titles above; REPORTS F2–F10 open the advisor screens; VIEW menu zooms; F1 opens the terrain-information popup (shared WOODPANL popup framework, `func_06F0F4`); click an own colony tile → colony screen; click a foreign colony → the sidebar trade variant.

7.4 Minimap

`func_066CD6`: panel box (0xF1, 8, 0x4F, 0x29) = **(241,8,79,41)**, byte-verified at 0x66CF4. Contents are **1 pixel per tile** — a 56×39 scrolling window over the map, not the whole map squashed. Per-tile dot colour from the DS:0x830 slot table: `[0x830]` ocean/coast, `[0x831]` land, `[0x832]` fog (fog byte & 0x80), `[0x833]` owned (& 0x20); the current-viewport rectangle is drawn in white (index 0x0F) — at zoom 0 a 15×12 rect, which independently confirms the zoom-0 span. The per-cell writer (`func_066968`) writes single bytes per tile at x = col − `[0x9CCC]` + 252, y = row − `[0x9CCA]` + 9, pitch `[0x2DAA]`.

7.5 Sidebar HUD

Status block (240,72,80,64), FONTTINY, white 0x0F: a three-line stack — **season + year** (season names from `@SEASONS`: Spring/Autumn; year global `[0x538A]`, banded `(year − 1500)/50` at 0x51F1F), **Gold: N** (power record +0x2A), **Tax: N %** (power record +0x01); label strings from the `@MISC`/`@INFO` sections via the string resolver. The line composer is `func_067700`; the per-line y-offsets are emitted through a runtime-installed far pointer (`[0xA644]` = 0x1A1F:0x0F10, installed at 0x7730C) — a true runtime indirection; the line stack itself is pixel-verified against the running game. Selected-unit panel below at y=136: unit sprite, then Moves / Locat / unit type / skill / orders / "(terrain)" lines (unit record stride 0x1C via `func_0672C8`; type and skill names from the `@UNIT`/`@JOB` sections; Locat shows plane coordinates, §7.1). Per-line coordinates beyond the stack order are (measured; not byte-cited): season/year at (244,58), Gold (244,66), Tax (290,66), unit sprite (244,80), Moves (270,82), Locat (270,92), type (244,104), skill (244,112), orders (244,120), terrain (244,128).

7.6 The colony screen scene panel — the same compositor at ×1.5

The colony screen's terrain vignette is the §6 map compositor rendering a **5×5 tile neighbourhood of the colony at native 16 px, then upscaled ×1.5 with an ordered dither** — there is no dedicated 24-px tileset. Chain (`func_026374` at 0x26374): scene latch `[0x18A]` = colony pointer (0x6891E); viewport forced 5×5, zoom `[0x184]`=0, pitch 16, scale 100, origin = colony − (2,2), screen offset 0 (0x67894..0x67912); the master loop `func_0685DC(cx−2, cy−2, 5, 5, power)` paints an **80×80** image on surface 0x839E — same painter, with map unit glyphs and the detail band suppressed by the scene latch. Colony markers (ICONS frames 1..4 + pennant 0x77+power, population number and name iff `[0x890]` == 0) and unit markers are drawn on the 80×80 *before* the upscale. Then `func_00531C` (0x263A9..0x263D6) stretch-copies (0,0,80,80) → **(200,8,120,120)**, duplicating every 2nd pixel and every 2nd row (exact 2→3 scaling) and passing every written pixel through the positional 4×4 dither `func_005296` (`(dstoff&3) + (row&3)·4`), which jitters the palette index within its 16-colour ramp for palette rows 0x10..0x87 — deterministic, no RNG. The **visible panel (224,32,72,72) is the central 3×3 tiles** of the 5×5 at 24-px pitch (the outer ring is overdrawn by the right-panel fill). Worker sprites are added *after* the upscale at x = cell·24 + 252, y = cell·24 + 60 (signed cell −2..+2), sprite `0x5A +` detail value, from PHYS0.SS.

PART III

Economy and colonies

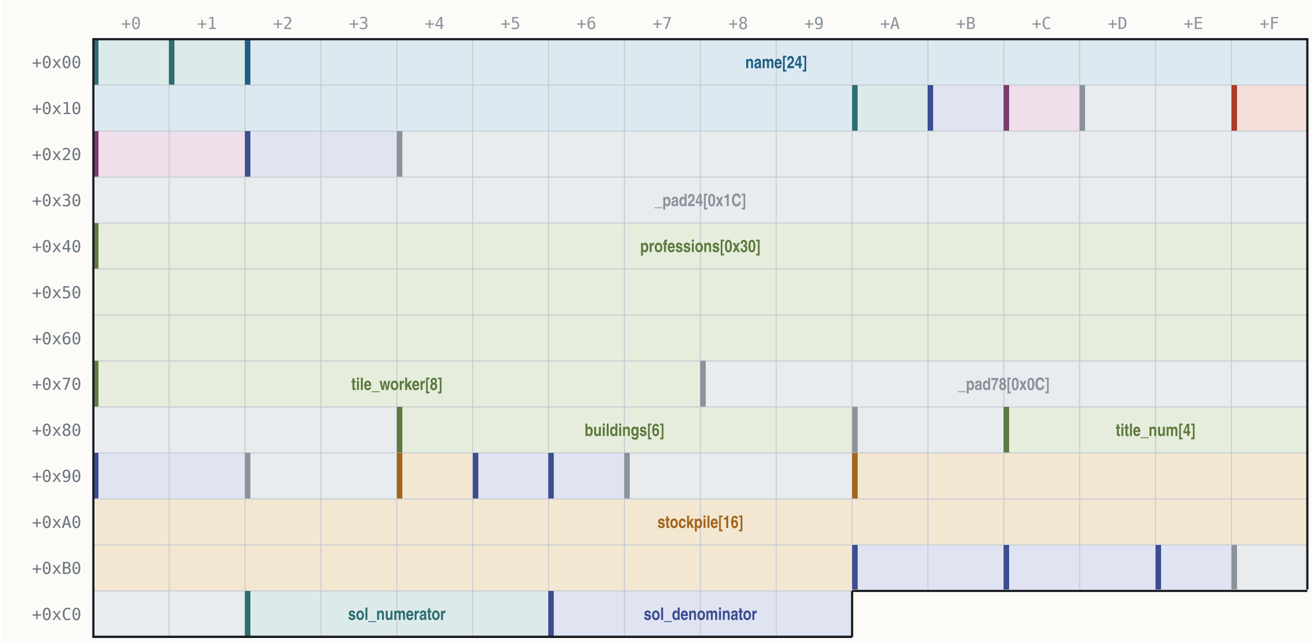
§8 Colonies

Every colony is a fixed-size 202-byte record in a single DGROUP array. The record carries the colony's position, name, owner, population, profession roster, constructed-building bitmask, warehouse stockpile, and the Sons-of-Liberty bell pool; everything the colony screen draws is derived from it plus a handful of BSS scratch tables rebuilt on screen entry. The most surprising subsystem — fully byte-verified and replay-validated — is that the on-screen *placement* of the colony's buildings among the 15 plots is produced by a deterministic RNG shuffle seeded from the colony's map coordinates.

8.1 ColonyRecord

The colony table lives at `DS:0x5D46`, stride `0xCA` (202 bytes); the live count is the word `[0x539E]`, capped at 48 (`cmp [0x539E],0x30` at 0x2EB82). The currently-active colony is the **near pointer `[0x8542]`**, written by the set-active-colony routine at 0x8302..0x830B (`imul bx,idx,0xCA; add bx,0x5D46; mov [0x8542],bx`). Slots are recycled: a razed colony's slot is reused for the next founded colony, and slot validity is determined by a non-empty name at `+0x02`. Live-verified in the running game: `*[0x8542]` → a record decoding as `(51,29) "Jamestown"`, exactly `0x5D46 + 4·0xCA`.

ColonyRecord sizeof = 0×CA (202)



202 bytes (0×CA) · mapped 202 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00	1	uint8_t	`map_x`	tile column (live-verified)
+0x01	1	uint8_t	`map_y`	tile row
+0x02..+0x19	24	char	`name[24]`	NUL-terminated colony name
+0x1A	1	uint8_t	`owner`	0 English 1 French 2 Spanish 3 Dutch;
+0x1B	1	uint8_t	`status_1B`	numeric prefix of the foreign-title
+0x1C	1	uint8_t	`flags_1C`	bit flags — colony-marker population-number
+0x1D..+0x1E	2	uint8_t	`_pad1D[2]`	unmapped (2 bytes)
+0x1F	1	uint8_t	`population`	colonist count (burn-loot formula 0x05DE1E;
+0x20..+0x21	2	uint16_t	`flags_20`	(measured; not byte-cited — observed 0x1002;
+0x22..+0x23	2	uint16_t	`state_22`	(measured; not byte-cited)
+0x24..+0x3F	28	uint8_t	`_pad24[0x1C]`	unmapped (28 bytes)
+0x40..+0x6F	48	uint8_t	`professions[0x30]`	..: one @JOB byte per colonist
+0x70..+0x77	8	uint8_t	`tile_worker[8]`	colonist index working each
+0x78..+0x83	12	uint8_t	`_pad78[0x0C]`	unmapped (12 bytes)
+0x84..+0x89	6	uint8_t	`buildings[6]`	42-bit constructed mask, bit i =
+0x8A..+0x8B	2	uint8_t	`_pad8A[2]`	unmapped
+0x8C..+0x8F	4	uint8_t	`title_num[4]`	four numeric fields appended by the
+0x90..+0x91	2	uint16_t	`field_90`	(measured; not byte-cited)
+0x92..+0x93	2	uint8_t	`_pad92[2]`	unmapped
+0x94	1	uint8_t	`cargo_94`	cargo-holds datum read by the colony
+0x95	1	uint8_t	`warehouse_level`	Warehouse Expansion counter (building id

OFFSET	B	TYPE	FIELD	NOTE
+0x96	1	uint8_t	capitol_level	Capitol Expansion counter (building id 0x1F)
+0x97..+0x99	3	uint8_t	_pad97[3]	unmapped
+0x9A..+0xB9	32	uint16_t	stockpile[16]	warehouse quantity per good,
		+0x9A Food +0x9C Sugar +0x9E Tobacco +0xA0 Cotton +0xA2 Furs +0xA4 Lumber +0xA6 Ore +0xA8 Silver +0xAA Horses +0xAC Rum +0xAE Cigars +0xB0 Cloth +0xB2 Coats +0xB4 Trade Goods +0xB6 Tools +0xB8 Muskets		
+0xBA..+0xBB	2	uint16_t	counter_BA	counter pair lo/hi (measured; not byte-cited)
+0xBC..+0xBD	2	uint16_t	counter_BC	(measured; not byte-cited)
+0xBE	1	uint8_t	marker_frame	map-marker frame byte (marker painter
+0xBF..+0xC1	3	uint8_t	_padBF[3]	unmapped
+0xC2..+0xC5	4	int32_t	sol_numerator	rebel bell pool (accumulated + clamped to
+0xC6..+0xC9	4	int32_t	sol_denominator	bell cap = decay + population·2

Sons of Liberty percent = `sol_numerator·100 / sol_denominator` (32-bit multiply 0x008557 + divide 0x00855E in `func_008524`; 0 if the denominator ≤ 0 at 0x008542), then +20 for a human-controlled colony (latch `add ax,0x14` at 0x00859F, gated on `owner<4`), clamped to 100 (0x0085A8..0x0085AD).

8.2 Building construction state and upgrade chains

The 42 building definitions of NAMES.TXT `@BUILDING` load into a stride-12 record table at `DS:0x8F82` (name pointer +0, prerequisite/predecessor +3, chain successor +4, category byte at `0x8F87+id·12`, loaded at 0x74D2F). A constructed building sets its bit in `ColonyRecord +0x84` (`func_092E0`, `or [bx],al` at 0x9308) **without clearing the predecessor's bit** — upgrade tiers coexist as bits and the highest tier present wins for render and production (build-complete handler at 0x02D19A confirms: set only, no clear). Two definitions have no bit at all: **Warehouse Expansion (id 0x10)** increments the counter `+0x95` and **Capitol Expansion (id 0x1F)** increments `+0x96`. School ids: Schoolhouse 0x0C, College 0x0D, University 0x0E.

Chain walking uses the `0x8F82` fields: prerequisite line shown when `byte 0x8F82[id]+3 ≥ 0`; the upgrade-chain loop steps `next = byte 0x8F82[b]+4` while ≥ 0 (Colonizopedia building/skill pages, 0x06A89C..0x06A947 and 0x06AD3C..0x06AD9F).

8.3 Worker/building byte tables

Two static DGROUP byte tables bind buildings to professions:

- **Building → job**: `DS:0x2CA` (file 0x1DC6A), 42 signed bytes, read through `func_009786` (`0x181F:0xACE`). Byte-verified entries: ids 3–5 → 0x0F Gunsmith, 9–11 → 0x11 Statesman, 21–23 → 0x0B Weaver, 27–29 → 0x09 Distiller, 35–36 → 0x0D Carpenter, 37–38 → 0x10 Preacher, 39–41 → 0x0E Blacksmith. Jobs 0x12 (Teacher) and 0x15 are skipped by the pedia header renderer.
- **Job → building**: `DS:0x2F4` (file 0x1DC94), 19 signed bytes, read through `func_008D9C` (`0x181F:0xB00`). Jobs 0–8 (the nine outdoor field jobs) → −1 (no workplace building); job 0x0D → 35 Carpenter's Shop, 0x0F → 3 Armory, 0x11 → 9 Town Hall, etc.

8.4 Colony-screen building placement — the RNG layout algorithm

The colony view has **15 fixed plots** in the upper-left town area. Which building occupies which plot is computed on every colony-screen paint by `func_025D34`, a deterministic random shuffle. The whole chain is byte-verified and was validated by exact replay: re-implementing the algorithm below reproduces the live game's Jamestown layout with every sprite pixel-exact (pixel-verified against the running game, 1994 binary under DOSBox).

Plot position table — static at `DS:0x266` (file 0x1DC06), 15 × (word x, word y); the renderer draws at `(x, y+8)` (the +8 applied once, at 0x02708B — applying it twice is a documented replay bug):

PLOT	X	Y (TABLE)	Y ON SCREEN	PLOT	X	Y (TABLE)	Y ON SCREEN
0	56	5	13	8	128	45	53
1	145	7	15	9	10	68	76
2	173	10	18	10	15	94	102
3	8	33	41	11	87	3	11
4	37	37	45	12	66	79	87
5	67	46	54	13	123	98	106
6	96	45	53	14	123	47	55

Plot categories — static counts `byte[0x224] = [7,4,2,1,1]` and bases `byte[0x22A] = [0,7,11,13,14]` (file 0x1DBC4/0x1DBCA): category 0 = plots 0–6, 1 = plots 7–10, 2 = plots 11–12, 3 = plot 13, 4 = plot 14. The per-plot category table `0x8D62` is rebuilt each open as `[0,0,0,0,0,0,0,1,1,1,1,2,2,3,4]` (0x025D7B..0x025DB8).

The RNG. Three byte-verified pieces:

```
seed:   func_009726 (0x181F:0xD62; sole caller func_025D34 @0x025D3A)
        seed32 = (colony_map_y << 8) + colony_map_x + dword[0x8D80]   @0x009736
        srand @0x103C2 keeps only the LOW 16 BITS of the seed:
        mov [0x28EE],ax ; mov word [0x28F0],0   -- effective seed space is 16-bit

rand:   @0x103D4 (MSC 6.0 LCG; bytes B8 FD 43 BA 03 00 ... 05 C3 9E / 83 D2 26)
        state = state·0x000343FD + 0x00269EC3    (32-bit, state at [0x28EE]/[0x28F0])
        return (state >> 16) & 0x7FFF
```

```
random_int(lo,hi):  func_00C322 (0x181F:0x4D4)
        r = rand(); return lo + ((r · (hi−lo+1)) >> 15)   @0x00C334
```

`[0x8D80]` is a **per-session dword** set at boot init: it read 0x2C55 in one live session and 0x5B7C in a fresh boot (its writer is unlocated — runtime-open). It is constant within a session, so a given colony always lays out the same way during play; the pure map-position seed alone does *not* reproduce a layout (exactly 2 of 65536 16-bit seeds reproduced the validated capture).

Registration groups. The building-definition registration block at 0x0746BC issues 42 calls (one per id) through the far trampoline at 0x76384 → `func_07464C`, assigning each id to one of **15 groups** — one screen slot per group:

GROUP	IDS	BUILDINGS	CAT
0	0–2	Stockade / Fort / Fortress	3
1	3–5	Armory / Magazine / Arsenal	1
2	6–8	Docks / Drydock / Shipyard	4
3	9–11, **30–31**	Town Hall ×3 + **Capitol** / **Capitol Expansion**	2
4	12–14	Schoolhouse / College / University	1
5	15–17	Warehouse / Warehouse Expansion / **Stable**	1
6	18	Custom House (alone)	0
7	19–20	Printing Press / Newspaper	0
8	21–23	Weaver's House / Weaver's Shop / Textile Mill	0
9	24–26	Tobacconist's House / Shop / Cigar Factory	0
10	27–29	Rum Distiller's House / Rum Distillery / Rum Factory	0
11	32–34	Fur Trader's House / Fur Trading Post / Fur Factory	0
12	35–36	Carpenter's Shop / Lumber Mill	1
13	37–38	Church / Cathedral	2
14	39–41	Blacksmith's House / Shop / Iron Works	0

The **category** of a group is NAMES `@BUILDING` numeric **column 3** of its first (representative) id, loaded to `0x8F87+id·12` at 0x74D2F. Over the 15 representatives the category histogram is exactly `[7,4,2,1,1]` — matching the plot counts. (The histogram over all 42 defs is 19/10/7/3/3 and does NOT match; an early decode tripped over that.) The group table is *not* `floor(id/3)`: Capitol 30/31 shares group 3 with Town Hall, Stable 17 sits with Warehouse in group 5, Custom House 18 is alone, and Fur Trader's House 32 opens group 11.

Placement loop (0x025DDB / 0x025DBF..0x025E07): after seeding, each of the 15 work-list slots (flattened category order: 7 cat-0 slots, 4 cat-1, 2 cat-2, 1 cat-3, 1 cat-4) picks a plot within its category block:

```
plot = base[cat] + random_int(0, count[cat]−1)
if plot already taken (0x8E92[plot] ≥ 0): retry     -- draw again, same range
0x8E92[plot] = slot                                 -- plot → slot map
```

i.e. a random permutation within each static category block. Then the 42 building defs are each mapped to their group's slot (0x025E0E..0x025E5A), and for every building the colony actually **has** — the present-gate query `0x181F:0x9FC` per id at 0x025E64 — the plot's def table gets `0x8E82[plot] = building id` (0x025E9F); unbuilt plots stay `0xFF`. **Id 0 (Stockade) is force-included** regardless of the query (0x25E70), so every colony renders something on the cat-3 plot. A later pass at 0x025E1A uses the `0x8F88` chain/produced-good column to assign goods — it plays no part in plot selection.

Frame selection (`func_026DD4`, 0x026DD4..0x026FF1): for an occupied plot the BUILDING.SS frame is `def_id + 1` in EXE-sheet space (`mov ax,[bp+6]; inc ax` at 0x026DE5..0x026DE9), blitted via `0x181F:0x254` at 0x026E4E. Overrides, all byte-read: `def_id==0` with build-query(0)==0 ⇒ frame **0x11** (0x026DEC); `def_id==0x0F` / `0x11` with garrison queries ⇒ frames **0x2F** / **0x30** (0x026E05..0x026E34). The Colonizopedia building page applies the same `id 0x11 → frame 0x2F` override (0x06AB62). Empty plots (`def < 0`) are drawn by `func_026FF2` with the terrain-decoration frame `byte[DS:0x260 + category]` (table `[45,44,43,0,46]` — category 3 draws nothing), skipped when the byte is 0. Live verification (Jamestown): 8 buildings at plots {2,3,4,5,6,10,12,13} with def-ids {0x20,0x1B,0x27,0x18,0x15,0x23,0x09,0x00}, every frame pixel-exact.

§9 Market and trade

The European market is per-power state inside the PowerRecord (stride 0x13C, European powers at `DS:0x8808` + power·0x13C; active-power pointer `[0x84FC]`). Prices are not a fixed table: the per-good base is random-seeded at game start and then driven by a per-turn decay plus per-transaction updates, with the four manufactured luxuries coupled through a shared supply pool.

9.1 Per-power money and market state

FIELD	TYPE	MEANING
`+0x01`	u8	**tax rate** (0..100) — read at 0x031043 for the Europe banner; raised by King events

FIELD	TYPE	MEANING
+0x1E	u16	artillery-bought counter (Europe artillery price escalation: `cost = base + count·100`, read ×100 at 0x035124/0x03527B, `inc` at 0x035282, zeroed at new game 0x03662F)
+0x20	u16	**boycott bitmask**, bit = good index. Test `func_030B38` (`(1<<good) & [bx+0x20]`); set at 0x34717 (Tea Party); back-tax lift at 0x3340C (`&= ~bit` after paying price×500 into the King fund `+0x22`); **Jakob Fugger** (Founding Father id 1) clears the whole word to 0 at 0x3BD45
+0x22	s32	King's REF fund (receives sale tax)
+0x26	s32	sales tally (net proceeds accumulator)
+0x2A	u32	**gold (treasury)** — every credit goes through the clamp helper at 0x8806: add s32, clamp to [0, 999999]
+0x4C	u8[16]	per-good **price level** (indexed by good; step-up `+=1` at 0x32272, step-down `-=1` clamp ≥0 at 0x3228D)
+0x5C	s16[16]	market pool (drift-only; never touched by the transaction path)
+0x7C	s32[16]	traded volume (cumulative value)
+0xBC	s32[16]	European supply per good
+0xFC	s32[16]	per-good trade accumulator (`@0x8904` in DGROUP terms) — summed by the drift function

Displayed bid/ask pair (Europe price strip, 16-good loop 0x38D40..0x38E3B): `sell = price_level[good] - 1` (accessor `func_030590`, clamp ≥ 0) and `buy = CARGO_row[good].col1 + price_level[good]` (accessor `func_030566`, `mov al,[bx-0x6900]` with bx=good·9 at 0x30575, `add ax,cx` at 0x30587). The on-screen spread is therefore the per-good constant `@CARGO` column 1 + 1: Food 1, Sugar 4, Tobacco 3, Cotton 2, Furs 4, Lumber 2, Ore 3, Silver 20, Horses 2, Rum/Cigars/Cloth/Coats 11, Trade Goods 2, Tools 2, Muskets 3.

9.2 Goods — @CARGO

The 16 goods — ICONS.SS frames good+0x17 NAMES @CARGO order · disk 22–37

Good id indexes every market array, the stockpile bar and the boycott bitmask; icons drawn at colony/Europe bars y=181.

Good ids 0..15, in NAMES `@CARGO` order (all market arrays use this index): 0 Food, 1 Sugar, 2 Tobacco, 3 Cotton, 4 Furs, 5 Lumber, 6 Ore, 7 Silver, 8 Horses, 9 Rum, 10 Cigars, 11 Cloth, 12 Coats, 13 Trade Goods, 14 Tools, 15 Muskets. Four **extended ids 16..19** are name-only production tokens: 16 Hammers, 17 Crosses, 18 Liberty Bells, 19 Flags — never traded, used for production display (icon remaps 0x0D→0x37, 0x10→0x39, 0x11→0x3F in the Colonizopedia product strips). Commodity icons are ICONS frames `good + 0x17` in EXE-sheet numbering.

9.3 Price drift

Per-turn driver: the end-of-turn processor `func_0755CC` invokes `func_036574` at 0x0757B0; it zeroes the per-power accumulators (`+0x5C/+0x7C/+0xBC/+0xFC` loop at 0x03670E) and runs a **4-power loop** calling the drift function `func_0305A8` (via the JMP-FAR trampoline at 0x368BD) once per power. **Per-transaction**: the SELL handler (`func_032914` at 0x32D99) and BUY handler (`func_0324F2` at 0x32902) each call `drift(good, 0)` — a single-good re-drift immediately after the trade.

The drift itself (`func_0305A8`, 0x0305A8..0x03064C):

```
for good in 0..15:                         # @0x305B3
    acc = price_seed[good]                 # word table DS:0x53EA (good·2)  @0x305B8
    for power in 0..3:                     # @0x305CE
        v = PowerRecord[power].accum_FC[good]   # @0x305D8
        if v < 0: v = 0                    # @0x305E0
        acc += v                           # 32-bit @0x305F0
    if driver-mode:                        # @0x305FF/0x30605
        price_seed[good] -= acc >> 8       # proportional decay @0x30618..0x30639
```

`price_seed[16]` at `DS:0x53EA` is **randomized at new-game init**: `func_07561C` fills each entry with `random_int(600, 1000)` (0x75645, loop ×16) — there is no fixed base-price table. Later phases of `func_0305A8` couple the luxuries: `S_pair = supply[9]+supply[10]+supply[11]+supply[12]` (Rum, Cigars, Cloth, Coats; 32-bit adds, clamp ≥ 1, at 0x030649), then for each finished good `target[i] = (S_pair·3)/supply[i]` (×3 at 0x03074F, 32-bit divide at 0x030759); the raw inputs 1..4 (Sugar/Tobacco/Cotton/Furs) use the same formula against their own supply (Furs halved, +1 if year<1700 and +1 if year<1600, at 0x0307C9). Dumping one luxury lowers its own price and nudges the other three up.

9.4 Buy/sell transactions

SELL (`func_032914`; args good, screen-idx, confirm): `gross = price·qty` (0x3249F); tax split at 0x32A4A..0x32A78: `tax = gross·tax_rate(+0x01)/100`, `net = gross - tax`; gold credit `+net` through the [0,999999] clamp helper (0x32A82); King fund `+0x22 += tax` (0x32A92); tally `+0x26 += net` (0x32A9C). **BUY** (six inline sites in the purchase pages, e.g. Muskets qty 0x32 at 0x526A2, Horses at 0x52790, Tools qty 0x64 at 0x52866): affordability check then inline debit `sub [bx+0x2A],ax; sbb [bx+0x2C],dx` — **buys are untaxed**.

Both paths call a mirror pair of accumulator updaters (good, qty):

FIELD	BUY FUNC_0322D0	SELL FUNC_03234A
EU supply `+0xBC`	`-= qty` @0x3231C	`+= qty` @0x323B4
accumulator `+0xFC`	`-= qty` @0x32324	`+= qty` @0x323BC
traded volume `+0x7C`	`-= price·qty` @0x32340	`+= price·qty·(100−tax%)/100` @0x32402
DGROUP pool `[bx−0x779C]` (4 × stride 0x9E)	`-= scaled_qty` ×4 @0x322FF	`+= scaled_qty` ×4 @0x32383 (4th power ×2/3)

`scaled_qty = ((price_level−2)·16·qty)/100` (helper 0x32294); the `@CARGO` "spread" column (field 4, `[bx-0x68FC]`) is a per-good left-shift exponent on qty inside these updaters (`shl dx,cl` at 0x322EA/0x32360), not the display spread.

On the Europe screen the market bar (0,179,320,21; 16 cells, pitch 0x13, icons `good+0x17`, bid price centred at y=194) routes clicks to the sell handler, which first tests the boycott bit and blocks with a message if set. Recruit gold cost comes from the recruit-pool slot word at `DS:0x978C + slot·6 + 4` (read 0x051E52/0x035114) — for Artillery (colonist type 0x0B) it escalates `base + artillery_bought·100`.

§10 The native economy

Each Indian settlement prices its trade through a single routine, `func_048F34` (file 0x048F34..0x0495FF), which fills two 16-word DGROUP arrays — `0x9E58[16]` per-good **demand** and `0x9E78[16]` per-good **supply**, in `@CARGO` order — for the active settlement. It runs in three phases and its outputs feed the village trade dialog, the food beg/gift events, and the haggle price. The routine also contains the cheat-menu "Supply and Demand (Indians)" debug dump, gated on debug bit `[0x894] & 4`.

Inputs: the active NativeSettlement record (pointer `[0x8D4A]` family; `+0x03` flags bit 4 = capital, `+0x04` population) and the active TribeData record (pointer `[0x8D4E]`; `+0x02` = tribe tier). `N = population + 1` (0x049242), `tier = tribe[+2]`.

10.1 Phase A — colony-claimed-tile mask (0x048F3C..0x049049)

A 25-byte mask marks which tiles of the settlement's 5×5 neighbourhood are worked by a European colony (those tiles contribute nothing). For each colony 0..`[0x539E]` (selected via `0x181F:0x9E6 → [0x8542]` at 0x04903A), each settlement-relative cell (a,b) is mapped to colony-relative coordinates (0x048F92..0x048FBA); if within the colony's 5×5 (0x048FCC..0x048FE2) the centre (2,2) is special-cased (0x048FE4) and otherwise the worked-slot test `func_008956` (`0x181F:0xCE0`, reading `ColonyRecord+0x70+slot`) decides; matches set `mask[a·5+b] = 1` (0x049002). **Original bug (byte fact)**: the in-bounds call at 0x048FC0 passes *relative* coordinates (x′−2, y′−2) to `func_005BFA`, which tests **absolute** bounds `1 ≤ x < map_w−1` — Phase B passes absolute coordinates correctly (0x04913D).

10.2 Phase B — 5x5 terrain point scan (0x04904A..0x049241)

Terrain id per tile via `0x181F:0x78C`. Contributions accumulated per tile:

TERRAIN	CONTRIBUTION	SITE
Mountains (27)	mountains counter +1	0x049190
Hills (28)	hills counter +1	0x049199
Arctic (24)	cold +4	0x0491A2
Forested 8..23	food/game point; base = t−8 (or t−16); base<3 ⇒ cold-forest counter, else warm-forest: sugar/tobacco/cotton +2	0x0491AB..0x04921F
Savannah	sugar +4	0x04909F
Swamp	sugar +2	0x0490A5
Grassland	tobacco +4	0x0490AF
Marsh	tobacco +2	0x0490B9
Prairie	cotton +4	0x0490C3
Tundra	ore +2	0x0490CD
Plains	cotton +1, food +2	0x0490D7
Ocean (25) / Sea Lane (26)	fish rate points; every 3 pts ⇒ food +2	0x049066..0x049090

10.3 Phase C — supply/demand arrays (0x049242..0x0495DC)

Both arrays zeroed at 0x049259..0x04926F, then per good (formulas exactly as decoded; where the manual gives only a site, the term exists but its algebra was not transcribed):

GOOD	SUPPLY	DEMAND
Food	`+= (tier+N)·food_pts/(7−tier)` @0x049271	`4·N²`, halved if tier ≥ 2 @0x04928F..0x0492A7
Silver	`tribe[+0xC]/K + 4·mountains` (K = per-tribe byte `[0x962A+idx]`) @0x0492B8..0x0492F0	—
Ore	`2·hills + mountains + tundra` (tier ≥ 1) @0x0492F4..0x04930B	—
Furs	`(2·coldforest + otherforest/2)/(tier+1)` @0x04930F..0x049328	—

GOOD	SUPPLY	DEMAND
Coats/Tobacco/Sugar/Cotton/Cloth	`supply terms @0x04932C..0x049386`	–
Tobacco	–	`(6−tier)·N + 2·cold + 5` @0x04938F
Cigars	–	`term @0x0493A5`
Coats	–	`8·cold + furs` @0x0493B5
Rum	–	`term @0x0493C2`
Trade Goods	–	`(tier+2)·(N+3) + 8` @0x0493D5
Tools	–	`(tier·N) << (cold/2 + 1)` @0x0493EA
Muskets	`supply = 0 @0x049434`	`4·(7 − tribe[+7] − tier)` @0x0493FC
Horses	`tribe[+0xA] / ([0x962A+idx]/2 + 1)` @0x04940D	`4·(9 − tribe[+8] − tier)` @0x049423

Then, in order:

1. **Demand clamp to [0, 50]** (0x32) via the clamp helper `0x181F:0x35C` (0x04943E..0x049460).
2. **Capital boost** (settlement flags `+0x03 & 4`, 0x049462..0x0494B5): demand[0..7] ×2, demand[13..15] ×1.5, supply[7..15] ×2.
3. **Tribe stock adjustment**: per-good tribe stock `tribe[+0xE+2g]` folded into both arrays (0x0494C0..0x0494D6 / 0x0495B9..0x0495D6).
4. **Mutual discount**: `supply -= demand/2; demand -= supply/2`, each floored at ≥ 1 (0x049537..0x049591). The debug dump sits between the two halves.

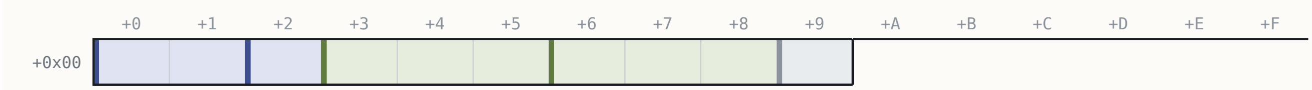
10.4 Consumers

- **Village trade** (`func_04A7CA`): zeroes last-bought goods (0x04A8C1..0x04A8F0), index-sorts the arrays (`0x191F:0xED0` at 0x04A8F7) and names the top goods in the "especially interested in …" line (0x04A91D/0x04A932).
- **Food events** (`func_056C3E`, the mission-village event handler and the sole caller of `func_048F34`, at 0x057093): a food *deficit* `demand[0]−supply[0]` gates the **INDIANBEGFOOD** popup (key pushed at 0x05716F); a surplus `supply[0]>demand[0]` gates **INDIANGIVEFOOD** (0x0573EB).
- **Haggle price** (`func_049600`): subtracts `supply[idx]·4` in the price computation (0x04A07A).

§11 Trade routes

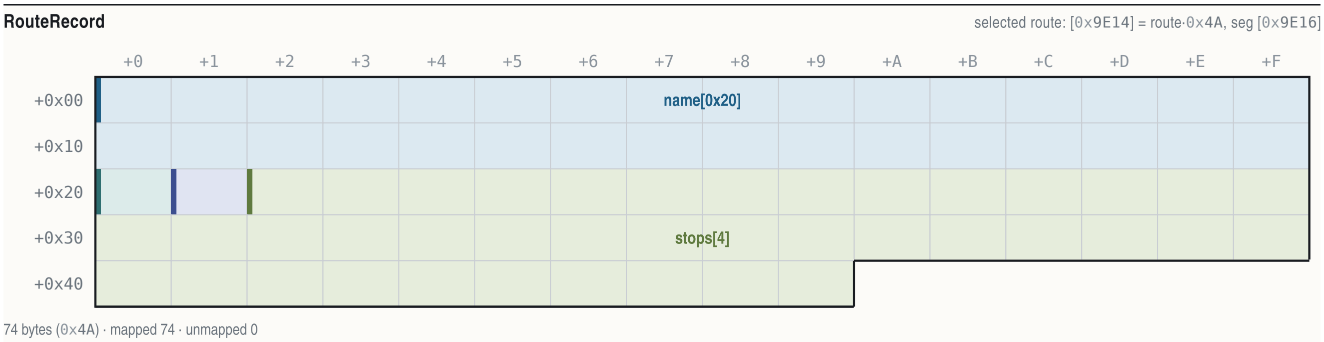
Trade routes are a small fixed array of 12 records, each holding a name, a sea/land type and up to four stops; each stop packs its destination and up to six load and six unload cargo types into nibbles. Units are attached to a route through two nibbles of their class byte, and the per-turn executor walks the stops under order code 2.

StopRecord nibble get/set func_0603DA / func_06040A



10 bytes (0xA) · mapped 10 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	uint16_t	dest	colony index; 0x3E7 (999) = Europe
+0x02	1	uint8_t	counts	lo nibble = unload count, hi nibble = load
+0x03..+0x05	3	uint8_t	load_nib[3]	load cargo ids, nibble-packed
+0x06..+0x08	3	uint8_t	unload_nib[3]	unload cargo ids, nibble-packed
+0x09	1	uint8_t	pad	unmapped (save-file diff pending)



OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x1F	32	char	name[0x20]	
+0x20	1	uint8_t	type	1 = sea, 0 = land
+0x21	1	uint8_t	stop_count	max 4
+0x22..+0x49	40	StopRecord	stops[4]	

```
//   (func_05FE60)
```

- **Commands**: menu ids 0x50 Edit / 0x51 Create / 0x52 Delete (MENU.TXT `@TRADE` row order) → `func_060FBC` / `func_0610B0` / `func_0612E6` (0x0238F2/0x0238EA/0x0238FC). Creating past 12 routes posts `@TRADEMANY` (cap check 0x0610B5).
- **Unit linkage**: unit byte `+0x17` (absolute 0x315B) — **low nibble = route id, high nibble = stop index** (accessors `0x181F:0x858/0x862/0x876/0x8B2`); the unit's order code is 2 ("Trade Route").
- **Assign** ("Begin Trade Route", `func_022D46`): `@TRADENONE` if no routes exist; the route menu is filtered sea-only for ships / land-only otherwise (`@TRADENONE2` if the filter empties); sets order 2 and steps immediately.
- **Create flow**: cap check → pick destination 1 → coastal test (`0x181F:0xD12`) → `@TRADETYPE` sea/land choice (or forced land) → default name = colony name + random `@TRADENAMES` word (collision appends " A") → `@TRADENAME` entry (max 0x1F chars) → stop count preset 2 → pick destination 2 (cancel aborts before `inc [0x53A0]`) → editor. The editor creates a phantom probe unit at (0xFF,0xFF) to filter reachable destinations, deleted on exit (0x0610A0). Delete compacts the array (`rep movsw` 0x25 words) and decrements higher route ids on all linked units.
- **Execution**: the per-turn executor for order 2 is `func_041080` (dispatcher jump table at file 0x24A38). A route with only one stop posts **@ROUTELOOP** (0x0413F7), verbatim:

```
Your Excellency, our "{%STRING0}" trade route
has only one port on its itinerary!
```

PART IV

Units and combat

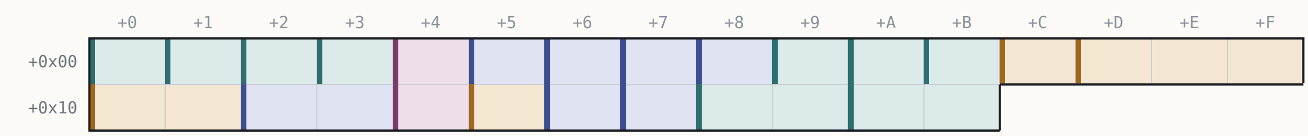
§12 Units

Up to 300 units live in a single 28-byte-stride array. A unit is its type (a row of the NAMES `@UNIT` table), an owner nibble, a position, an order code, cargo nibbles and a handful of per-turn scratch fields; the type indexes a 14-byte runtime stat table loaded from `@UNIT` at boot.

12.1 UnitRecord

Base `DS:0x3144`, stride 0x1C. (Field addresses below are absolute DGROUP; the record-relative offset is the low nibble progression from +0x00.)

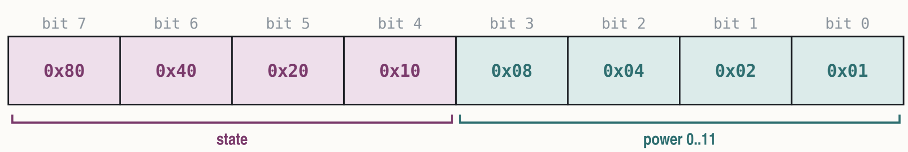
UnitRecord sizeof = 0x1C (28)



28 bytes (0x1C) · mapped 28 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00	1	uint8_t	map_x	(0x3144): renderer @0x03A63, placer @0x06958
+0x01	1	uint8_t	map_y	(0x3145)
+0x02	1	uint8_t	unit_type	(0x3146): @UNIT row 0..22; 694 refs
+0x03	1	uint8_t	owner	(0x3147): low nibble = power 0..11
+0x04	1	uint8_t	flags	(0x3148): transient bit register — 0x80
+0x05	1	uint8_t	moves_spent	(0x3149): move credits spent this turn
+0x06	1	uint8_t	timer	(0x314A): countdown (init 0xFF, dec)
+0x07	1	uint8_t	ai_state	(0x314B): persistent AI state letter
+0x08	1	uint8_t	order	(0x314C): order code 0..0x0C (dispatchers
+0x09	1	uint8_t	goto_x	(0x314D): Go-To target / route next stop
+0x0A	1	uint8_t	goto_y	(0x314E): (writer @0x22D38)
+0x0B	1	uint8_t	heading	(0x314F): 8-way facing 0..7, 8 = none
+0x0C	1	uint8_t	cargo_count	(0x3150): goods in hold (@0x0B2AB)
+0x0D..+0x0F	3	uint8_t	cargo_ids[3]	(0x3151..0x3153): nibble-packed
+0x10..+0x11	2	uint8_t	cargo_qty[2]	(0x3154..0x3155): per-slot
+0x12..+0x13	2	uint16_t	timer_16	(0x3156): overloaded — natives: snapshot of
+0x14	1	uint8_t	turn_flag	(0x3158): per-turn land-unit boolean
+0x15	1	uint8_t	tools	(0x3159): pioneer tools 0..100
+0x16	1	uint8_t	work_counter	(0x315A): turns-in-activity
+0x17	1	uint8_t	profession	(0x315B): colonist profession 0x13..0x1C;
+0x18..+0x19	2	uint16_t	occ_back	(0x315C): per-tile occupancy back link
+0x1A..+0x1B	2	uint16_t	occ_next	(0x315E): next link (placer @0x06968/@0x06976)

UnitRecord +0x03 — owner byte set_unit_owner @0x738E



power 0..11 — 0..3 European, 4..11 tribes

state — high-nibble unit state

UnitRecord +0x04 — flags byte

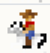
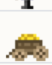
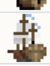
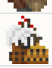
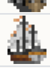
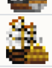
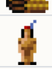
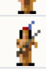
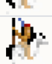
transient per-pass bit register

bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0
0x80	0x40	0x20	0x10	0x08	0x04	0x02	0x01
draw	cargo	merch	path	dirty	class	fortif	

draw — draw-active / Damaged display pair (set @0x069923; @ARTILLERY @0x05B6F6)
cargo — ship-carrying-cargo (@0x02F37A)
merch — Merchantman tag (@0x04CE44)
path — path >= 8 hops (@0x05106E)
dirty — tile-dirty (@0x0481B0)
class — ship-cargo class (@0x04CDDC)
fortif — was-fortifying (@0x04CEC9)

12.2 The @UNIT stat table

The loader at 0x074EC3 parses the 23 `@UNIT` rows into a runtime table at `DS:0x5230`, stride 14: +0 name ptr, +2 icon, **+4 moves stored ×3** (`shl al,1 / add al,cl` at 0x074F04; road cost = 1/3), **+5 combat (defense)**, **+6 attack**, +7 cargo holds, +8 move class (99 = naval), +9 hull, +0x0A size, **+0x0B guns**, **+0x0C ai-value** (the ship-combat pair), +0x0D flags. Values verbatim from NAMES.TXT:

ID	UNIT		ICON	MV	ATK	CMB	CRG	CLS	HULL	SIZE	GUNS	AI	FLAGS
0	Colonists		101	1	0	1	0	1	1	0	0	0	01000000
1	Soldiers		103	1	2	2	0	1	2	0	0	0	00011100
2	Pioneers		102	1	0	1	0	1	2	0	0	0	01000000
3	Missionaries		106	2	0	1	0	1	1	0	0	0	00100000
4	Dragoons		105	4	3	3	0	1	3	0	0	0	00111100
5	Scouts		104	4	1	1	0	1	2	0	0	0	01100100
6	Regulars		126	1	5	5	0	1	3	0	0	0	00011100
7	Cont. Cav.		130	4	5	5	0	1	3	0	0	0	00011100
8	Cavalry		127	4	6	6	0	1	4	0	0	0	00011100
9	Cont. Army		129	1	4	4	0	1	3	0	0	0	00011100
10	Treasure		17	1	0	0	0	6	4	0	0	0	00000000
11	Artillery		10	1	7	5	0	1	6	4	0	0	00011000
12	Wagon Train		9	2	0	1	2	99	1	0	0	0	00000000
13	Caravel		6	4	0	2	2	99	4	4	0	4	10100010
14	Merchantman		7	5	0	6	4	99	6	8	1	8	10000010
15	Galleon		8	6	0	10	6	99	10	10	4	20	10000010
16	Privateer		15	8	8	8	2	99	8	12	4	12	00000001
17	Frigate		16	6	16	16	4	99	16	20	12	32	10000001
18	Man-O-War		128	5	24	24	6	99	32	90	32	64	10000001
19	Braves		110	1	1	1	0	0	1	0	0	0	00111000
20	Armed Braves		111	1	2	2	0	0	2	0	0	0	00111000
21	Mtd. Braves		112	4	2	2	0	0	2	0	0	0	00111000
22	Mtd. Warriors		113	4	3	3	0	0	3	0	0	0	00111000

Ship types are `0x0D`..`0x12` (13 Caravel .. 18 Man-O-War) — this range gates ship logic everywhere (pathfinder, combat, docks). The Artillery "Damaged" display pair is attack +2 / damaged −2 (delta = col atk − col cmb, 0x069B39).

12.3 Professions — @JOB

28 rows (id, base name, expert form, tier = minimum school level 1=Schoolhouse / 2=College / 3=University / 4=not school-taught, and the expert's gold value): 0 Farmer (Expert Farmers, 1, 1100), 1–3 Sugar/Tobacco/Cotton Planter (Master …, 2, −1), 4 Fur Trapper (1, −1), 5 Lumberjack (1, 700), 6 Ore Miner (1, 600), 7 Silver Miner (1, 900), 8 Fisherman (1, 1000), 9 Distiller (2, 1100), 10 Tobacconist (2, 1200), 11 Weaver (2, 1300), 12 Fur Trader (2, 950), 13 Carpenter (1, 1000), 14 Blacksmith (2, 1050), 15 Gunsmith (2, 850), 16 Preacher (Firebrand, 3, 1500), 17 Statesman (Elder, 3, 1900), 0x12 Teacher (4, −1), 0x13 Colonist (Free Colonists, 4, −1), 0x14 Pioneer (Hardy, 1, 1200), 0x15 Soldier (Veteran, 2, 2000), 0x16 Scout (Seasoned, 1, −1), 0x17 Dragoon (Veteran, 2, −1), 0x18 Missionary (Jesuit, 3, 1400), and the specials **`0x19` Ind. Servant, `0x1A` Criminal, `0x1B` Convert** (all tier 4, −1) plus the pseudo-profession **`0x1C`** used as a display variant and remapped to 0x13 by the context-help dispatcher (0x02BD83). The runtime job table is `DS:0x8EA2`, stride 8 (+0 name, +2 expert plural).

Unit-type → expert-job table at `DS:0x30E` (file 0x1DCAE), bytes `13 15 14 18 17 16` (hex): type 0 Colonists → 0x13 Colonist, 1 Soldiers → 0x15 (Veteran Soldiers), 2 Pioneers → 0x14 (Hardy Pioneers), 3 Missionaries → 0x18 (Jesuit Missionaries), 4 Dragoons → 0x17 (Veteran Dragoons), 5 Scouts → 0x16 (Seasoned Scouts).

12.4 Orders

Order codes live at `UnitRecord +0x08` (0x314C); the per-turn dispatcher at 0x249CB jump-tables codes 2..9 (table at file 0x24A38). The `@ORDERS` section carries 13 rows of `name, key-letter`; the accelerator/status-letter array is built into `DS:0x54DE` (writer 0x074F96) and read as the on-map status glyph indexed by the order code (0x0391D) — the letters are exactly `- S T G L F F B P R - - -`:

CODE	ORDER	KEY	INIT WRITE	PER-TURN EXECUTOR
0	No Orders	`-`	@0x21ED7	— (auto-activate)
1	Sentry	`S`	@0x21FEB	— (skip)
2	Trade Route	`T`	@0x22E05	func_041080
3	Go To	`G`	@0x22D2D	func_040E22
4	Live In Village	`L`	no store site exists (menu row only)	passive
5	Fortify	`F`	@0x22105	func_04101C → writes code 6 @0x41024
6	Fortified	`F`	@0x41024	passive (+50% defense)
7	Build Colony	`B`	@0x2279E	func_040C1E
8	Clear/Plow	`P`	@0x22324	func_040656
9	Build Road	`R`	@0x2250E	func_0409D6
0xA–0xC	No Orders (AI reserved)	`-`	AI-only	AI

Status-glyph overrides (renderer at 0x0386A): ships not owned by the viewer show their cargo count as an ASCII digit (`+0x30` at 0x0393F); 'X' when hidden; the AI state letter `+0x07` for AI units, replaced by 'E' when ≥ 0x80.

§13 Movement and pathfinding

Terrain movement cost is data-driven: the `@TERRAIN Movement` column is loaded to `terrain·16 + 0x2F76` and charged as `Movement·3` against a budget stored ×3, so a road (cost 1/3) costs exactly one stored point. Short-range pathfinding is a bounded 16×16-window BFS with a cost cache, decoded in full below together with its three debug overlays.

13.1 The short-range path-step finder — func_061F02 (file 0x061F02)

Register args: `ax` = target_x, `dx` = target_y, `bx` = cost bound; returns `ax` = best direction 0..7 *at the target toward the path start*, −1 on failure (init at 0x06205A).

- **Window**: start tile = (`[0xA14E]`,`[0xA14C]`); window origin = start − 8 (0x061F2F/0x061F38); 16×16 tiles.
- **Cost cache**: 256 bytes at `DS:0xA270`, memset at 0x061FBA..0x061FC7; BFS queues x at `DS:0xA372` / y at `DS:0xA472`, write/read indices `[0x2D16]/[0x2D18]`, capacity 0xE1 (0x062055). The cache is reused when the origin (`[0x2D1A]/[0x2D1C]`) is unchanged (0x061F83..0x061FA9).
- **Neighbour tables** (file 0x1DA54/0x1DA5E): `DS:0xB4` dx = {0,1,1,1,0,−1,−1,−1}, `DS:0xBE` dy = {−1,−1,0,1,1,1,0,−1} — directions 0..7 = N, NE, E, SE, S, SW, W, NW (applied 0x06210D/0x06211B).
- **Ship gating**: moving unit type `[0x1DD2]` in 13..18 (0x061F41/0x061F48) = the ship range; tile is water iff terrain id == 0x19 (25 Ocean) or 0x1A (26 Sea Lane) (0x062174..0x06217C); land/water mismatch allowed only at the endpoints (0x0621BE..0x0621E5); water-water extra checks via the helpers at 0x06219D/0x0621AF.
- **One-move units**: if the type's stored moves ≤ 3 (`byte[0x5234+type·14] ≤ 3` at 0x061F6B) every step costs a flat 3 (0x0622F4).

- **Step costs** (in priority order): road/plow layer mask `0x0A` at both ends → **+1** (`0x0622A7..0x0622C0`); river bit `0x40` on a cardinal step → **+1** (`0x0622CC..0x0622EC`); otherwise **terrain Movement ×3** — `byte[terrain·16+0x2F76]·3` (`0x062300..0x06230C`). NAMES values: open land 1, forests 2, Hills 2, Arctic 2, Mountains 3, Ocean/Sea Lane 1.
- **Occupancy/power rules**: occupying power at the tile must be −1 or the moving power `[0x1DD6]` (`0x062217`); a second ownership probe rejects power ≥ 4 (`0x062245`) and AI-controlled powers, else adds +8 (`0x06225E`); native units (type ≥ 19, `0x0621E8`) reject rumor tiles (`0x0621F5`).
- **Phase 2**: picks the minimum-cost neighbour of the target (cost init 99 = `0x63` at `0x06206B`), **tie-break by distance** (`func_00493C` at `0x06259F`), pruned against the bound `[0xA370]` (`0x06202C`/`0x062062`).

13.2 The movement debug overlays

Three of the seven DEBUG.TXT `@OPTIONS` bits in the cheat bitfield `[0x894]` (builder `func_02356C`, exactly 7 checkbox items) are movement overlays: **bit `0x10` "Close Moves"** (gate at `0x061F14`), **bit `0x20` "Far Moves"** (sibling `func_06295E`, `0x062975`, format `"Far: %d(%d,%d)…"`), **bit `0x40` "All Movement"** (`func_062D84`, `0x062D94`, sets the latch `[0x1DF2]` that Close Moves honours). The Close-Moves overlay draws over the live map view (full redraw via `0x181F:0xE1C(1)`):

- **Per-tile cost numbers**: for each nonzero cache cell, `func_078068` prints the cost in white (`0x0F`): px = `(x+[0x832A]−[0x8328])·[0x5AD4]`, py = `(y+[0x832C]−[0x832E])·[0x8326]+8`, nudged `+7>>zoom, +6>>zoom` with zoom = `[0x184]` (`0x078081..0x0780E3`); at zoom 0 a backing rectangle is drawn behind the digits (`0x07810C`).
- **Summary line**: format `"(%d,%d)-(%d,%d) %d == %d"` (DS:`0x1DD8`, file `0x1F778`) filled with start x/y, target x/y, bound, best direction; colour 12 (red); drawn at **(5,190)** directly to the 320×200 VGA surface (`0x062683..0x0626B1`).
- **Keys**: 'Z' zoom in (`[0x184]--`, clamp ≥0), 'X' zoom out (clamp ≤3), each redraws; ESC clears the latches and exits; any other key exits (`0x0626D0..0x062705`).

13.3 Go-To orders

Order code 3 with the destination at `UnitRecord +0x09/+0x0A` (writer `0x22D38`/`0x22D2D`); per-turn executor `func_040E22`. The destination is chosen with the shared picker (`func_060026 / func_022CDC`): headers `@SAILPORT` (ship) / `@TRAVELPLACE` (land unit); rows are eligible own colonies filtered by water/land region match, plus a Europe row for ships only (choosing Europe issues set-sail); pages of 10 with "(More)".

§14 Combat

Combat resolves one attack with a single roll over modified strengths. The base strength comes from the unit stat table; a chain of byte-cited multipliers (veterancy, terrain, colony, fatigue, difficulty, Sons of Liberty) modifies it; the optional Combat Analysis dialog itemizes exactly those modifiers before the result is applied.

14.1 Base strength — func_007C2A

The root strength calculator writes the per-side base at `[col·2+0x8D06]` (`0x007CA5`): `base = stat_table[type·14 + 0x5235]` (the +5 combat column); **carriers add the +6 attack column** (`+0x5236`); a **damaged ship** (flag bit) takes **−2**. Modifier-flag words per side: primary `F = [col·2+0x8D00]`, secondary `S = [col·2+0xA156]` (cleared at `0x05CB3A`); producers include Veteran at `0x007CD6`, Drake at `0x007CF0`/`0x007D09`, Fatigue (`[0x8D01]|=1`) at `0x05CB9B`, Bombard (`[0x8D01]|=0x80`) at `0x05CF7D`.

14.2 The Combat Analysis modifier table

The dialog (see §14.4) renders one row per set flag bit; the same bits drive the strength math. Complete row table (labels are LABELS.TXT `@MISC` lines):

FLAG	ROW LABEL	EFFECT
`F&0x200`	unit shown under its veteran-profession name	base strength
`F&0x400`	Muskets (good 15 name + icon `0x26`)	"+1" (value semantics unresolved, @`0x05EBDA`)
`F&2`	Veteran (types 1/4, profession `0x15`)	+50%
`F&4`	Cargo	−12.5% per used hold (cargo·100/8, @`0x05ED65`)
`F&0x100 / S&8`	Fatigue	−33% / −66%
`F&1`	Attack Bonus	+50%
`F&0x8000`	Bombard	+50%
`S&2`	Tory Unrest	−(100 − SoL%)
`S&4`	Rebel Unrest	+SoL% (SoL from `func_008524`, `0x181F:0xC86`)
`F&0x80`	Ambush (attacker) / Terrain (defender) — draws the target tile	+terrain_defense·25% (row skipped if 0)
`F&0x40`	Colony	+(fort_level+1)·50% (`0x181F:0x9D2`)
`F&8 (+0x10/+0x20)`	colony-structure row (building name, table DS:`0x9634`)	+n·50%, n = 1/2, doubled by F&`0x20`
`F&0x800 (S-side)`	Artillery In Open	−75%
`S&1`	Artillery Vs. Raid	+100%
`F&0x2000`	Fortified	+50%

FLAG	ROW LABEL	EFFECT
`F&0x1000`	Spain Bonus	+50%
`F&0x4000`	Drake (privateers, Founding Father 13 owned)	+50%
`[0x5383]&0x20 (cheat)`	extra rows: final strengths + the raw roll vs att+def	—

Terrain defense values are the `$TERRAIN` "Defensive" column (byte-verified): open land 0, Marsh/Swamp 1, forests 2 (Rain 3), Hills 4, Mountains 6. The defense-bonus filler `func_007D3E` accumulates into `[0x8D04]`: colony +2 (0x7D8D), fortified building (level ≥ 2) +4 with a doubling condition (0x7DBC/0x7DD1), river/road +(n+1)·2 (0x7E12), open terrain + the per-terrain byte at `terrain·16+0x2F77` (0x7E63).

14.3 The strength chain and the roll — func_05CA7E

Attacker strength `[bp-0x90]` (defender `[bp-0xA6]`) is built from the stat columns and modified, in order:

1. terrain/fort bonus: `strength·([0x8D04]+4)/4 · 3/2` (0x05CE05, `·3/2` chain at 0x05CE16);
2. **difficulty handicap**: a human-controlled combatant gets `strength += (4 − difficulty)` on **both** sides (attacker 0x5CE35, defender 0x5CE54; +4 at Discoverer down to +0 at Viceroy);
3. generic terrain multiplier `·[bp-0x96]/3` (0x05CE69);
4. colony present on the defending tile → +50%, flag `[0x8D01]|0x10` (0x05CF43);
5. War-of-Independence bombardment (endgame flag `[0x5382]&1`, REF defender) → +50%, flag `[0x8D01]|0x80` (0x05CF7D);
6. Sons-of-Liberty scaling `strength·SoL%/100` (0x05CF98);
7. difficulty scaling `strength·difficulty/20` (0x05CFFC);
8. a further strength **doubling at 0x05D0D9 gated on the difficulty byte `[0x53A6]`** — its exact condition is an open item (runtime).

Fatigue is offered *before* the roll via GAME.TXT `@HALF` — attacking with tired troops fights at reduced strength (the −33%/−66% rows above); text verbatim:

```
Your Excellency, these men are tired.  If we force
them to attack this turn, they will fight at {%NUMBER0/3
strength}.

"Charge!"
"Then let them rest."
```

The roll (act mode): `roll = random_int(1, ATK+DEF)` via `func_00C322` (`0x181F:0x4D4`) at 0x05D188; **the attacker wins iff `roll ≤ ATK`** (`cmp roll, [bp-0x90]; jg` ⇒ loss). Evaluate mode instead returns the odds score `(ATK·8)/(DEF+1)` (0x05D032) for AI ranking. The naval unit-vs-unit roll in the consequence applier `func_05B2C2` uses the **raw** ship stat pair `+0x0B/+0x0C` (`0x523B/0x523C`, no modifier scaling): `roll = random_int(1, A+D)` at 0x05B844, with independence-war special cases (0x05B87D/0x05BA2D).

14.4 When the Combat Analysis dialog shows

Gate at 0x05D221 inside `func_05CA7E`, *after* the roll is computed but before resolution renders: Game Options bit **`[0x5383]&2`** ("Combat Analysis" checkbox) AND (attacker human OR defender human OR full-view `[0x53A2]≠0`). Sole call site 0x05D291, 13 arguments (both unit indices, both positions, both owners, both strengths, and the roll). The dialog itself (`func_05E9B0`, page 0x11) is a two-pass measure/draw modal: frame x=53, w=214, h=rows·20+6, vertically centred; title "COMBAT ANALYSIS" (`@MISC` 75); attacker column x=56, defender +80; row pitch 20; values right-aligned at column+0x50; each cell dual-drawn dark/light for a drop shadow; modal-wait terminator.

14.5 Naval prompts

- `@HALF` — the pre-attack fatigue prompt above (also flags the −33% row).
- `@EVASIVE` — posted at 0x05D469 when a ship evades; text verbatim:

```
{%STRING0 %STRING1} evades {%STRING2 %STRING3}.
```

PART V

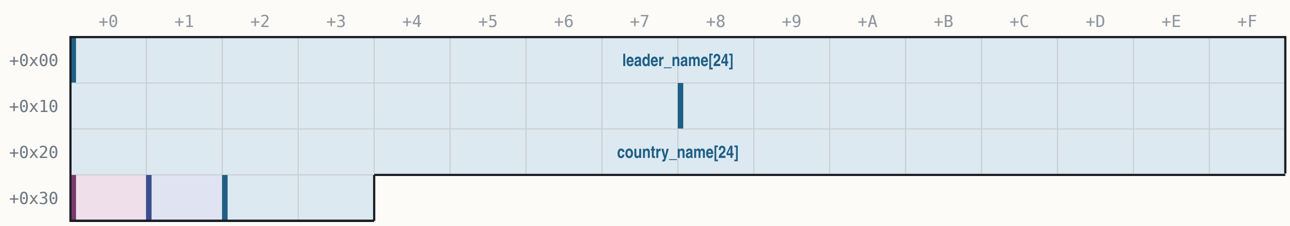
Politics and powers

§15 Powers and relations

Four European powers (0 = English, 1 = French, 2 = Spanish, 3 = Dutch) share the New World with eight native tribes (power ids 4–11) and, late in the game, the King's Royal Expeditionary Force. Per-power state is split across two parallel record arrays — a small 52-byte "personality" record holding names and control flags, and a large 316-byte record holding the economy, diplomacy and Congress state — plus a handful of per-power scalar tables. Everything in this section is the substrate the diplomacy, Congress and revolution machinery of sections 16–18 reads and writes.

15.1 The 52-byte personality record (base 0x540E, stride 0x34, count 4)

AIPersonality DGROUP:0x540E + power*0x34 (runtime BSS; loaded from NAMES.TXT / save)



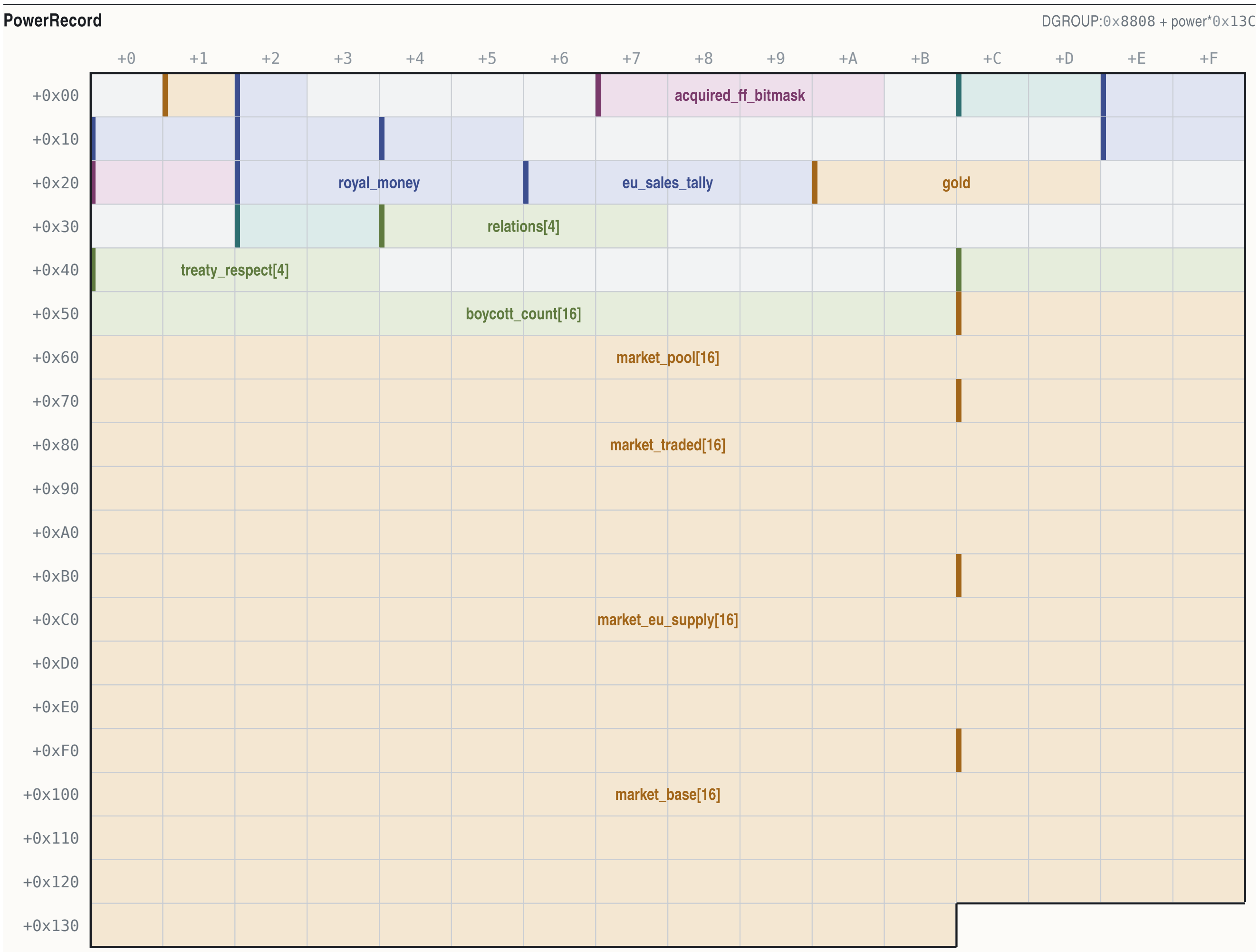
52 bytes (0x34) · mapped 52 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x17	24	char	leader_name[24]	NUL-terminated ("Walter Raleigh"); default for the @LEADERNAME entry box
+0x18..+0x2F	24	char	country_name[24]	region name ("New England"); %STRING source for diplomacy popups
+0x30	1	uint8_t	event_flags	one-time-event bitfield; bit 0x40 test-and-set at 0x61870/0x61877 (burial-ground anger)
+0x31	1	uint8_t	controller	(abs 0x543F+p*0x34): 0 = human, 1 = AI, 2 = eliminated (set at 0x3C91A)
+0x32..+0x33	2	uint16_t	colony_name_seq	default-colony naming counter (INC at 0x4056E, zeroed at 0x75930)

The controller byte at `[0x543F + power*0x34]` is the single most-tested per-power flag in the binary (~218 references): the turn loop skips inactive powers on it (0x57C5/0x58AA), the parley dispatcher's human-only gate reads it (0x57F8C), the Founding-Father cost formula branches on it (0x3C29C), and hot-seat multiplayer writes it from the `@MULTI` checkbox dialog (section 18.6).

15.2 The 316-byte power record (base 0x8808, stride 0x13C, count 4)

Active record far pointer: `[0x84FC]`. The whole 4×0x13C block is serialized verbatim as save block #8 (section 20.3).



316 bytes (0x13C) · mapped 282 · unmapped 34

OFFSET	B	TYPE	FIELD	NOTE
+0x01	1	uint8_t	**tax_pct**	royal tax rate 0..75 (clamp at 0x3434F)
+0x02	1	uint8_t	**rebel_sentiment_pct**	Sons-of-Liberty % (F3 display)
+0x07..+0x0A	4	uint32_t	**acquired_ff_bitmask**	bit f = Founding Father f owned (byte base 0x880F; reader func_00BC10 @0xBC10)
+0x0C..+0x0D	2	uint16_t	**bells_toward_next_ff**	liberty-bell pool; resets on each acquisition
+0x0E..+0x0F	2	uint16_t	**bells_per_turn**	bells produced last turn (zeroed at 0x2F23F each production phase)
+0x10..+0x11	2	uint16_t	**crosses_per_turn**	immigration points per turn
+0x12..+0x13	2	int16_t	**ff_in_progress**	father id being worked toward; 0xFFFF = none (cleared at 0x3BD37)
+0x14..+0x15	2	uint16_t	**founding_father_count**	owned-father count (cost-curve input)
+0x1E..+0x1F	2	uint16_t	**artillery_bought**	Europe artillery escalation counter (read*100 at 0x35124)
+0x20..+0x21	2	uint16_t	**boycott_bitmask**	bit g = good g boycotted after a Tea Party (set 0x34717, cleared 0x33423 / 0x3BD45)
+0x22..+0x25	4	int32_t	**royal_money**	the King's REF fund; +=(8*diff+10) per turn, era-doubled; buys a REF unit at 1800
+0x26..+0x29	4	int32_t	**eu_sales_tally**	cumulative net European sales (0x32A9C)
+0x2A..+0x2D	4	uint32_t	**gold**	treasury, clamp 0..999999 (add helper 0x181F:0xABA)
+0x32..+0x33	2	uint8_t	**home_x, home_y**	sea-lane spawn/arrival coordinates (read 0x58D72; writers 0x418D0/0x65CCB/0x74D74)
+0x34..+0x37	4	uint8_t	**relations[4]**	relation-bit row vs powers 0..3 (matrix base 0x883C; see 15.3)
		+0x34 England +0x35 France +0x36 Spain +0x37 Netherlands		
+0x40..+0x43	4	uint8_t	**treaty_respect[4]**	treaty-respect counters (base 0x8848; see 15.4)
		+0x40 England +0x41 France +0x42 Spain +0x43 Netherlands		
+0x4C..+0x5B	16	uint8_t	**boycott_count[16]**	per-good back-tax accumulator (INC 0x32271 / DEC 0x32288); also market sensitivity load
		+0x4C Food +0x4D Sugar +0x4E Tobacco +0x4F Cotton +0x50 Furs +0x51 Lumber +0x52 Ore +0x53 Silver +0x54 Horses +0x55 Rum +0x56 Cigars +0x57 Cloth +0x58 Coats +0x59 Trade Goods +0x5A Tools +0x5B Muskets		
+0x5C..+0x7B	32	int16_t	**market_pool[16]**	European supply/demand imbalance
		+0x5C Food +0x5E Sugar +0x60 Tobacco +0x62 Cotton +0x64 Furs +0x66 Lumber +0x68 Ore +0x6A Silver +0x6C Horses +0x6E Rum +0x70 Cigars +0x72 Cloth +0x74 Coats +0x76 Trade Goods +0x78 Tools +0x7A Muskets		
+0x7C..+0xBB	64	int32_t	**market_traded[16]**	cumulative traded volume
		+0x7C Food +0x80 Sugar +0x84 Tobacco +0x88 Cotton +0x8C Furs +0x90 Lumber +0x94 Ore +0x98 Silver +0x9C Horses +0xA0 Rum		

OFFSET	B	TYPE	FIELD	NOTE
	+0xA4 Cigars +0xA8 Cloth +0xAC Coats +0xB0 Trade Goods +0xB4 Tools +0xB8 Muskets			
+0xBC..+0xFB	64	int32_t	market_eu_supply[16]	European stock
	+0xBC Food +0xC0 Sugar +0xC4 Tobacco +0xC8 Cotton +0xCC Furs +0xD0 Lumber +0xD4 Ore +0xD8 Silver +0xDC Horses +0xE0 Rum +0xE4 Cigars +0xE8 Cloth +0xEC Coats +0xF0 Trade Goods +0xF4 Tools +0xF8 Muskets			
+0xFC..+0x13B	64	int32_t	market_base[16]	drift accumulator (buy += at 0x323BC, sell -= at 0x32324)
	+0xFC Food +0x100 Sugar +0x104 Tobacco +0x108 Cotton +0x10C Furs +0x110 Lumber +0x114 Ore +0x118 Silver +0x11C Horses +0x120 Rum +0x124 Cigars +0x128 Cloth +0x12C Coats +0x130 Trade Goods +0x134 Tools +0x138 Muskets			

15.3 The relations matrix (PowerRecord +0x34, base 0x883C)

Relations between every pair of European powers are one byte at `0x883C + subject*0x13C + target` (the code addresses it as `[bx+si-0x77C4]`, the wrapped-displacement form of 0x883C — read at 0x58A72). Accessors: `func_007F34` get, `func_007F96` symmetric set, `func_008000` symmetric clear (error strings "Treaty on/off error").

BIT	MEANING	EVIDENCE
0x01	resolved/normalised relationship	set 0x5318F
0x02	at war	set 0x58A7B / 0x59A61 / 0x3F0E8; cleared 0x5DE98
0x08	grievance pending	set 0x3F0D7 / 0x59AE9; per-turn transition to bit 0x01 when the +0x40 timer expires and `random_int(0,3)==0` (0x53165)
0x10	parley cooldown (16 turns; stamp word `[0x53C8+power*2] = turn+0x10` at 0x58075 / 0x5914C)	
0x20	met / contacted	
0x40	peace treaty in force	set both ways 0x59139
0x80	privateer hidden attribution — a privateer (unit type 0x10) attack sets this instead of the war bit (guard 0x3F092, set 0x3F0A1); cleared/revealed 0x58BE1	

15.4 The treaty-respect counter (PowerRecord +0x40, base 0x8848)

A plain byte counter, **not** a second bit matrix. On signing a treaty it is seeded `2*(6 - difficulty)`, halved if the power has Benjamin Franklin (0x59B00–0x59B31); the AI↔AI treaty handler `func_057DC0` writes it 1/0 (0x57EC5 / 0x57F2D). While nonzero, an AI aborts planned attacks on its treaty partner (`func_03ECF0` @0x3F163). The decrement site is unmapped (runtime).

15.5 Per-power scalar tables

ADDRESS	SHAPE	MEANING
0x940C + p	byte	AI **attitude** toward action (parley willingness input; `(attitude>>2)` vs demand at 0x58C24; target eligibility needs ≥ 8 at 0x57B1A)
0x941C + p*2	word	per-power **grievance/finance** word (grievance-score read `[bx-0x6BE4]` in the war-bit path; census writer 0x42245; REF-intervention and succession strength term)
0x942C + p	byte	per-power **strength/coastal-cargo** census total (succession ranking and SMITE-factor input)
0x53C8 + p*2	word	parley-cooldown timestamp (turn + 16)
0x53DA/0x53DC/0x53DE/0x53E0	words	REF Regulars / Cavalry / Man-O-War / Artillery counts
0x53D2	word	the King/REF (or withdrawn) power id; negative = none (tested 0x23930)
0x5394 / 0x5396 / 0x5398	words	current power / render power / human (rebel) power

Power ids 0–3 are the Europeans; ids 4–11 are the tribes in `@TRIBES` order (4 = Inca, 5 = Aztec, 6 = Arawak, 7 = Iroquois, 8 = Cherokee, 9 = Apache, 10 = Sioux, 11 = Tupi — section 19.1). The King's power is not a fifth record: it reuses an eliminated European slot, its id held in `[0x53D2]`.

§16 European diplomacy

All power-to-power diplomacy runs through a single 7-kilobyte dispatcher, `func_057F4E` (file 0x057F4E, `enter 0xD6`), driving a 48-section GAME.TXT family: 42 conversation popups (width 220, rival-leader portrait), 6 announcements/guards (width 190, advisor portrait) and 5 support list-sections. Section names are built at runtime by strcpy/strcat from a fragment pool at file 0x1F250+ ("MEEK" 0x1F250, "MANLY" 0x1F255, "HELLO" 0x1F267, "AHOY" 0x1F26D, "FIRST" 0x1F272, "USA", …), which is why the full names never appear as string literals. Conversations are emitted through `func_06F61C` (0x1A1F:0x688), which sets speaker channel 3 (`[0x1F60]` = power B → portrait sheet MYR0..MYR3.SS); announcements go through `func_06F5F2` (0x181F:0x652, advisor channel `[0x1F5E]` → MSS1/MSS2 portraits).

16.1 Entry chain

```
unit moves onto a foreign unit/colony tile
  func_03ECF0 (unit-vs-tile confrontation resolver)  @0x3F82B ──┐
  func_046FFA (movement processor, unit flag [0x3148]&8) @0x481CB ─┤
                                                                  ▼
  func_059B90 (contact evaluator; sole dispatcher call @0x5A2CE)
                                                                  ▼
  func_057F4E(humanA, powerB 0..3, unit, neighbor table, force)
    human-only gate @0x57F8C (cmp [bp+6],4; controller [0x543F+p*0x34]==0)
    if side A is AI → silently delegates to the AI↔AI ticker func_057DC0 @0x57FA4
```

First European-to-European contact also fires woodcut 10 ("MEETING FELLOW EUROPEANS") at 0x57FDF, and every WAR*/`@MERCENARY` emission is preceded by the war fanfare `func_005108(4)`; first-contact fanfare id is 0x8020+power. String helper `func_057A3A` fills `%STRINGn` from `@GREATKINGS`/`@GREATDEEDS`/`@GREATLEADER`/ `@GREATLEADER2`[power]; `func_057AA2` picks `@MEEKNESS` row 1 "request" / row 2 "demand". Leader name comes from `0x540E+p*0x34`, region name from `0x5426+p*0x34`, and the player's title from the difficulty-rank pointer `[0x8394+diff*2]`. Benjamin Franklin (Founding Father 19) — tested via `func_00BC10(p,0x13)` — halves demands and prices and cancels AI hostility at 6 sites in this dispatcher (e.g. 0x5834E).

16.2 Greetings (@HELLO*)

Key = `"HELLO"` + (not-met ? (ship ? `"AHOY"` : `"FIRST"`) : tone `"MEEK"`/`"MANLY"`); an independent (post-revolution) counterpart selects `"HELLOUSA"`. Key pushes at 0x588CD–0x58923, emit 0x58939. Representative body (`@HELLOFIRST`):

> "Greetings, %STRING0, and welcome to {%STRING1}. We have justly claimed all of this land in the name of {%STRING2}, and we are here to %STRING3. Please do not interfere with this God-given mission."

16.3 Subfamily outcome tables

Row numbering is the 1-based dialog return. "Row 2" is always the second option line of the section text quoted.

Third-party demands — `@APOSTATES` (attack a European treaty partner) / `@HEATHEN` (attack a tribe), key push 0x58989 / 0x589C0, emit 0x58A30:

OPTION	STATE WRITES
row 1 "Never! …"	none (relations with B unchanged; B's attitude worsens via the demand bookkeeping)
row 2 "Yes! We shall crush …"	European target: treaty bit 0x40 cleared + war bit 0x02 set vs the third party (0x58A6A/0x58A7B). Tribe target: tension hit `func_045DF2(tribe, A, +100, 0)` at 0x58A91

Protests — `@PIRACY(USA)`, push 0x58B0D, emit 0x58B45:

OPTION	STATE WRITES
row 1 "What pirates? We have NEVER condoned piracy!"	`hidden-attribution bit 0x80 stays set`
row 2 "Very well, we shall withdraw our privateers to Europe."	`every privateer recalled to Europe and bit 0x80 cleared (0x58B7D–0x58BE1)`

Protests — `@SIEGES(USA)`, push 0x58C99, handler 0x58CD8: row 2 withdraws all units adjacent to B's colonies. **Latent bug 1:** `@SIEGESUSA`'s two option lines are textually swapped (withdraw is row 1 in the text) but the handler acts on row 2 for both sections — so answering "Our forces … shall stay" to an independent power actually executes the withdrawal.

Extortion — `@TRIBUTE(USA)` / `@WANTSTUFFUSA` / `@PROVOKE` / `@WARMANLY` / `@RID(USA)`: the AI accumulates a demand from forces-near-colonies, scaled by difficulty (16.5). Paying transfers gold at 0x58ED0; a goods demand moves colony stock rows at 0x58FB4; refusal escalates to `@PROVOKE`/`@WAR*` ("Prepare for WAR!") and the war bit. **Latent bug 2:** for a non-independent extorter the code builds the key `"WANTSTUFF"` at 0x58F56, but GAME.TXT contains no `@WANTSTUFF` section — only `@WANTSTUFFUSA` exists; the lookup misses.

Treaty and standing-peace menu — `@WORTHY` (demarcation-treaty proposal, 0x5911D→0x59120), `@GIVECASH` (0x591D5), `@PEACE*`/`@OLDPEACE*`/`@PEACEUSA` (0x59150/0x59283/0x59356 → emit 0x59395): the peace menu carries four fixed rows:

OPTION (@PEACEMEEK TEXT)	OUTCOME
"Go in peace, {%STRING1} brothers."	`end parley; treaty set both ways (bit 0x40, verb 0x181F:0xA06 @0x59139) + siege stand-down func_057CE0 + 16-turn cooldown`
"First you must withdraw your forces from our colonies!"	`withdraw branch → @WITHDRAW / @NOTWITHDRAW / @NOTHINGWITHDRAW / @MAYBEWITHDRAW (0x593B7–0x5949B). Withdrawal price = 25*(difficulty+2)*forces, minimum 100, doubled at war, −50 per unit, halved by Franklin`
"How much do you value your worthless lives, heathen swine?"	`threat branch → @GIFTS ("a gift of {%NUMBER0$} in exchange for your continued forbearance", 0x59700) or @THREATS ("We laugh at your feeble threats", 0x59755), possibly @PROVOKE war (0x5974C)`
"We suggest an alliance."	`@MILITARY dynamic-row menu (rows built at 0x5976D over lea 0x19FA, shown via func_06E3D0 @0x59848) → per-target @NOCONTACT / @ALREADYSMITE / @SMITEINDIANS / @SMITEEUROPE / @UNFORTUNATE (0x5989D–0x59A0F). Purchase: B declares war on target T (bits at 0x59A49–0x59A71), player pays B (0x59AC7), @MERCENARY announcement "The {%STRING0} declare war on the {%STRING1}." (0x59AB3). @UNFORTUNATE fires when the treasury (+0x2A/+0x2C, 32-bit compare 0x58E1F/0x58E29) cannot cover the promise`

16.4 The AI↔AI ticker (func_057DC0)

Runs every 3rd turn per met pair when the human dispatcher delegates. Peace resolution emits `@SIGNTREATY` ("The {%STRING0} and {%STRING1} have signed a peace treaty.", 0x57E86, MSS2 advisor), sets bit 0x40 both ways symmetrically (0x57EC5/0x57ED0) and seeds treaty-respect = 1; war emits `@DECLAREWAR` (0x57F18). Willingness gates: turn ≥ 0x28 (40) at 0x57B10 and at least one of the pair's attitude bytes `[0x940C+p] ≥ 8` (0x57B1A/0x57B24). **Latent bug 3:** the had-a-treaty branch pushes the key `"CANCELTREATY"` at 0x57F10, which has no GAME.TXT section (only `@CANCELPEACE` exists) — the announcement is silently lost.

16.5 Demand accumulation and difficulty scaling (inside func_057F4E)

```
grace period   : no AI war/refusal before turn 10*(10-diff)          (0x58374)
demand value   : value * 10*(diff+8) / 100        (×0.8…×1.2)        (0x583A0)
flat surcharge : += 500*(diff+1)                                     (0x5842B)
roll term      : (diff+1)*value >> 3 feeds a 0..400 roll             (0x58409)
attitude term  : += (diff-2)*meeting_value                           (0x58580)
action gate    : random_int(1,1000) < 200*diff + 100  (10%…90%)      (0x58315)
no-action gate : decline when (attitude>>2) > demand AND
                 (demand <= 12 OR random_int(0,4) != 0)              (0x58C24)
afford gate    : demand vs 32-bit gold +0x2A/+0x2C                   (0x58E1F)
```

16.6 Attacking a treaty partner, movement guards, succession

- Human attacker on a treaty partner (`func_03ECF0`): `@HAVETREATY` ("We have signed a peace treaty… / Cancel Action. / Break Treaty.") at 0x3F130 — row 2 sets the war bit (0x3F298), clears the treaty (0x3F2A5) and continues, then the `@CANCELPEACE` announcement (0x3F22F). AI attacker → `@DECLAREWAR` (0x3F262); human victim → `@SNEAK` "Sneak attack by the treacherous {%STRING0}!" (0x3F1B4). A second `@HAVETREATY` site exists in the order-issuing flow `func_021FF2` @0x220CE (UI trigger unmapped).
- Movement guards: `@NOWARSDURINGREV` ("Foreign colonies cannot be attacked during the {War of Independence}.") emitted at 0x5A912 inside the foreign-colony attack handler `func_05A862`, reached only under the war-declared gate `test [0x5382],1` @0x5A8C8; `@TRADEATWAR` at 0x5A458 and the Jan-de-Witt gate `@TRADEMERCANTILISM` (Founding Father 4) at 0x5A469 in the foreign-colony trade entry `func_05A40E`.
- `@SUCCESSION` (War of the Spanish Succession, `func_03C638` @0x3C76A, MSS2 advisor): the whole-map power merge of section 18.7. Skipped in multiplayer (gate 0x3C63D).

§17 Congress, bells, and Founding Fathers

Liberty bells produced by colonies accumulate per power toward the next session of the Continental Congress, which appoints one of 25 Founding Fathers. Fathers are permanent: nine apply a one-time effect on acquisition; the rest are continuous gates tested at each affected mechanic via the owned-bit reader `func_00BC10`. The F3 advisor report ("CONTINENTAL CONGRESS ACTIVITIES") shows the running totals.

17.1 Bell accrual

The per-power production phase `func_02F052` first zeroes `bells_per_turn` (PowerRecord +0x0E := 0 at 0x2F23F), then loops all colonies owned by the power and runs the colony-turn processor `func_02D658` on each. Its sole Congress call site, 0x2D6A7, invokes the driver `func_03C322(nation, bells)`: bells accrue into `[0x84FC]+0x0C` (pool) and +0x0E (per-turn display). If no candidate is selected (`+0x12 < 0`) the pick dialog runs; when the pool reaches the cost the acquisition runs.

17.2 The bell-cost formula (func_03C282, file 0x03C282..0x03C322)

```
diff = [0x53A6]; year = [0x538A]; ff = PowerRecord[power]+0x14

if power < 4 and controller [0x543F+power*0x34] == 0:   # human European
    cost = (diff + 3) * 16                               # 0x3C29C
else:                                                    # AI
    cost = (14 - diff) * 8                               # 0x3C2A9
for gate in (1600, 1650, 1700, 1750):                    # 0x640/0x672/0x6A4/0x6D6
    if year >= gate: cost += cost >> 1                   # each gate compounds x1.5
cost = (ff + 1) * cost + 1                               # 0x3C302
if ff == 0: cost >>= 1                                   # first father half price (0x3C30B)
if [0x5382] & 1:                                         # after declaring independence
    cost = diff * 1500 + 2000                            # 0x3C30D (diff*0x5DC + 0x7D0)
```

Cross-check: a human at difficulty 1 holding one father, pre-1600, gives `(1+1)*((1+3)*16)+1 = 129` — the observed "Brewster next = 129". The F3 subtitle "(NN in MM)" is `NN = cost - pool`, `MM = cost` (F3 body call at 0x37B08); there is no graphical progress bar anywhere in the game.

17.3 The pick dialog (@WHICHFREEDOM)

Candidate build `func_03BFD2`: the father table is the runtime array 0x9652, stride 6, loaded from NAMES.TXT `@FATHERS` (25 rows: +0 name id, +2 category, +3/+4/+5 era-weight bytes). The era band is year <1600 / 1600–1699 / ≥1700 (`func_03B95A`, year gates at 0x3B963). For each of the 5 categories (Trade/Exploration/Military/Political/Religious, names from NAMES `@FOUNDING`) one candidate is drawn by weighted random over the un-owned fathers with nonzero current-era weight — `budget = random_int(1, Σweights)`, subtract-walk until ≤ 0 (0x3BFFC–0x3C046). An empty category produces no row.

The dialog (width 190) lists up to five rows "FATHERNAME (Category Adviser)"; row id = category+1. It **cannot be cancelled** — a result ≤ 0 re-shows the dialog (0x3C231). Right-click/help (`[0x1F68]`) opens the Colonizopedia FATHER page for the candidate (0x3C24E), then re-shows. The choice is stored to `PowerRecord +0x12` (0x3C269). The AI picks its category via `func_03BA5A` (internals unmapped).

17.4 Acquisition flow (func_03BC42)

On reaching the cost: the owned bit — bit `(f & 7)` of `[0x880F + nation*0x13C + (f>>3)]` — is set, and the first-owner byte `[0x53A9 + f]` records which power got the father first. Player flow: the `@FREEDOM` popup ("%STRING1 Founding Fathers announce that {%STRING0} has joined the Continental Congress!") → the congress splash `func_03BB4A`: full-screen CCBKGD.PIK (asset id 0x1253 pushed at 0x3BB6A, no frame/title/OK chrome), portraits drawn **without** the new father, present, then the bit is set and the screen redrawn — the new portrait "lights up" — with sound 8 and a wait-key (0x3BC14); then the pedia FATHER page (0x3BD26). Bookkeeping: `+0x14++`, `+0x12 := 0xFFFF`. Portraits are the 25 sheets CC-00..CC-24.SS (1:1 with `@FATHERS` order); each owned portrait is blitted at the coordinates baked into its own sheet frame-0 descriptor (`es:[bx+0x46/0x48]`) — positions live in the art, not the code.

Per-father **instant** effects applied by the acquisition dispatcher:

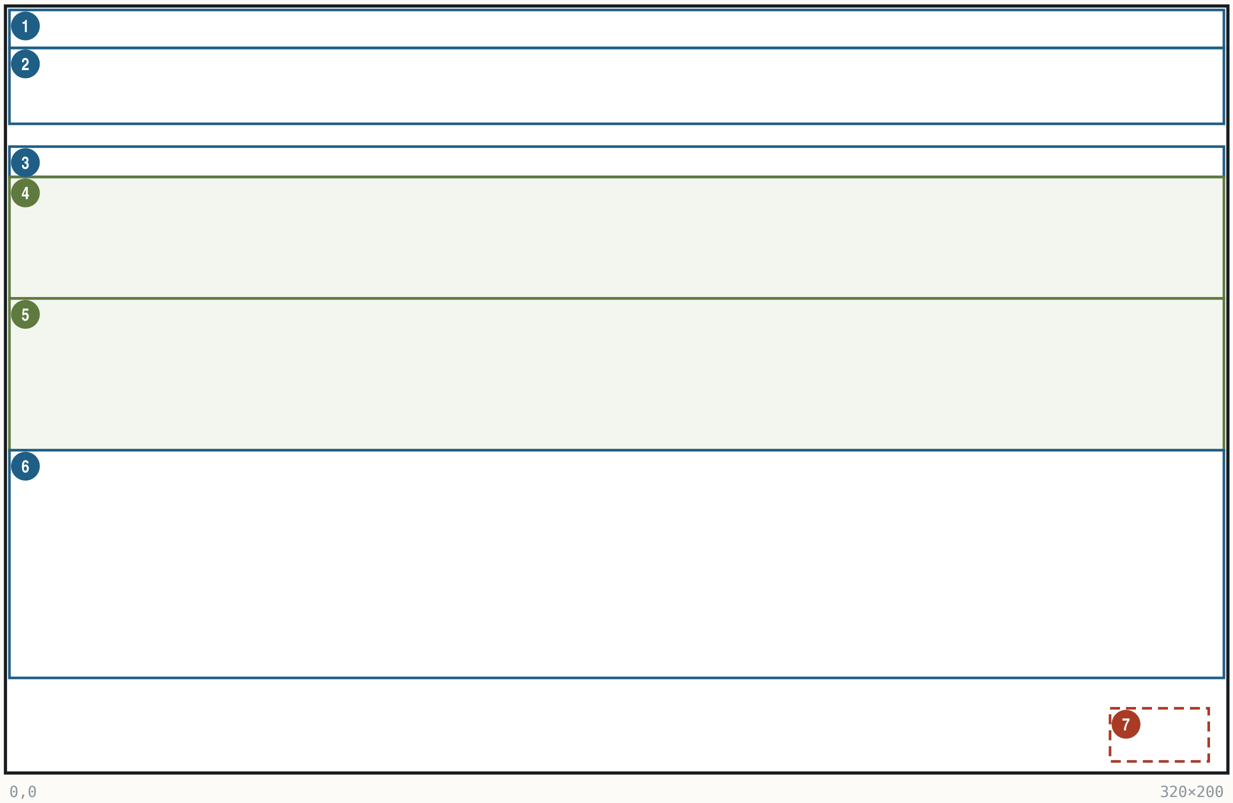
ID	FATHER	ONE-TIME EFFECT (SITE)
1	Jakob Fugger	`clear all boycotts: PowerRecord +0x20 := 0 (0x3BD45)`
6	Francisco Coronado	`reveal every colony on the map (0x3BF54)`
9	Sieur de La Salle	`free Stockade for own colonies of size ≥ 3 (0x3BD4A)`
14	John Paul Jones	`spawn a free Frigate, unit type 0x11 (0x3BD8B)`
16	Pocahontas	`reset all native attitudes to content (0x3BDDD)`
18	Sim n Bol var	`revolution meter [0x53D0] += 20, cap 100 (0x3BE64)`
20	William Brewster	`dock pool: Petty Criminals/Indentured Servants → Free Colonists (0x3BF85)`
22	Jean de Br beuf	`all own missions become expert: settlement +0x05 |= 0x10 (0x3BE77)`
24	Bartolom de las Casas	`all own Indian Converts (class 0x1B) → Free Colonists 0x1C (0x3BEB2)`

All other fathers are continuous gates tested at their own mechanic (e.g. Washington auto-promotion 0x5C758, Franklin's six diplomacy sites, Penn crosses ×1.5 at 0xA16B, Jefferson bells +50% via owned-bit 0x0F).

17.5 The F3 Continental Congress screen

Reached from REPORTS → F3 (menu letter 'B') and as the post-acquisition report. Body `func_037A20` (file 0x37A10..0x3807D); overlay on CCBKGD.PIK.

The F3 Continental Congress screen 320×200 · Mode 13h · band rects (measured; not byte-cited), text params byte-cited



#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×10	text	Title: CONTINENTAL CONGRESS ACTIVITIES	`fill (0,0,320,5) c=0x90; centered`
2	(0,10) 320×20	text	Next Session subtitle: (<FF>) (NN in MM)	`text-only progress`
3	(0,36) 320×8	text	Rebel/Tory Sentiment strip	`PowerRecord +0x02`
4	(0,44) 320×32	art	Bell row (bells/turn)	`sprite 0x3F filled / 0x38 empty, span 300`
5	(0,76) 320×40	art	REF rows (2 x 4-column count badges)	`counts 0x53DA..0x53E8; icons runtime BSS [0x52xx]`
6	(0,116) 320×60	text	Founding Fathers list (plain text)	`owned-father names, marker sprite 0x61`
7	(290,184) 26×14	hit	OK	`dismiss`

Fonts and inks: whole body FONTTINY; title color `0x90` (pale yellow), body `0x92` (bright yellow) against the CCBKGD palette; left margin x=4, y seed 25 (`0x37A4E`/`0x37A49`), line pitch = glyph height 6 + 2 = 8 px (`0x37BD1`–`0x37BDB`). The bell row uses the shared proportional count-strip verb `0x181F:0x236` (`func_002EE4`): `stride = (300 - sprite_w)/(count-1)` clamped to `[1, sprite_w+1]` — many bells overlap toward 1 px pitch; fullness is filled-vs-empty sprites, never a gauge. REF land badges at `0x37E1C` (counts `[0x53DA]`/`[0x53DC]`/`[0x53E0]`/`[0x53DE]` — the screen draws Artillery before Man-O-War), war-stage naval badges at `0x37EFE` (counts `[0x53E2]`..`[0x53E8]`); each row is a 4-column proportional badge layout (open `0x181F:0x218`, flush `0x22C`, span 300). No US-flag sprite is drawn anywhere in the F3 body or the reveal popup (byte-verified negative).

§18 Revolution, the King, and multiplayer

The endgame pivots on one meter, one flag word and one power id: the national Sons-of-Liberty meter `[0x53D0]` (0..100), the game-state bits `[0x5382]`, and the King/REF power `[0x53D2]`. Declaring independence flips `[0x5382]` bit 0, turns the Crown's standing Expeditionary Force into an on-map power, and is won by attrition, not by timer.

18.1 The revolution meter [0x53D0]

Initialized 0 at new game (`0x75620`); Bol var adds +20 capped at 100 (`0x3BE64`); the king's tax-severity score reads it (`0x361F9`). The endgame dispatcher at `0x2391C` clamps it to 75 (`0x2392A`) and routes: below 75 with `[0x53D2] < 0` → the Spanish-Succession arm; at/above 75 → the revolution handlers. In hot-seat multiplayer (`[0x5381]&0x80`) the pre-war state is held at this 75 cap and the auto-revolution arms are suppressed.

18.2 Declaring independence

EVENT_ID	STRING_KEY	TRIGGER	CONDITION	OPTIONS	OUTCOMES	ARMS
declare-1	`@ALREADYREVOLUTION`	Declare-Independence command → `func_03E984`	`[0x5382]&1` already set (0x3E988)	—	none (return)	—
declare-2	`@TOOTORY`	same	`[0x53D0] < 50` (cmp 0x32 at 0x3E99E)	—	shows "Only {%NUMBER0%%} of the colonists support the independence movement… We cannot start a rebellion against the King until the {majority} is behind us."; return	—
declare-3	`@MULTIREV`	same, hot-seat only	`[0x5381]&0x80` (0x3E9C5)	"Declare independence" / "Never Mind"	on confirm: `and [0x5381],0x7F` (0x3E9D3) — the game demotes to single-player exactly as the text warns	falls through to declare-4
declare-4	`@DECLARE`	same	SoL ≥ 50	"Never! That would be treasonous! God save the King!" / "Yes! Give me liberty or give me death!"	rebel power `[0x5398] := [0x5394]` (0x3E9D8); on row 2 → `func_03DE46` (0x3EA06)	declaration
declare-5	`@INDEPENDENCE`	`func_03DE46`	–	—	`[0x5382] |= 1` (0x3E031); declaration year to `[0x53A7]/[0x53A8]` (0x3DE65); initial REF dispatch	war

18.3 Game-state bits [0x5382] (word; save block #3)

BIT	MEANING
0x0001	War of Independence declared (stage 1)
0x0002	foreign intervention active (stage 2; set by `func_03D948` @0x3DA22)
0x0008	independence WON (set 0x2F55A; gates the score bonus)
0x0010	REF-arrival phase flag (set 0x2FAE0); also gates the Congress driver off
0x0020	forced/end-stage flag (set by the dispatcher once SoL≥75+declared+intervention, and by `@FORCED` stage d)
0x0080..0x8000	the Game Options word (high-byte rows: Tutorial Hints 0x0080, Water Cycling 0x0100 inverted, Combat Analysis 0x0200, Autosave 0x0400, End of Turn 0x0800, Fast Slide 0x1000, cheat master 0x2000, Show Foreign 0x4000, Show Indian 0x8000)

Victory: the per-turn resolver at 0x2F464 (runs while bit 0 set, bit 3 clear) counts surviving King-owned units of types {6, 8, 0xB}; when the count falls below the threshold (1 normally, 8 when bit 0x40 is set; 0x2F4D2–0x2F4E5) and the intervention tally clears, `or [0x5382],8` (0x2F55A) — the rebels win, the message built from the rebel record `[0x5398]*0x34+0x540E` (0x2F510). If independence is won and was declared before 1780, the score gains `(1780 - declaration_year) * 2` (0x3A609).

18.4 The King's power and the @FORCED staged advancer

The REF exists pre-war only as the four count globals (0x53DA..0x53E0), seeded at new game by `func_0755CC` (`8d+15 / 5d+5 / 3d+2 / 6d+2` for difficulty d) and grown by `func_03E162` (royal fund +0x22 accrues `(8*diff+10)*2^era` per turn; at ≥ 1800 buy one unit, subtract 1800, slot by ratio). At the war transition the Crown becomes a real on-map power whose id lands in `[0x53D2]`.

Cheat id 0x68 "Advance Revolution Status" (handler 0x2391C, DEBUG.TXT `@FORCED`) advances one stage per invocation and documents the staging exactly:

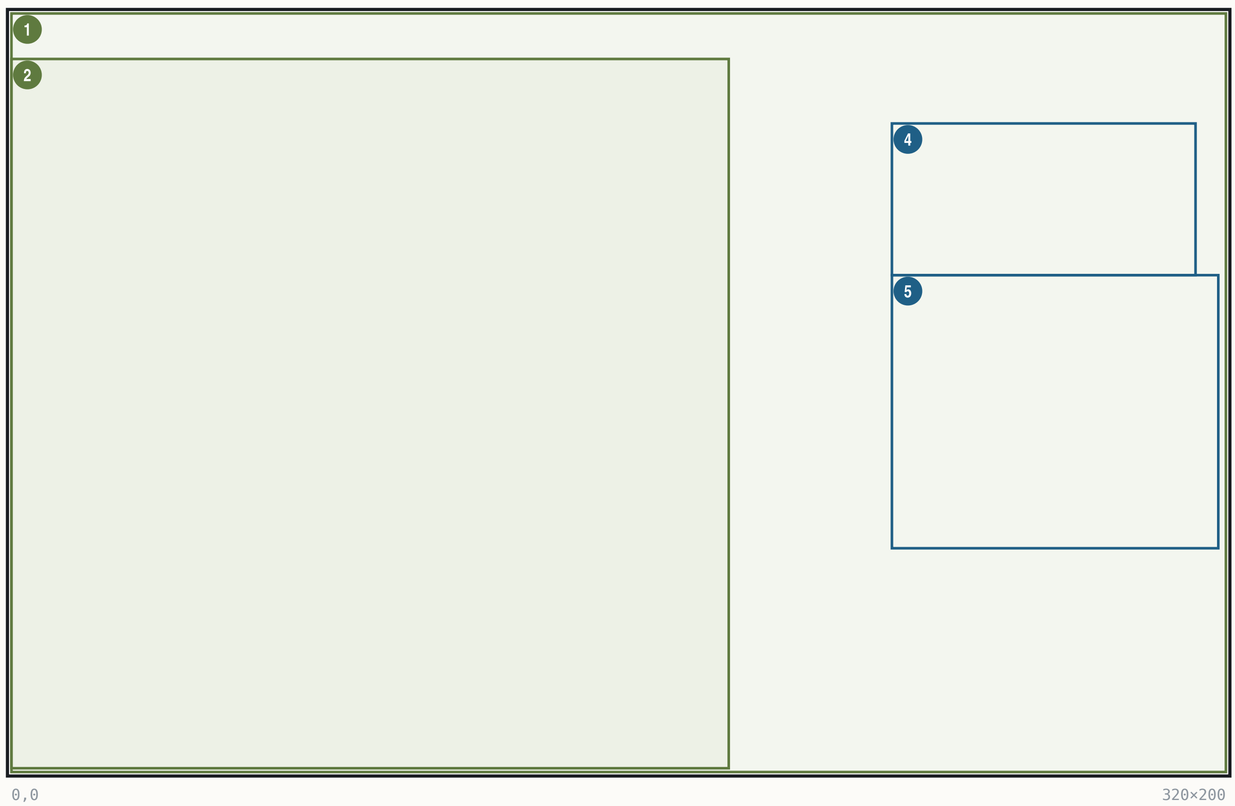
STAGE	CONDITION	ACTION
a	`[0x53D0] < 75`	set meter to 75 + create the REF power if none (0x191F:0x364 → `func_03C638`)
b	–	declare independence (0x191F:0x356 → `func_03DE46`, `[0x5382]|=1`)
c	–	next war stage (0x191F:0x348 → `func_03D948`, `|=2`)
d	–	`[0x5382] |= 0x20` + show the `@FORCED` text

Stages b–d are blocked while `[0x5381]&0x80` (multiplayer).

18.5 The King audience screen

One renderer, `func_075352` (file 0x075352), paints the audience/tax-demand, the loss and the win screens. Assets: backdrop **KINGLSS1.PIK** (throne room, empty chair, blank scroll); outcome-selected foreground sheet **KING1.SS** (audience) / KINGLOSE / KINGWIN — the king-and-dog figure, 189×187 — plus the nation banner sheet (nation stem + digit, e.g. ENGLND1.SS, the throne-canopy banner). Variant select at 0x75430 from the two stack args; nation prefix from `[0x5398]` (switch 0x753BB: ENGLND/FRANCE/SPAIN/DUTCH). Callers: audience sequencer `func_075594` (pushes 1,1 → KING1), and the King-event orchestrator `func_02F3A2` (loss at 0x2F552, win at 0x2F6A8).

The King audience screen 320×200 · Mode 13h



#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×200	art	KINGLSS1.PIK throne room	full-screen backdrop
2	(0,12) 189×187	art	KING1/KINGLOSE/KINGWIN.SS king	bottom-anchored to row 199 (frame-descriptor anchor)
3	(32,0) ·×· R	art	ENGLND<d>.SS canopy banner	nation-selected; size from sheet
4	(232,29) 80×40	text	speech header, 4 lines	per-line centered on x=271.5, tops y=29..61 (measured; not byte-cited)
5	(232,69) 86×72	text	speech body, 9 lines	left x=232, pitch 8 = FONTKING H+1 (measured; not byte-cited)

FONTKING is loaded at 0x754F6; the pen stores (242,47) at 0x75526/0x7552C are engine register values, not the on-screen origin — the glyph runner re-lays the text under mode flags `[0x1F56]|=0x18` (0x75538). Body text is runtime-built by `func_02F3A2` from the GAME.TXT tax family; the pen/ink is restored to the FONTINTR color on exit (0x75576). Fade-in via the palette verb at 0x75553.

Tax-event branch keys (GAME.TXT, `@width=190`, speaker channel `[0x1F5C]=8` → KING1.SS): `@KINGTAX` ("...we have graciously decided to raise your tax rate by {%NUMBER0%%}. The tax rate is now {%NUMBER1%%}. If you wish, you may kiss our royal pinky ring."), `@KINGRAISE` (punitive raise for demanding lower taxes), `@KINGLOWER`, `@KINGNOTHING` ("...You may, however, kiss our royal pinky ring."), `@MERCANTILISM`, `@PURCHASETAX`, pretexts `@KINGWIFE`/`@KINGWAR`/`@KINGNAVACT`/ `@KINGSTAMPACT` (severity-selected, section on taxation), options `@TAXOPTIONS` ("Kiss pinky ring." / "Hold '{%STRING3 Party}.'") and `@TEAPARTY`.

18.6 Hot-seat multiplayer

- **Unlock:** environment `SET COLONIZE=MULTI` (`getenv` pair at DGROUP 0x2064/0x206D) → `[0x201E]=1` (0x70EDD–0x70EFB; also a CLI switch 0x70D0E), adding a 5th game-start menu entry (mode 4, 0x70B9F).
- **`@MULTI`** (new-game setup `func_07431E` @0x74531): "Select powers to be controlled by human players." — a checkbox dialog; each checked power gets controller `[0x543F+p*0x34] = 0` (read from the checkbox bitmask `[0x1F54]`); more than one human sets the hot-seat flag `[0x5381] |= 0x80` (0x745D5); none checked defaults to England.
- **`@MULTINEXT`** (turn loop @0x597C): between human turns the screen blanks and shows "^ ^ {%STRING0} Player Turn ... Press any key for {%STRING0} player's turn." (advisor 2), then the view power switches and re-centers.
- Consumers of `[0x5381]&0x80`: rebel sentiment clamped to 75 with auto-revolution suppressed (0x2391C region); `@FORCED` stages b–d blocked; the Spanish-Succession merge skipped (0x3C63D); `@MULTIREV` demotes the game to single-player on a confirmed declaration (18.2).

18.7 The War of the Spanish Succession merge

Handler `func_03C638` (event id 0x68 in the master event dispatcher; gate: `[0x53D0] < 75` clamped, `[0x53D2] < 0`, single-player only). It ranks the four powers by `3*[0x9418+p] + 2*[0x9298+p] + [0x9410+p]` (0x3C655–0x3C66B, sort 0x3C68E), picks the weakest eligible AI as ceding and the strongest as beneficiary, emits `@SUCCESSION` ("War of the Spanish Succession ends in Europe! {%STRING0}, ravaged by war, agrees to cede %STRING1 to the {%STRING2}. Treaty of Utrecht specifies that all {%STRING3} possessions in the New World now fall under {%STRING2} rule.", 0x3C76A), then rewrites every map-tile owner nibble (0x3C7AF–0x3C80C), every unit owner (0x3C81D), every colony owner (+0x1A, 0x3C8A0), and a third owner-nibble table (0x3C8CA–0x3C8FE). Aftermath: the ceding power's controller := 2 "eliminated" (0x3C91A) and its id stored to `[0x53D2]` (0x3C922); the power thereafter renders as "(Withdrawn from New World)" (LABELS `@MISC`).

§19 Natives

Eight tribes populate the map with individually tracked villages. The engine keeps three layers of native state: an 78-byte per-tribe record, an 18-byte per-village record, and two per-power anger signals (a per-village alarm word and a 0..100 tension meter). Village interaction — trade, missions, training, tribute, war — flows through a 10-entry action menu and the GAME.TXT `@CHIEF*`/`@VILLAGE*`/`@INDIAN*` families.

19.1 Tribe records and ids

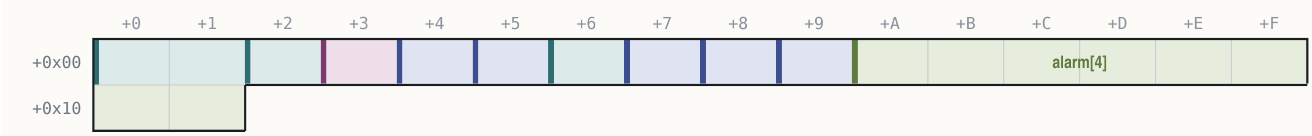
TribeData: base 0x5AD6, stride 0x4E (78), 8 entries, populated at game init from NAMES.TXT `@TRIBES`. Field +2 is the settlement-size/level factor (CHIEFKILL byte-trace at 0x4AB24); the byte at +3 carries the **tribe-dead bit 0x80** — the cheat-menu tribe list tests `[0x5AD9 + i*0x4E] & 0x80` to grey out dead tribes. Village owner ids 4..11 follow `@TRIBES` order:

ID	TRIBE	GIFT GOOD	LEVEL	SPRITE
4	Incas	Jewelled Relics	3	97
5	Aztecs	Gold Bars	2	149
6	Arawaks	Bone Jewelry	1	54
7	Iroquois	Wood Carvings	1	87
8	Cherokee	Turquoise	1	67
9	Apache	Beads	0	111
10	Sioux	Beads	0	118
11	Tupi	Gems	0	71

(Reserve name-only tribes — Maya, Toltecs, Kiowa, … — follow in `@TRIBES` but never instantiate.) Tribe indices 0/1 = Inca/Aztec select the special first-contact woodcuts below.

19.2 The village array (base 0x54EC, stride 0x12)

NativeSettlement DGROUP:0x54EC + v*0x12



18 bytes (0x12) · mapped 18 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	uint8_t	**map_x, map_y**	
+0x02	1	uint8_t	**owner**	tribe/power id 4..11 (scans at 0x37638/0x45D11/0x46078)
+0x03	1	uint8_t	**flags**	bit 0x04 capital (set 0x66225, doubles value 0x7DCA);
+0x04	1	uint8_t	**population**	size (CHIEFKILL input)
+0x05	1	uint8_t	**mission**	0xFF none; low nibble = owning power; bit 0x10 expert-mission
+0x06	1	uint8_t	**growth_counter**	(runtime cross-ref; no static reader)
+0x07	1	uint8_t	**trespass**	escalation counter (0xFE on trespass 0x4A337; bumped on trade 0x5C3F2)
+0x08	1	uint8_t	**last_bought**	cargo id of last good bought
+0x09	1	uint8_t	**last_sold**	write-only in the static image (init 0xFF at 0x46EB6)
+0x0A..+0x11	8	uint16_t	**alarm[4]**	per-European-power anger words (indexed settlement*9+power @0x4734E)
	+0x0A England +0x0C France +0x0E Spain +0x10 Netherlands			

NativeSettlement +0x03 — flags

bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0
0x80	0x40	0x20	0x10	0x08	**0x04**	**0x02**	0x01
					capital	taught	w/o

capital — set @0x66225; doubles value @0x7DCA
taught — already taught (set @0x4A78A)
w/o — write-only bit

NativeSettlement +0x05 — mission byte 0xFF = no mission

bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0
0x80	0x40	0x20	**0x10**	**0x08**	**0x04**	**0x02**	**0x01**
			expert	owning power			

owning power — European power holding the mission
expert — expert-mission doubler (Brebeuf; set @0x48C81, tested @0x57300)

The per-power **anger words** at `+0x0A + power*2` drive the war state at alarm ≥ 128 (raid scan 0x4734E; placement gates 0x4CAD7, 0x53D4E). A parallel 0..100 **tension table** at 0x5B1C (stride 39 words per village row, only columns 0..3 used) is written solely by the applier `func_045DF2`: `tension += delta`, clamped [0,100], with positive deltas halved for the French power (0x45E21) and for Pocahontas owners (0x45E30); thresholds 75 = hostile, 100 = war (0x45E9E/0x45EB2). Notable deltas: ±1 per-turn drift, +1/+2/+3 trespass, −4 successful trade (0x5C41E), +100 incite/burial-ground desecration (0x486F8/0x61B84), mission established negative (clamped so tension ≤ 70).

19.3 First contact

First contact with a tribe (handler `func_056C3E` @0x56DA6) shows woodcut 3 "MEETING THE NATIVES" — or woodcut 4 "THE AZTEC EMPIRE" (tribe 1) / woodcut 5 "THE INCA NATION" (tribe 0), with tune cues 0x33/0x35/0x36 — then the `@INDIANWELCOME` treaty offer:

> "The {%STRING0} tribe welcomes you. We are a glorious nation of {%NUMBER0 %STRING1}. To celebrate our friendship, we generously offer you the land you now occupy as a gift. Will you accept our treaty and live with us in peace as brothers?" — Yes / No

19.4 Village visits

Entering a village (woodcut 7 on the first) offers the NAMES `@ACTIONS` menu: Trade With Village · Enter Hostile Village · Establish Mission · Denounce Heresy of %Fs Mission · Live Among The Natives · Ask to Speak With Chief · Incite Indians · Demand Tribute · Attack Village · Cancel Action.

- **Supply/demand**: the trade pricing, the "especially interested in …" line and the `@INDIANBEGFOOD`/`@INDIANGIVEFOOD` food events are all driven by the village supply/demand routine `func_048F34` — section 10 documents it in full (phases, capital ×2 boost, consumers).
- **Training** ("Live Among The Natives"): only outdoors skills are learnable; Petty Criminals are refused (`@LEARNCRIMINAL`); masters are refused (`@LEARNMASTER`); each village teaches once — the grant at 0x4A782 writes the profession into the unit's expertise byte and stamps the village flag +0x03 bit 0x02 (`@LEARNALREADY`). Unskilled colonists succeed on `random_int(1,1000) ≥ 200*difficulty + 100` (0x4A72C), i.e. 90/70/50/30/10 %.
- **Chief audience** (`@CHIEFHOWDY`/`@CHIEFGIFT`/`@CHIEFAREA`/`@CHIEFGUIDES`): gift beads scaled by tribe, map-area reveal for scouts; `@CHIEFKILL` is the taboo execution outcome of razing.

19.5 Missions

`Establish Mission` places a mission: village byte +0x05 = owning power, with bit 0x10 (expert) set when the founding power has Jean de Br beuf (`has_father(0x16)` gate at 0x48C71 → `or [bx+5],0x10` at 0x48C81); acquiring Br beuf retroactively upgrades all own missions (0x3BE77). Conversion (`func_0572E6`, "INDIANSCONVERT"): each eligible turn rolls `random_int(0,15)` against `threshold = TribeData[+2] + 2`, doubled by the expert bit — success spawns an Indian Convert (class 0x1B) at the colony (0x57374). A destroyed mission or expelled missionary applies a computed positive tension delta (0x57267).

19.6 Attitude and anger displays

The village attitude phrase is built from a colonial-presence score banded at cutoffs −5 / 0 / 10 into Content / Uneasy / Restless / Angry (builder 0x48B62–0x48B90); War is the separate alarm ≥ 128 state. The per-power European attitude byte `[0x940C+p]` is a distinct diplomacy signal (section 15.5). Pocahontas resets all village attitudes to content on acquisition and halves subsequent tension rises. Debug bit 0x01 of `[0x894]` ("Anger & Friction Levels") overlays the live anger word — white number `[v*0x12 + 0x54F6 + 2*viewpower]` at village pixel (+2,+9) (0x4241) and appends eight per-tribe rows to the map info panel (0x44303).

19.7 Village destruction (and the Kill-Indians cheat)

Razing (`func_04A7CA`) rolls a village-escape check `random_int(0, 40*scout+100)` (scout = Seasoned-Scout attacker bonus) with re-rolls biased by size; on a raze the treasure is `(Σ 3×random_int(0,10-diff)) * random_int(0,6) * 4 * (tier+1)`, credited straight to the attacker's gold (+0x2A add/adc at 0x4AB66). The cheat-menu item 0x67 "Kill Indians" (0x23BDC) exposes the same internals: it builds the live tribe list from TribeData (stride 0x4E, dead bit 0x80 at +3) and calls `func_046FC2`, which destroys every village whose owner equals tribe+4 — confirming the owner-id convention and the destroy path used by combat.

§20 Turn flow and persistence

A turn is one pass of the resident loop `func_005760` (file 0x5760): for each of the four European powers in strict index order it runs King → Orders → Production → Diplomacy → Periodic, then a once-per-turn year-advance and autosave tail. Natives are not a separate top-level pass — their AI runs inside the per-power processing. Saves are verbatim memory dumps: 43 raw DGROUP blocks behind a "COLONIZE" magic, no compression, no reordering.

20.1 The per-power phase chain

PHASE	FUNCTION	CONTENTS
1 King/mercenary	`func_03E664` (call 0x58E2; gated `[0x5382]&1==0`)	peacetime mercenary roll (0x3E707) and King events
2 Orders/movement	`func_024A48` (0x58E7)	per-unit orders pump; REF fund accrual `func_03E162` rides here (via 0x24B42); AI unit moves `func_04E2D6` with the contact evaluator `func_059B90` firing diplomacy on unit-vs-tile encounters (section 16.1)
3 Production	`func_02F052` (0x59EA)	zeroes bells/turn (+0x0E at 0x2F23F); per-colony turn processor `func_02D658` — yields, food/starvation/spoilage report popups, school teaching, bell accrual into the Congress driver `func_03C322` (0x2D6A7)
4 Diplomacy	`func_052F7E` (0x5A37)	king-action dispatch `func_034C24` (tax raises are event-driven here, not periodic); AI diplomacy
5 Periodic/congress	`func_02F3A2` (0x5AE5)	colony stats `func_042138`, Founding-Father congress `func_03B2F8` (gated `[0x5382]&0x10==0`), King defeat/victory screens (0x2F552/0x2F6A8)

The **market drift** is an end-of-turn phase of its own: the end-of-turn processor `func_0755CC` calls the drift driver `func_036574` at 0x757B0, which clears the per-power 16-good accumulators and runs the four-power loop into the drift function `func_0305A8` (price base relaxes by `(base + Σ clamped trade)/256` per good). Immigration crosses (`func_035D9A`) run immediately after the price recompute (0x363E2), followed by the religious-unrest arrival chain (`@UNREST`).

Year cadence (loop tail 0x5A9D–0x5ACC): `inc [0x538E]` every turn; before 1600 one turn = one year; from 1600 the season word `[0x538C]` toggles Spring/Autumn and the year steps every second turn; start 1492, forced-end check at 1725 (0x5BB5 sets `[0x82B]=1`).

20.2 Autosave

Gated by Game-Options bit 0x0400 (row 5 "Autosave") and suppressed while the autoplay/suppressor flag `[0x826]` is nonzero (gate 0x5AD7). Turn-loop consumers 0x58D7/0x5A29 call the helper at 0x5642: a **rolling autosave to slot 9 every turn**, plus a **decade autosave to slot 8** when the year is divisible by 10 — matching the manual's "most recent save in the last slot, previous decade save beside it". A further end-game save fires near the forced 1725 end (0x5BDB, gated `[0x82B]`). Manual slots are chosen in the `@SAVEGAME` dialog; filenames are COLONY<slot>.SAV (stem at file 0x1FA82, extension at 0x1FA89).

20.3 The save file (serializer func_0734F8, loader func_073BB0)

Header: the 8 bytes `"COLONIZE"` + 0x1A, mode "wb". Body: **43 raw DGROUP blocks**, each a single `fwrite(base,1,size)` — on-disk offset = sum of the preceding block sizes; within a block, layout = the runtime struct. The loader is a 1:1 mirror (43 `fread`s), proving no compression or field reordering. Principal blocks:

#	BASE	SIZE	CONTENT
1	`[0x081A]`	2	save format/version word
2	0x853A	4	map width + height
3	0x5380	0x8E	**game-globals block**: `[0x5382]` game flags + Game Options word, `[0x5384]` colony-report options, `[0x5386]` sound mirror, season/year/turn 0x538A–0x538E, counts 0x539A–0x539E, difficulty 0x53A6, revolution meter 0x53D0, REF power 0x53D2, REF counts 0x53DA–0x53E0
4	0x540E	0xD0	4× AIPersonality (stride 0x34)
6	0x5D46	n·0xCA	ColonyRecords
7	0x3144	n·0x1C	UnitRecords
8	0x8808	0x4F0	4× PowerRecord (stride 0x13C)
9	0x54EC	n·0x12	NativeSettlements

#	BASE	SIZE	CONTENT
10	0x5AD6	0x270	TribeData
13	0x9298	4	per-power colony count
15–23	0x940C … 0x942C	4–8 each	per-power attitude / AI / economy word tables (incl. 0x940C, 0x941C, 0x942C)
39–43	0x8540 / 0x853E / view words	2 each	current colony, map cursor, viewport/scroll

Because block 3 carries all three option words, every options dialog survives save/load; the sound toggles `[0xA0]/[0xA2]/[0xA4]` are re-expanded from `[0x5386]` on load (0x74249). The debug bitfield `[0x894]` is in **no** block — debug options are session-only. No configuration file exists.

20.4 The music scheduler

The background rotation pump `func_004EE6` runs from the input-idle loops each turn-idle: it skips unless background music `[0xA2]` (or a one-shot `[0x9E]`) is on, polls the driver ("playing?" id 8), honors a forced-next tune `[0x94]`, then seeds the RNG from the tick clock `[0x83A8]` and rolls a tune index inside a state window — **peace** (`[0x5382]&1` clear): folk tunes 1–12 with a 1-in-9 excursion into 13–23; **War of Independence**: independence/military tunes 13–18 with a 1-in-5 excursion back to folk. Event classes requested by game code (war fanfare, native themes 0x33/0x35/0x36, etc.) preempt via `[0x9A]`. The index→id map is `func_004DF8`; a re-roll avoids repeating the current tune `[0x96]`.

§21 Random numbers

Every roll in the game comes from one Microsoft C 6.0 linear congruential generator in the resident image, wrapped by a range-scaling helper that the overlays reach through a single thunk. With the seed and the call sequence, all game randomness is exactly reproducible — the colony-screen building layout is replayed from its seed on every screen open.

21.1 The generator

```
srand(seed)      @0x103C2 : stores only the LOW 16 BITS of the argument —
                            mov [0x28EE],ax ; mov word [0x28F0],0
                            (effective seed space is 16-bit)
rand()           @0x103D4 : seed32 = seed32 * 0x343FD + 0x269EC3    ; MSC 6.0 constants
                            return (seed32 >> 16) & 0x7FFF          ; AND AH,0x7F
                            seed dword at DGROUP:0x28EE/0x28F0
random_int(lo,hi)@0x0C322 : r = rand()                              ; 15-bit
                            return lo + ((r * (hi - lo + 1)) >> 15) ; inclusive range
                            reached via overlay thunk 0x181F:0x4D4
```

The multiplier 0x343FD (= 214013) occurs exactly once in the binary (byte pair `FD 43` at 0x103D5); the `>>15` is implemented as a byte swap plus seven SAR/RCR pairs.

21.2 Known seeded subsystems

SUBSYSTEM	SEEDING / ROLL	SITE
Colony building placement	per-colony deterministic seed `srand((colony_y<<8) + colony_x + dword[0x8D80])` (seed helper `func_009726`, 0x181F:0xD62, sole caller 0x25D3A); then per-plot `random_int(0, count[cat]-1)` shuffle with occupied-retry — a colony always lays out identically; `[0x8D80]` is a per-session boot value (runtime)	0x9736 / 0x25DBF
Combat resolution	single inclusive roll `random_int(1, ATK+DEF)`; attacker wins if roll ≤ ATK; separate 50% ambush coin `random_int(0,1)`	0x5B849 / 0x5D188
Music shuffle	pump re-seeds from the tick clock `[0x83A8]`, rolls the tune-index window, re-rolls on repeat	`func_004EE6`
Founding-Father pick	weighted walk `budget = random_int(1, Σ era-weights)`, subtract until ≤ 0	0x3C0DB / 0x3C035
Trade-route default name	new route's default name = colony name + a random `@TRADENAMES` word (collision appends " A")	`func_0610B0`
King tax attempt	difficulty roll `random_int(1, diff+1)` gate on the raise	0x34B25
Native raid outcome	gate `random_int(1,12)` + base outcome `random_int(1,4)`	0x5BEFD / 0x5BF35
Tory uprising	fires when `random_int(0, diff+1) != 0` — probability `(diff+1)/(diff+2)`	0x3CADD
Mission conversion	`random_int(0,15)` vs `tier+2` (doubled by the expert bit)	0x5730A
Lost City Rumor gold	ruins `10*3d8` (0x61770); big treasure `2*4d10` (0x617C0)	`func_061454`

PART VI

Events and messages

§22 The string files

Every word the game displays lives in eleven plain-text `.TXT` resources shipped beside the executables, all sharing one section format: an `@KEY` line opens a section, the following lines are its body, and `@;` lines are comments. GAME.TXT holds the dynamic message templates (with `%`-substitution slots), LABELS.TXT the static UI labels, NAMES.TXT the game-data taxonomy whose *row order is the runtime id*, PEDIA.TXT the encyclopedia, MENU.TXT the pull-down menu tree, DEBUG.TXT the cheat/debug dialogs, WOODCUT.TXT the event-screen captions, and MAPEDIT.TXT/MAPMENU.TXT the map editor's text. This chapter inventories all of them and specifies the template grammar and the engine that renders it.

22.1 File inventory

FILE	@-SECTIONS	ROLE
GAME.TXT	499	dynamic message/dialog templates (semantic catalogue counts 510 sections; the raw file also carries valueless directive lines)
LABELS.TXT	7	static UI labels (`@INFO` 4 · `@MISC` 221 · `@ROUTE` 9 · `@CMISC` 3 · `@CTITLE` 10 · `@CMESSAGE` 19 · `@EUROLABEL` 4)
NAMES.TXT	31	data taxonomy — row index = runtime id
PEDIA.TXT	166	Colonizopedia articles + category index
MENU.TXT	8	in-game menu bar (`@GAME @VIEW @ORDERS @REPORTS @TRADE @CUP @PEDIA @END`)
DEBUG.TXT	20	cheat/debug dialogs (4 sections dead)
WOODCUT.TXT	1	`@WOODCUT`, 17 caption lines (0–16)
MAPEDIT.TXT	19	map-editor dialogs (MAPEDIT.EXE only)
MAPMENU.TXT	5	map-editor menu bar (`@GAME @VIEW @CUP @HELP @END`)
OPENING.TXT / CLOSING.TXT	3 / 2	intro/outro cinematic scripts (`@CREDITS`/`@OPENING`/`@CLOSING` timing rows + `@MESSAGES` "Loading Game...")
COLONY.TXT / TRIBE.TXT	5 / 9	colony-name pools per nation; native-settlement coordinate lists per tribe

22.2 GAME.TXT — the message bank

499 sections. The major key families:

FAMILY	KEYS	NOTES
Boot & options dialogs	`@BEGINMENU @AMERICA @GAMEOPTIONS @COLONYOPTIONS @SOUNDOPTIONS @SAVEGAME @LOADGAME* @PICKNATION @DIFFICULTY @LEADERNAME @LANDHO`	checkbox/options dialogs of Part V
Music pickers	`@PICKMUSIC @PICKINDEPENDENCE @PICKMILITARY @PICKINDIAN`	see §24.2
European diplomacy	48 sections: 42 `@width=220` conversations + 6 `@width=190` announcements/guards + 5 support lists (`@GREATKINGS @GREATDEEDS @GREATLEADER* @MEEKNESS @FRIEND`)	keys are **built at runtime** from a fragment pool at file `0x1F250`+ ("MEEK" `0x1F250`, "MANLY" `0x1F255`, "HELLO" `0x1F267`, "AHOY" `0x1F26D`, "FIRST" `0x1F272`, "USA"...), which is why full names never appear as string literals
Tutorials	`@TUTORIAL1..19` + `@TUTNOLUMBER @TUTNOSPACES`	see §23.2
Intro caption cards	`@BUILD1..10`	one per LEVN000n.PIK card, rendered at pen (14,54) during world generation
Nation flavour	`@NATION0A/0B..3A/3B`	two briefing pages per power
Hot-seat multiplayer	`@MULTI @MULTINEXT @MULTIREV`	unlocked by `SET COLONIZE=MULTI`
Trade routes	`@TRADESTART @TRADETYPE @TRADENAMES @TRADENAME @TRADENONE @TRADENONE2 @TRADESELECT @TRADEDELETE @ROUTELOOP @TRADEWITH @TRADENOCARGO @TRADENOWANT`	`@TRADENAMES` = "5 / Run / Ferry / Cargo / Transport / Triangle" (count + 5 name stems)
Unit-option menus	`@UNITOPTIONS @SHIPOPTIONS @ARMOPTIONS @EUROPESHIPOPTIONS @EUROPESHIPCLICK @EUROPEARM`	dock/unit right-click order menus
King & tax	`@KINGTAX @KINGRAISE @KINGLOWER @KINGNOTHING @KINGNAVACT @KINGSTAMPACT @KINGWAR @KINGWIFE @MERCANTILISM @PURCHASETAX @TAXOPTIONS @TEAPARTY` + audience/war keys `@KINGRECRUIT @KINGFUND @KINGGALLEON2/3 @KINGFRIGATE @KINGNEWWAR @KINGVICTORY @KINGMERCY @KINGBUY @KINGMOBILIZE @KINGLOSE @KINGWIN @KINGBLESS @KINGLAUGH @KINGNO @KINGWELCOME0`	see §23.4
Lost City	`@LOSTCITY0..9 @BURIAL1..3 @SCREWED @VANISH`	see §23.5
Native events	`@INDIAN*` (welcome/treaty/gifts/war), `@RAID*` (6-key block at file `0x1F52A` + orphan `@RAIDSCALP`), `@CHIEF*`, `@EXTORT*`, `@VILLAGE*`, `@LEARN*`, `@MISSION0..3`, `@HERESY0/1`	see §23.6

FAMILY	KEYS	NOTES
Revolution	@DECLARE @INDEPENDENCE @TOOTORY @ALREADYREVOLUTION @MOBILIZE* @UPKEEP @WARN1..3 @INTERVENTION @INTERVENE @CONSIDER @SUCCESSION @SEIZURE* @INVASION @REFIT + guards @NOWARSDURINGREV @NOCOLONIESEITHER @NOMAYORSDURINGREV @EUROPENOTAVAIL @FOREIGNNOTAVAIL	see §23.7

Layout directives across the whole file: the `@width` histogram is {190: 336 sections, 220: 99, 300: 11, 310: 10, 160: 8, …}; only 21 sections carry a literal `@x`/`@y` (menus, tutorials, and the King-audience trio `@VICEROY` x=232/y=21, `@KINGLOSE` x=232/y=31, `@KINGWIN` x=202/y=125). No gameplay event popup is pinned — they all centre.

22.3 LABELS.TXT — the @MISC string-ID table

LABELS.TXT carries no substitution slots; it is pure label text. Six of its sections are consumed positionally (`@INFO` unit-info panel, `@CMISC`/`@CTITLE`/`@CMESSAGE` colony screen, `@EUROLABEL` Europe buttons, `@ROUTE` trade-route editor). The 221-line `@MISC` section is special: it feeds a runtime **string-ID table**.

- **Loader** (at file 0x75226–0x7523C, in the page-0x1A boot loader cluster): opens file "LABELS", selects section "MISC" (via 0x191F:0x928), then loops idx 0..220 (`cmp 0xDD` @0x75237) storing one interned value per line with `mov [bx+0x2DBA],ax` (`shl bx,1`).
- **The slots hold integer string IDs, not pointers.** Live values run **327..547 sequential**; they are resolved to text through the fetch verb **0x181F:0x22**. (A snapshot oracle that dereferenced slot value 537 as an address landed mid-string — the ID model is the byte-proven one.)
- **Slot formula:** any DGROUP word at offset O in 0x2DBA..0x2F72 holds `@MISC line = (O − 0x2DBA)/2`.

Notable slot map (line = slot index):

LINE	TEXT	LINE	TEXT
0/1	"a" / "an"	106	"and"
2	End of Turn	108	ENCYCLOPEDIA OF COLONIZATION
29/30/37/49–52/93	the advisor-report titles (F9/F2/F3/F4/F5/F6/F7/F8)	109/110	(More) / (Exit) — pedia pager
61–64	Ship / Cargo / Location / Destination (F7 headers)	129	Artillery Vs. Raid
65	Veteran	132/133	Tory Unrest / Rebel Unrest
75	COMBAT ANALYSIS	161–176	setup labels ("Click Here When Finished", Choose/Difficulty Level/Level, Easiest..Toughest, Select/European Power/Power, Immigration/Cooperation/Conquest/Trade)
76–84	Fatigue, Attack Bonus, Ambush, Terrain, Colony, Fortified, Spain Bonus, …, Artillery In Open	179–188	pedia stat labels: Combat, Attack, Cargo Holds, Moves, Plow, River, Coast, Move Cost, Defense or Ambush Bonus, Prerequisite
90	Drake	200/201	Prime / Damaged
95–100	F8 strength rows: Colonies, Population, Average Colony, Military Power, Naval Power, Merchant Marine	203/204	Bid Price / Ask Price
104	Bombard	210	Exit
–	—	211–220	placeholder numerals "211".."220"

`@ROUTE` (9 lines, the trade-route editor): EDIT TRADE ROUTE · Route Name: · Route Type: · Sea · Land · Destination · Unload Cargo · Load Cargo · (Delete Destination).

22.4 NAMES.TXT — sections and column legends

31 sections; row order is the runtime id everywhere (e.g. `@UNIT` row = unit type byte, `@COUNTRY`/`@TRIBES` order = power index 0..3 / 4..11). Column legends, verbatim from the file's own comment headers:

SECTION	LINES	COLUMNS (LEGEND)
@SEASONS	2	Spring / Autumn
@UNFORESTED / @FORESTED	8+8	name; Movement, Defensive, Improvement, Value; Yield (Farmer, Planter(s), Planter(t), Planter(c), Trapper, Lumberjack, Ore Miner, Silver Miner, Fisherman)
@OTHER	5	same columns — Arctic, Ocean, Sea Lane, Mountains, Hills (ids 24..28)
@OTHER_NAMES	5	Forest, River, Major River, Minor River, Unexplored
@RESOURCE	14	"Special resource squares & values" — name, value
@COUNTRY	4	"Country names (color => must be 9-15)" — name, colour
@NATIONALITY / @NATIONABBREV / @HOMEPORT / @COLONYNAME / @INDEPENDENT / @MISSION	4 each	positional per-power strings
@LEADERNAME	4	"Leaders: a) aggressive/friendly b) expansionist/perfectionist c) civilize/militaristic" — name + AI-bias triplet (loaded at 0x547A1 into the per-power table at DGROUP 0x9566)
@DIFFICULTY	5	Discoverer..Viceroy
@CLASS	8	"European classes: Name, transportation costs"
@BUILDING	42	"name, cost, tools(*10), size, min_colony, upkeep"
@SCENARIO	2	"map file (do not change), start, end, x0, y0, x1, y1, x2, y2, x3, y3"

SECTION	LINES	COLUMNS (LEGEND)
`@JOB`	28	"name, student level (4 = unlearnable), cost in europe" (+ expert name)
`@CARGO`	20	"start1, 2, low, high, burden, rise, fall, attrition, volatility" — the market model; stats loaded only for rows 0–15, rows 16–19 name-only
`@UNIT`	23	"icon, movement, attack, combat, cargo, size, cost, tools, guns, hull, role; AI Role(binary) => Invade Settle Explore Attack Defend Escort Transp Naval"
`@ORDERS`	13	order name + status letter
`@ACTIONS`	10	village-visit action labels (incl. "Denounce Heresy of %Fs Mission")
`@VALUES`	4	quality grades low quality/good/fine/excellent
`@ATTITUDE` / `@ATTITUDINAL`	5+5	Content..War; Extremely..Slightly
`@LEVELS`	5	tribe tech levels: Semi-Nomadic/Camp, Agrarian/Village, Advanced/City, …
`@TRIBES`	26	"Indian tribe info: tech-level, color" — first 8 rows full 5-column tribes (name, adjective, treasure good, level, sprite), rows 9+ name-only reserve pool
`@FOUNDING`	6	father categories (Trade/Exploration/Military/Political/Religious/Independence)
`@FATHERS`	25	"type, weight 1500-1600, weight 1600-1700, weight 1700+"
`@COLORS`	1	"Text Colors: basic, hilite, grey, enhance, shadow, select, border 0, 1, 2" = 68, 149, 8, 128, 47, 138, 134, 128, 138

22.5 PEDIA.TXT — 166 surfaces

The Colonizopedia's article bank: seven category renderers, each keyed `<KEY><idx>`:

CATEGORY (`@PEDIA` LINE)	KEY	ENTRIES
Cargo Type	`@CARGO0..15`	16
Unit Type	`@UNIT0..23`	24
Terrain Type	`@TERRAIN0..28`	29 (spans the unforested+forested+other id space)
Colonist Skill	`@JOB0..27`	28
Colony Building	`@BUILDING0..41`	42
Founding Father	`@FATHER0..24`	25
Game Concept	*(see quirk)*	12 titles
index sections	`@PEDIA` (7 category labels) + `@MISCELLANEOUS`	2

Total = 166 top-level sections. **The `@MISC0..11` quirk:** the Game Concept category loads its 12-entry index from `@MISCELLANEOUS` (line 0 = count "12", then Disband, Fortify, Plowing, Roads, Sentry, Trade Route, Veteran Units, Prices, Taxes, Liberty Bells, Crosses, Hammers — loader at 0x07530B–0x07534B, count to `[0x846]`, line pointers to `[0x935C+2i]`), and the article renderer then builds the key `"MISC"+n` — but **no `@MISC0..11` sections exist in PEDIA.TXT**. What the engine renders when `menu_lookup_run` misses the section is unresolved (likely a header-only page); unmapped.

22.6 MENU.TXT, DEBUG.TXT, WOODCUT.TXT, editor files

MENU.TXT is flat: 8 sections, each one dropdown (line 0 = the bar title). Grammar: `~` precedes a hotkey/underline letter (`~GAME`, `~F~1`, `~F~0~1`, `~s~p~a~c~e~ bar`), `#` marks a separator/value-fill cell (e.g. "Zoom In# ~Z"). `@CUP`'s title line is `~CHEAT` (the cheat menu, hidden until the Alt-W/I/N combo); `@PEDIA`'s is `~COLONIZOPEDIA`; `@END` is the empty terminator. Row texts are given verbatim in Part V §menus.

DEBUG.TXT: 20 sections. Sixteen are live cheat/debug dialogs (`@MEMORY @CREATE @CREATE2 @CSHIP @FOREIGN @FOREIGN2 @SETVIEW @SETHUMAN @SETAUTO @SETREPORT @SETEUROPE @DANGER @SOUND @OPTIONS @FORCED @TEST`); **four are dead** — `@MOTD`, `@MOTD2`, `@BADGUYS`, `@END` have no referencing string in any shipped EXE (`@END` is empty anyway). `@DANGER` ("DANGER, WILL ROBINSON!") is the AI assertion box, reachable in the shipping binary from 37 call sites.

WOODCUT.TXT: one section `@WOODCUT`, 17 caption lines 0–16 ("A NEW WORLD" … "INDIAN RAID"); lines 14–16 are placeholders with no art (see §23.1).

MAPEDIT.TXT (19 sections: `@MAPTOLOAD @MAPTOEDIT @SAVE @LOAD @ERROR @EXIT @SAVEAS @CREATENOW @NEWNAME @XS @YS @CONTINENTS1 @CONTINENTS2 @HELP1..5 @ABOUT`) and **MAPMENU.TXT** (5: `@GAME @VIEW @CUP @HELP @END`) serve MAPEDIT.EXE only; none of the 19 sections uses any `@`-directive.

22.7 The substitution grammar

Token inventory across GAME.TXT (502 scanned sections): `%STRING0` ×360, `%STRING1` ×211, `%STRING2` ×87, `%STRING3` ×49, `%STRING4` ×6; `%NUMBER0` ×124, `%NUMBER1` ×35, `%NUMBER2` ×11, `%NUMBER3` ×2; `%COUNTRY` ×7; `%YEAR` ×1. DEBUG.TXT adds `%HEXn` (`@MEMORY`'s "PSP at = %HEX4"). `%%` is a literal percent; `$` after a number renders as currency. The substituter uses longest-digit-run matching with trailing alpha kept literal (`%STRING0catraz` → "Alcatraz").

Slot storage:

- **VICEROY.EXE**: `%NUMBERn` values live in the slot array at DS:0x9CB0. `%STRINGn` slots are registered immediately before each emit via the two resident setters `func_06C220` (thunk 0x181F:0x416) and `func_06C23C` (thunk 0x181F:0x438, slot 0) — e.g. TUTORIAL12 registers the colony name at 0x2C7A7 just before its emit at 0x2C7B1.
- **MAPEDIT.EXE** (a compiled twin of the same engine): `%STRINGn` slots at **DS:0x634E + 64·n**, set by `_popup_say_string`; `%NUMBERn`, `%HEXn`, `%%` likewise.

Line and span directives (parser state machine, byte-cited in the MAPEDIT twin at 0x7C82 and in VICEROY's dialog engine):

SYNTAX	EFFECT
(blank line)	separates the text block from the option block; option lines get ids 1..n in read order
`^`	raw (non-wrapped) line
`^^`	centred line
(plain)	word-wrapped paragraph text
`{…}`	highlight span — `{` / `}` toggle the hilite latch `[0x1F62]` (glyph loop `func_06C388` @0x06C3C4/@0x06C478); ink = the hilite entry of the dialog ink record
`|`	truncates — ends the visible span of the line
`~x`	accelerator: underlines/hot-keys the following character (menus, checkbox rows); `~F~1`-style sequences bind F-keys
`#`	separator / value-fill placeholder in menu rows

`@`-directives inside a section body — the parser `func_06F0F4` @0x06F0F4 (keyword table at file 0x1F967) recognises exactly **10 live directives**:

DIRECTIVE	HANDLER	EFFECT
`@OPTIONS`	`mode switch`	following lines are option rows
`@PROMPT`	`mode switch`	text-entry mode
`@TEXT`	`@0x6F1D8`	back to body-text mode
`@SMALLFONT`	`@0x6F207`	copies the current font latch `[0x89E]/[0x8A0]` into the dialog — it does **not** load FONTSMAL.FF (never loaded by the engine)
`@X=` / `@Y=`	`@0x6F266 / @0x6F21E`	literal popup origin (−1 = centre sentinel)
`@WIDTH=`	`@0x6F2B0`	pixel content-width **floor** (never a clamp; keyword "WIDTH\0" at file 0x1F989)
`@LENGTH=`	`@0x6F302`	text-entry max length → `[0xA5B6]`
`@CHECKBOX`	`@0x6F350`	checkbox dialog (`FLAGS |= 5`)
`@DEFAULT=`	`@0x6F374`	pre-highlighted row index (an index, not a colour)

An 11th keyword string, `TEXTCOLR` (file 0x1F9AA), is **vestigial as a directive** — the parser never compares it. The string is instead the sheet name for the TEXTCOLR.SS colour-table load (`func_06F6DA` @0x06F6F0), whose sprite pixels seed the dialog ink globals `[0x1F3C..0x1F4E]`. There is no per-popup text-colour override.

22.8 Template-engine key facts

- **`menu_lookup_run` = 0x181F:0x998 = `func_06F51A`.** Calling convention: **AX = section-name pointer, BX = file-name pointer, DX = preselect row; returns the 1-based selected row**. The GAME-file wrapper 0x181F:0x3FE (@0x06F594) hardwires file "GAME" (DS:0x87C) and takes the section in BX.
- Build chain: section reader 0x191F:0x928 (`func_06F8FA`) → template parser 0x191F:0x182 (`func_06F0F4`) → geometry finalize `func_06D316` (centred on (160,100) unless `@x/@y`) → modal pump 0x191F:0x16A (`func_06E3D0`), which returns the row.
- Checkbox channel: bitmask word `[0x1F54]` — reset 0x191F:0x26E, pre-seed 0x262, read-back 0x306.
- **ESC returns 0xFFFF** (byte-cited in the editor twin's event loop `@popup_exec` @0x6F5E: Up/Down move skipping greyed rows with wrap, Enter/Space select, hotkey match, mouse row select; entry mode appends printables to maxlen with backspace).
- Text entry lands in the popup text buffer (editor twin: DS:0x4B64); dead wrapper `@popup_ask_number` has zero callers in MAPEDIT.EXE.

§23 The event catalogue

This chapter is the game's event book: every interrupting event, one row per event, in the schema *event_id / string_key / trigger / condition / options / outcomes (state writes) / arms (downstream)*. All string keys are GAME.TXT sections (verbatim bodies quoted where load-bearing); all popups render through the shared centred-dialog engine of Part V with a speaker channel (`[0x1F5C]` king/tribe, `[0x1F5E]` advisor, `[0x1F60]` missionary), all three reset to 0xFFFF after close at 0x06EE6B. Where a probability or write was not byte-decoded it is marked unmapped rather than estimated.

23.1 Woodcut event screens (17)

One renderer, `func_06B722` @0x06B722 (`show_woodcut(n)`): black clear, WOODFRAM frame 1 centred, title `"<year>: <CAPTION>"` from `@WOODCUT` line n, NAMEPLAT strip at y=162, caption at y=165 in FONT-NP (ink LUT palette indices 0x5C/0x5D/0x5E), WDCUT art blit, staged fade, modal wait. The wrapper `func_00543C` (0x181F:0x524) enforces **once-only** per game via the shown-bitmask at `[0x540A]` and fires the sound cues. Art = WDCUT01..WDCUT13.SS (no 00/14/15/16); a missing-file check @0x06B79F makes the art-less numbers unshowable. The caller scan is exhaustive: exactly 10 call sites.

N	CAPTION (STRING = @WOODCUT LINE N)	TRIGGER	SOUND CUE	ARMS
0	A NEW WORLD	**`no caller`** `(latent save-under popup mode)`	`music class 2 wired`	—
1	DISCOVERY OF THE NEW WORLD	`first landfall — func_020EFE @0x020F00 (sole caller func_03FDDE, after [0x543E]|=0x80)`	`music class 2`	tutorial T2 tail

N	CAPTION (STRING = @WOODCUT LINE N)	TRIGGER	SOUND CUE	ARMS
2	BUILDING A COLONY	first colony — build executor `func_040C1E` @0x040E00, human only	sfx 0x54	—
3	MEETING THE NATIVES	first tribe contact, tribe ≥ 2 — `func_056C3E` @0x056DA6	tune 0x33	then `@INDIANWELCOME`
4	THE AZTEC EMPIRE	same site, tribe 1 (Aztec)	tune 0x35	then `@INDIANWELCOME`
5	THE INCA NATION	same site, tribe 0 (Inca)	tune 0x36	then `@INDIANWELCOME`
6	DISCOVERY OF THE PACIFIC OCEAN	**no caller** — sound cue wired @0x0054A2 but never hooked	tune 0x39 (dead)	—
7	ENTERING INDIAN VILLAGE	first village entry — `func_04B308` @0x04B56C (human)	–	village-visit dialog
8	THE FOUNTAIN OF YOUTH	Lost City outcome 1 — `func_061454` @0x0618F9	after tune 0x37	recruit prompt `@LOSTCITY0`
9	CARGO FROM THE NEW WORLD	first cargo arrival in Europe — `func_041EEA` @0x0420EF	music class 2	—
10	MEETING FELLOW EUROPEANS	first power-to-power contact — `func_057F4E` @0x057FDF	contact fanfare 0x8020+p (§24.4)	`@HELLO*` greeting
11	COLONY BURNING	colony burned — `func_05CA7E` @0x05DADC (with `@BURNED`) and @0x05DFCB	sfx 0x53 + tune 0x32	—
12	COLONY DESTROYED	**no caller**	–	—
13	INDIAN RAID	natives attack a human colony — `func_05CA7E` @0x05D219	–	raid outcome popup (§23.6)
14–16	placeholders	unreachable — no caller *and* no .SS art	–	—

23.2 Tutorial overlays (@TUTORIAL1..19)

All 19 are ordinary GAME.TXT popups emitted through 0x181F:0x652 = `func_06F5F2(name, advisor)` (sets the advisor portrait channel `[0x1F5E]` → MSS.SS) or the 0x3FE wrapper. Gate: Game-Options bit 0x80 "Tutorial Hints" (T18 is ungated). Each step is **event-driven and idempotent**: its site does `test [0x5386/7],bit; jne skip → emit → or [0x5386/7],bit`; new-game init pre-marks `[0x5386]=0x0E`. Sections with literal placement: T1 (10,40), T4 (x=10), T12 (y=5), T16 (5,10 smallfont), T17/T18 (y=10 w=300 smallfont); the rest centre.

The unit-focus dispatcher `func_020F50` @0x020F50 (called from the end-of-move handler @0x021E63 and the map idle loop @0x024AC6 after a ~30-tick wait) serves T1, T3, T8–T11, T13–T15, T19 from an if/else chain over the selected unit:

#	TRIGGER SITE	CONDITION	ADVISOR
T1	dispatcher	first turn (%STRING0 = unit-type name)	0
T2	@0x020F43	land discovered (tail of `func_020EFE`; no once-flag)	0
T3	dispatcher	pioneer on a ≥5-resource site (%STRING0 = signature good)	3
T4	@0x02C73F	colony open: better job available from the terrain ring (%STRING0/1 = current/alternative goods)	5
T5	@0x036514	religious-unrest immigration — chained after `@UNREST`	4
T6	@0x02EA41	goods ready for export at end of turn (%NUMBER0 qty, %STRING0..2 goods/colony/port)	0
T7	@0x028D36	colony pop ≥ 3 and no stockade	1
T8	dispatcher	petty-criminal/servant near a training village	5
T9	dispatcher	pioneer on unroaded forest/hills near a colony	3
T10	dispatcher	pioneer on a plowable/clearable colony ring tile	3
T11	dispatcher	idle ship, turn < 20	–
T12	@0x02C7B1	colony open with a ship at the tile (%STRING0 = colony name)	5
T13	dispatcher	pioneer before the first colony	3
T14	dispatcher	soldier selected	1
T15	dispatcher	colonist on a colony tile (%STRING0 = colony name)	5
T16	@0x0286F6	colony food deficit (red-X corn counters)	–
T17	@0x035C22	Europe screen first open	–
T18	@0x032760	Europe buy: cannot afford 100 units — **ungated** (no hints bit, no once-flag)	–
T19	dispatcher	Indian convert selected	4

Related conditional warnings from the found-colony validator `func_022542` (fire only at difficulty < 2, i.e. `[0x53A6] < 2` @0x22763): `@TUTNOSPACES` when adjacent productive squares < 4 (@0x22772), `@TUTNOLUMBER` when forested squares = 0 (@0x2278A); both are two-option confirms and the build proceeds only on row 2.

23.3 European diplomacy (the 48-section family)

One dispatcher owns the family: **`func_057F4E`** (page 0x0F, 7,151 bytes), entered from the contact evaluator `func_059B90` when a unit meets a foreign power (unit-vs-tile resolver @0x03F82B; movement processor @0x0481CB). AI-to-AI meetings delegate silently to the ticker `func_057DC0` @0x057FA4 — popups run only for the human. Conversations emit via 0x1A1F:0x688 = `func_06F61C` (speaker channel `[0x1F60]` = power B → MYR0..MYR3.SS portrait; returns the 1-based row); announcements via 0x181F:0x652 (advisor portraits MSS1/MSS2). Relation state = the 4×4 matrix at PowerRecord+0x34

(DGROUP 0x883C, row stride 0x13C): bits 0x02 war · **0x08 pending grievance** · 0x10 parley cooldown (16 turns) · 0x20 met · 0x40 peace treaty · **0x80 privateer hidden attribution**. PowerRecord+0x40 is the **treaty-respect counter** (plain byte, seeded `2·(6−difficulty)`, halved with Franklin; a nonzero value makes an AI abort attacks on its treaty partner @0x03F163; the decrement site is unmapped). `%STRING` slots are filled from `@GREATKINGS/@GREATDEEDS/@GREATLEADER*[power]` by `func_057A3A`; `@MEEKNESS` supplies "request"/"demand". Franklin (FF #19) halves demands/prices and cancels AI hostility at 6 cited sites; a war fanfare (`func_005108(4)`) precedes every WAR*/MERCENARY emit.

EVENT / KEY(S)	TRIGGER (KEY-PUSH → EMIT)	OPTIONS & OUTCOMES (STATE WRITES)	ARMED BY / ARMS
Greeting `@HELLOFIRST/@HELLOAHOY/@HELLOMEEK/@HELLOMANLY/@HELLOUSA`	key = "HELLO" + (not-met ? ship ? "AHOY" : "FIRST" : tone "MEEK"/"MANLY"); USA for an independent power; @0x0588CD–0x058923 → @0x058939	greeting only; leads into the parley menu	first contact also fires woodcut 10 @0x057FDF + fanfare 0x8020+p
Third-party demand `@APOSTATES` (+USA)	AI asks the player to attack its treaty partner; @0x058989 → @0x058A30	row 2 accepts → player's treaty with the target cleared + war bit 0x02 set @0x058A6A/@0x058A7B	—
Third-party demand `@HEATHEN` (+USA)	AI asks the player to attack a tribe; @0x0589C0 → @0x058A30	row 2 accepts → tribe tension +100 vs the target tribe (`func_045DF2(t,A,100,0)` @0x058A91)	—
Protest `@PIRACY/@PIRACYUSA` — "*%STRING0 is most displeased with the {%STRING1 pirates} lying in wait off the coast of %STRING2…*"	fires when the war-matrix **privateer bit 0x80** is set for the pair; @0x058B0D → @0x058B45	row 1 "What pirates? We have NEVER condoned piracy!" — denial; row 2 recalls **all** privateers to Europe and clears bit 0x80 @0x058B7D–0x058BE1 (Europe is the engine's destination sentinel **999/0x3E7**, the same value used by trade-route stops; a ship-type-guarded scan against it sits @0x6013C–0x60146)	armed by `func_03ECF0` @0x03F0A1: a Privateer attack (unit type 0x10 guard @0x03F092) sets 0x80 *instead of* the war bit
Protest `@SIEGES/@SIEGESUSA`	player units besieging B's colonies; @0x058C99 → @0x058CD8	row 2 withdraws the besieging units. **Latent bug:** `@SIEGESUSA`'s rows are textually swapped but the handler acts on row 2 for both — answering "our forces shall stay" to an independent power executes the withdrawal	—
Extortion `@TRIBUTE/@TRIBUTEUSA`	demand accumulated from forces-near-colonies, difficulty-scaled (`value·10·(diff+8)/100`, surcharge `+500·(diff+1)`); @0x058E7D → @0x058EB8	pay → gold transfer @0x058ED0; refuse → escalation into the WAR keys below	grievance: bit 0x08 set when the grievance score crosses its threshold @0x3F0D7/@0x59AE9
Extortion `@WANTSTUFFUSA` — goods demand	@0x058F56 → @0x058F95	accept → colony stock rows moved to B @0x058FB4. **Latent bug:** the non-USA key "WANTSTUFF" is built @0x058F56 but **has no GAME.TXT section** (only `@WANTSTUFFUSA` exists)	—
War declarations `@WARMEEK`/`@WARMANLY` — "*You reject our generous offer? Then in the name of %STRING0 we shall wipe you from the face of the New World. Prepare for WAR!*"	refusal outcomes of the demand tree; @0x059222/@0x059516 → @0x059274/@0x05957E	war bit 0x02 set for the pair	war fanfare class 4 first
Ultimatum `@RID/@RIDUSA`, provocation `@PROVOKE`	@0x059077 → @0x0590A3; @0x058FEE/@0x05974C	leave-or-war ultimatum; `@PROVOKE` = "*We can no longer tolerate your foul*	—

EVENT / KEY(S)	TRIGGER (KEY-PUSH → EMIT)	OPTIONS & OUTCOMES (STATE WRITES)	ARMED BY / ARMS
		provocations. Prepare for WAR!"	
Treaty menu `@WORTHY` → `@PEACEMEEK/@PEACEMANLY/@OLDPEACE*/@PEACEUSA`, `@GIVECASH`	standing-peace proposals; @0x05911D–@0x059395	treaty set both ways @0x059139 (0x181F:0xA06, bit 0x40) + siege stand-down `func_057CE0`; respect counter set 1	16-turn parley cooldown stamp `[0x53C8+p·2]`
Withdraw family `@WITHDRAW/@NOTWITHDRAW/@NOTHINGWITHDRAW/@MAYBEWITHDRAW`	@0x0593B7–@0x05949B	withdraw price = `25·(diff+2)·forces` (min 100, ×2 at war, −50/unit, Franklin ÷2)	`@GIFTS` @0x059700 / `@THREATS` @0x059755 side outcomes
Alliance `@MILITARY` → `@NOCONTACT/@ALREADYSMITE/@SMITEINDIANS/@SMITEEUROPE/@UNFORTUNATE/@MERCENARY`	dynamic row list (lea 0x19FA @0x05976D, shown via the modal pump @0x059848); smite family @0x05989D–@0x059A0F	purchase → B declares war on target T (bits @0x059A49–0x059A71) + player pays B @0x059AC7; `@MERCENARY` = "*The {%STRING0} declare war on the {%STRING1}.*"	war fanfare class 4
AI↔AI ticker `@SIGNTREATY` / `@DECLAREWAR`	`func_057DC0`, every 3rd turn per met pair; @0x057E86 / @0x057F18	peace → treaty bit 0x40 both ways + respect := 1; war → `@DECLAREWAR`. **Latent bug:** the had-treaty branch pushes key "CANCELTREATY" @0x057F10 which **has no GAME.TXT section** (only `@CANCELPEACE` exists)	—
Attacking a treaty partner `@HAVETREATY` → `@CANCELPEACE`; `@SNEAK`	human attacker → `@HAVETREATY` @0x03F130 (row 2 "Break Treaty." continues) → `@CANCELPEACE` @0x03F22F; AI attacker → `@DECLAREWAR` @0x03F262; human victim → `@SNEAK` @0x03F1B4	war bit set @0x03F298, treaty cleared @0x03F2A5	second `@HAVETREATY` site @0x0220CE (order-issuing flow; its UI trigger is unmapped) — that path sets the war bit @0x220E6, clears the treaty via 0x181F:0xA10, plays **SFX 0x58** @0x220F9 and issues attack order 5 before the attack-execution call
`@SUCCESSION` — "*War of the Spanish Succession ends in Europe! {%STRING0}, ravaged by war, agrees to cede %STRING1 to the {%STRING2}…*"	`func_03C638` @0x03C76A (MSS2 advisor), scheduled while the SoL meter `[0x53D0] < 75` and no power has seceded (`[0x53D2] < 0`)	whole-map owner-bit rewrite @0x03C799–0x03C7D1 — the weakest AI power is absorbed	skipped in hot-seat multiplayer @0x03C63D
Movement guards `@NOWARSDURINGREV` / `@TRADEATWAR` / `@TRADEMERCANTILISM`	`@NOWARSDURINGREV` @0x05A916 (also enforcement @0x5A912 in the attack handler, only inside the WoI-declared gate @0x5A8C8: emits and sets the cancel flag, skipping the attack call @0x5A92C); `@TRADEATWAR` @0x05A458 and the **Jan de Witt gate** (FF #4) `@TRADEMERCANTILISM` @0x05A469, both in the foreign-colony trade entry `func_05A40E`	attack/trade cancelled; no state change	—

23.4 King and tax events

The per-turn tax-demand driver is `func_036138`. Cadence: nothing before turn 30; then a demand fires when `turn % interval == 0`, interval 18 shrinking to 15/12/9 as the year crosses 1600/1700/1750, further reduced by `(diff−2)` for the human; skipped once tax > 85. Speaker channel `[0x1F5C]=8` → KING1.SS.

The **pretext** is chosen by a composite severity score

```
sev = random_int(1,1000) + (2·SoL[0x53D0] − tax)·5 + gold_term(+0x2A,100)
    + per_player_const[0x9410+p] + turn/30            ; @0x361CC..0x36221
```

EVENT_ID	STRING_KEY	CONDITION (SEV)	MESSAGE OPENING
KING-WIFE	`@KINGWIFE`	`< 0x28A (and [0x53A7] < 0x1E)` @0x362C7	"In honor of our recent wedding to our %STRING2 wife…"
KING-WAR	`@KINGWAR`	`< 0x3B6` @0x362FA (+`random_int(1,8)` war number)	"Because of recent developments in our ongoing war with %STRING2…"
KING-NAVACT	`@KINGNAVACT`	`< 0x44C` @0x36348 (+`random_int(3,4)`)	"…impose a new {Navigation Act}…"
KING-STAMPACT	`@KINGSTAMPACT`	else @0x36371 (+`random_int(5,8)`)	"…teach them proper respect… by imposing a new {Stamp Act}…"

The core demand `@KINGTAX` (width 190): "*It is essential that the Crown receive proper recompense for its efforts on your behalf. Therefore we have graciously decided to raise your tax rate by {%NUMBER0%%}. The tax rate is now {%NUMBER1%%}. If you wish, you may kiss our royal pinky ring.*" Options `@TAXOPTIONS`: "**Kiss pinky ring.**" (accept — tax applied, hard-clamped to 75 at 0x03434F) / "**Hold '{%STRING3 Party}.'**" (refuse). Refusal fires `@TEAPARTY` — "*{%STRING3 Party}! Sons of Liberty throw {%NUMBER0} tons of %STRING0 into the sea at %STRING1! … %STRING0 cannot be traded in %STRING2 until boycott is lifted.*" — and sets the per-good boycott bit `PowerRecord+0x20 |= (1<<good)` @0x034717. The boycott is lifted per-good by paying back-tax = `count × 500` gold (count = `PowerRecord[+0x4C+good] + base_table[0x9700+good·9]`, clamped ≥0; payment @0x03340D moves the gold into the royal fund and clears the bit @0x033423), or wholesale by acquiring Jakob Fugger (FF id 1: `mov [bx+0x20],0` @0x03BD45).

Related rows:

EVENT_ID	STRING_KEY	TRIGGER / CONDITION	OUTCOME
KING-RAISE	`@KINGRAISE`	player *demands lower taxes* and fails	punitive raise ("Your DARE to demand lower taxes!…")
KING-LOWER / KING-NOTHING	`@KINGLOWER` / `@KINGNOTHING`	outcome of the lower-taxes petition	tax −%NUMBER0 / unchanged
KING-MERCANTILISM	`@MERCANTILISM`	building a profit-taking manufactory	tax raise, same options
KING-PURCHASETAX	`@PURCHASETAX`	use of Crown resources (Royal University etc.)	tax raise
KING-GALLEON	`@KINGGALLEON2/3`, `@CASHTREASURE`, `@LOOTCASH`	treasure unit with no Galleon; accept → Crown ships it	cut% = tax (with Cort s, FF #10) else `max(5·diff+50, 2·tax)` clamped ≤ 90; player receives gross − cut
KING-NEWWAR	`@KINGNEWWAR`	Crown declares war on a rival and orders the player in ("…we shall provide you with {%NUMBER0$}…")	peace arrangement cancelled. Portrait = **KING1.SS** (no "KING2.SS" exists anywhere in the binary — byte-refuted)
Tax-level gate	—	`tax ≥ 60` (0x3C, @0x034A1B) branches the king message flow; 75 (0x4B) is the hard cap	—

23.5 Lost City rumors (func_061454)

Trigger: a unit enters a rumor tile. Rumor presence is **procedural** — predicate `func_006188` @0x6188 computes it from a coordinate hash against the map seed `[0x190]` (`((x>>2)·0x13 + (y>>2)·0x11 + seed + 8) & 0x1F − (y&3)·4 == (x&3)`), gated on terrain ≠ ocean/sea-lane/arctic and feature nibble = "none". Outcome index `n = max(anti_streak_floor, random_int(1,9))` — the floor rises by 1 per rumor and caps at 3, so the good low outcomes are only reachable on the first rumors; a quality roll `random_int(1,100) + scout·10` against thresholds 10/25 demotes/refines; per-game caps `[0x1DC6]/[0x1DC7]` limit Fountain and Cibola to one each; with debug bit `[0x5382]&1` the outcome is forced to 2. `s` = Seasoned-Scout bonus (unit type 5, class 0x16). The key is built literally as `"LOSTCITY"+n` (itoa append @0x618D1).

N	STRING_KEY	OUTCOME (STATE WRITES)	REWARD ROLL
1	`@LOSTCITY1` — "*You have discovered a {Fountain of Youth}!…*"	**8 free immigrants** queued on the Europe docks; recruit prompt `@LOSTCITY0`; **woodcut 8** @0x0618F9 after **tune 0x37** @0x618ED; promotes to 2 if `[0x5382]&1`	—
2	`@LOSTCITY2` — Seven Cities of Cibola	Treasure unit created (type 0xA); value stored /100 in its class byte; sound 0x3C	`%NUMBER1 = 100·(10·(s+2) + 1d20)`
3	`@LOSTCITY3` — ruins of a lost civilization	gold credited to PowerRecord+0x2A @0x61C4C	`10·(3d8)`, scaled `·(s+2)/2`
4	`@LOSTCITY4` — burial mounds	options: "Let us search for treasure!" / "Stay clear of those!"; search → sub-dispatch `@BURIAL1` (empty) / `@BURIAL2` gold `10·(3d8)` / `@BURIAL3` treasure `200·(1d8+2s+10)`; a **human** desecrating a **hostile** tribe's grounds appends `@SCREWED` ("*…You have trespassed on sacred land. Now you must die!*") and the unit is lost; desecration raises that tribe's tension **+100** (`func_045DF2` @0x61B84 → war footing); one special path per power (flag bit 0x40 in `[0x543E]` @0x6186B)	—
5	`@LOSTCITY5`	expedition **vanishes** — triggering unit destroyed (downgrades to 6 when disallowed)	—
6	`@LOSTCITY6`	nothing but rumors	—

N	STRING_KEY	OUTCOME (STATE WRITES)	REWARD ROLL
7	`@LOSTCITY7` — small friendly tribe	chief's gift of gold	`2·(4d10)`
8	`@LOSTCITY8`	trespass near holy shrines — tribe displeased	—
9	`@LOSTCITY9` — desperate survivors	colonist(s) spawn and join the nation @0x61809	—

23.6 Native events

Speaker channel `[0x1F5C]` = tribe index (0=Inca … 7=Tupi) → INDA.SS portrait, read from the settlement's owner byte.

EVENT_ID	STRING_KEY	TRIGGER / CONDITION	OPTIONS	OUTCOMES / ARMS
NAT-WELCOME	`@INDIANWELCOME` — *"The {%STRING0} tribe welcomes you. We are a glorious nation of {%NUMBER0 %STRING1}… Will you accept our treaty and live with us in peace as brothers?"*	first contact with a tribe (`func_056C3E`, after woodcut 3/4/5)	Yes / No	No → `@INDIANSHUN` ("…Prepare for WAR!"); related `@INDIANBOW`/`@INDIANTREATY`/`@INDIANPEACE`/`@INDIANCOME`
NAT-GIVEFOOD	`@INDIANGIVEFOOD`	supply/demand model in `func_056C3E`: the tribe's per-good **supply array [0x9E78] exceeds demand [0x9E58]** for food @0x05737C, and the player's stores are low — emit @0x0573EB	—	+%NUMBER0 food gifted
NAT-BEGFOOD	`@INDIANBEGFOOD` — "…*Will our brothers of {%STRING1} share the bounty of their harvests…*"	food **deficit** `[0x9E58]−[0x9E78]` @0x057098–0x05709F — emit @0x05716F	"I'm sorry, we gave at the office." / "We offer you {%NUMBER0} of our {%NUMBER1 food}…"	refusal/gift affect tension (delta site unmapped)
NAT-GIVESTUFF / CONVERT	`@INDIANGIVESTUFF`, `@INDIANSCONVERT`	goodwill gift; mission conversion `func_0572E6` — P(convert) = `(TribeData[+2]+2)/15`, doubled by Jean de Brebeuf (FF 0x16)	—	convert unit created at the colony, class 0x1B
NAT-RAID	6-key block `@RAIDWREAK` `@RAIDSTORES` `@RAIDBURN` `@RAIDSHIP` `@RAIDGOLD` `@RAIDNOTHING` (contiguous in the EXE at file 0x1F52A) — e.g. "*Spies report: {%STRING0} raiding party wreaks havoc in the {%STRING3} colony of {%STRING1}.*"	raid handler `func_05BE84`: gate roll `random_int(1,12)−1` (+`diff−2` vs a human European) vs threshold `3·K+1`; base outcome `random_int(1,4)` adjusted by turn (`turn < 40·(2−diff)` downgrades) and availability gates; 5-way dispatch @0x5C026	—	1→`@RAIDSTORES` (loot cargo, sfx 0x4F), 2→`@RAIDWREAK`, 3→`@RAIDGOLD` (sfx 0x4E), 4→`@RAIDBURN`/`@RAIDSHIP`, 0→`@RAIDNOTHING` (raiders wiped out, sfx 0x5B). `@RAIDSCALP` exists as a section but is **not** in the 6-key block — an orphan, not a 7th outcome. Raid on a human colony also fires **woodcut 13** @0x05D219
NAT-WARPATH	`@INDIANWARPATH` `@INDIANWARPATH2` `@INDIANWARFARE` `@INDIANWAR` `@INDIANGRUDGE` `@INDIANSURPRISE`	warpath handler `func_04B036` (sets `[0x1F5C]` = tribe owner); `@INDIANGRUDGE` = the Tory-side war-council entry during the revolution	—	war footing; alarm ≥ 128 (per-settlement per-power word at +0x0A+2·p) is the raid state; the parallel tension table (0..100) turns hostile at 75, war at 100
NAT-EXTORT	`@EXTORTSTUFF` `@EXTORTPOOR` `@EXTORTLAUGH` `@EXTORTNO`	player Demands Tribute (`func_04AC00`)	—	gold clamped to `[10, min(3·tribe_wealth+10, 100)]`, moved from settlement to player
NAT-VILLAGE	`@VILLAGEHAPPY` `@VILLAGEMEDIUM` `@VILLAGESAVAGE` `@VILLAGEBAD` `@VILLAGEWAR`; `@MADATSHIPS` `@MADATWAGONS` `@DONTKNOWSHIPS`	scout enters village (`func_04B308`; attitude words from NAMES `@ATTITUDE`, banded at score cutoffs −5/0/10; War = alarm ≥ 128)	—	display; ship/wagon anger blocks trade
NAT-RAZE	`@CHIEFKILL` `@INDIANGOLD` `@INDIANBURN`	player attacks a settlement (`func_04A7CA`); `@CHIEFKILL` = taboo execution	—	raze gold = `(Σ3·random(1,10−diff)) · random(1,6) · 4 · (tribe+1)` → +0x2A; no woodcut fires here (the old "WDCUT12" gloss is byte-refuted — WDCUT12 has no caller)
NAT-TENSION	*(silent)*	tension applier `func_045DF2` (33 call sites): trespass	—	deltas halved for France and with Pocahontas (FF 16); clamp [0,100]

EVENT_ID	STRING_KEY	TRIGGER / CONDITION	OPTIONS	OUTCOMES / ARMS
		+1/+2/+3, successful trade −4, mission established − (clamped), burial desecration +100, incite ±100, per-turn drift ±1		

23.7 Revolution events

EVENT_ID	STRING_KEY	TRIGGER / CONDITION	OPTIONS	OUTCOMES / ARMS
REV-DECLARE	`@DECLARE` — "*Shall we declare our independence from {%STRING0}…? This will end our turn and place us at war with our King!*"	GAME menu "DECLARE INDEPENDENCE" → `func_03E984`; refused with `@ALREADYREVOLUTION` if already at war, or `@TOOTORY` (+%NUMBER0 = SoL) while the national SoL meter `[0x53D0] < 50` (@0x3E99E)	"Never! That would be treasonous!…" / "Yes! Give me liberty or give me death!"	yes → `func_03DE46`: `[0x5382]|=1`, rebel power `[0x5398]:=[0x5394]`, declaration year stored, initial REF dispatch
REV-INDEPENDENCE	`@INDEPENDENCE` — "*Continental Congress signs {Declaration of Independence}! … General %STRING0 calls for volunteers for new Continental Army!*"	emitted by `func_03DE46` @0x3E104	—	war begins; veteran soldiers promote (`@MOBILIZE`)
REV-WARN	`@WARN1/2/3` — "*…the King's forces control all but %NUMBER0 of the ports in %STRING0!…*" / "*…all but %NUMBER1 of our colonies!…*" / "*… %NUMBER2%% of the %STRING0 population. If he ever controls 90%%, the Continental Congress will be unable to continue the war…*"	wartime status warnings (ports / colonies / population thresholds); emit sites unmapped	—	surrender conditions foreshadowed
REV-CONSIDER	`@CONSIDER` — "*%STRING0 is considering intervention on our behalf… If we can generate %NUMBER0 liberty bells, they will join us.*"	pre-intervention notice	—	arms the intervention watch
REV-INTERVENTION	`@INTERVENTION` (+ ally names from `@FRIEND`: British General Cornwallis / French General Lafayette / Spanish Generals / Dutch Admiral de Ruyter)	intervention declaration `func_03D948` — picks the strongest eligible foreign ally, sets `[0x5382]|=2` @0x3DA22; arrival waves `func_03D510` land at a weighted colony pick `random_int(1, Σ weights)` @0x3D57E	—	Intervention Force joins the rebel side
REV-TORY	`@TORYUPRISING`	per-turn roll in `func_03CAC6`: `random_int(0, diff+1) ≠ 0` @0x3CADD ⇒ probability `(diff+1)/(diff+2)`	—	Tory uprising spawns loyalist units
REV-END	`@KINGVICTORY` / win path	per-turn resolver @0x02F464: rebels **win** when surviving REF combatants fall below the threshold (1, or 8 with `[0x5382]&0x40`) → `[0x5382]|=8` @0x2F55A	—	score bonus `+2·(1780 − declaration_year)` if declared before 1780
REV-MULTI	`@MULTIREV` — "*The Revolution does not function in multi-player mode…*"	declaring in hot-seat (`[0x5381]&0x80`) @0x3E9C5	Declare independence / Never Mind	confirm clears the multiplayer flag (`and [0x5381],0x7F` @0x03E9D3) — game continues single-player
REV-GUARDS	`@NOWARSDURINGREV` `@NOCOLONIESEITHER` `@NOMAYORSDURINGREV` `@EUROPENOTAVAIL` `@FOREIGNNOTAVAIL`	action guards while `[0x5382]&1` (see §23.3 last row for the byte-cited `@NOWARSDURINGREV` enforcement)	—	action cancelled

23.8 Europe arrival and immigration chain

EVENT_ID	STRING_KEY	TRIGGER	OUTCOME / ARMS
EUR-UNREST	`@UNREST` — "*Religious unrest in %COUNTRY causes increased emigration. Colonists ({%STRING1}) now available in %STRING0.*"	crosses-driven immigration event (market/king phase; emit @0x3651F region)	**arms tutorial T5** (chained immediately after, advisor 4, @0x036514); recruit variant `@RECRUITCHOOSE` presents the dock choice
EUR-ARRIVE	*(banner, not a popup)*	ship reaches the Europe port: the header banner is composed by `func_030F76` into the top text band (band rect (x=320,y=7,w=0,h=0) set by `set_text_box` @0x035B2D) from the dock-state strings of LABELS `@MISC` lines 5–8 ("Sailing For" / "Inbound From" / "Now Arriving In" / "Docks At") plus port, season/year and tax/gold state	first cargo sold in Europe fires **woodcut 9** @0x0420EF
EUR-SAILHOME	`@SAILHOME` — "*We have reached the {high seas}… Shall we sail for Europe?*" (default row 1)	ship enters the sea-lane column	yes → Europe screen on arrival; `@SAILAWAY`/`@SAILPORT` are the return prompts

§24 Music and sound

All music and effects are driven through an external, load-time sound driver; VICEROY.EXE itself contains **no `.XMI` filenames and no tune names beyond the picker menus** — a tune id is an opaque byte handed to the driver. Three master switches (Background Music `[0xA2]`, Event Music `[0xA0]`, Sound Effects `[0xA4]`) gate everything, persisted via the save-side mirror `[0x5386]`.

24.1 The tune-id table (0x20..0x3E)

Byte-verified row↔id mapping from the Pick-Music jump tables (names verbatim from the `@PICKMUSIC` family):

ID	NAME	ID	NAME
0x20	Bird Song	0x30	Morelli's Lesson
0x21	Smoky Tune	0x31	To Arms
0x22	Cornwall	0x32	Indian Victory
0x23	Shady Grove	0x33	Natives
0x24	Fiddler's Dance	0x34	*(event-only; no picker row)*
0x25	Jine the Cavalry	0x35	Tenochtitlan
0x26	Joe Clark	0x36	Pizarro at Cuzco
0x27	Little Fiddle	0x37	*(event-only — Fountain of Youth, requested @0x0618ED)*
0x28	*(unnamed, independence-class scheduler-only)*	0x38	Bonny Morn
0x29	Love Forever	0x39	Hornpipe
0x2A	York Fusiliers	0x3A	Hole In The Wall
0x2B	Washington Artillery March	0x3B	Nightingale
0x2C	Road to Boston	0x3E	*(event-only; requested @0x02F30A/@0x05C93D/@0x07544B)*
0x2D	Independence Way		
0x2E	The Reveille		
0x2F	Successful Campaign		

The id→XMI resolution happens inside the driver binary (`?SOUND.COL`), so the names of 0x28/0x34/0x37/0x3E are unmapped. One further id outside the table, **0x3F**, is played exactly once — at the intervention-force arrival (`mov ax,0x3F` @0x3D7B1); it has no picker row and its name is likewise unmapped.

24.2 The four pickers (func_023344 @0x023344)

One function drives all four GAME.TXT picker sections (PICKMUSIC / PICKINDEPENDENCE / PICKMILITARY / PICKINDIAN; section-name strings at file 0x1E428–0x1E450). Reached from the GAME menu, command 4 (@0x023617). Behaviour:

- **Preselect**: the current tune `[0x96]` maps to a picker row via a 28-entry jump table at file 0x0233E4 (ids 0x20..0x3B; 0x34/0x37 have no row — event-only).
- **Main menu**: rows 1–12 = the 12 folk tunes; rows 13/14/15 open the Independence/Military/Indian sub-pickers (each run via 0x181F:0x3FE) and offset the returned row: 13 → id = sel+0x28, 14 → sel+0x2D, 15 → sel+0x31 with a skip over 0x34 (@0x02351A).
- **On pick** (selection→id jump table at file 0x02353A): `mov [0x96],ax` then gated play via 0x181F:0x4C0. There is **no persistent lock** — normal rotation resumes when the tune ends.

The Sound Options dialog (`@SOUNDOPTIONS`, `func_0232AE` @0x0232AE) is the standard 3-row checkbox: row 1 → `[0xA2]` Background Music, row 2 → `[0xA0]` Event Music, row 3 → `[0xA4]` Sound Effects; results mirrored into `[0x5386]` @0x023301–0x023322; turning an option off immediately sends **driver command 1 (stop)** @0x023339.

24.3 The background scheduler (func_004EE6 @0x004EE6)

Pumped from the input-idle loops (verb 0x181F:0x470). Per pump:

1. Skip unless background music `[0xA2]` is on or a one-shot `[0x9E]` is pending; poll the driver with **command 8 ("playing?")** and return while a tune is still sounding.
2. **Forced-next** `[0x94]` wins if set. The queue-tune API `func_0050BC` (0x181F:0x48E) sets `[0x94]` and sends a stop so the pump switches immediately.
3. **Class requests** `[0x9A]` (set by events via 0x181F:0x498/0x4A2/0x4AC/0x4B6 = `func_0050F0/0050FC/005108/00513C`, plus the woodcut wrapper `func_00543C`) map through the jump table @0x005008: **1** → folk window A, **2** → folk window B, **3** → independence tunes, **4** → military tunes, **5** → tune 0x33 once, **6** → 0x35, **7** → 0x36.
4. Otherwise the RNG (seeded from `[0x83A8]`) picks a tune-index window:
5. **peace** (`[0x5382]&1` clear): indices 1–12 (folk), with a **1-in-9** crossover into 13–23 (the war/period tunes);
6. **War of Independence**: indices 13–18, with a **1-in-5** crossover back into folk.
7. Index→id via `func_004DF8` (table @0x004EAC); a result equal to the current `[0x96]` is re-rolled; playback through the gate `func_00518E`.

24.4 Event cues

CUE	WHERE
Woodcuts 0/1/9	music **class 2** request (folk window B) in the wrapper `func_00543C`
Woodcuts 3/4/5/6	direct tunes 0x33 / 0x35 / 0x36 / 0x39 (the 0x39 cue @0x0054A2 is wired but its woodcut has no caller)
Woodcut 11 (colony burning)	**sfx 0x53 + tune 0x32** ("Indian Victory") @0x05DFCB
Build first colony (woodcut 2)	**sfx 0x54** @0x040E00
Fountain of Youth	**tune 0x37** @0x0618ED, then woodcut 8
Cibola treasure	sound 0x3C
Native raid outcomes	sfx **0x4F** (stores looted), **0x4E** (gold seized), **0x5B** (raid wiped out)
First European contact	**fanfare id 0x8020+power** (high-bit ids address driver fanfare banks; `mov ax,0x8020` @0x58040)
Native contact / live-among-natives	fanfare **0x8024** @0x48C41 / @0x48EB7; native raze/massacre **sfx 0x53** @0x48EE6
Treaty-break attack (order flow)	**sfx 0x58** @0x220F9, right after the war-bit write @0x220E6 (§23.3)
War declaration / mercenary	war fanfare — class-request 4 via `func_005108(4)` before every WAR*/MERCENARY emit
Revolution	independence-class (3) requests: declaration @0x3DE88, intervention @0x3D790/@0x3D9A3, REF phase @0x3E2EF; intervention arrival also plays the unnamed id **0x3F** @0x3D7B1
Lost City sub-events	tune 0x33 queued @0x61910 (native outcome); tune 0x24 queued @0x61ABB (burial treasure); tune 0x32 queued @0x61B42 (desecration → war footing); class requests 1/2 @0x61BCE/@0x61920
First cargo in Europe (woodcut 9)	tune **0x24** queued @0x4208C alongside the class-2 request
Colony burn music	class-2 requests @0x5C8AC/@0x5CA2B (with the sfx 0x53 + tune 0x32 pair above)
New-game start	plays tune **0x39** @0x756E4 and queues **0x25** @0x759A0 in the page-0x1A init path
Turn-loop event tune 0x3E	requested @0x02F30A / @0x05C93D / @0x07544B (name unmapped)

24.5 Driver architecture

- **Load**: at boot `func_07845A` (called @0x0762E6) builds the driver filename from the template **`"#SOUND.COL"`** (file 0x1FD5A) — the **`#`** replaced by the sound-config byte `[0x2608]` — and loads it via DOS **int 21h AX=4B03** (load-overlay) in `func_01287A`; the driver image is identified by the tag **`"$sound$ "`** (file 0x2004B). Seven config words are fed to it; the writers of `[0x2608]` / `[0x260A..0x2616]` (setup-program output) are unmapped.
- **Vectors**: the driver header exports **5 entry vectors**, installed to DGROUP 0xA654–0xA667 by `func_012928`, reset @0x012976. Vector 1 = play/query command entry; vector 2 = shutdown flush; vectors 3/4 = ISR service entries.
- **Dispatch** (@0x01299A): a lock byte `[0x26C5]` guards re-entry — unlocked commands `ljmp [0xA658]` straight into the driver; locked ones queue (8 deep, ring at `[0x26B4]`, count `[0x26C4]`).
- **Clocking**: the timer ISR @0x00C6D9 calls **vector 4 every tick and vector 3 every 5th tick**. The exit path sends stop (id 1), polls id 8 until silent, then calls vector 2.
- **Command gate** `func_00518E` (AX = id): ids `< 0x10` are driver commands and always pass; **bit 0x20 ids (tunes) are gated on `[0xA0]`** (Event Music); **bit 0x40 ids (SFX 0x40–0x5F) on `[0xA4]`**; the surviving id goes `lcall 0x1059:0xA` into the driver. Command ids seen: **1 = stop**, **8 = query-playing**.
- **Sound Test cheat** (MENU `@CUP` cmd 0x69 → @0x023D86): numeric-entry dialog from DEBUG.TXT `@SOUND` ("Play what sound #?"), result `[0x9CC8]` → gated play — arbitrary id playback against the live driver.

PART VII

User interface

§25 UI engine — draw primitives, dialog framework, popups, menus

Every screen in the 1994 binary is painted through one resident draw-verb library (the far-call window `0x181F:NNNN`, resolved through the RTLink thunk table at file 0x1A5F0–0x1B5EF; a type-B thunk resolves as `target_file_offset = 0x2400 + (jmpf_seg<<4) + jmpf_off`). On top of the verbs sit four engine layers: the `@`-directive dialog framework (every GAME.TXT template dialog, plaque, and list menu), the gameplay-popup engine (speaker portraits + modal wait), the pulldown-menu engine of the in-game map bar, and the mouse/keyboard input pipeline. This section documents each layer to rebuild precision; §26 then applies them screen by screen.

25.1 The draw-verb vocabulary (0x181F:NNNN)

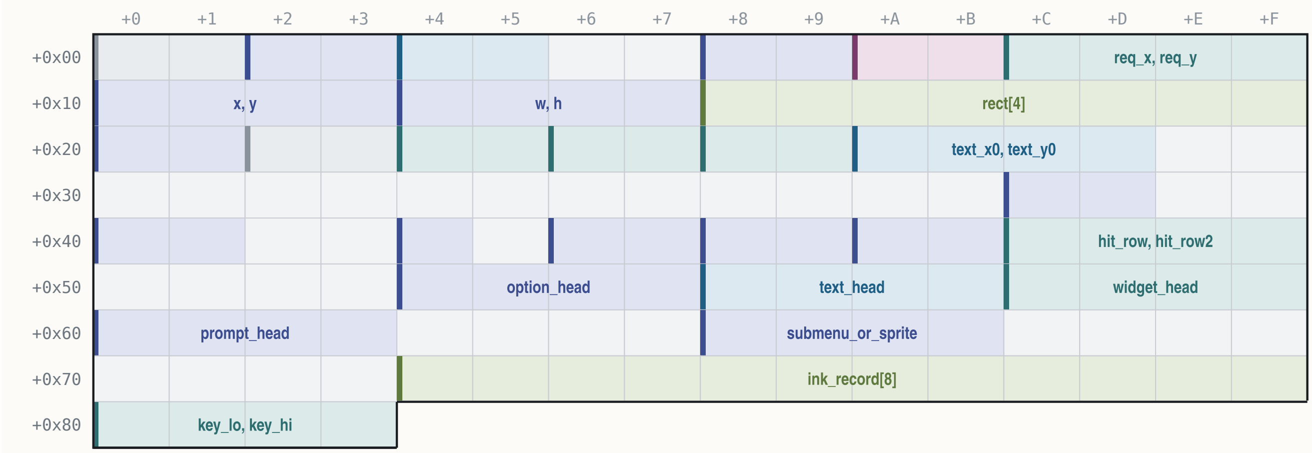
Text verbs split into two colour families: the **global-colour** family (`func_002Axx`) takes no colour argument — the pen colour is the screen-level latch byte `[0x830]` (with `[0x831]` as the hilite slot); the **explicit-colour** family (`func_002Bxx`) takes the colour as a push argument. Both bottom out in the string rasteriser core `func_00E51C` (`0x181F:0x1FA`), whose 2-bpp ink levels map through the 4-entry LUT at `[0x269E]:[0x26A0]`.

THUNK 0X181F:	TARGET (FILE)	CLASS	FUNCTION
`0x1FA`	0x0E51C	text core	proportional string rasteriser (`ch−1` glyph lookup, 2 bpp → ink LUT); all text bottoms out here
`0x204`	0x0E6A6	measure	pixel width of a string (Σ glyph widths) — the width source for every centred/right-aligned element
`0x22`	0x02462	fetch	**string-fetch-by-id**: walks N NUL-terminated strings in the heap at `[0x2D42:0x2D44]` and returns a far ptr to string #N. Draws nothing (an early "fill_rect" gloss is wrong)
`0x100`	0x02BC8	text	**centred** text-in-box (args: colour, y, box-w, x)
`0x13C`	0x02B38	text	draw text at explicit (x,y) — explicit-colour
`0x132`	0x02AFE	text	draw text at (x,y) — global-colour `[0x830]`
`0x150`	0x02B72	text	**right-aligned** draw (anchor x minus measured width; mechanism at 0x002B9F — the Colonizopedia "(Exit)" uses it)
`0x182`	0x029DE	build	append a **decimal number** to the working string buffer
`0x1A0`	0x02A06	build	append a zero-padded number
`0x16E`	0x02992	build	strcat into the working buffer
`0x114`	0x02AC6	measure	measure/justify helper (right-align maths)
`0x11E` / `0x128`	0x02922 / 0x02932	build	append literal `(` / `)` (glyph pool at file 0x1D9F0+, DGROUP 0x5E/0x60)
`0x1BE`	0x028F2	build	append `": "` (DGROUP 0x55)
`0x146` / `0x15A`	0x02962 / –	build	append `+` (DGROUP 0x66) / `–` (DGROUP 0x68)
`0x178`	0x028B0	build	append `" "` (DGROUP 0x50)
`0x196`	–	build	append `" "` × N (repeat-space)
`0xE2`	0x0DB3A	sprite/present	**clipped sprite/cell blit** from sheet ctx `[0x2DA8]` — used both for element frames and as the "present composed rect" step (`push h; push w; push x`). NOT a line rule
`0x254`	0x0E76A	sprite	**blit ONE sprite**. Convention: `AX` = frame (bit 15 = H-mirror), `DX` = x, stacked args = y + far sheet handle, `BX` = &surface-ctx (`0x2DA8` screen / `0x839E` work surface). Reads frame header `[bx]` =w−1, `[bx+2]` =h−1
`0x24A` / `0x1468` / `0x18F8`	0x0380C / 0x0D9E0 / 0x0DB80	sprite	blit variants
`0x2BC`	0x0386A	sprite	**per-unit info panel** composite (unit figure + condition) — map sidebar, Europe ship rows, F6/F7 reports, pedia UNIT page
`0x222` / `0x22C` / `0x218`	0x033F2 / 0x03104 / 0x03193	sprite row	enqueue icon+count (`0x222`, row counter `[0x2CE0]++`), open row (`0x218`), **flush** the accumulated row left-to-right into a fixed span (`0x22C`, `push 4` columns, span `bx=0x12C` =300)
`0x236`	0x02EE4	sprite row	**proportional filled/empty icon strip**: `count` filled sprites (AX) tiled across a span, pitch `(span−w)/(count−1)` clamped `[1, w+1]`, remainder = empty sprite 0x38 (hard-coded at 0x002FA5). The game's only "gauge" — there are no fill bars
`0x3C0`	0x04A80	modal	**wait-for-key/click loop** (~120-tick timeout at 0x4ADD, mouse region `[0x826]` / `[0x7F4]`). Draws NOTHING — the dialog builder paints the box first. Returns the key in DI
`0x444`	0x0DCF6	fill	rect block-fill / 2-D copy
`0x484`	0x0DCD4	fill	horizontal solid-colour span fill
`0xBA`	0x0DDEA	fill	**solid bar fill** (args `ax` =x, `dx` =y, `bx` =w; stack colour, h, sheet) — dialog interiors, highlight bars
`0xCE`	0x0E0A2	rect	**1-px HOLLOW rectangle outline** (h/v-span helpers `0xBBC:0xC` / `0xBC3:6`) — selection boxes, dropdown row highlight. Never a filled cell
`0xC4`	0x0E350	fill	**tiled fill** from a 4-word tile record (§25.2 fill-record system)
`0x510`	0x0531C	blit	src→dst stretch blit; sole call site 0x0263D6 is the colony-scene ×1.5 upscaler (§26.8)
`0x590` / `0xDAE`	0x0BCEA / 0x0BCAA	fill	span-fill + VRAM variants
`0x44E` / `0x438`	(overlay 0x76B9E)	load	**load_PIK by name** (appends ".PIK") / paired asset blit-by-name
`0x4C0`	–	sound	**gated sound play** (id gated on the sound-option globals; ids <0x10 = driver commands always pass)
`0x998`	0x6F51A	engine	**menu_lookup_run**: given a section name + file ctx `[0x87C]`, builds the `@`-template via `0x191F:0x182` (parser) and runs it via `0x191F:0x16A` (modal pump), returning the 1-based chosen row in AX
`0x3FE`	0x6F594	engine	**GAME wrapper**: hardwires file `"GAME"` (`[0x87C]`), section name in BX, then runs the same `0x998` core — the verb behind `@BEGINMENU`, the options dialogs and every plain GAME.TXT popup
`0x652`	0x6F5F2	engine	**advisor popup**: as `0x3FE` but first sets the advisor portrait channel `[0x1F5E]` → `MSS<n>.SS` (tutorials, announcements); the companion `0x1A1F:0x688` = 0x6F61C sets the third channel `[0x1F60]` → `MYR<n>.SS` for conversations
`0x3CA`	0x04B16	hit	point-in-rect against mouse `[0x7E8]/[0x7EA]` (args x,y,w,h)
`0x35C`	0x048CC	util	`clamp(v,lo,hi)` — NOT a draw (a mis-cite that propagated; corrected)

25.2 The dialog framework (the @-template layout engine)

One engine on overlay page 0x17 lays out and runs every GAME.TXT `@KEY` dialog, popup body, boot plaque and list menu: construct `func_06C520` (0x06C520) → template parser `func_06F0F4` (0x06F0F4) → finalize/layout `func_06D316` (0x06D316) → modal pump `func_06E3D0` (0x06E3D0), with row records appended by `func_044D16` (0x044D16, thunk `0x1A1F:0x33E`).

Dialog dialog struct (far ptr; les bx,[bp+4] in func_06D316)



132 bytes (0x84) · mapped 99 · unmapped 33

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	uint16_t	pad0	
+0x02..+0x03	2	uint16_t	option_count	option-row count (inc @0x06CA2B via func_06C850)
+0x04..+0x05	2	uint16_t	text_count	text-line count (inc @0x06CB87 via func_06CA82)
+0x08..+0x09	2	uint16_t	prompt_count	third item-class count (func_06CB94)
+0x0A..+0x0B	2	uint16_t	flags	0x10=borderless, 0x40=off-screen, 0x20=sibling, |=5 checkbox
+0x0C..+0x0F	4	int16_t	req_x, req_y	requested X/Y from @x/@y (−1 = centre sentinel)
+0x10..+0x13	4	int16_t	x, y	final on-screen X/Y (centre/clamp resolved)
+0x14..+0x17	4	uint16_t	w, h	box W/H
+0x18..+0x1F	8	uint16_t	rect[4]	final absolute rect (stores @0x06D5B9)
+0x20..+0x21	2	uint16_t	longest_px	longest-line pixel width
+0x22..+0x23	2	uint16_t	pad	= 4: option-row x-INDENT component (NOT outer width)
+0x24..+0x25	2	uint16_t	content_x0	= (flags&0x10)?0:3; option row x = +0x24+inset+pad = box_x+9
+0x26..+0x27	2	uint16_t	row_y_seed	= inset'(3)+border(3) = 6; += border+text_h if text present
+0x28..+0x29	2	uint16_t	width_floor	content-width floor: init 0x50 (80), overridden by @WIDTH
+0x2A..+0x2D	4	uint16_t	text_x0, text_y0	= 3 / 6; text line x = +0x2A+inset = box_x+5
+0x3C..+0x3D	2	uint8_t	fill_c1, fill_c2	fill pair ← [0x1F3C]/[0x1F3E] (TEXTCOLR.SS spr 1/2 pixels)
+0x40..+0x41	2	uint8_t	sel_c1, sel_c2	selection-band pair ← [0x1F40]/[0x1F42] (boot 0x37)
+0x44	1	uint8_t	ring2_c	ring-2 frame colour ← [0x1F44]
+0x46..+0x47	2	uint16_t	border	= (flags&0x10)?0:3
+0x48..+0x49	2	uint16_t	inset	= (flags&0x10)?0:2
+0x4A..+0x4B	2	uint16_t	content_cursor	content-height cursor (init 0; item y = border+cursor)
+0x4C..+0x4F	4	uint16_t	hit_row, hit_row2	cursor: the row under the pointer (pump stores)
+0x54..+0x57	4	void far *	option_head	option-row list head
+0x58..+0x5B	4	void far *	text_head	text-line list head
+0x5C..+0x5F	4	void far *	widget_head	child/widget list (pump loop B; painter func_06DE6E)
+0x60..+0x63	4	void far *	prompt_head	prompt list head
+0x68..+0x6B	4	void far *	submenu_or_sprite	attached submenu; on widget nodes = the ELEMENT
+0x74..+0x83	16	uint16_t	ink_record[8]	5 ink words + font ptr, built by func_06C296:
+0x80..+0x83	4	uint16_t	key_lo, key_hi	identity key / @SMALLFONT font latch copy

Template parser `func_06F0F4` (ENTER 0x168): reads section lines, blank line = paragraph, `@`-directive test `cmp byte [bx],0x40` at 0x06F193. Exactly ten live directives (keyword pool at file 0x1F967, DGROUP base delta 0x1D9A0): `OPTIONS` (mode 2 — option rows), `PROMPT`, `TEXT` (mode 1 — body), `SMALLFONT` (copies the current font latch `[0x89E]/[0x8A0]` into `+0x80` — it does NOT load FONTSMAL, which is never loaded by the binary), `Y` → `+0x0E`, `X` → `+0x0C`, `WIDTH` (pixel content-width **floor** → `+0x28`; "WIDTH\0" at file 0x1F989), `LENGTH` (text-entry max → `[0xA5B6]`), `CHECKBOX` (`flags |= 5`), `DEFAULT` (pre-highlighted row index — an index, not a colour). The eleventh pool string `TEXTCOLR` (file 0x1F9AA) is never compared by the parser — it is the sheet name for the **TEXTCOLR.SS colour-table load** in `func_06F6DA` (0x06F6F0): sprite-pixel reads there seed the ink globals `[0x1F3C..0x1F4E]`.

Box geometry `func_06D316` (0x06D316..0x06D889):

```
content_w = max(@WIDTH[+0x28], longest_line_px[+0x20], [+0x34])         ; clamp @0x06D392
box_w     = content_w + 2*border(3)                                     ; @0x06D4D0/@0x06D4E5 — NO pad term
box_h     = 2*content_cursor[+0x4A] + border[+0x46], item-driven        ; @0x06D35F..0x06D369
```

```
X = (req_x == -1) ? 160 - W/2 : req_x                          ; @0x06D522
Y = (req_y == -1) ? 100 - H/2 : req_y                          ; @0x06D53B
clamp: right > 0x140 shift left @0x06D563 ; bottom > 0xC8 shift up @0x06D571
negative left/top -> error logger 0x181F:0x772 @0x06D5AD
```

Vertical layout: text block pens from `+0x2C` (box-relative 6) with **text-line pitch = glyph_h + 1** (0x06D07E); if text is present the option seed bumps `+0x26 += border(3) + text_h` (0x06D440). Option rows sit at x = box_x+9, text/title lines at box_x+5; option text draws at row-pen+1 (0x06DB8C). **Row pitch = clamped_glyph_h + border**: the pitch helper `func_06CD66` (0x06CD66) returns the font cell height **clamped 6→5 on bordered dialogs** (`[0x1F8A]==0`, latched at pump entry 0x06E3F6), so a bordered FONTTINY dialog has pitch 5+3 = **8 px** (borderless: 6). Boot-menu check: `@y=91`, 1 title line → title top 91+6 = 97; first option top 91+6+3+6+1 = 107. Return AX=0 laid out, AX=1 empty-item bail.

Modal pump `func_06E3D0` (thunk `0x191F:0x16A`): hit-tests the frame bbox `+0x10..+0x16` against mouse `[0x7E8]/[0x7EA]` with gates `[0x7F6]/[0x7F0]`; loop A walks the option list from `+0x54` at y-seed `+0x26` with the 8-px pitch above (disabled rows — node flag bit 0 — are skipped; the hit row lands in `+0x4C/+0x4E`); loop B walks widgets from `+0x5C` (row top = dialog_y + inset + node[0], height = node[+2], action dispatch `call 0x3D26`). There is **no universal x/y constant** — both are per-dialog struct state.

Row records (`func_044D16`, 0x16-byte nodes): `+0` flags (bit 0 = empty ⇒ skipped), `+2` string-derived scalar, `+4` command id the row fires, `+6/+8` row text far-ptr, `+0x0E/+0x10` NEXT, `+0x12/+0x14` PREV. Nodes carry **no screen coordinates** — row (x,y) is computed at pump/paint time.

Selection bar (`func_06D9CC`, hit row == `+0x4C`): filled band at **(box_x+4, option_top−1, content_w−2, glyph_h+2)** — boot menu: (81, 106, 158, 7) — colour `+0x40` ← `[0x1F40]` (boot 0x37; tiled instead if the byte is 7). Pixel-confirmed: the extra 2 px once measured on the left was the box's own bevel column sharing palette index 0x37.

Box paint chain (driver `func_06E2DE` → box painter `func_06E0C8`, all chrome skipped when flags&0x10): 1-px **black outline** (colour 0 pushed immediate, `0x181F:0xCE`); ring 2 inset 1 colour `[0x1F44]`; ring-3 **bevel** at inset 2 — light `[0x1F46]` top+right, dark `[0x1F48]` left+bottom; interior fill (x+3, y+3, w−6, h−6) colour `[0x1F3C]/[0x1F3E]` via `func_06C18C`.

Fill-record system: `func_06C18C` (0x06C18C) takes the **tiled** path only when `[0x1F6C] != 0` AND fill colour 1 == **7** (the wood-tile sentinel) — then `0x181F:0xC4 = func_00E350` tiles the 4-word record at near ptr `[0x1F6C]`, phase-anchored at the **box origin** (`phase = |fill_xy − box_xy| mod (tile_w, tile_h)` at 0x00E371..0x00E3A2); otherwise a flat `0x181F:0xBA` fill. The boot loader builds three 32×24 tile records: `[0x93F0]` ← WOODTILE.SS spr 1 (0x07620F), `[0x93F8]` ← PARCH.SS spr 1 (0x07624D), `[0x9400]` ← OPENTILE.SS spr 1 (0x07627C; literal "opentile" at file 0x1FD51); default `[0x1F6C]=0x93F0`. Two mode setters: **in-game** at 0x073474 (inks from `[0x830..0x839]`, WOODTILE) and **boot/title** at 0x0734BC (text `[0x1F4A]=0xFE`, gold hilite `[0x1F4E]=0xFC`, disabled `[0x1F4C]=8`, ring `[0x1F44]=0x2E`, bevel light `[0x1F46]=0xFD` / dark `[0x1F48]=0x37`, selection `[0x1F40]=[0x1F42]=0x37`, **OPENTILE** `[0x1F6C]=0x9400`), invoked from the title composer at 0x075C52 right before the `@BEGINMENU` run. `[0x1F4E]=0xFC` at the boot menu is additionally confirmed by a live RAM read (0x95 in-game — state-dependent).

Ink selection per glyph (`func_06C346`): disabled → `+0x74+4`; hilite when `[0x1F62]!=0` → `+0x74+6`; else normal `+0x74+2`. `{` / `}` in any string toggle the hilite latch `[0x1F62]` 1/0 (`func_06C388`); `|` ends the visible span.

Frame sprite: the dialog element painter is `func_06D938` — it blits the sprite far-ptr stored at widget-node `+0x68/+0x6A` via `0x181F:0x254`, taking h/w from the sprite header. The WOODFRAM/NAMEPLAT chrome sprites are pre-loaded handles bound by the builder; the dialog overlay pushes no asset-name string itself. **Save-under**: dialogs and menus save the screen under their rect with mode `ax=0xFFF8` via `0x1A1F:0x364` → `func_078640` and restore via `0x1A1F:0x38A` → `func_0786FE`.

25.3 The popup engine (gameplay event dialogs)

Every gameplay popup (~30 GAME.TXT event templates: king demands, raids, Lost City, combat outcomes…) is the §25.2 engine plus a **speaker-portrait channel system**. Three DGROUP channel words select the portrait sheet; `0xFFFF` = no sprite:

CHANNEL	GLOBAL	BUILDER	SHEET BUILT FROM THE CHANNEL VALUE
King / tribe	`[0x1F5C]`	`func_06BE92` (0x06BE92)	`0..7 → IND<n>A<pose>.SS (tribe order = NAMES @TRIBES: Inca, Aztec, Arawak, Iroquois, Cherokee, Apache, Sioux, Tupi); >7 → KING<n>.SS (split cmp 7/jle at 0x06BE96; only KING1 exists — KING2 is byte-absent)`
Advisor	`[0x1F5E]`	`func_06BF12` (0x06BF12)	`0..5 → MSS0..MSS5.SS`
Missionary	`[0x1F60]`	`func_06BF3C` (0x06BF3C)	`0..3 → MYR0..MYR3.SS`

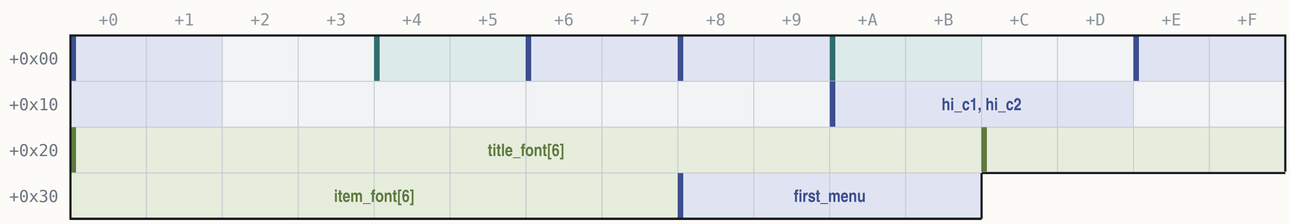
Popups reach the engine through per-channel emit wrappers: plain body = the `0x3FE` GAME wrapper; advisor-voiced = `0x181F:0x652` (`func_06F5F2`, sets `[0x1F5E]` before the run); conversation = `0x1A1F:0x688` (`func_06F61C`, sets `[0x1F60]` and returns the chosen row); king/tribe events set `[0x1F5C]` directly (e.g. `mov [0x1F5C],8` for `@KINGTAX`). The blitter `func_06BF66` copies the loaded sheet handle into a 0x14-byte cel and blits it via the page-27 graphics overlay (`0x1A1F:0x372`, file 0x76642) to the back-buffer 0xA000:0xFC00, clipped to the popup rect `[0x839E..0x83A4]`. **There is no coordinate literal** — the landing pixel is computed inside the blitter from the sheet handle and returned in AX/DX (stored back to cel `+0xC/+0xE`); the .SS directory carries no per-cel anchor field, so a specific frame's pixel is runtime state. After a popup closes all three channels reset to `0xFFFF` at file **0x06EE6B**.

Placement: gameplay popups are **centred** (no `@x/@y`); across all of GAME.TXT the `@width` histogram is {190: 336 sections, 220: 99, 300: 11, 310: 10, 160: 8, …}. Standard event popups (`@KINGTAX`, `@RAIDWREAK`, `@LOSTCITY0..9`, `@FOODLOW`, `@SHIPCOMBAT`, `@LANDFALL`, `@BURNED`, `@DECLARE`, `@INVASION` …) are `@width=190`; the wide set (`@TEAPARTY`, `@CASHTREASURE`, `@INTERVENTION`, `@INDEPENDENCE`, `@SONSUP`, `@SMITEINDIANS`) is `@width=220`. Only 21 sections carry a literal `@x/@y` (menus, tutorials, `@VICEROY` x=232/y=21, `@KINGLOSE` x=232/y=31, `@KINGWIN` x=202/y=125). Dismissal = any key/click via the modal wait `0x181F:0x3C0`; there is **no OK/Cancel button sprite anywhere** — the strings "OK"/"Cancel" do not exist in the binary as button labels; options are `@OPTIONS` text rows. Body font = FONTTINY (the engine default latch `[0x89E]`).

25.4 The pulldown-menu engine (in-game map menu bar)

The map bar (GAME / VIEW / ORDERS / REPORTS / TRADE / CHEAT / COLONIZOPEDIA) is a separate module on overlay page 0x0A. The bar object at `[0x896]` is built once by `func_072090` (0x072090) from the `game menu` data sections (`"game"/"menu"` opened at 0x0720BE via reader `0x191F:0x928`; 7 add-pulldown calls → `func_044B7A`, 91 add-item calls → `func_044D16`). The interaction core is **`func_0452D4`** (0x0452D4, 1559 B, page 0x0A) — the modal open/navigate/select tracker (an earlier attribution of the dropdown to the dialog pump `func_06E3D0` was overturned; that engine serves the `@`-directive dialogs).

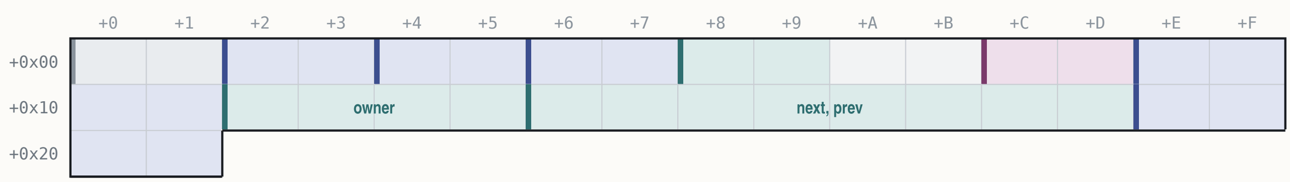
Menubar menubar object at [0x896] (created by func_044836)



60 bytes (0x3C) · mapped 46 · unmapped 14

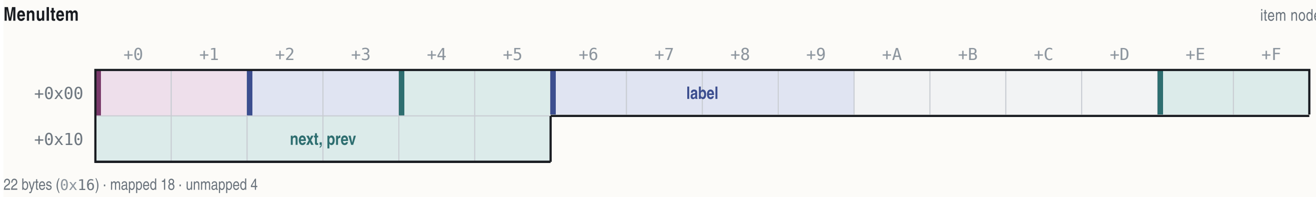
OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	uint16_t	result_cmd	selected command id (write @0x045895; 0 = none)
+0x04..+0x05	2	uint16_t	bar_y	= 1
+0x06..+0x07	2	uint16_t	title_gap	= 0x0C (12) between titles
+0x08..+0x09	2	uint16_t	item_leading	= 3
+0x0A..+0x0B	2	uint16_t	title_xpad	= 1
+0x0E..+0x11	4	uint16_t	bar_c1, bar_c2	bar colours ← [0x149C]/[0x149E] (runtime-filled,
+0x1A..+0x1D	4	uint16_t	hi_c1, hi_c2	highlight colours ← [0x14A8]/[0x14AA]
+0x20..+0x2B	12	uint16_t	title_font[6]	title font descriptor (string at +0x28)
+0x2C..+0x37	12	uint16_t	item_font[6]	item font descriptor (string at +0x34)
+0x38..+0x3B	4	void far *	first_menu	

Menu menu node (0x22 bytes, alloc @0x044BD9)



34 bytes (0x22) · mapped 32 · unmapped 2

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	uint16_t	pad0	
+0x02..+0x03	2	uint16_t	x	= prev menu x+width + gap; FIRST TITLE x = 0x0C (12)
+0x04..+0x05	2	uint16_t	title_w	
+0x06..+0x07	2	uint16_t	panel_inner_w	(init 0x0A, grown by add-item)
+0x08..+0x09	2	uint16_t	hotkey	title accelerator char
+0x0C..+0x0D	2	uint16_t	flags	bit0 = disabled
+0x0E..+0x11	4	void far *	title	
+0x12..+0x15	4	void far *	owner	owner menubar
+0x16..+0x1D	8	void far *	next, prev	
+0x1E..+0x21	4	void far *	first_item	



OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	uint16_t	`flags`	bit0 disabled, bit1 hidden
+0x02..+0x03	2	uint16_t	`shortcut`	item accelerator key
+0x04..+0x05	2	uint16_t	`command_id`	returned on commit
+0x06..+0x09	4	void far *	`label`	(empty first byte = separator row)
+0x0E..+0x15	8	void far *	`next, prev`	

Layout (`func_044FA4`): panel x = menu.x; panel y = bar_y + title-text-height + 3; width = `menu.+6` + 2; height = n_visible·(item_font_h + leading) + leading + 2; item x = panel_x + 1; first item y = panel_y + leading + 1. **Screen clamps**: right edge ≥ 0x13E → shifted so right = **0x13D** (317); bottom ≥ 0xC6 → shifted so bottom = **0xC7** (199). Bar draw (`func_044E7C`): full-width fill (0, 0, 320, title_h + bar_y + 1); selected title gets a highlight box in colours `+0x1A/+0x1C`; title text at menu.x + pad, y = bar_y (=1). Save-under before open (`0x1A1F:0x364`), restore + blit on close. Interaction: Alt-tap (`[0xB96]`) or a bar click opens; '8'/0x148 up, '2'/0x150 down (skipping hidden/disabled/separators, wrapping), 0x14B/0x14D prev/next menu, Enter accepts, Esc cancels, any other key is scanned against item `+0x02` shortcuts; releasing Alt closes. The result command id is switch-dispatched by the executor `func_0235D6` (0x0235D6, `switch [bp+6]`). The **CHEAT menu is built always** but hidden unless the cheat bit is set (`test [0x5383],0x20` at 0x072A8B → menu-record hidden bit; see §27.2 for the Alt-W-I-N combo).

25.5 Mouse / keyboard pipeline (summary; bindings in §27)

Mouse: one resident module (file 0xC980+, segment 0xA58) wraps `int 0x33` in exactly 8 call sites; in mode 13h the driver cursor is suppressed and a **16×16 software cursor** (transparent colour 0xFF, screen stride 320) is drawn by an installed AX=0x14 event handler (handler at 0xCB87). The poll/edge-detector at 0xD106 publishes: `[0x7E8]/[0x7EA]` cursor x/y, `[0x7E6]` raw buttons, `[0x7EC]` down-edge, `[0x7F4]` release-edge, `[0x7F6]` any-button-down, and `[0x7E4]` = `!(buttons & 1)` — **0 = left click, 1 = right click**, written only on a fresh press. There is no central hit-test table: each screen compares the globals against its own rects via `0x181F:0x3CA`.

Keyboard: 100 % BIOS INT 16h polling — `kbhit` at 0xD272 (AH=1) and `getch` at 0xD286 (AH=0); no INT 09h ISR is ever installed. `getch` normalises: printable key → AH zeroed (clean ASCII); extended key (AL==0) → scan code kept in AH, so callers distinguish letters from arrows/F-keys by the high byte. Wait helpers: `wait_for_keypress` (0x4A5C), `wait_keyOrClick` = `0x181F:0x3C0` (0x4A80), `drain_keyboard_buffer` (0x4AFA), interruptible `idle_poll` (0x4D1E, abort codes 0x110/0x12D → `[0x828]=1`). Dispatch is table-driven: the key indexes a normalisation/flag table at DS:0x27ED (bit 1 = case-fold −0x20) and then per-screen action tables; the executor is `func_0235D6`.

§26 Screens — geometry, fonts, keys, state

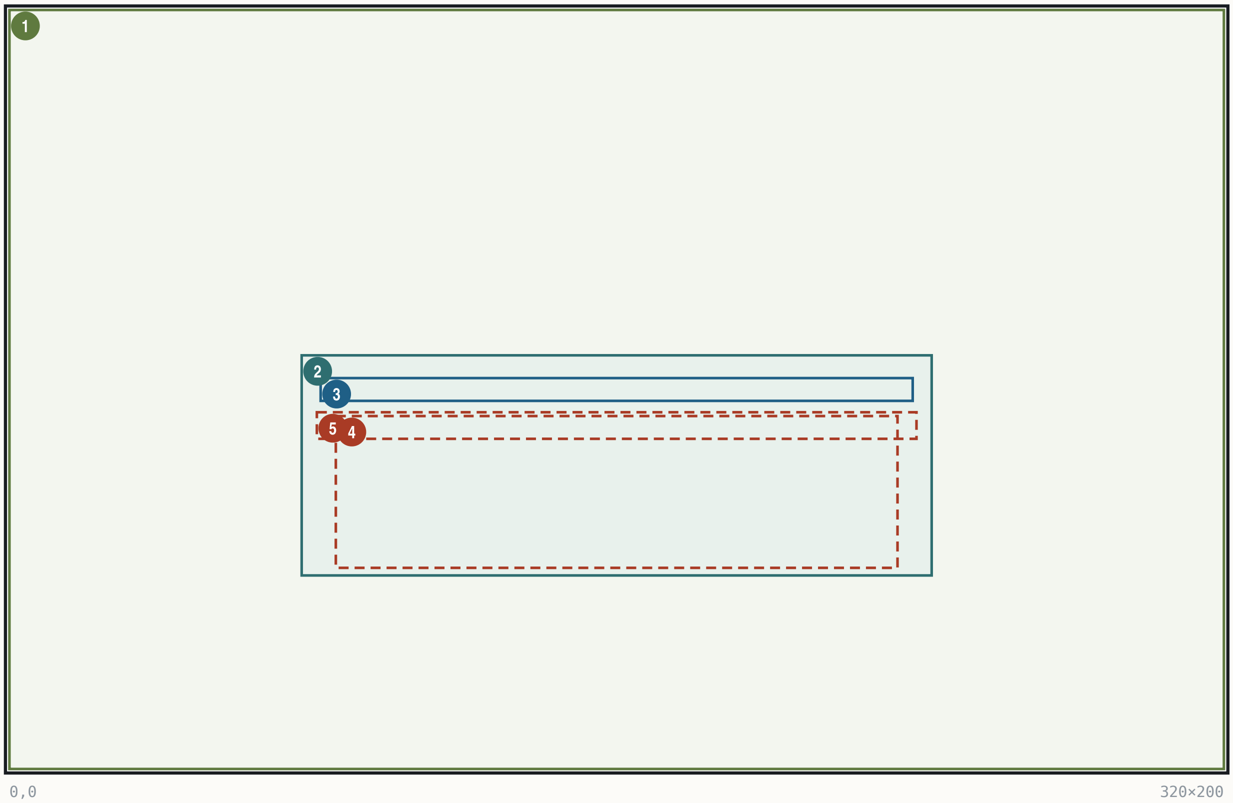
Every screen below is native 320×200 (mode 13h). Coordinates are byte-cited EXE immediates unless marked "(measured; not byte-cited)" — pixel-verified against the running game, 1994 binary under DOSBox. Palette indices refer to the screen's loaded PIK/gameplay palette.

26.1 Boot / main menu

The title screen's plaque menu: OPENMENU.PIK backdrop, one wood-framed dialog run from GAME.TXT `@BEGINMENU` (`@options @width=160 @y=91 @smallfont`), painted by the §25.2 engine under the boot-mode ink setter (0x0734BC). The three `0x1A1F:0xDF8` calls before the menu are full-screen **palette-index remaps** (7→6, 8→9, 15→14 — `func_00E146`), a no-op on OPENMENU.PIK; there is no "OPENBORD sprite" pass.

Boot / main menu

320×200 · Mode 13h



#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×200	art	OPENMENU.PIK backdrop	load_PIK @0x075AE4; composited over OPENING.PIK
2	(77,91) 166×58	panel	Menu plaque box	w=160+2*3 byte-derived; x=160-166/2; h item-driven (measured 58)
3	(82,97) 156×6	text	Title line	'{COLONIZATION} Version ...' x=box+5, top=box+6
4	(86,107) 148×40	hit	Option rows x5	x=box+9; tops y=107+8k, k=0..4 (pitch 8)
5	(81,106) 158×7	hit	Selection bar	(box_x+4, option_top-1, 158, 7) tracks the hit row

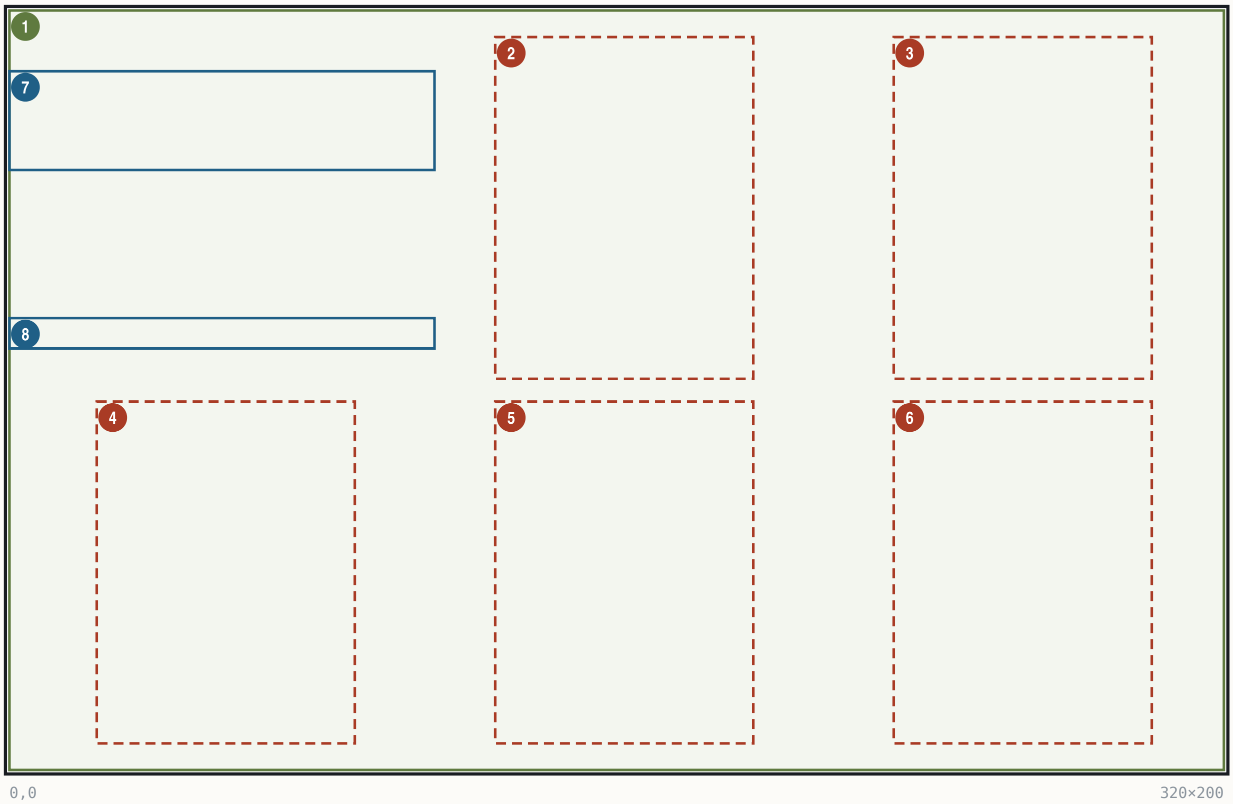
REGION	BOUNDS	KIND	CONTENT SOURCE	STATE BINDING
Plaque box	(77,91,166,58)	panel	dialog engine; OPENTILE.SS 32×24 tile fill phase-anchored at box origin (`[0x1F6C]=0x9400`, sentinel colour 7)	–
Box chrome	outline+2 rings	panel	black outline (idx 0), ring `[0x1F44]`=0x2E, bevel light 0xFD top/right, dark 0x37 left/bottom	boot-mode setter 0x0734BC
Title	y=97	text	`@BEGINMENU` title; `{...}` span in gold	hilite latch `[0x1F62]`
Options ×5	y=107+8k, x=86	hit	"Start a Game in NEW WORLD / Start a Game in AMERICA / CUSTOMIZE New World / LOAD Game / View Hall of Fame"	`dec ax` ladder at 0x075C6D
Selection bar	(81,106,158,7)	hit	flat fill colour `[0x1F40]`=0x37	hit row `+0x4C`

Fonts/inks: **FONTTINY** (the `@smallfont` directive copies the FONTTINY latch; pixel-proven — FONTINTR's 9-px cells cannot fit these spans). Text 0xFE, `{}`-hilite gold **0xFC** (live-confirmed), disabled 8. Keys: arrows move the bar, ENTER(13) selects, ESC(27) cancels, SPACE(32), digit + first-letter hotkeys. Dispatch (`dec ax` at 0x075C6D): 1=exit branch, 2=load/AMERICA sub-picker, 3=setup/scenario list, 4=new game → `begin_game` 0x072578. State: returned 1-based row in AX from `0x181F:0x3FE` run at 0x075C60.

26.2 Difficulty select

Full-screen DIFFICUL.PIK (five conquistador figures baked into the art); code adds only the labels, the finish prompt, and a 1-px hollow selection outline. Painter `func_070494` / `func_070580`, cell helper `func_0702C0`.

Difficulty select 320×200 · Mode 13h



#	BOUNDS	KIND	REGION	NOTE
1	`(0,0) 320×200`	art	DIFFICUL.PIK backdrop	load_PIK 'DIFFICUL' @`0x0705A8`; own embedded palette
2	`(128,7) 68×90`	hit	Discoverer cell	grid: x=(idx%3)*105+23, y=(idx//3)*96+7, idx=n+1
3	`(233,7) 68×90`	hit	Explorer cell	
4	`(23,103) 68×90`	hit	Conquistador cell	
5	`(128,103) 68×90`	hit	Governor cell	
6	`(233,103) 68×90`	hit	Viceroy cell	
7	`(0,16) 112×26`	text	Title 'Choose / Difficulty Level'	y=16/29, centred over the left column (measured)
8	`(0,81) 112×8`	text	Finish prompt	'(Click Here When Finished)' y=81 (measured)

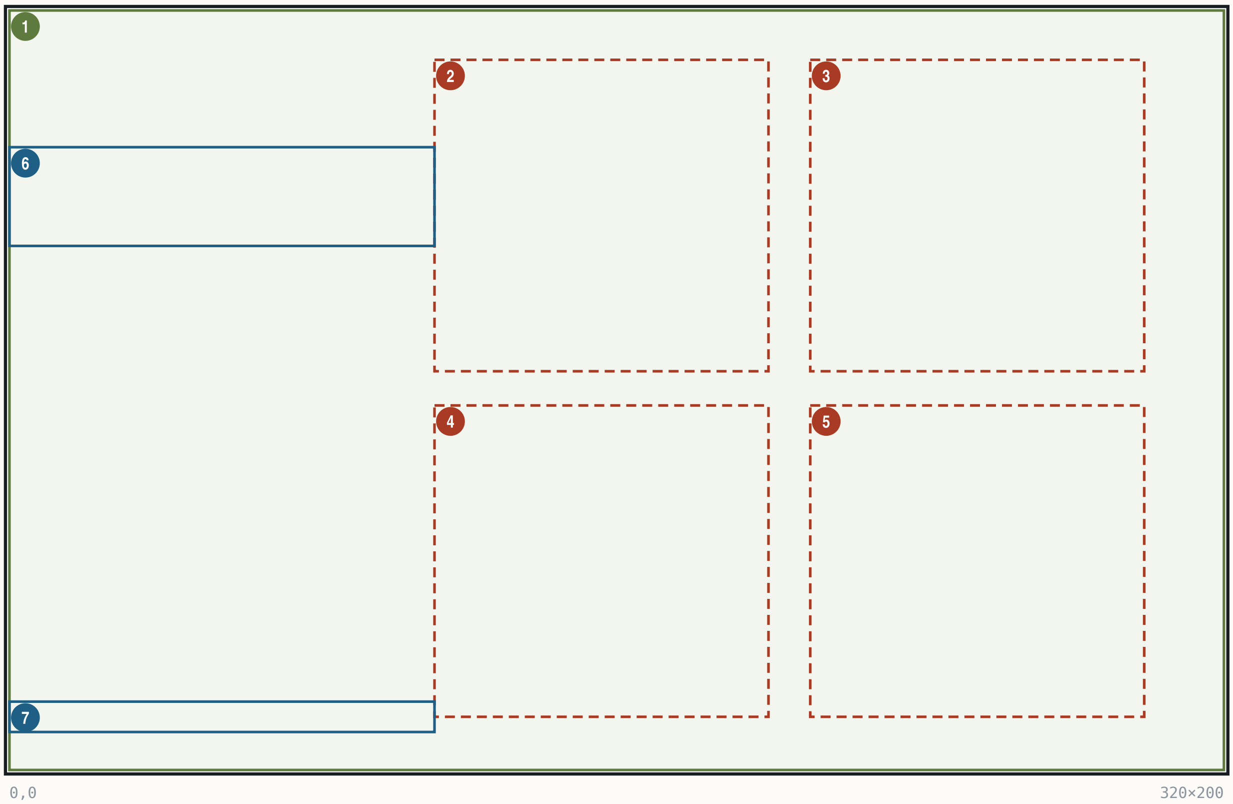
REGION	BOUNDS	KIND	CONTENT SOURCE	STATE BINDING
Cell i (0..4)	(col·105+23, grp·96+7, 68, 90)	hit	level name = NAMES `@DIFFICULTY` (Discoverer/Explorer/Conquistador/Governor/Viceroy); sub-label = LABELS `@MISC` 165–169 (Easiest..Toughest)	`selected → [0x53A6]`
Selection outline	selected cell	rect	`0x181F:0xCE` 1-px hollow; per-row colour byte {`0xA`, 9, `0xE`, `0xD`, `0xC`}; captured state shows ink 9 (blue)	`[0x53A6]==row`
Titles	y=16/29	text	`@MISC` 162/163 "Choose"/"Difficulty Level", FONTINTR, black shadow at (1,0),(0,1),(1,1) (measured)	–
Finish prompt	y=81	text/hit	`@MISC` 161 + parens, FONTTINY ink 254; commit zone = click with mouseY<103 & mouseX<128 (`0x07073A`)	`exit`

Fonts/inks: FONTINTR labels, inks level1=254 / level2=253 / level3=0 (measured; not byte-cited). Keys: up = (level+4)%5, down = (level+1)%5 (`0x070692`/`0x0706C8`), ESC exits. Name/description drawn centred in the cell for the selected row only. PIK-load failure degrades to the `@DIFFICULTY` text list via `0x181F:0x998`.

26.3 Nation select

Full-screen NATIONS.PIK (four flag plaques baked in), 2×2 grid, twin of §26.2. Painter `func_07092E` / `func_070A1A`, cell helper `func_070782`; menu-mode `[0x1F5C]=4`.

Nation select 320×200 · Mode 13h



#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×200	art	NATIONS.PIK backdrop	load_PIK 'NATIONS' @0x070A42
2	(112,13) 88×82	hit	England cell	grid: x=(i%2)*99+112, y=(i//2)*91+13
3	(211,13) 88×82	hit	France cell	
4	(112,104) 88×82	hit	Spain cell	
5	(211,104) 88×82	hit	Netherlands cell	
6	(0,36) 112×26	text	Title 'Select / European Power'	y=36/49, left-column-centred (measured)
7	(0,182) 112×8	text	Finish prompt	y=182 (measured)

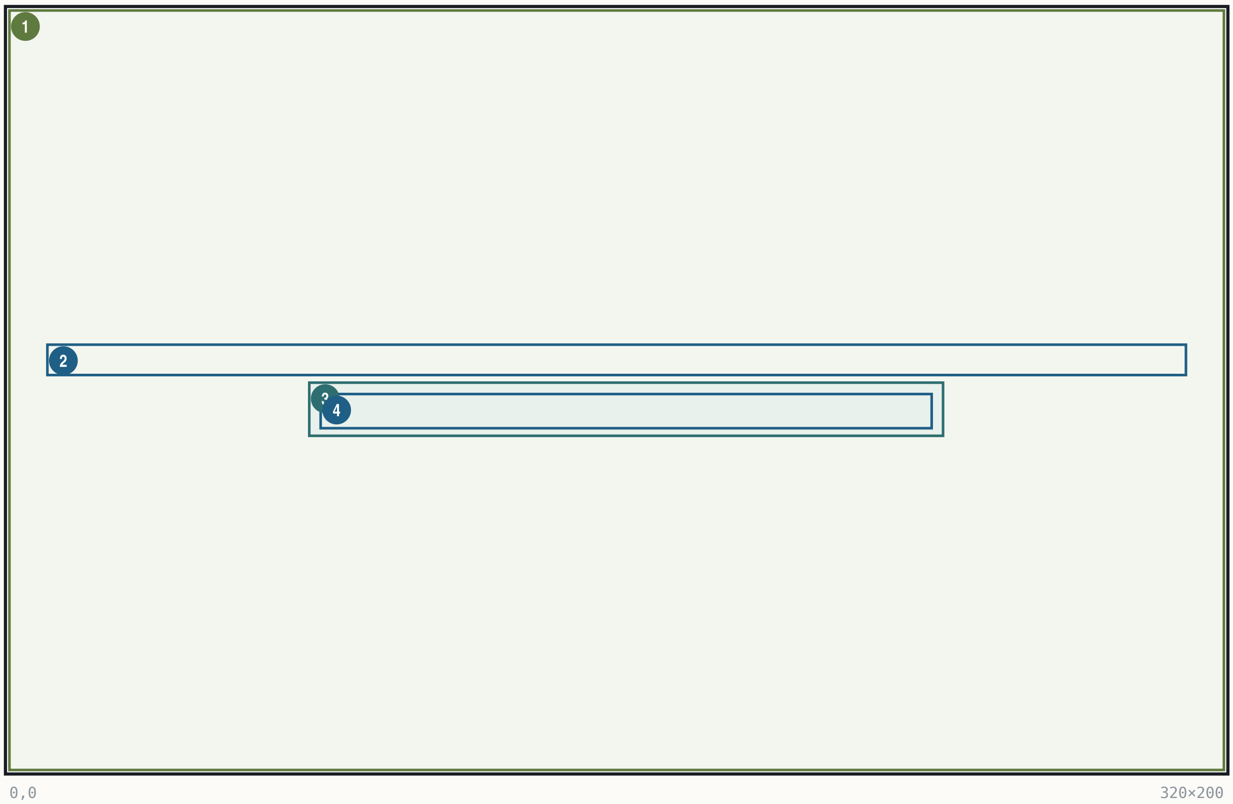
REGION	BOUNDS	KIND	CONTENT SOURCE	STATE BINDING
Cell i (0..3)	(col·99+112, row·91+13, 88, 82)	hit	NAMES `@COUNTRY` England/France/Spain/Netherlands; leaders `@LEADERNAME`; trait label at cell bottom = `@MISC` 173–176 (Immigration/Cooperation/Conquest/Trade)	`selected → [0x5398]`
Selection outline	selected cell	rect	`0x181F:0xCE` 1-px hollow, colour = per-nation flag byte `[bx+0x848]`; captured state ink 12 (red)	`[0x5398]==row`
Titles	y=36/49	text	`@MISC` 170/171 "Select"/"European Power", FONTINTR	–
Commit	click & mouseX<112	hit	left-margin zone (0x070BFC)	`exit, returns [0x5398]`

Fonts/inks: FONTINTR, same ink triplet as §26.2. Keys: arrows rotate mod 4 (0x070B40), ESC. Fallback: `@PICKNATION` text list. The per-nation flavour pages `@NATION0A..3B` follow (§26.5).

26.4 Leader-name entry

A WOODPANL-backed text-entry dialog (`@LEADERNAME`, `@width=300`, maxlen 23) shown after nation select; default text = the nation's leader name from the AIPersonality record 0x540E + n·0x34.

Leader-name entry 320×200 · Mode 13h



#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×200	art	WOODPANL.PIK backdrop	wood-panel background
2	(10,88) 300×8	text	Prompt line	@LEADERNAME body, centred; y=88 (measured)
3	(79,98) 167×14	panel	Entry field outline	green 1-px outline (measured; not byte-cited)
4	(82,101) 161×9	text	Entry text + cursor	default 'Walter Raleigh' + '_' cursor; X=160-W/2 centring

REGION	BOUNDS	KIND	CONTENT SOURCE	STATE BINDING
Prompt	y=88 centred	text	GAME `@LEADERNAME`	–
Field	(79,98,167,14)	panel	green outline + text-bbox background fill (measured)	–
Entry text	(82,101)	text	FONTINTR, proportional glyph advance; `_` cursor	`result copied to leader-name field 0x540E+n·0x34; @LENGTH max → [0xA5B6]`

Font: **FONTINTR** (this dialog carries no `@smallfont`; the FONTTINY rule applies only to `@smallfont` dialogs). The `@width=300` box draws no visible frame on this screen. Keys: printable chars append (proportional FONTTINY-style advance per glyph), ENTER accepts, ESC cancels.

26.5 Nation briefings (@NATION<n>A/B)

Two full-width text plaques shown once, at game start, immediately after leader-name entry — history page A then gameplay-bonus page B. Sole invoker: new-game setup `func_07431E`, which builds `"NATION0A"`, patches digit `buf[6] += [0x5398]` (0x07444F), shows page A, then `inc buf[7]` ('A'→'B', 0x0744A8) and shows page B. Both sections are `@width=300`, centred, run by the standard §25.2 engine over the WOODPANL backdrop.

Nation briefings (@NATION<n>A/B) 320×200 · Mode 13h

#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×200	art	WOODPANL.PIK backdrop	
2	(7,·) 306×· R	panel	Briefing dialog	@width=300 => box_w=306, x=160-306/2=7; y centred, h item-driven

Fonts/inks: FONTTINY body per the engine defaults; keys: any key/click dismisses (modal wait `0x181F:0x3C0`). State: nation index `[0x5398]` selects the section pair.

26.6 Intro caption cards (@BUILD1..10)

A self-advancing slideshow over world generation: ten full-screen LEVN PIK plates, each with a GAME.TXT caption block. Renderer `func_004B72` (resident): builds `"LEVN00"+n`, loads via `0x181F:0x44E` (card 1 blanks the screen and latches the 768-byte palette), then renders section `"BUILD"+n` with `^^`-centred lines; staged present `func_005160(8)`.

Intro caption cards (@BUILD1..10) 320×200 · Mode 13h

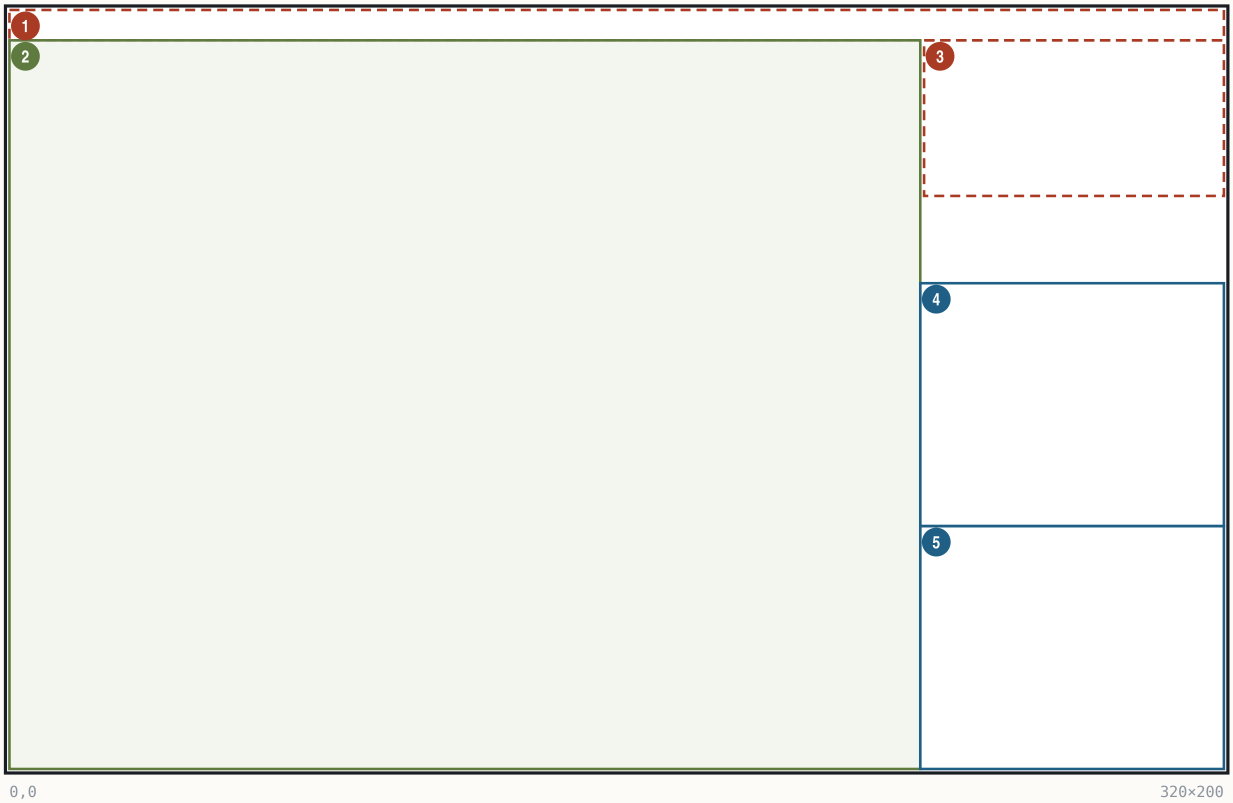
#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×200	art	LEVN000n.PIK plate	one per card, full-screen
2	(14,54) 292×· R	text	Caption text block	pen (14,54); ^ ^-centred lines

ITEM	VALUE	SOURCE
Text pen	`(14, 54)`	`func_004B72`; inks `[0x1F4A]=0x0E`, `[0x1F50]=0x36`, restored after
Card 2 substitutions	`%STRING0 = difficulty rank [0x8394+2·diff], %STRING1 = leader name (0x540E+p·0x34)`	byte-cited
Card 3	`%STRING0 = home port [0x838C+2·nation]`	byte-cited
Card 4	`%STRING0 = nation name, %STRING1 = @MYLEADER[nation] ("King/King/King/Stadtholder")`	byte-cited
Advance	`one card per 0x`**`23A ticks`**` via counter [0x8C] (sequencer func_004D1E = 0x181F:0x3AC, 34 call sites in world-gen)`	byte-cited
Skip / abort	`any key or click ([0x8A]=1); Alt-X / Alt-Q exits to DOS (exit code 3)`	byte-cited

26.7 Map view (main gameplay screen)

The default in-game screen: tile viewport left, wood sidebar right, pulldown bar on top. Tile rendering itself (terrain decode, zoom compositor, fog, units) is specified in Part II; this entry gives the screen geometry and bindings.

Map view (main gameplay screen) 320×200 · Mode 13h



#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×8	hit	Pulldown menu bar	bar fill h=title_h+2; titles y=1, first x=12, gap 12
2	(0,8) 240×192	art	World viewport	render_frame_setup func_06787C; 15x12 tiles @16px (zoom 0)
3	(241,8) 79×41	hit	Minimap panel	func_066CD6, panel box @0x66CF4; 1px/tile 56x39 window
4	(240,72) 80×64	text	Sidebar B: season/gold/tax	x-origin [0x8550]=240 (@0x071039)
5	(240,136) 80×64	text	Sidebar C: unit panel	sprite + @INFO labels; foreign-colony hover variant

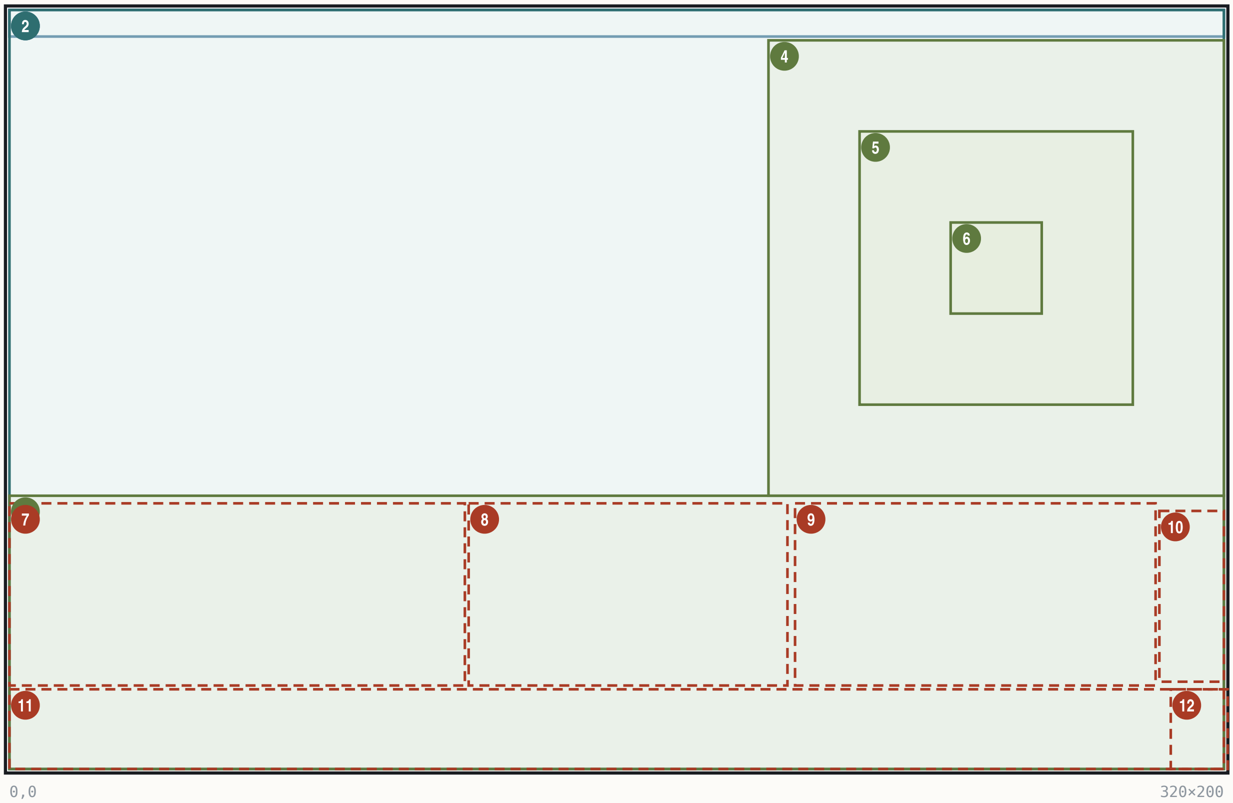
REGION	BOUNDS	KIND	CONTENT SOURCE	STATE BINDING
Menu bar	(0,0,320,8)	hit	7 titles from MENU sections `@GAME @VIEW @ORDERS @REPORTS @TRADE @CUP @PEDIA`; per-title x from glyph widths (mechanism byte-cited; exact x's not)	Alt-tap `[0xB96]`; result → `func_0235D6`
Viewport	(0,8,240,192)	art	zoom `[0x184]`: spans `0xF<<z` × `0xC<<z`, tile px `0x10>>z` → 15×12@16 / 30×24@8 / 60×48@4 / 120×96@2	cursor `[0x853E]/[0x8540]`
Minimap	(241,8,79,41)	hit	owner-dot colours from `[0x830..0x833]` = NAMES `@COLORS` bytes (68,149,8,128…, palette indices); viewport rect idx 0x0F	click recentres
Sidebar B	(240,72,80,64)	text	season NAMES `@SEASONS` + year `[0x538A]`; gold = PowerRecord+0x2A; tax = +0x01; FONTTINY white 0x0F	live
Sidebar C	(240,136,80,64)	text	unit sprite (`0x181F:0x2BC` panel), `@INFO` "Moves:/Locat:", type NAMES `@UNIT`, skill `@JOB`, orders, terrain name	selected unit `[0x5392]`

Sidebar per-line stack (single-frame measurement, approximate): season/year (244,58), Gold (244,66), Tax (290,66), unit sprite (244,80), Moves (270,82), Locat (270,92), type (244,104), skill (244,112), orders (244,120), terrain (244,128) — 8-px FONTTINY lines (measured; the per-line y is emitted through a runtime-installed printer pointer `[0xA644]`). Keys: see §27.1. Click own colony → colony screen (entry chain via set_active_colony at file 0x82DC).

26.8 Colony screen

The colony management screen: composer `func_028592` (0x028592) draws 12 ordered steps — terrain scene first, then a full-screen WOODTILE region fill composited over it, then title, panels, buildings. COLONY.PIK is a 320×72 town-scene strip blitted at y=128 (no embedded palette; renders on the gameplay palette). Entry: screen id 0x2C; active colony ptr `*[0x8542]` (ColonyRecord base 0x5D46, stride 0xCA; `+0` =cx, `+1` =cy, `+2` =name).

Colony screen 320×200 · Mode 13h



#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×7	text	Title strip	name+season+year+gold; paint 0x181F:0xB0, origin runtime (text-box globals [0x2CC6..])
2	(0,0) 320×200	panel	WOODTILE region fill	step 4 func_02633E; composited over the scene, from (0,0)
3	(0,128) 320×72	art	COLONY.PIK town strip	320x72 strip at y=128
4	(200,8) 120×120	art	5x5 scene (x1.5 upscale)	80x80 render stretch-copied via 0x181F:0x510 + 4x4 dither
5	(224,32) 72×72	art	Visible 3x3 scene window	central 3x3 of the 5x5; 24px tiles; outer ring overdrawn
6	(248,56) 24×24	art	Colony-centre tile	field-production centre cell
7	(0,130) 120×48	hit	Left panel: colonist plaza	region id 0; rows left-aligned at origin+2
8	(121,130) 84×48	hit	Middle panel: cargo dock	region id 8; 6 crate slots (ICONS disk 122) or centred caption
9	(207,130) 95×48	hit	Right panel: SoL/cargo/msg	x207..301 on screen (fill push was 211,91); mode [0x337]
10	(303,132) 17×45	hit	Nation flag panel	region id 3; ICONS 0x44 at +3, frame=[0x337]/[0x339]
11	(0,179) 320×21	hit	Stockpile bar	16 cells pitch 19; icons y=181; digits (9+19i,194)
12	(306,179) 15×21	hit	Exit caption/zone	region id 9; string-id slot [0x2F5E]=210 -> @MISC 'Exit'

REGION	BOUNDS	KIND	CONTENT SOURCE	STATE BINDING
Title	(0,0,320,7)	text	`colony name (rec+2) + NAMES @SEASONS[0x538C] + year [0x538A] + gold (PowerRecord+0x2A via money formatter); green FONTTINY`	live
Buildings ×15	table DS:0x266 (stride 4: x@+0, y@+2)	art	`BUILDING.SS frame =` **`def_id+1`** `(EXE-sheet; specials: def 0 + no-stockade → 0x11; def 0xF/0x11 garrison → 0x2F/0x30; blit 0x181F:0x254 at 0x026E4E); empty plot (byte 0x8E82[i]==255) → frame DS:0x260[category]-1`	def table 0x8E82, categories 0x8D62 = [0,0,0,0,0,0,0,1,1,1,1,2,2,3,4]; plot assignment = per-colony 16-bit-seeded RNG shuffle (`func_025D34`)
Plot positions	(56,13)(145,15)(173,18)(8,41)(37,45)(67,54)(96,53)(6,14)(128,53)(10,76)(15,102)(87,11)(66,87)(123,106)(123,55)	art	`the DS:0x266 table values as printed (no extra +8)`	—
Scene	(200,8,120,120)	art	`map compositor: 5x5 neighbourhood at 16 px from TERRAIN.SS [0x16C] + PHYS0.SS [0x174], colony/unit markers on the 80×80, then ×1.5 stretch (2→3 duplication) with positional 4x4 ramp dither (func_00531C / func_005296) — deterministic, no dedicated 24-px tileset`	colony (cx,cy)−2 origin
Scene workers	x=cell·24+252, y=cell·24+60	art	`PHYS0 sprites, drawn after the upscale; cells signed −2..+2 from DS:0xC8/0xDE`	colony+0x329 count

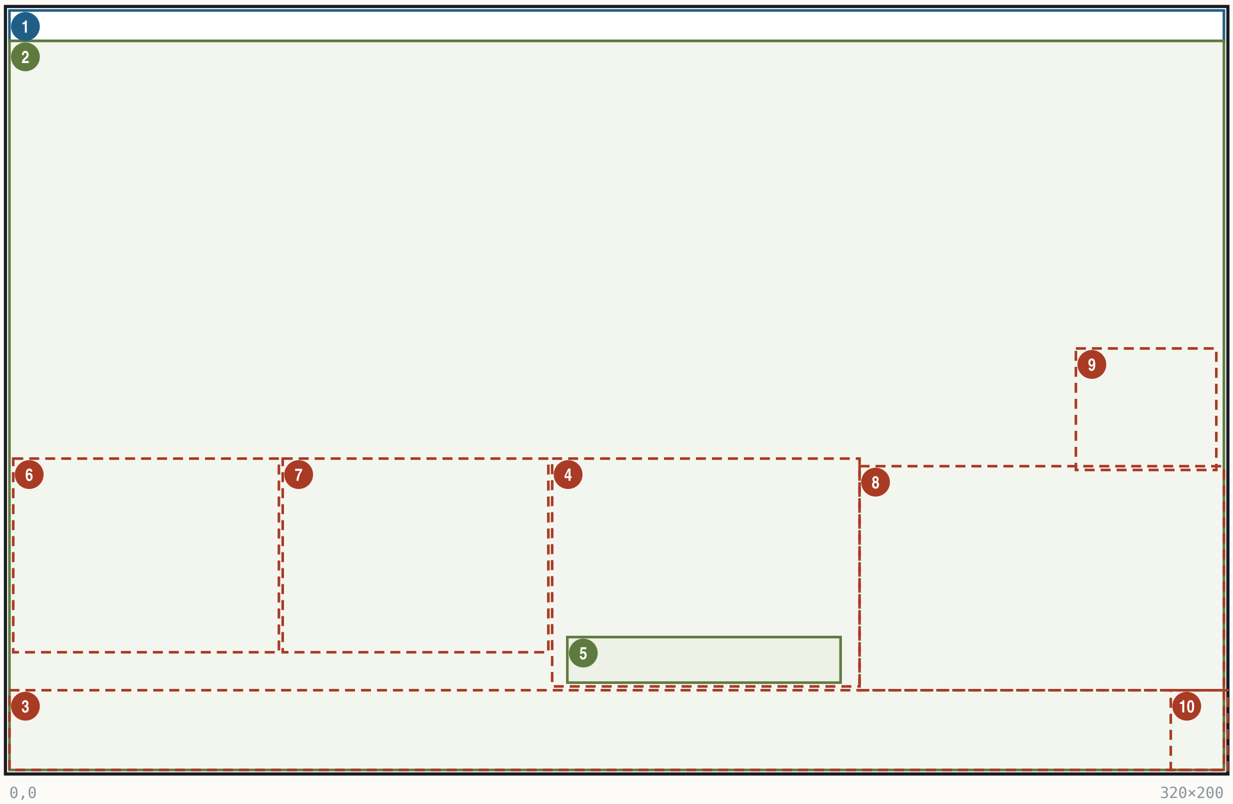
REGION	BOUNDS	KIND	CONTENT SOURCE	STATE BINDING
Plaza row	(0,130,120,48)	hit	colonist sprites, rows left-aligned at panel origin+2; pitch = sprite width + adaptive gap (2→0, fit-to-96px, 0x02715C)	count = colony+0x1F + [0x8D72]
Middle panel	(121,130,84,48)	hit	6 cargo-crate slots (ICONS disk frame 122) or centred caption via string-id slot + 0x181F:0x22 / 0x100	[0x33C]
Right panel	(207..301,130,48)	hit	[0x337] 3-way: 0 = SoL/garrison icon bar (0x181F:0x222 rows); 1 = cargo + caption [0x939A]; 2 = cargo + caption + hammer strip (0x236 sprite 55)	SoL% = (colony+0xC2·100)/colony+0xC6 (+20 human latch, clamp 100 — func_008524)
SoL band text	"100% (1)" at (75,133) white; "No Ships In Port" caption (118,130)	text	digit in parens (not letter I); caption = @MISC string id	live
Stockpile bar	(0,179,320,21)	hit	16 cells pitch 19 (0x13); icon = ICONS good+0x17 (EXE) at y=181; qty digits centred at (9+19i, 194), white 0x0F, red 0x0C over warehouse cap; order Food first	colony+0x9A 16×u16
Carpenter overlay	colonist ICONS 81 at (42,111); green box #55ff55 x39..50 y112..127; hammers ICONS 54 ×production at (15,104), (22,104),(29,104)	art	production count = live building-production state (strip verb 0x181F:0x236)	per-turn
Exit	(306,179,15,21)	hit	FONTTINY white "Exit" (string-id 210 of the 221-entry @MISC id table at 0x2DBA)	region id 9

Fonts/inks: FONTTINY throughout; title green (screen latch); digits white 0x0F / red 0x0C. Click regions (hit-tester at 0x299A0, ids): title 0xA, scene-left 2 (0,8,199,120), scene-right 1 (200,8,120,120), plaza 0, minimap 8, SoL 4, flag 3, stockpile 5 (0,179,305,21), exit 9, default 0x14. Keys (manual-sourced; routed through the multi-function display, no per-letter compare in the static image): Tab view-to-view, arrows within view, Enter jobs menu, L/=/+ load, U/−/_ unload, M toggle views, 1/2/3 production/units/construction, N numbers, C construction menu, B buy, F1 info, ESC exit.

26.9 Europe screen

The home-port harbour (screen-view id 0x2B, EUROPE.PIK key 0x0FBA; composer func_031E4C, 9-step chain). The dock town, market grid and the red "E" are baked PIK art; the engine draws the title band, market prices, dock contents, captions and the recruit menu.

Europe screen 320×200 · Mode 13h



#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×8	text	Title band	text y=1 centred on x=160; band rect (320,7,0,0) via set_text_box @0x035B24
2	(0,8) 320×192	art	Play-area fill over PIK	func_030D86 @0x031E4C
3	(0,179) 320×21	hit	Market bar	16 cells stride 19; icons x=1+19i y=181; bid/ask pairs y=194
4	(143,118) 81×60	hit	Dock rect	id 1; 'Loading: <ship>' centres in this rect
5	(147,165) 72×12	art	Dock ship slots x6	x=147+12k, y=165, 10x12 each; crate frame (disk 0x7A)
6	(1,118) 70×51	hit	Loading panel	id 3; caption slot 336 'Docks At'/Loading
7	(72,118) 70×51	hit	Bound For panel	id 2; caption slot 337/338
8	(224,120) 96×59	hit	Expected panel	id 4
9	(281,89) 37×32	hit	Recruit/Purchase/Train	id 5; rows (281,89+11r,37,9)
10	(306,179) 15×21	hit	Exit zone	id 0xB; white 'Exit' + red 'E' at (308,187)

REGION	BOUNDS	KIND	CONTENT SOURCE	STATE BINDING
Title band	(0,0,320,8), text y=1	text	"London, England. Spring, 1500. Tax:0% Gold: 1000$" — banner builder `func_030F76`; centred within box (320,7,0,0); the gold suffix is the FONTTINY '$' glyph; literal "Tax:0%" has no space	ink state-dependent: idle green **68**, arrival-banner gold **149**; good `[0x9E12]`, year `[0x538A]`, tax PowerRecord+0x01
Market icons	x=1+19i, y=181	art	ICONS `good+0x17` (EXE), 16 goods in NAMES `@CARGO` order	boycott: the good's own icon redrawn as the marker (gate 0x031A73)
Market prices	cell-centred, y=194	text	**bid/ask pairs**, FONTTINY ink 0x2F; bid = nibble of the per-good market record (base 0x3150 stride 0x1C); ask = bid + `@CARGO`.Burden + 1	live prices; click = buy/sell (sell handler 0x32914)
Selected-good outline	around the active cell	rect	1-px outline, yellow 14 / green 10 (runtime `[0x9E12]`-driven; measured)	`[0x9E12]`
Dock slots	(147+12k, 165, 10, 12)	art	crate sprite (`mov ax,0x7B` engine = disk frame 122)	ships in port `[0xFA2]`
Ship status rows	y=146 / 137 / 132	text/art	sail-state 1/2/3 bands (`func_031298` jump-table); bar width `0x64>>state`; type icon `@UNIT[type]`	per-ship sail distance
Captions	"Expected Soon" (16,120); "Bound For"/"New England" (87,120/127); "Loading:"/"Caravel" (150/186,120) (measured x/y)	text	fixed per-panel `@MISC` string-id slots: 336 Docks At, 337 Expected Soon, 338 Bound For, 339 No Ships In Port	in-port loop selects only colour (0xA/0xF)

REGION	BOUNDS	KIND	CONTENT SOURCE	STATE BINDING
Recruit menu	rows (281, 89+11r, 37, 9)	hit	LABELS `@EUROLABEL` "RECRUIT/PURCHASE/TRAIN/x"; rows horizontally centred; bevel colours 57/48 (measured), accelerator first letter yellow; row pitch 11 (measured — the "glyphH+2" formula does not reproduce it)	ink 0x0F / 0x00 by selection
Exit	(306,179,15,21)	hit	white "Exit" caption; the red "E" at (308,187) is PIK art	generic screen-view close; keys `x`/ESC/E

State fields: treasury PowerRecord+0x2A (u32), tax +0x01, boycott bitmask +0x20, price_level +0x4C[16], vol_accum +0x5C[16]. Keys (manual-sourced except the byte-corroborated `x`): Tab, arrows, Enter dock/harbor options, L/= buy full, + buy some, U/−/_ sell, R/1 recruit, P/2 purchase, T/3 train, F1 info, ESC/E exit. Water animation is pure palette cycling (indices 54–60), gated by the Water Color Cycling option (§27.4).

26.10 Combat Analysis dialog

A modal two-column modifier breakdown shown inside land-combat resolution (`func_05CA7E`), after the roll is computed and before resolution renders, when Game-Options bit 0x0200 is set and a human is involved. The dialog is `func_05E9B0` (page 0x11; thunk `0x1A1F:0x704`), called once at 0x05D291 with 13 args.

Combat Analysis dialog 320×200 · Mode 13h

0,0 320×200

#	BOUNDS	KIND	REGION	NOTE
1	(53,·) 214×· R	panel	Analysis box	x=53, w=214; h=rows*20+6, vertically centred (y=100-h/2)
2	(56,·) 80×· R	text	Attacker column	labels at x=56; values right-aligned at 56+0x50
3	(160,·) 80×· R	text	Defender column	labels at x=160; values right-aligned at 160+0x50

ELEMENT	VALUE	SOURCE
Frame	`0x1A1F:0x710`, x=53, w=214, h=rows·20+6, vertically centred; row pitch 20	byte-cited
Title	"COMBAT ANALYSIS" = LABELS `@MISC` line 75 (slot `[0x2E50]`)	byte-cited
Columns	attacker x=56, defender x=160; labels colour `[0x830]`, values right-aligned at col_x+0x50 colour `[0x831]`; per-column unit sprite + info panel (`0x181F:0x2BC`)	byte-cited
Inputs	per column: flag word `F=[col·2+0x8D00]`, secondary `S=[col·2+0xA156]`, base strength `[col·2+0x8D06]`	byte-cited
Dismiss	modal wait `0x181F:0x3C0`	byte-cited

Row table (flag → label → value): F&0x200 veteran-name row (base strength); F&0x400 Muskets "+1"; F&2 Veteran +50%; F&4 Cargo −12.5% per used hold; F&0x100/S&8 Fatigue −33%/−66%; F&1 Attack Bonus +50%; F&0x8000 Bombard +50%; S&2/S&4 Tory/Rebel Unrest −(100−SoL%)/+SoL%; F&0x80 Ambush (att) / Terrain (def, draws the target tile) +terrain_def·25%; F&0x40 Colony +(fortlevel+1)·50%; F&8(+0x10/0x20) colony-structure row +n·50%; F&0x800 Artillery In Open −75%; S&1 Artillery Vs. Raid +100%; F&0x2000 Fortified +50%; F&0x1000 Spain Bonus +50%; F&0x4000 Drake +50%. With the cheat bit (`[0x5383]&0x20`) extra rows show the final strengths and the raw roll.

26.11 Colonizopedia

The in-game encyclopedia: an alphabetized 3-column browser plus seven category entry-page renderers on overlay page 0x16 (one function per `@PEDIA` category). Reached from the eight `@PEDIA` pulldown items (commands 0x70..0x77 → `func_06B398`) and from per-screen context help (dispatcher `func_02BC72` on selection type `[0x32E]`).

Browser / index pager (`func_06B398` / `func_06B02A`):

Colonizopedia 320×200 · Mode 13h

#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×15	text	Title strip	'ENCYCLOPEDIA OF COLONIZATION' centred y=5, colour 0xF
2	(5,5) 100×7	hit	(More)	left, only when count>72; hover recolour [0x831]
3	(215,5) 100×7	hit	(Exit)	right-aligned to x=315, y=5 via 0x181F:0x150
4	(0,15) 320×185	hit	Index grid	3 cols x 24 rows; x=col*100+5 (5/105/205), y=row*7+25, pitch 7

ELEMENT	VALUE	SOURCE
Grid cell	x = col·100+5, y = row·7+25 (25..186); text at (x+2, y+1); cell hit 100×7	`func_069156` 0x069182/0x069190
Inks	normal `[0x830]`, highlighted `[0x831]`; highlight bar `0x181F:0xBA` w=textW+4 h=fontH+1 colour `[0x835]`; browser title colour 0xF literal	byte-cited
List	capacity 216; category sizes 16/23/21/27/38/25/12 (skips: terrain 0x10..0x17, Teacher, 4 buildings); forested terrains get " Forest" suffix; gnome-sorted alphabetically	byte-cited
Keys	Up '8'/0x148, Down '2'/0x150 (±1 mod count); Left '4'/0x14B −24 (wrap); Right '6'/TAB/0x14D +24; ENTER/SPACE open; ESC exit; "(More)" pages forward 3 columns cyclically	byte-cited
Font	FONTTINY (`[0x89E]`)	byte-cited

Shared entry-page skeleton (identical opcode sequence in all seven): WOODPANL.PIK backdrop (fallback fill colour 8); screen title `[0x2E92]` "ENCYCLOPEDIA OF COLONIZATION" centred y=5 colour `[0x831]`; entry header ": " centred at y = font_h+7; body seed y = header_y + font_h + 0xE

(JOB page: +font_h+3), x = 10; article = PEDIA section `<KEY><idx>` via menu_lookup_run (`0x181F:0x998`), text-window y-cursor `[0x1F5A]`; terminator: present `0xE2` (0,320,200) + modal wait. Sheets: ICONS.SS `[0x83E]`, BUILDING.SS `[0x842]`.

Per-page layouts (all byte-cited):

PAGE (FN)	LAYOUT FACTS
Cargo (`func_0694AE`)	production-chain rows, pitch y+=0x14; per row: job figure ICONS `job+0x52` at (10, y−2), cargo icon `cargo+0x17` then 6 more copies at x+=4 (7-icon stack), text " With " at (x, y+4) colour `[0x830]`
Unit (`func_0696C6`)	temp preview UnitRecord (destroyed at exit); figures via `0x181F:0x2BC`, pitch 18 px, x home 8; type 0 = 25-figure profession gallery, 17/row, wrap y+=0x14; name line at fig_y+6; stat line "Combat/Moves(÷3)/Cargo Holds" at (8, y) — type 0xB's "(and Damaged " is missing its ")" in the shipping binary; article `@UNIT0..23`, `[0x1F5A]` = stat_y+0xC
Terrain (`func_069D8C`)	header "(: Terrain Type)"; **3×3 tile preview** framed (7,y0)–(0x3A,y0+0x33) (52×52 double rect colours `[0x839]`/`[0x837]`), 16-px cells at x=9/25/41; base ground from the boot-rasterized 12-tile TERRAIN array `[0x16C]`; forest overlay PHYS0 `0x41+M[c+3r]`, M=[5,7,6,13,15,14,9,11,10]; hills 0x31+M, mountains 0x21+M; rivers (0,1)→0x17 (0,2)→0x1B; roads (2,0)/(2,1); centre resource `0x5A+R[id]` (word table at file 0x1DB32); stat rows right column x=63 pitch 0x10 — figure `0x52+j` + ": N" at (75, row+6) colour `[0x831]` + bonus lines colour `[0x830]`; movement/defence line "M / +25·defence%" at (63, row+6); article `@TERRAIN0..28`
Skill (`func_06A700`)	workplace building chain (BUILDING frame b+1, x += frame_w+3); job figure `idx+0x52` at (10, y+bldg_h/2−7); product strip icons `idx+0x17`, x+=0x10
Building (`func_06AA88`)	big picture BUILDING `rec+1` at (10, y) (idx 0x11→0x2F, 0x10/0x1F none); header at (10+w+3, h/2−7+y+6) colour `[0x831]`; worker figure + product; prerequisite line at (10, y) colour `[0x830]`, y+=0x14
Father (`func_06AE08`)	text-only: name (table 0x9652 stride 6), y += font_h+0xE, article `@FATHER0..24`
Concept (`func_06AF1C`)	text-only: 12 names from the PEDIA `@MISCELLANEOUS` loader, article `"MISC<n>"`

Inks: `[0x830]` = 0x44, `[0x831]` = 0x95 static inits (runtime rewrites possible).

26.12 Continental Congress (F3) + advisor-report geometry

The F3 Activities screen: full-screen REPORT3.PIK backdrop, text/sprite body `func_037A20` (0x037A10..0x3807D). Progress toward the next Founding Father is **text only** — "(NN in MM)" — the game has no fill bars anywhere.

Continental Congress (F3) + advisor-report geometry 320×200 · Mode 13h

#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×5	text	Title fill + centred title	fill colour 0x90; 'CONTINENTAL CONGRESS ACTIVITIES' (@MISC 37)
2	(4,25) 312×150	text	Body line stack	x=4, y-seed 25, pitch 8 (FONTTINY h6+2); colour 0x92
3	(4,·) 300×· R	art	Bell gauge row	0x181F:0x236: filled 0x3F / empty 0x38, span 300
4	(4,·) 300×· R	art	Rebel/Tory strip	sprites 0x7C x rebels + 0x7D x tories, span 300
5	(4,·) 300×· R	art	REF rows x2	0x222 enqueue x4 -> 0x22C flush, 4 columns, span 300
6	(4,·) 312×· R	text	Founding Fathers grid	cols x={4,82,160,238}, step 0x4E, colour 0x61, 4/row

State: PowerRecord +0x02 rebel%, +0x0C bells_current, +0x0E bells/turn, +0x12 FF-in-progress, +0x14 FF count; REF counts `[0x53DA/DC/E0/DE]` + naval `[0x53E2..E8]`; threshold computed by `func_03C282` (base `(diff+3)·2` human / `14−diff` AI, ×8, +50 % per year band 1600/1650/1700/ 1750, ×(FF+1)+1, halved at 0 FF; endgame override `diff·0x5DC+0x7D0`). The FF-acquisition **reveal popup** is a different path: full-screen CCBKGD.PIK (`func_03BB4A`), owned portraits CC-00..24.SS blitted at each sheet's own baked frame-descriptor coordinates, two-phase light-up, key/ESC dismiss — no frame, title, or OK widget.

Advisor reports F1–F10 — summary geometry (all bodies at file 0x37xxx–0x3Axxx; shared frame: REPORT.PIK, centred title in fill (0,0,320,~5) colour 0x90, footer sprite y=200, OK = `@MISC` 46 via the modal wait; body font FONTTINY, row flow = glyph_h+2):

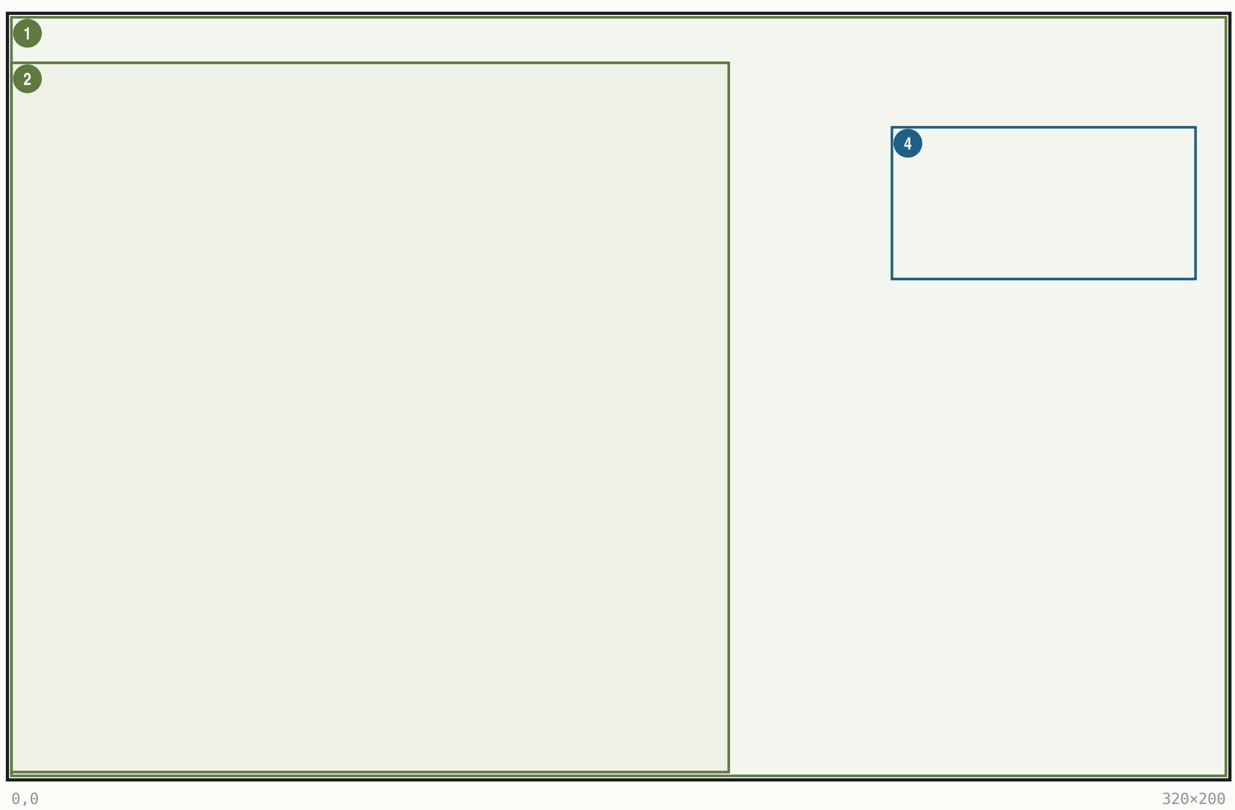
REPORT (BODY @FILE)	STATIC GEOMETRY
F1 Terrain (0x3744A)	rows y=0xA x=0x19; advance y+=0x1E then font+2; icon sprite terrain+0x72; right-justified counts at x=0x136−textW
F2 Religious (0x37958)	crosses gauge `0x236` X=10 Y=25 span 300, filled 0x39/empty 0x38; optional text x=10 y=25 colour 0xF
F3 Congress (0x37A10)	above
F4 Labor (0x38418)	matrix: name x=2, y-base 42, pitch 8, colour 0x92; count at +0x27 colour 0x61; dark-red (0x77) separator line x=2..311
F5 Economic (0x38A50)	headers x=76/170/220 y=25; commodity table x=2 stride 17; value cols x=250/150 stride 12; y 25/33 pitch 8
F6 Colony (0x39218)	rows base (2,20) pitch 17, 9/page; name colour 0x92 at +0x17; 4 centred captions y=27 at (2/82/162/242, box 80/80/80/76)
F7 Naval (0x3954C)	4-col table: first row y=42, pitch 20, 7 ships/page; name LEFT x=26 colour 0x61; cargo sprite row; Location centred box (162,80); Destination centred box (242,76)
F8 Foreign (0x39888)	gate `[0x5382]&1` clear = draws; labels x=2 colour 0x91; power value columns x=13/80/160/240; full-width 0x77 separators

REPORT (BODY @FILE)	STATIC GEOMETRY
F9 Indian (0x39EE2)	rows from village table 0x54EC stride 18; columns x=16, +72, +20; y-start 24, second block 150; text colour = [0x830] (@COLORS basic)
F10 Score (0x3A9C0)	WOODPAN2 + SCORE plate, panel = largest i with i²/3 ≥ scaled score; FONTTINY labels + FONTINTR figures

26.13 King audience / King-defeat screens

One renderer, `func_075352` (0x075352), paints the King audience (tax demands), the player-wins (`@KINGLOSE`) and player-loses (`@KINGWIN`) plates. Backdrop = **KINGLSS.PIK** (throne room, empty chair, blank scroll); the outcome-selected **foreground** sheet (KING1.SS mocking king + dog / KINGLOSE.SS crying / KINGWIN.SS triumphant) and the nation banner (ENGLND1/2, FRANCE…, stem + digit) land by the .SS frame-descriptor anchor convention (descriptor stores centre-x / bottom-y; on-screen $x = ax - \lfloor w/2 \rfloor$, $y = ay - h + 1$).

King audience / King-defeat screens 320×200 · Mode 13h



#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×200	art	KINGLSS<n>.PIK throne room	load_PIK @0x0753A9
2	(0,12) 189×187	art	KING1.SS figure	desc (94,198) -> (0,12), bottom-anchored to row 199
3	(32,0) ·×· R	art	ENGLND1.SS canopy banner	desc (118,121) -> (32,0); nation stem + digit
4	(232,29) 80×40	text	Scroll header, 4 lines	per-line centred on x~271.5, tops y=29..61 (measured)
5	(232,·) 80×72 R	text	Scroll body, 9 lines	left-aligned x=232, pitch 8 = FONTKING h+1 (measured)

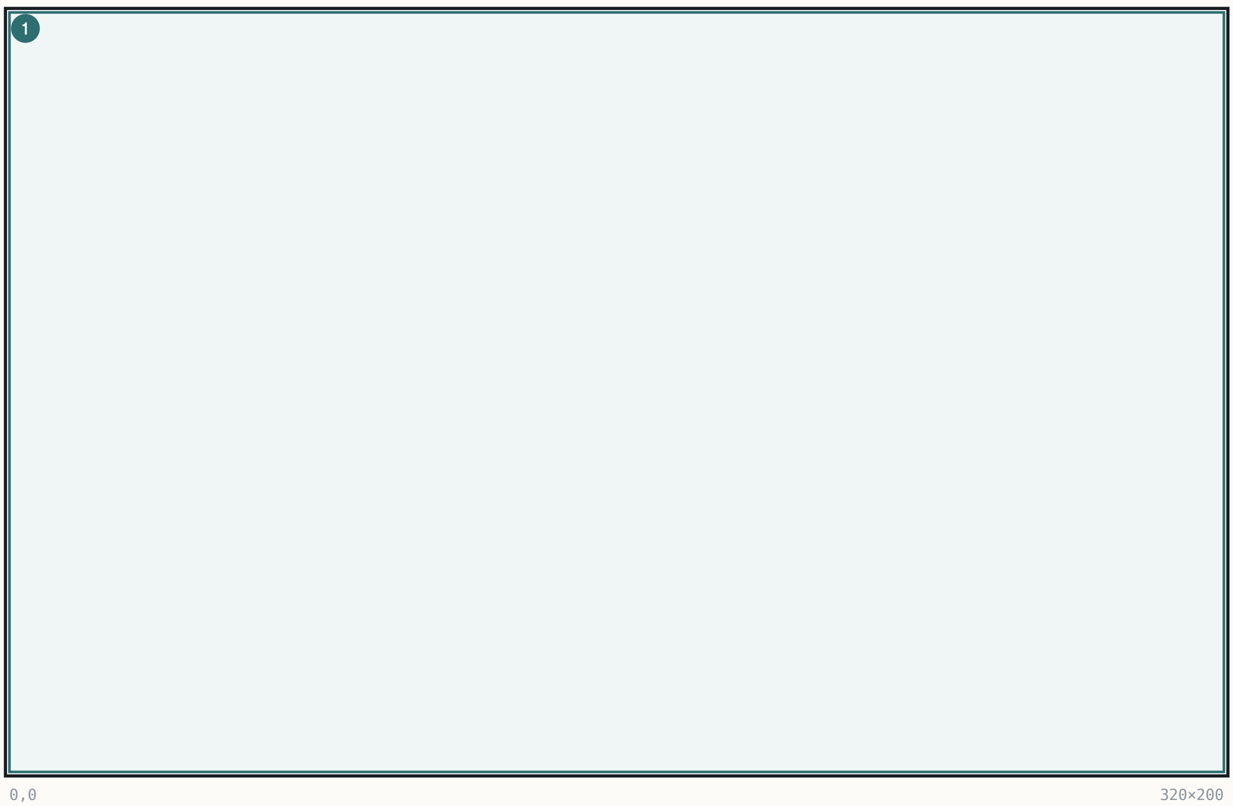
ELEMENT	VALUE	SOURCE
Variant select	(bp+6,bp+8): (1,1)→KING1 (audience); (1,other)→KINGLOSE (player WINS); (2,·)→KINGWIN (player LOSES); nation prefix switch on [0x5398]	0x075430 / 0x0753BB
Font	**FONTKING** (sole user in the binary; loaded 0x0754F6, dialog font latch [0x1F9E]/[0x1FA0]; falls back to FONTTINY)	byte-cited
Pen stores	[0x1F4A]=242, [0x1F50]=47, flags [0x1F56]|=0x18 — register values, NOT the on-screen origin; the glyph runner re-lays-out under the 0x18 flags	0x075526/0x07552C
Text layout	header per-line centred x≈271.5, tops y=29..61; body x=232, pitch 8 (measured; FONTKING metrics pixel-perfect)	measured
Body strings	GAME @KINGLOSE (@width=68 @x=232 @y=31) / @KINGWIN (@width=90 @x=202 @y=125); audience bodies built by the King-event orchestrator func_02F3A2	byte-cited (GAME.TXT directives)
Dismiss / choice	the @-menu run at 0x075540 (king's option list); page-flip + palette restore 0x075553	byte-cited

Ink: FONTKING is 2-bpp; level-3 measures black (idx 0) on this palette (measured).

26.14 Woodcut event screens

Full-screen carved-wood event plates (WDCUT01..13.SS) with a caption strip, shown once per event. Renderer `func_06B722` (0x06B722, `0x181F:0x52E`); once-only wrapper `func_00543C` (`0x181F:0x524`, shown-bitmask `[0x540A]`, per-woodcut music cues).

Woodcut event screens 320×200 · Mode 13h



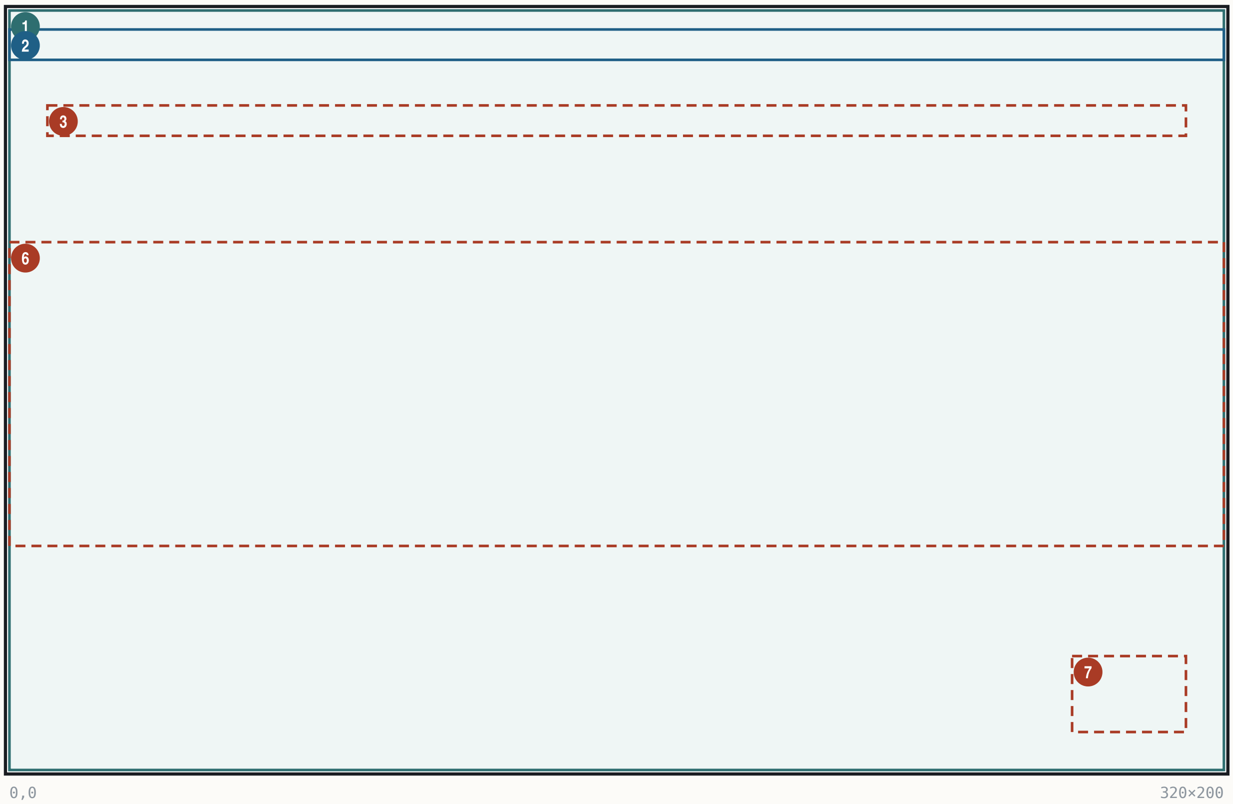
#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×200	panel	Black clear	
2	(·,·) ·×· R	art	WOODFRAM frame 1	centred from sheet-header words
3	(·,·) ·×· R	art	WDCUT<n> art	plate blit inside the frame
4	(·,162) ·×· R	art	NAMEPLAT strip	left cap + N mid tiles + right cap, centred on x=160, y=162
5	(·,165) ·×· R	text	Caption	'<year>: <CAPTION>' centred y=165, FONT-NP

Caption = line n of the single `@WOODCUT` section, prefixed `"<year>: "` from `[0x538A]`; ink LUT palette indices 0x5C/0x5D/0x5E; staged present/fade `func_005160(8)`; modal wait; palette restore. Frame numbering is 1-based over disk descriptors. Trigger table (caller scan exhaustive): 1 DISCOVERY (first landfall), 2 BUILDING A COLONY (first colony, human), 3/4/5 MEETING THE NATIVES / AZTEC / INCA (first contact by tribe), 7 ENTERING INDIAN VILLAGE, 8 FOUNTAIN OF YOUTH (Lost City outcome 1), 9 CARGO FROM THE NEW WORLD (first Europe cargo), 10 MEETING FELLOW EUROPEANS, 11 COLONY BURNING, 13 INDIAN RAID; 0/6/12/14–16 have no caller (unreachable in the shipping binary).

26.15 Trade-route editor

The Edit Trade Route screen (page 0x12; commands 0x50 Edit / 0x51 Create / 0x52 Delete → `func_060FBC`/`func_0610B0`/`func_0612E6`; painter `func_06083A`). Data: RouteRecord stride 0x4A, max 12 (`[0x53A0]`): +0 name[0x20], +0x20 type (1=sea), +0x21 stop count (max 4), +0x22 stops[4] stride 0xA (dest word — 0x3E7 = Europe; load/unload cargo nibbles).

Trade-route editor 320×200 · Mode 13h



#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×200	panel	Clear colour 0x22	
2	(0,5) 320×8	text	Title	'EDIT TRADE ROUTE <n+1>' centred y=5, colour 0x0F @0x060898
3	(10,25) 300×8	hit	Route Name row	'Route Name:' + name at (10,25); click = @TRADENAME entry
4	(10,·) 300×8 ^{R}	text	Route Type row	'Route Type:' + Sea/Land at (10, glyph_h+0x1B)
5	(10,·) 310×8 ^{R}	text	Column headers	Destination / Unload / Load at y=0x37-glyph_h; x=w('0. ')+10 / 0x7D / 0xD0
6	(0,61) 320×80	hit	Stops table (5 bands)	rows y=0x3D..0x8D pitch 0x14; separators x=0x73 and x=0xC6
7	(280,170) 30×20	hit	OK button	box (0x118,0xAA)-(0x135,0xBD), label 'OK' (@MISC)

REGION	BOUNDS	KIND	CONTENT SOURCE	STATE BINDING
Stop row n	y = 0x3D + n·0x14	hit	"N. " at (10, rowY+8); unload icons from x=0x7D, load icons from x=0xD0 — ICONS `cargo+0x17`, advance sprite_w+2	route stops[4]; row=(y−0x3D)/0x14; x<0x73 destination, <0xC6 unload, else load
Cargo cells	icon columns	hit	click icon = remove (shift left); click space = `@CARGOUNLOAD` / `@CARGOLOAD` 16-row menu (width 120), max 6	nibble get/set `func_0603DA` / `func_06040A`
Destination picker	shared dialog	hit	`@TRADESTART` header; rows = eligible own colonies; Europe row for ships only (per-nation port name `[0x838C]`)	`func_060026` (also serves Go-To orders)
OK / exit	(0x118,0xAA,0x1E,0x14)	hit	y≥0xAA exits; Enter/Esc exit	commit

Labels from the runtime `@ROUTE` table `[0x93DE..0x93EE]` (9 entries). The editor creates a phantom probe unit at (0xFF,0xFF) to filter reachable destinations (deleted at exit). Create flow: cap check → dest 1 → coastal test → `@TRADETYPE` → default name = colony + random `@TRADENAMES` word → name entry → dest 2 → editor. Delete compacts the array and fixes unit links (unit byte +0x17: lo nibble route, hi nibble stop).

26.16 Founding-Father pick dialog

The `@WHICHFREEDOM` dialog (width 190, centred, standard engine), posted by the colony-turn update when no FF candidate is in progress. Rows = one weighted-random candidate per each of the 5 categories: "FATHERNAME (Category Adviser)" — category names from NAMES `@FOUNDING` (Trade/Exploration/Military/Political/Religious), "Adviser" from `@MISC`; row id = category+1.

Founding-Father pick dialog 320×200 · Mode 13h

0,0 320×200

#	BOUNDS	KIND	REGION	NOTE
1	(62,·) 196×· R	panel	Pick dialog	@width=190 => box_w=196, x=160-196/2=62; y centred, h item-driven

Bindings: **cannot cancel** (result ≤0 re-shows, 0x03C231); right-click/help (`[0x1F68]`) opens the pedia FATHER page for the candidate, then re-shows; result → `[0x84FC]+0x12` = father id. Acquisition fires the `@FREEDOM` popup then the CCBKGD reveal (§26.12). Candidate weights = NAMES `@FATHERS` era-weight bytes (era bands <1600 / 1600–1699 / ≥1700).

26.17 Tutorial / popup placement notes

Tutorial hints (`@TUTORIAL1..19`) are ordinary GAME.TXT popups through the §25.2/§25.3 engine (portrait channel `[0x1F5E]` → MSS.SS), gated by Game-Options bit 0x80 (T18 ungated). Placement is either centred or a fixed GAME.TXT literal — never unit-relative:

SECTION	LITERAL DIRECTIVES
`@TUTORIAL1`	`@x=10 @y=40`
`@TUTORIAL4`	`@x=10`
`@TUTORIAL12`	`@y=5`
`@TUTORIAL16`	`@x=5 @y=10 @smallfont`
`@TUTORIAL17` / `@TUTORIAL18`	`@y=10 @width=300 @smallfont`
`@VICEROY`	`@x=232 @y=21`
`@KINGLOSE` / `@KINGWIN`	`@x=232 @y=31` / `@x=202 @y=125`
all §25.3 gameplay popups	centred, `@width` 190 or 220

Once-flags live in save bytes `[0x5380]/[0x5386]/[0x5387]`; the unit-focus dispatcher `func_020F50` (0x020F50) drives T1/T3/T8–T11/T13–T15/T19 from the selected unit `[0x5392]`.

§27 Input, cheats, and options

The complete keyboard surface of the shipped binary, the hidden Alt-W-I-N cheat system, and the three persisted option dialogs. Facts here fall into three trust classes: literal compare sites in the EXE (exact codes), `~`-marked accelerator letters in the TXT menu data (exact text; the per-row handler binding is data-driven), and printed-manual key lists whose engine wiring is runtime table dispatch — each class is labelled where it matters.

27.1 Keyboard maps

Dispatch model. The command executor is `func_0235D6` (`switch [bp+6]` on a normalised command/key id). Only the F-key report ladder is a literal compare chain (0x023843–0x02390B); the single-letter accelerators are data-driven — the `~`-marked letter parsed from each menu row (live-verified in RAM: the menu nodes carry the `MENU.TXT` labels verbatim) is matched against the typed key by the menu engine, and the matched row's command id dispatches. So the letters below are exact (string-table + manual-corroborated); the per-row handler binding is runtime menu-node state, not a static per-key compare.

Map view — global single-letter commands (from the `~` accelerators of the ORDERS/VIEW rows):

KEY	ACTION	KEY	ACTION
Arrows	move active unit / cursor (8-way with the keypad; menus and the pedia also accept ASCII '8'/'2'/'4'/'6' aliases of scancodes 0x148/0x150/0x14B/0x14D)	`G`	Go to port / place
`A`	Activate unit	`O`	Dump cargo overboard
`W`	Wait for next unit	`L` / `U`	Load / Unload cargo
`Space`	No orders (skip)	`T`	Begin trade route
`F`	Fortify	`Shift-D`	Disband unit
`S`	Sentry	`E`	Return to Europe
`B`	Build / Join colony	`V` / `M`	View mode / Move mode
`P`	Clear forest / Plow	`H`	Show hidden terrain
`R`	Build road	`Z` / `X`	Zoom in / out
`C`	Centre view	`ESC`	exit (confirm)
Alt+letter	open that pulldown (Alt-G/V/O/R/T…)	right-click	info popup

8-way movement / view scroll. Cursor movement is handled by the keyboard movement dispatcher `func_023F1C` (0x023F1C..0x0241CE) on the last scan code `[0x981E]`: the arrow/numpad arms bump the cursor `[0x17C]/[0x17E]` by ±1 (or the page step `[0x188]`), clamped to the map bounds `[0x853A]/[0x853C]`; the extended-scancode block 0x147..0x151 selects a **direction code 0..7** through the 9-entry jump table at file 0x024170 (`jmp cs:[bx·2+0x3290]`) — full 8-way movement including the keypad diagonals — which then either issues the unit move or scrolls the view (`0x181F:0xDA4`).

F-key reports (explicit `cmp [bp+6],code` ladder — all byte-cited):

KEY	REPORT	CODE	THUNK	BODY @FILE
F1	Terrain Information	0x48	`0x191F:0x41A`	0x3744A
F2	Religious Adviser	0x41	`0x191F:0x40C`	0x37958
F3	Continental Congress	0x42	`0x191F:0x3FE`	0x37A10
F4	Labor Adviser	0x43	`0x191F:0x3F0`	0x38418
F5	Economic Adviser	0x44	`0x191F:0x3E2`	0x38A50
F6	Colony Adviser	0x45	`0x191F:0x3D4`	0x39218
F7	Naval Adviser	0x46	`0x191F:0x3C6`	0x3954C
F8	Foreign Affairs	0x47	`0x191F:0x3B8`	0x39888
F9	Indian Adviser	0x49	`0x191F:0x3AA (gated, see §27.2)`	0x39EE2
F10	Colonization Score	–	`score path`	0x3A9C0

Colony screen (manual-sourced — the keys drive the multi-function display, no static per-letter compare): Tab view-to-view; arrows within view; Enter jobs menu; `L`/`=` load all, `+` load some; `U`/`-` unload all, `_` unload some; `M` toggle views; `1`/`2`/`3` production/units/construction; `N` numbers on/off; `C` construction menu; `B` buy; `F1` info; `ESC` exit.

Europe screen (manual-sourced; `x`/ESC byte-corroborated): Tab; arrows; Enter dock/harbor options; `L`/`=` buy full, `+` buy some; `U` sell, `-`/`_` sell all/some; `R`/`1` recruit, `P`/`2` purchase, `T`/`3` train; `F1` info; `ESC`/`E` exit.

Dialogs / popups: arrows move the highlighted row; Enter/click confirms; Space/Enter/click dismisses a message; row highlight = the `0x181F:0xCE` 1-px hollow outline. Boot menu: arrows, ENTER 13, ESC 27, SPACE 32, digits + first letters. Pulldowns: '8'/'2' or arrow scancodes navigate, 0x14B/0x14D switch menus, item-shortcut letters fire rows, releasing Alt closes. Internal abort codes 0x110/0x12D in the idle poll set the abort flag `[0x828]`.

27.2 The Alt-W-I-N cheat system

- **Master flag** = bit 0x20 of `[0x5383]`. Cleared at new-game init (`mov word [0x5382],0xC600` at 0x0755E5); survives load (`and word [0x5382],0x207F` at 0x02306A).
- **Enable combo: Alt-W, Alt-I, Alt-N** in the map key handler `func_023F1C` — sequence state `[0xB92]`, key compares 0x111/0x117/0x131 at 0x023FA9/0x023FB9/0x023FD0 → `xor byte [0x5383],0x20` + un/hide the CHEAT menu + redraw. No CLI or file enable exists.

- The CHEAT pulldown (`@CUP`, header "~CHEAT") is always built as menu 6 with hard-coded command ids, hidden while the bit is clear (`test [0x5383],0x20` at 0x072A8B).
- **Anti-cheat**: with cheat mode on, F10 Colonization Score is refused with a **beep** (`test [0x5383],0x20` at 0x0238D1 diverts the dispatch); F9 uses the same gate bit.

Cheat-menu command table (row order per `@CUP`; ids 0x62..0x6F):

ID	ITEM	EFFECT (BYTE-CITED)
0x62	F01 Create Unit	DEBUG `@CREATE` (peace) / `@CREATE2` (war): unit spawner at the map cursor (`[0x853E]/[0x8540]`); rows → unit types (row 9 → `@CSHIP` Caravel..Man-O-War; rows 10–13 Indians owned by the village under the cursor, or war remaps to Continental/King's forces; row 14 → `@FOREIGN`/`@FOREIGN2` creating-power picker)
0x63	F02 Debug Info Flags	the `[0x894]` checkbox dialog (§27.3)
0x65	F04 Reveal Map	`@SETVIEW`: view-as-power `[0x53A4]`, row 5 Complete Map `[0x53A2]=1` + clears the cheat bit
0x66	F05 Set Human Player	`@SETHUMAN`: all powers AI, picked power human (`[0x5398]/[0x5394]/[0x5396]`); "None" → `@SETAUTO` autoplay `[0x826]=1`
0x67	F06 Kill Indians	runtime tribe menu → `func_046FC2` destroys every village (base 0x54EC stride 0x12) owned by tribe+4
0x68	F07 Advance Revolution Status	`@FORCED`, staged: (a) rebel meter `[0x53D0]`=75 + create REF power; (b) declare independence (`[0x5382]|=1`); (c) next war stage (`|=2`); (d) `|=0x20` + text
0x69	Sound Test	DEBUG `@SOUND` numeric dialog ("Play what sound #?") → `[0x9CC8]` → gated play `0x181F:0x4C0` — arbitrary sound-id playback
0x6A	Memory Check	DEBUG `@MEMORY` display-only: far-heap / menu-arena / near / stack free + PSP segment
0x6B	F08 Show Strategy	`func_02165E`: plots the 64×4-byte AI strategy slots per power (BSS 0x98B0+power·0x100, {x,y,?,type}); pass 2 prints 14 counter rows at (5, i·7+10) colour 0xF
0x6C	F09 Show Colony Sites	`func_021602`: per-tile site desirability (low nibble of the tile flag byte) drawn over the map
0x6F	F010 Test Routine	`and [0x5382],0xF4` + dialog with unit count `[0x539C]` / colony count `[0x539E]`

Ungated debug: the `@DANGER` AI-assertion box (`func_078142`, 37 call sites) fires in the shipping binary on AI sanity-check failure.

27.3 The [0x894] debug bitfield (7 bits; session-only; default 8)

Builder `func_02356C` (cheat id 0x63): checkbox dialog over DEBUG `@OPTIONS` (7 rows), preset `and dx,[0x894]`, rebuild `or [0x894],ax`. The field lives in **no save block** — session-only — and boots with **bit 0x08 already set** (live-verified at boot and in-game; invisible without the cheat bit).

BIT	ROW	TESTER	EFFECT
0x01	Anger & Friction Levels	0x004241 / 0x044303	white anger number at village px+2/py+9; info panel appends 8 tribe rows
0x02	Indian AI movement	0x0470A3	shows AI moves + tile flash when visible to the human
0x04	Supply and Demand (Indians)	0x0494DA/0x0495DE	16-good supply/demand dump, x=1, y=8·(g+1), colour 0x0F, blocking getch
0x08	Foreign AI planning modes	0x003971 (ALSO requires the cheat bit)	AI units' map letters become their plan-mode char (≥0x80 → 'E')
0x10	Close Moves	0x061F14	per-tile path-cost overlay, red summary (5,190), Z/X zoom
0x20	Far Moves	0x062975	"Far: %d(%d,%d)…" overlay
0x40	All Movement	0x062D94	sets latch `[0x1DF2]` honoured by the Close-Moves renderer

27.4 The three options dialogs

All are standard §25.2 checkbox dialogs over GAME.TXT sections; checkbox channel = bitmask word `[0x1F54]` (reset `0x191F:0x26E`, pre-seed `0x262`, read-back `0x306`).

Game Options (`@GAMEOPTIONS`, width 190, `func_022FD6`; state word `[0x5382]`, clear mask `and 0x207F` — deliberately preserving the 0x2000 cheat-master bit):

ROW	OPTION	BIT	POLARITY
1	Show Indian Moves	0x8000	direct
2	Show Foreign Moves	0x4000	direct
3	Fast Piece Slide	0x1000	direct (slide step 8 vs 10, zoom-shifted)
4	End of Turn	0x0800	direct
5	Autosave	0x0400	direct (rolling slot 9 each turn + slot 8 on decades)
6	Combat Analysis	0x0200	direct (gate at 0x05D221)
7	Water Color Cycling	0x0100	**INVERTED** — bit set = cycling OFF; side-effect: `[0x372]` master + vblank-synced full DAC restore when disabling
8	Tutorial Hints	0x0080	direct

Colony Report Options (`@COLONYOPTIONS`, width 220, `func_02311A`; state word `[0x5384]`, clear `and 0xFC00`). **All 10 bits are INVERTED — a set bit means "suppress"**:

ROW	OPTION	BIT	ROW	OPTION	BIT
1	Labels on buildings	0x0002	6	Report tools needed	0x0010
2	Labels on cargo and terrain	0x0001	7	Report inefficient government	0x0008
3	Report when colonists trained	0x0080	8	Report new cargos available	0x0004
4	Report food shortages	0x0040	9	Report Sons of Liberty membership	0x0100
5	Report raw materials shortages	0x0020	10	Report rebel majorities	0x0200

Sound Options (`@SOUNDOPTIONS`, `func_0232AE`): row 1 `[0xA2]` Background Music, row 2 `[0xA0]` Event Music, row 3 `[0xA4]` Sound Effects — mirrored into the persisted flag word `[0x5386]`; turning an option off sends driver command 1 (stop). **Pick Music** (`@PICKMUSIC` + 3 sub-pickers, `func_023344`): rows 1–12 = the folk tunes, rows 13/14/15 open the Independence/Military/Indian sub-lists; selection → tune id `[0x96]` (ids 0x20..0x3B) + gated play; no persistent lock — normal rotation resumes when the tune ends.

Persistence: `[0x5382]` (game options), `[0x5384/5]` (colony options) and `[0x5386]` (sound mirror) all live in **save block #3** (base 0x5380, size 0x8E), written by `func_0734F8` and restored by `func_073BB0` — all three dialogs survive save/load; `[0xA0]/[0xA2]/[0xA4]` and the water-cycling master `[0x372]` are re-derived on load. The debug bitfield `[0x894]` is in no save block — session-only. No configuration file exists.

PART VIII

The editor and appendices

§28 The map editor (MAPEDIT.EXE)

Colonization ships a stand-alone map editor, MAPEDIT.EXE (145,292 bytes), which shares the MADS engine and the game's asset files (VICEROY.PAL, TERRAIN.SS, PHYS0.SS, ICONS.SS, WOODTILE.SS, FONTINTR, FONTTINY, CURSOR.SS) with the main executable. Uniquely among the shipped binaries it retains a CodeView NB02 debug block, so every function in this section is cited by its real, compiler-emitted name together with its file offset. The editor reads and writes the 3-layer `.MP` map format (6-byte header `width u16, height u16, version u16 = 4`, then terrain / feature / continent layers of width×height bytes each; AMER2.MP is 12,534 = 6 + 3·58·72 bytes).

28.1 The CodeView symbol trove

The debug block at file offset 0x1BE09 carries **1,071 public symbols**; a symbol's file offset is `segment·16 + offset + 0x1600` (0x1600 = MZ header size). The symbols partition the binary into named modules: the editor's own code in `mapedit.obj` (138 symbols), `popup.obj` (84), `map.obj` (76, plus `map_2/map_5/map_6/map_9/map_a`), `menu.obj` (38), `write.obj` (30), `text.obj` (14), `me_mini.obj` (13), `tile.obj`, `stuff.obj`, `strings.obj`, `env_1.obj`, `compass.obj`; and the MADS engine library (`mouse_1/mouse_2`, `mem_*`, `pal_1`, `ems_1/ems_2`, `xms_1`, `himem_1`, `pack_5`, `pfabcomp` (the FAB compressor), `loader_1`, `timer_1/timer_3`, `keys_4`, `sound_1/2`, `font_1`, `cycle_1`, `mcga_7`, `sprite_e`, `matte_0`, `heap_1`, `error_1`, `dos\crt0.asm`). These names anchor the whole section and — because the engine modules are shared — resolve otherwise-anonymous code in VICEROY.EXE (§28.8).

28.2 Startup and initialisation

`_main` @0x3ED8 scans the command line. Arguments beginning `-` or `/` go through the per-character `_flag_parse` @0x3E72:

FLAG	EFFECT
`-c`	`_create_me_now` = 1 — force the create-new-map path
`-m:file`	`_map_name` ← file, `_map_selected` = 1
`?` (any arg starting `?`)	`_show_flags` @0x3DF6 — prints the usage text at DS:0x3AA..0x487 and exits
any other letter	ignored

Otherwise control passes to `_viceroy_game` @0x3B16. On exit, a non-zero `_exit_value` prints `"Exit value: %d\n"`. Initialisation is strictly ordered and each failure sets a distinct exit code:

STEP	ASSET / ACTION	EXIT CODE ON FAILURE
video mode 0x13 (MCGA 320×200) @0x3B2E	–	–
palette	VICEROY.PAL	0x13
two 320×200 work buffers `_scr_work` / `_scr_orig`	–	0x14
`_font_inter`	FONTINTR	0x15
`_menu_font`	FONTTINY	0x16
`_load_terrain_tiles` @0xB152 → `_terrain_1`	TERRAIN.SS as a flat 12-frame 16×16 array	0x321 / 0x322
cursor	CURSOR.SS	0x17
`_tiles` / `_tiles2`	PHYS0.SS	0x18
icons	ICONS.SS	0x19
`_scr_back` 32×24 tile	WOODTILE.SS	0x1A / 0x1B

then `_get_tile_colors` (mini-map colour table, §28.5), `_load_data` @0x3936 (NAMES.TXT), a **12,000-byte undo buffer** `_map_undo_memory` (`_undo_available` = 1) @0x3D6E, `_start_new_game` @0x3A7A, and the main loop `_turn_control_loop` @0x38B0. The TERRAIN.SS-as-base-ground load order is one of the independent proofs that TERRAIN.SS is the ground sheet composited under the PHYS0.SS overlays.

28.3 Text-data loads

`_load_data` @0x3936 reads **NAMES.TXT**: `@UNFORESTED` fills terrain records 0..7, `@FORESTED` fills 8..15 and records 16..23 are `memcpy` aliases of 8..15 (@0x39B1–0x39CD), `@OTHER` fills 24..28 (Arctic/Ocean/Sea Lane/Mountains/Hills — the same authority order the game uses: 25 = Ocean, 26 = Sea Lane), `@OTHER_NAMES` fills `_terrain_names` ("Forest / River / Major River / Minor River / Unexplored"), and `@COLORS` fills nine palette-index globals (`_basic_color`, `_hilite_color`, `_grey_color`, `_enhance_color`, `_shadow_color`, `_select_color`, `_border0/1/2`, DS:0x92..0x9B), propagated to the popup engine by `_popups_normal` @0x1618. Each terrain record is 16 bytes (appendix A). The 13th NAMES numeric column is never read.

MAPMENU.TXT feeds `_construct_mapedit_menu` @0x1796: sections `@GAME` ("Editor"), `@VIEW`, `@CUP` ("Map"), `@HELP`. Menu items receive hard-coded event ids at the `_menu_add_item` call sites (0x13 Save As, 0x14 New, 0x1A Save, 0x1B Load, 0x1F Exit, 0x24–0x2B zooms, 0x4A–0x4E map operations, 0x51+ help). Dialog wording comes from the 19 sections of MAPEDIT.TXT.

28.4 Session flow

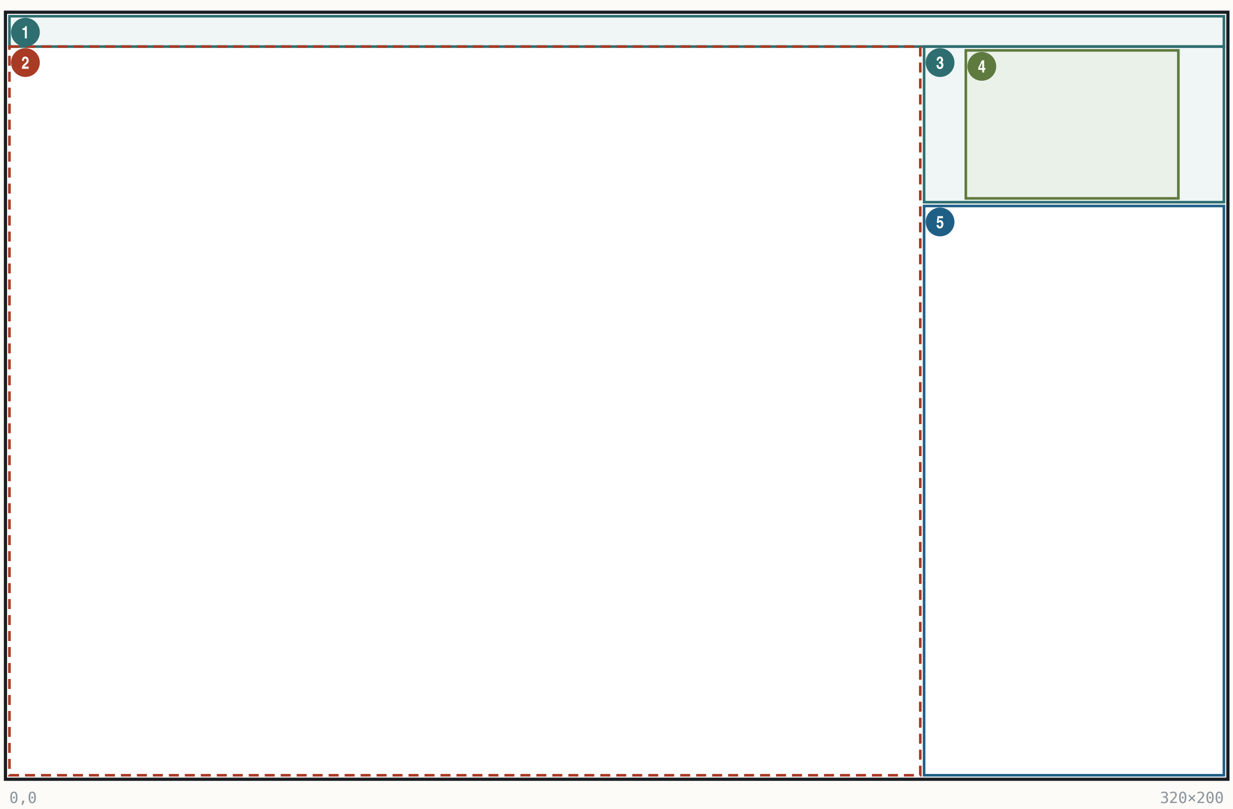
`_start_new_game` @0x3A7A:

- With no `-m`/`-c`: the **file picker** `_file_menu("MAPEDIT", "MAPTOEDIT", "*.MP")` @0x3A8C ("Select Map File to Edit / (ESC to create new map)"); ESC or cancel falls through to the create path.
- Map defaults `w=58, h=72` are set @0x3AB5; `@map_startup` allocates four 0x2EE0-byte (= 58·72 = 4,176-tile, 12,000-byte-rounded) layer buffers: terrain, feature, continent, plus a memory-only `_site` layer.
- **Create**: `_create_me` @0x2BFC — a name-entry popup (sections "MAPEDIT"/"NEWNAME", default `UNTITLED.MP`, max 0x14 chars) → `_create_blank_map(58,72)`: size hard-coded, every tile filled with **Ocean (0x19)**, version = 4, then `_map_changes`=1. **There is no map-size picker and no procedural map generation anywhere in MAPEDIT.EXE** — arbitrary sizes can only enter via files.
- **Load**: `_load_map_file` @0xB700 (header/size validation), then **`_forest_fix` @0x16B6**, which normalises forest alias ids 16..23 down to 8..15 and strips the forest id from tiles carrying the mountain/hill overlay bit.
- Both paths centre the cursor and the view at (w/2, h/2).

`_file_menu` @0x1B6A runs DOS findfirst/findnext over the pattern into 13-byte name slots at DS:0x64F0, pages of 10, with "(More)" pager rows (codes 0x61/0x62); the result goes to `_file_select`.

28.5 The main editor screen

The main editor screen 320×200 · Mode 13h · drawn into two offscreen 320x200 buffers _scr_work/_scr_orig



#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×8	panel	Menu bar	WOODTILE fill; FONTTINY titles (_main_screen_refresh @0x2317)
2	(0,8) 240×192	hit	Map viewport	fixed 240x192 px (_map_pixel_size 0xF0/0xC0 @0xB633/@0xB639; hit test _mouse_area @0x31CA)
3	(241,8) 79×41	panel	Mini-map panel	frame rect (251,8)-(308,48) (_show_mini @0xCF14)
4	(252,9) 56×39	art	Mini-map	1 px per tile, max 56x39 (@0xCF8D)
5	(241,50) 79×150	text	Info window	border (240,49)-(320,200) (_info_window_clear @0x1DBD); click opens Tile Select

Zoom (`@compute_view_parameters` @0xBA76): scale 0..3, visible tiles = (15<<scale) × (12<>scale. The four fixed levels:

KEY	SCALE	VISIBLE TILES	PX/TILE	SPRITE SCALE
F4	0 (startup default)	15×12	16	100%
F3	1	30×24	8	50%
F2	2	60×48	4	25%

KEY	SCALE	VISIBLE TILES	PX/TILE	SPRITE SCALE
F1	3	120×96	2	12%

The view corner is clamped to [1, dim−view−1]; maps smaller than the view are centred via `_map_tile_inset`.

Info window (FONTTINY, ink `_basic_color`, origin x=242 y=51, line pitch fontH+1; content painters @0x1F4E–0x22D2), top to bottom:

1. `Size: (w, h)`
2. `Curs: (x, y)`
3. `Terrain at cursor:` + name — with `" Forest"` appended for ids 8..0x17 and `(Major River)`/`(Minor River)` decoded from bits 0x40+0x80
4. `Selected:` + a 16×16 tool swatch (ground tile + PHYS0 forest overlay; or PHYS0 frames 4 / 0x14 / 0x24 / 0x34 for the river/river/mountain/hill tools)
5. shift-click help lines
6. `Fill radius: N`
7. `Coast Protect: ON|OFF`

Mini-map: 1 pixel per tile, window = clamp(centre −28, −19). Colour = `_terrain_colors[id]`, where ids 0..23 sample **pixel (8,8) of the tile's TERRAIN.SS frame** and Mountains/Hills sample PHYS0 frames 0x21/0x31 (`_get_tile_colors` @0xCBCC). A white (palette 0x0F) rectangle marks the current view (@0xCFB4).

Cursor: ICONS.SS frame **0x13+scale** (a 16/8/4/2-pixel box matching the tile size), blinking with a 20-tick (~1/3 s) period (@0x29C6/@0x372E).

28.6 The "Map Tile Select" screen

`_selection_screen` @0x2826 (menu id 0x4B, accelerator M, or a click on the info window) is a custom full-screen picker on black. Items are 16×16 on a **17-pixel pitch**:

- row 0, y=1: the 8 unforested base tiles;
- row 1, y=18: the forested variants (base tile + PHYS0 frame 0x41 overlay; Desert's forest uses the dedicated Scrub ground tile via id 0x11);
- bottom row, y=48: Arctic / Ocean / Sea Lane tiles, then PHYS0 sprite items frame 4 = Major River, 0x14 = Minor River, 0x24 = Mountains, 0x34 = Hills.

A white selection box is drawn at (x−1,y−1)–(x+16,y+16); the selected tool's name is labelled at (160,10). Arrow keys move ±1/±8 across the 24 slots; any other key or a click-release exits. The startup default tool is Ocean. Each tool compiles to a (sel, and, or, rmc) mask set (`_parse_spot` @0x26CC) applied to the terrain byte:

TOOL	SEL	AND	OR	RMC
terrain id t (rows 0/1, Arctic)	t	0xFF	0	0
Ocean / Sea Lane	0x19 / 0x1A	0x40	0	0
Major / Minor River	0	0x1F / 0x3F	0xC0 / 0x40	1
Mountains / Hills	0	0x1F / 0x5F	0xA0 / 0x20	1

28.7 Paint interaction

- Screen→tile mapping: `tx = mx/tsz − inset_x + corner_x`, `ty = (my−8)/tsz − inset_y + corner_y` (@0x35C4–0x35F1).
- **Plain click** recentres the view (`_set_center`). **Shift+left** paints (`_fill_map`): a square brush of side 2r+1 where r = `_fill_radius` 0..2, i.e. **1×1 / 3×3 / 5×5** (menu "Fill Mode Change", id 0x4C, cycles r=(r+1)%3 @0x3104); painting is continuous while dragging, with one undo snapshot taken at stroke start. **Shift+right** is context-sensitive: with a terrain tool it is an **eyedropper** (picks up the tile under the cursor as the current tool); with a feature tool it removes the feature (applies the and-mask only). Keyboard: Enter/Space = paint at cursor, Backspace = pick-up/remove.
- `_change_map` @0x31E0 enforces three guards: the **1-tile border ring is immutable**; while **Coastline Protect** is ON (`_coastline_protect` [DS:0x4E], toggled by menu id 0x4D) painting terrain onto Ocean/Sea-Lane tiles is skipped (@0x3265–0x327D); and the mountain/hill or-bit is **always** refused on water (@0x328B). Any accepted write sets `_map_changes`.
- **Undo** is a single slot covering the terrain layer only: a 12,000-byte copy taken at stroke start; `_perform_undo` @0x1D7E restores it (menu id 0x4E / key U, gated by `_undo_available`/`_undo_active`). The mini-map is rebuilt on every draw (`_small_map_needs_update` exists but is never referenced).
- Cursor movement: numpad 1–9 and Home/Up/PgUp/Left/Right/End/Down/PgDn give 8-way movement (jump table at file 0x346E), clamped to [1, dim−2], with auto-scroll when within 2 tiles of the view edge (`_possibly_center` @0x2A5E).

28.8 Menus and dialogs

Engine identity. MAPEDIT's `menu.obj` is **the same pulldown module as VICEROY's in-game menu bar, built from the same source**: the dropdown screen clamps @0x008E97/@0x008EA6 (`cmp [bp-8],0x13e` / `cmp [bp-2],0xc6` — right edge ≤ 317, bottom ≤ 199) are instruction-identical, with the same locals and constants, to VICEROY `func_044FA4` @0x04505F/@0x04506E, and the node shapes match (bar's menu list at +0x38, a menu's item list at +0x1E, `~`-hotkey extraction, pending-command word at struct offset +0).

Menu bar (`_construct_mapedit_menu` @0x1796): `_menu_create(0x800, FONTTINY)` @0x17AA → `_menu_bar` at DS:0x78. Constants (@0x86C3–0x86E6): bar y=1, bar gap 12, drop-row pad 3, bar text pad 1, drop text pad 4. Bar and dropdown backgrounds use fill colour 7,7, the sentinel that selects the **WOODTILE fill** — the pre-rendered 32×24 `_scr_back` tile (menu helper @0x83C2, popup helper @0x4D1A). Bar items read positionally from MAPMENU.TXT (`_text_get`, one line per call, `_` → space): "Editor", View, "Map", Help (Help right-justified at x = 0x140−width−12 @0x8B0B). First bar item x=12, then prev.x+prev.w+12; hotkey characters (after `~`) draw in the hilite colour @0x84FF–0x8567. Dropdowns: x = bar-item x, y = barFontH+4, width = maxItemW+2 (min 0xA), height = (fontH+3)·visible+5, 1-px border in `_menu_border_color`, wood interior, empty rows drawn as centred 1-px separator rules, save-under id 0xFFF8.

Command dispatch (`_execute_menu_event` @0x2DE0, jump table at file 0x2DFC):

ID	ITEM	ACTION
0x1A	Save	confirm popup `@SAVE` → `_write_map_file`; failure → `@ERROR`; success clears `_map_changes` @0x2F8E
0x13	Save As	strip path, string popup `@SAVEAS` (14 chars), force extension "MP", write @0x2EAC
0x1B	Load	if dirty confirm `@LOAD`; `_file_menu(@MAPTOLOAD,"*.MP")`; load; `_forest_fix`; recentre @0x2FD4
0x14	Create	if dirty confirm `@CREATENOW`; `_create_me`; recentre + `_new_mini` @0x2F24
0x1F	Exit	if dirty, 3-way `@EXIT` (exit unsaved / save+exit / cancel) @0x305E
0x24/0x25	Zoom In ~Z / Out ~X	`_set_zoom_level(_map_scale ∓ 1)` @0x30C2/@0x30D2
0x26–0x29	F1..F4 zoom rows	`_set_zoom_level(0x29-id)`, clamp 0..3 @0x30D8/@0x2B8B
0x2B	~Center View	`_set_center(cursor, 1)` @0x30E0
0x4A	Find Continents	`_continent_check` @0x2C70: `_map_find_continents` @0xB242 — two flood passes labelling continents 1..15 into the layer-3 low nibbles; >15 land regions → `@CONTINENTS1` popup, >15 water → `@CONTINENTS2`; then a full-screen continent-id view (Ocean/Sea-Lane blanked), any key restores
0x4B	~Map Tile Select	`_selection_screen` @0x2826
0x4C	~Fill Mode Change	`_fill_radius` = (r+1)%3 @0x3104
0x4D	Coastline Protect	toggle `_coastline_protect`
0x4E	~Undo Last Change	`_perform_undo` @0x3128
0x51–0x54	Help rows 1–4	`@popup_box("HELP1".."HELP4")` @0x3140–@0x3164
0x5F	"How To Use Maps" row	`@popup_box("ABOUT")` @0x3170 — shipped off-by-one, §28.9
0x21, 0x6A	*(no menu row)*	`_set_view_mode` / `_memory_check` @0x30BA/@0x317C — dead, nothing emits these ids

Popup/dialog engine (`popup.obj`, segment 0x33D). `_popup_create` @0x50AE: frame inset 3; bevel borders from the `@COLORS` border0/1/2 palette indices (outer ring + inset−1 rect in border0, top/left in the border-down colour, bottom/right in border-up, @0x6CB3–0x6DA1); wood (WOODTILE) interior; FONTINTR text, switched to FONTTINY by a `@SMALLFONT` directive. Minimum width 0x50; auto-centre x=160−w/2, y=100−h/2, clamped to 320×200; item rows fontH+3, entry rows fontH+8. The section parser `@popup_start_box` @0x7C82 is a small state machine: a blank line separates the text block from the option block; `^^` = centred line, `^` = raw line, plain lines word-wrap; `{…}` = hilite span, `|` truncates; option items get ids 1..n in read order; directives `@OPTIONS/@PROMPT/@TEXT/@SMALLFONT/@X=/ @Y=/@WIDTH/@LENGTH/@CHECKBOX/@DEFAULT` are parsed (none is used by the 19 MAPEDIT.TXT sections). Substitutions: `%STRINGn` (DS:0x634E+64n, set by `_popup_say_string`), `%NUMBERn`, `%HEXn`, `%%`. The event loop `@popup_exec` @0x6F5E: Up/Down move (skip greyed, wrap), Enter/Space select → popup word 0 = item id, **ESC** → **`0xFFFF`**, hotkey match, mouse rows with release-select; entry mode appends printable chars to maxlen, accepts on Enter into `_popup_text_buffer` DS:0x4B64. `@popup_ask_number` has zero callers (dead), as do `_menu_read_colors` ("MENUCOLR.SS") and `_popup_read_colors` ("TEXTCOLR"). HELP1–4/ABOUT are plain text popups (no buttons, no scrolling; a popup taller than 200 px aborts with error 0xFFAF).

Keyboard accelerators (`_human_interface_loop` @0x3724; keys upcased, tried in order): (1) bar hotkeys E/V/M/H open the dropdowns (`@menu_bar_key_parse` @0x97C2); (2) global item accelerators (`@menu_key_parse` @0x9856, firing without opening a menu): S=Save, A=Save As, L=Load, Z/X=zoom in/out, F1–F4=zoom levels, C=Center, M=Tile Select, F=Fill Mode, P=Coastline Protect, O=Find Continents, U=Undo; (3) `_parse_main_keys` @0x33B6: ESC/Ctrl-Q/Ctrl-X/Alt-Q/Alt-X → exit flow, Space/Enter paint, Backspace pick-up, numpad/arrows move. `_shift_key` (BIOS 0x417 & 3) gates paint-vs-move on mouse strokes.

28.9 The five shipped bugs

All five are byte-verified in the shipped binary:

1. "**Memor~y check**" **never appears.** The `@CUP` (Map) menu section has 6 item rows but the construction code issues only 5 `_text_get` reads — the last row is never read, and its handler id 0x6A is dead. (The handler itself, `_memory_check` @0x2BCC, would show a popup from the file "DEBUG", section `@MEMORY`.)
2. **Help off-by-one.** `@HELP` has 6 rows but 5 reads, assigned ids 0x51, 0x52, 0x53, 0x54, 0x5F. The row labelled "**How To Use Maps**" **therefore fires id `0x5F`, which opens the `@ABOUT` popup**; the "About Map Editor" row is never read and the `@HELP5` section is unreachable.
3. **Dead `@XS`/`@YS`.** No referencing string for either section exists anywhere in the EXE — the map size is hard-coded 58×72; the size-entry dialogs the text file provides for were never wired up.
4. **Dead command ids `0x21` and `0x6A`.** Both have live handlers in the dispatch jump table (`_set_view_mode`, `_memory_check`) but no menu row or key ever emits them.

5. **Load→save is not byte-preserving.** `_forest_fix` runs on every load, folding forest alias ids 16..23 to 8..15 and stripping forest under mountain/hill overlays — so round-tripping a file that contains ids 16..23 (AMER2.MP does) rewrites those bytes. (The main game performs the same fold in its own loader, so the meaning is unchanged.)

28.10 Renderer frame map and engine parity

The editor's tile renderer (helper @0xC3A2) is the same compositor scheme as the game's: ground from the 12-tile TERRAIN.SS array, PHYS0.SS overlays on top, BDARK.SS never referenced. Ground selection (`_tile_id` @0x46CE): ids 0..7 → frames 0..7; ids 9 and 0x11 → frame 8 (Scrub); 0x18/0x19/0x1A → frames 9/10/11 (Arctic/Ocean/Sea Lane). Frame constants below are **engine frame numbers, 1-based over the disk descriptors** (disk sprite = engine frame − 1):

OVERLAY	ENGINE FRAMES	RULE
forest	`0x41+mask`	mask = N8|S4|W2|E1 over neighbouring forests
mountains	`0x21+mask`	neighbours connect only when `byte&0xA0` is equal
hills	`0x31+mask`	same adjacency rule
major river	`0x01..0x10`	4-neighbour mask; isolated → mask 0xF
minor river	`0x11..0x20`	same
river mouths	`0x8D+dir / 0x91+dir`	on water tiles adjoining a land river
unexplored	`0x95`	
straight coasts	`0x97..0x9A (disk 150–153)`	by edge class
beach-halo quadrants	`0x6D+quad+4·code`	8×8 sub-tiles at 8×8 sub-offsets
scale-0 extras	`0x5A+idx resources, 0x68 lost city, 0x51/0x52+dir roads`	inert on fresh maps (feature layer empty)

Engine frame 0x96 (disk 149), the feature-bit wave/hatch overlay, is present but dormant in the editor (@0xC550). The identity of these constants with VICEROY's in-game map renderer — same coast adder at VICEROY file 0x06850D, same halo adder at 0x0684E8 — plus the instruction-identical menu clamps of §28.8, make MAPEDIT a second, independently shipped witness to the game's rendering rules.

§29 Verification

Everything in this manual was established by three mutually checking methods against the shipped 1994 binaries and data files: static disassembly with byte-level citation, live memory reads of the running game under emulation, and pixel-level render-and-diff in which whole screens were rebuilt purely from the documented facts and compared against captures of the real game. This section records what each method contributed and — just as importantly — what remains unproven.

29.1 Static disassembly

The primary evidence layer is the disassembly of VICEROY.EXE (an RTLink overlaid MZ executable: resident segments plus 31 overlay pages reached through thunk tables), MAPEDIT.EXE (with its CodeView symbols, §28.1), and the OPENING/CLOSING cinematic executables. Every load-bearing number in this manual is cited to a file offset in one of these binaries, to a field of the NAMES.TXT/GAME.TXT-family data files, or to a recorded ruling; where a value is computed at runtime it is marked as such rather than guessed. Cross-checks internal to this layer repeatedly caught errors: for example, the sprite-frame numbering convention (engine frame = disk descriptor + 1) was proven from descriptor counts — TERRAIN.SS holds exactly 12 disk descriptors while the engine loads "frames 1..12", and PHYS0.SS holds 154 (disk 0..153) while engine code references frame 154 — with no subtraction anywhere in the draw verb (the offset lives in the in-memory record layout, appendix A).

29.2 Live-RAM reads under emulation

The running game (DOSBox 0.74-3) is the top of the evidence order. Memory snapshots were taken at known moments and the game's data segment located by **DGROUP anchoring**: scanning the dump for known DGROUP string constants (e.g. "WOODPANL" at DS:0x2189) pins the physical base (0x1CFD0 in the verification sessions), after which every documented DGROUP offset can be read directly. This validated record layouts end-to-end (the active colony pointer `[0x8542]` → bytes `33 1D "Jamestown"` = (51,29) exactly as the ColonyRecord head specifies), exposed state the disassembly could not decide (the debug bitfield `[0x894]` defaults to 8, not 0; the boot text ink latch `[0x1F4E]` reads 0xFC at the boot menu and 0x95 in-game), and killed a systematically wrong oracle: the word `[0x2F5E]` = 537 had been dereferenced as a string pointer yielding "Sons of Liberty"; the live table walk showed it is an integer **string id** (slot 210 of a 221-entry id table, resolving to "Exit") — consistent with the pixel evidence in both screens where the wrong reading had propagated.

29.3 Render-and-diff

Twelve-plus screens were rebuilt from scratch using only the documented facts (geometry, fonts, palette indices, sprite frames, formulas) and diffed pixel-by-pixel against captures of the running game (pixel-verified against the running game, 1994 binary under DOSBox):

SCREEN	REBUILT-VS-LIVE RESULT
difficulty select	99.96% identical
nation select	99.86% identical

SCREEN	REBUILT-VS-LIVE RESULT
boot (main) menu	98.7% identical; four documented claims falsified and corrected in the process
leader-name entry	pixel-identical (residual: the mouse cursor)
King audience	pixel-identical (residual: the mouse cursor)
Europe screen, idle state	**100.00%** outside declared dynamic-sprite masks
Europe screen, ship-arriving state	**100.00%**
colony screen	structurally exact: every element matched or produced a recorded correction; **all 15 building plots pixel-exact from an exact replay of the placement RNG chain** (16-bit seed → LCG → 15-group category shuffle → frame select)
map view (ocean window)	**100.0000%** — 45,056/45,056 non-overlay pixels
land window (Jamestown region)	**100.0000%** of 41,540 non-overlay pixels
crafted-test-map windows (5 viewports over a purpose-built 58×72 map exercising hills, mountains, rivers, mouths, lakes, forest aliases)	**100.0000%** non-overlay in all five (raw including live-object overlays: 95.15–98.00%)

The method is deliberately adversarial: a mismatch is treated as a falsifier of the documentation, not of the capture. It refuted, among others, a claimed sprite-blit chain on the boot menu (actually a palette-index find-and-replace, a no-op on that background), transposed width/height argument labels on the frontend cell grids, a wrong market-bar y, and the "Sons of Liberty" string described above; and it discovered mechanisms no static read had found — the beach-halo **ground substitution** (a coastal water tile is grounded with its last cardinal land neighbour's terrain, coast frames drawn over it, and water backfilled through the frames' zero-holes) and the colony scene panel's deterministic ×1.5 dithered upscale of the shared 16-px map compositor.

Non-overlay means: engine object sprites (units, villages, the view cursor, a ≤2-px sprite overhang) are masked from the diff, since they are game objects, not tile-compositor output; every masked region is declared.

29.4 Capture pipeline

Comparisons must model the capture chain or they fail for the wrong reasons. Three facts matter: (1) the captures are **2× frames** — each native pixel is a 2×2 block, recovered by sampling every second row and column; (2) the emulator framebuffer is **RGB565**: the 6-bit VGA palette entry is expanded `(v<<2)|(v>>4)` and then floored to 5/6/5 bits, and the renderer must apply the same quantisation before diffing; (3) **palette cycling**: the sea-lane sparkle is a VGA palette rotation over indices 120–127, so each capture's cycle phase (0..7) is fitted before the diff, and the Europe harbour water indices 54–60 are likewise pure palette animation with zero pixel-index changes. None of these steps touches the documented render rules; they model only the measurement instrument.

29.5 What remains unexercised or unmapped

Stated honestly, in the open:

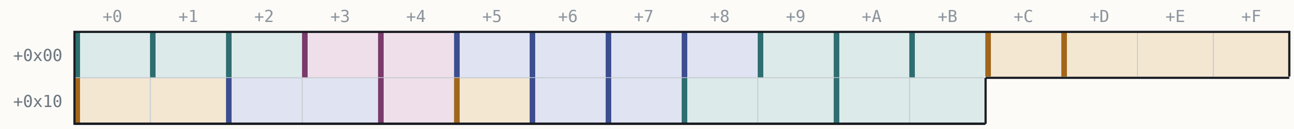
- **Shore-hatch 0x96, roads, and the feature-resource bit have byte-cited draw gates but no pixel test.** The game's `.MP` loader discards layer 2 (features) entirely and rebuilds the plane at runtime, so no crafted map can exercise them; the draw sites (land/water shore gates at VICEROY file 0x6834F/0x68354, the road walker with its per-direction frames 0x52+d) are decoded from the disassembly only. Exercising them needs an organic in-game state with pioneer-built roads. (Hills, rivers, river mouths, lake coasts, forest aliases, fog blends — all previously on this list — are now live-confirmed by the crafted-map and land-window tests.)
- **AIPersonality tail bytes** +0x30/+0x32/+0x33 are unlabelled.
- **ColonyRecord** retains unmapped runs; two locations (+0x24, +0x99) have provably **no static accessor** in the entire EXE and read zero in live dumps.
- The boot-time writer of **`[0x8D80]`** (a session-constant term mixed into the colony building-placement seed; live values 0x2C55, 0x5B7C in two sessions) is unlocated.
- **`func_003E40`** — the map unit-marker sprite mapping drawn into the colony scene panel — is undecoded.
- The live value of **`[0x890]`** on the colony screen (it gates the marker name/population text inside the scene panel) has not been read.

§A Appendix — data structures

This appendix collects every record layout established for the shipped binaries, as C structs with byte offsets. Only fields actually mapped are named; gaps are declared. Unless noted, addresses are DGROUP-relative in VICEROY.EXE; per-field citations are the decisive read/write sites. "(runtime-verified)" marks fields whose meaning rests on live-memory observation of the running game rather than a static code citation.

A.1 UnitRecord — 0x1C bytes, base DGROUP:0x3144, 300 slots

UnitRecord fully mapped



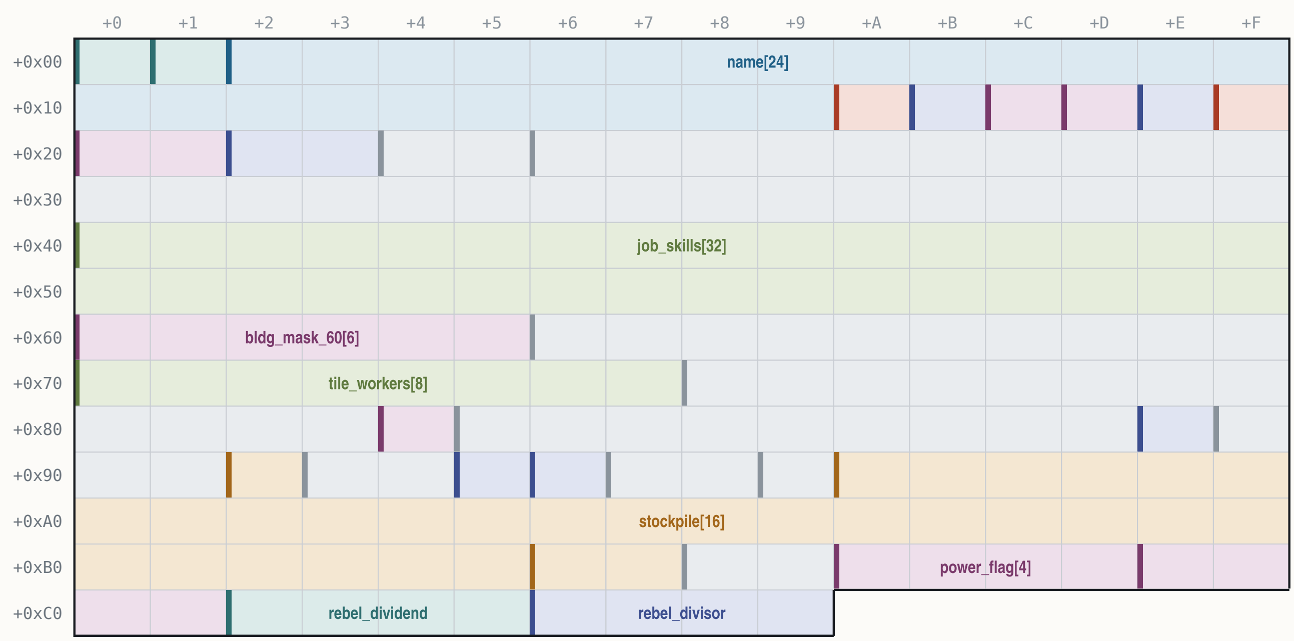
28 bytes (0x1C) · mapped 28 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00	1	uint8_t	map_x	drawn position (renderer @0x03A63, placer @0x06958)
+0x01	1	uint8_t	map_y	
+0x02	1	uint8_t	unit_type	NAMES @UNIT row 0..23 (dispatcher @0x51D6B; 694 refs)
+0x03	1	uint8_t	owner_flags	low nibble = power 0..11, high nibble = state (setter @0x738E)
+0x04	1	uint8_t	scratch_bits	per-pass flag register: 0x08 tile-dirty (@0x0481B0), 0x80 draw/AI marker (@0x069923), 0x20 Merchantman tag (@0x04CE44), 0x10 path>=8 hops (@0x05106E), 0x02 was-fortifying (@0x04CEC9), 0x04 ship-cargo class (@0x04CDDC)
+0x05	1	uint8_t	moves_spent	AI move-credits spent this turn (reset @0x005872; +3/step @0x05CAE2; gate @0x03EE95)
+0x06	1	uint8_t	countdown	timer, init 0xFF, dec (@0x2EF17)
+0x07	1	uint8_t	ai_state	persistent AI state letter ('X','0','1','G','E','R','V',...; init @0x06D84)
+0x08	1	uint8_t	order	order code 0..0x0C = NAMES @ORDERS row (dispatch @0x249CB)
+0x09	1	uint8_t	goto_x	goto / trade-route next-stop target (writer @0x22D38)
+0x0A	1	uint8_t	goto_y	
+0x0B	1	uint8_t	heading	facing 0..7, 8 = none (xor-4 reverse @0x047AA8; bound @0x0516F0)
+0x0C	1	uint8_t	cargo_count	goods in hold (@0x0B2AB)
+0x0D..+0x0F	3	uint8_t	cargo_ids[3]	nibble-packed good ids, up to 6 (@0x0B2CB)
+0x10..+0x11	2	uint8_t	cargo_qty[2]	per-slot quantities (@0x0B2FB)
+0x12..+0x13	2	uint16_t	timer	overloaded: AI/native = snapshot of [0x538E] (@0x06DB3); player = byte 0xFF then rand 0..0x13 (@0x06DA3/@0x50C75)
+0x14	1	uint8_t	moved_flag	per-turn land-unit boolean; read only for Wagon Trains (@0x04968D/@0x04F730/@0x0507E1; exact label runtime-open)
+0x15	1	uint8_t	tools	pioneer tools 0..100, -20/action (@0x4060F)
+0x16	1	uint8_t	work_counter	turns in clear/road/fortify activity (@0x04071D)
+0x17	1	uint8_t	class_prof	colonist profession 0x13..0x1C; on route units: low nibble = route id, high = stop idx (@0x5B60E / @0x0075D4)
+0x18..+0x19	2	uint16_t	occ_back	per-tile occupancy list back link (@0x06976)
+0x1A..+0x1B	2	uint16_t	occ_next	next link (@0x06968)

A.2 ColonyRecord — 0xCA bytes, array head DGROUP:0x5D46, ~50 slots

Reached via the active-colony far pointer `[0x8542]` (0 at boot, real after founding; slot free when the name at +0x02 is empty; slots are recycled).

ColonyRecord stride 0xCA; serialized to save-games



202 bytes (0xCA) · mapped 140 · unmapped 62

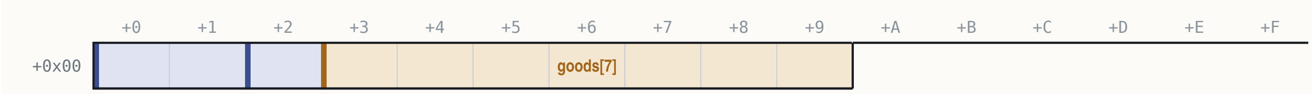
OFFSET	B	TYPE	FIELD	NOTE
+0x00	1	uint8_t	map_x	
+0x01	1	uint8_t	map_y	

OFFSET	B	TYPE	FIELD	NOTE
+0x02..+0x19	24	char	name[24]	NUL-terminated
+0x1A	1	uint8_t	owner_power	0..3 (colony-burn trace)
+0x1B	1	uint8_t	foreign_status	(runtime-verified; semantics open)
+0x1C	1	uint8_t	status_flags	per-colony status byte
+0x1D	1	uint8_t	flags_1d	bit 0x80 only (test @0x551D8, set @0x55C20, clear @0x55A2F)
+0x1E	1	uint8_t	countdown_1e	gated by +0x8E (@0x4D9C7, 14 sites)
+0x1F	1	uint8_t	population	size (burn-loot formula @0x05DE1E)
+0x20..+0x21	2	uint16_t	flags_20	(runtime-verified; foreign-marker byte at low half)
+0x22..+0x23	2	uint16_t	state_22	packed state (runtime-verified)
+0x24..+0x25	2	uint16_t	unused_24	NO static accessor exists; 0 in live dumps
+0x26..+0x3F	26	–	–	unmapped(26 bytes)
+0x40..+0x5F	32	uint8_t	job_skills[32]	1 byte per colonist, live length = population (runtime-verified); declared span to +0x5F
+0x60..+0x65	6	uint8_t	bldg_mask_60[6]	buildings-constructed bitmask (runtime-verified observation)
+0x66..+0x6F	10	–	–	unmapped(10 bytes)
+0x70..+0x77	8	uint8_t	tile_workers[8]	colonist idx per surrounding tile, NW..SE, 0xFF empty (runtime-verified)
+0x78..+0x83	12	–	–	unmapped(12 bytes)
+0x84	1	uint8_t	constructed_mask	building mask (static accessor cite)
+0x85..+0x8D	9	–	–	unmapped(9 bytes)
+0x8E	1	uint8_t	gate_8e	gates the +0x1E countdown
+0x8F..+0x91	3	–	–	unmapped(3 bytes)
+0x92	1	uint8_t	hammers	build progress (paired with +0xB6)
+0x93..+0x94	2	–	–	unmapped(2 bytes)
+0x95	1	uint8_t	warehouse_level	
+0x96	1	uint8_t	counter_96	inc/dec counter (@0x2C244/@0x5C474)
+0x97..+0x98	2	–	–	unmapped(2 bytes)
+0x99	1	uint8_t	unused_99	NO static accessor; 0 in live dumps
+0x9A..+0xB9	32	uint16_t	stockpile[16]	per-good cargo, NAMES @CARGO order (runtime-verified against the in-game bar)
		+0x9A Food +0x9C Sugar +0x9E Tobacco +0xA0 Cotton +0xA2 Furs +0xA4 Lumber +0xA6 Ore +0xA8 Silver +0xAA Horses +0xAC Rum +0xAE Cigars +0xB0 Cloth +0xB2 Coats +0xB4 Trade Goods +0xB6 Tools +0xB8 Muskets		
+0xB6..+0xB7	2	uint16_t	hammers_b6	build-progress pair of +0x92
+0xB8..+0xB9	2	–	–	unmapped(2 bytes)
+0xBA..+0xBD	4	uint8_t	power_flag[4]	per-power byte flags, init 1 in the colony-reset loop (@0x2ED7A)
		+0xBA England +0xBB France +0xBC Spain +0xBD Netherlands		
+0xBE..+0xC1	4	uint8_t	power_flag2[4]	paired array, init 0 (@0x2ED7F)
		+0xBE England +0xBF France +0xC0 Spain +0xC1 Netherlands		
+0xC2..+0xC5	4	int32_t	rebel_dividend	Sons-of-Liberty numerator (runtime-verified)
+0xC6..+0xC9	4	int32_t	rebel_divisor	denominator; SoL% = dividend/divisor

A.3 RouteRecord (0x4A) and StopRecord (0x0A) — trade routes

Routes live in their own segment 0x1B22, base offset 0, max 12 (`select_route` = `func_05FE60`; count `[0x53A0]`, cap @0x610B5; delete shifts 0x4A bytes @0x605DB).

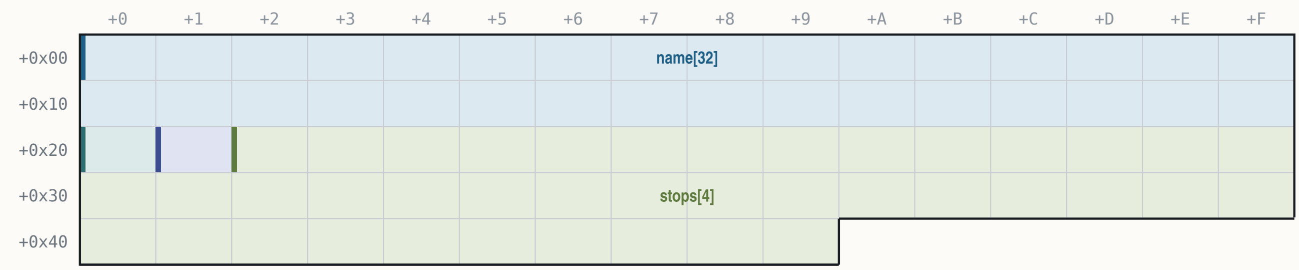
StopRecord 0x0A bytes



10 bytes (0xA) · mapped 10 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	uint16_t	destination	colony id (record = id*0xCA + 0x5D46), 0x3E7 = Europe, 0x3E8 = none (@0x05FEE1)
+0x02	1	uint8_t	counts	low nibble = UNLOAD count (lanes +0x06..), high = LOAD count (lanes +0x03..) (@0x060382)
+0x03..+0x09	7	uint8_t	goods[7]	nibble-packed good ids, 2 per byte: +0x03..+0x05 load lanes, +0x06..+0x08 unload lanes (@0x603DA)

RouteRecord 0x4A bytes; unit binding = UnitRecord +0x17 nibbles



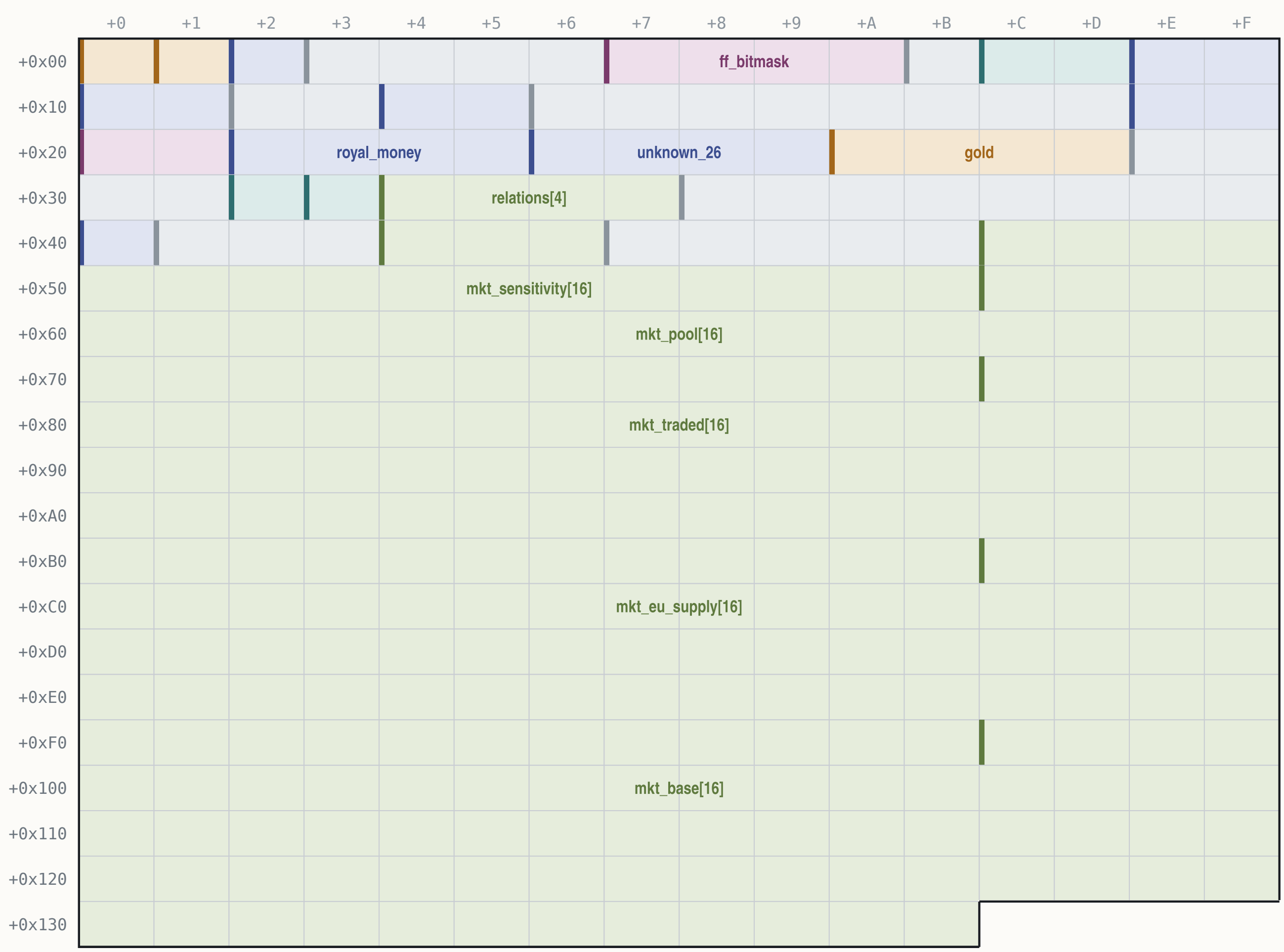
74 bytes (0x4A) · mapped 74 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x1F	32	char	**name[32]**	route name (memcpy @0x61273; uniqueness strcmp @0x611FF)
+0x20	1	uint8_t	**type**	0 = sea, 1 = land (@0x61282)
+0x21	1	uint8_t	**cursor**	current-stop cursor (init 2 @0x61286; inc @0x60C7A)
+0x22..+0x49	40	StopRecord	**stops[4]**	up to 4 stops (set_stop_ptr @0x05FE7A)

A.4 PowerRecord — 0x13C bytes, base DGROUP:0x8808, 12 entries

Entries 0..3 are the European powers, 4..11 the native tribes. (Earlier field cites off a base of 0x8809 are the same bytes: that table's +0x21 gold / +0x25 loot / +0x29 treasury are +0x22/+0x26/+0x2A here.)

PowerRecord ends exactly at +0x13C



316 bytes (0x13C) · mapped 281 · unmapped 35

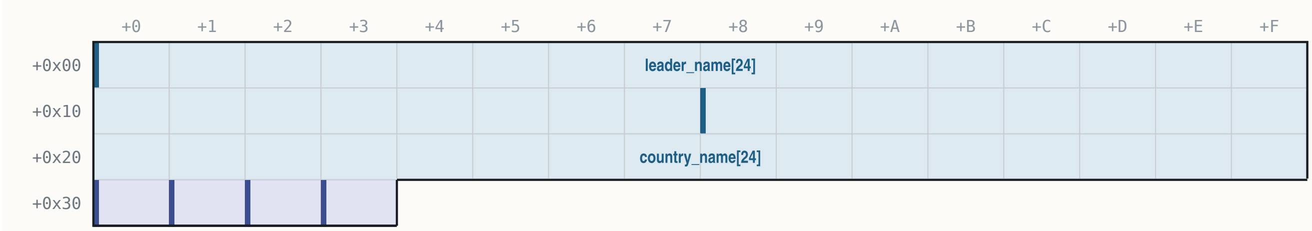
OFFSET	B	TYPE	FIELD	NOTE
+0x00	1	uint8_t	**treasure_pool**	(SMITE multiplier trace)
+0x01	1	uint8_t	**tax_pct**	0..100 (@0x034AE0 chain)
+0x02	1	uint8_t	**rebel_sentiment**	0..100 (runtime-verified vs display)

OFFSET	B	TYPE	FIELD	NOTE
+0x03..+0x06	4	–	–	unmapped(4 bytes; +0x06 = attribute-bitfield start)
+0x07..+0x0A	4	uint32_t	ff_bitmask	acquired Founding Fathers, bit = FF idx (reader func_00BC10 @0x00BC10)
+0x0B	1	–	–	unmapped(1 byte)
+0x0C..+0x0D	2	uint16_t	bells_next_ff	bells toward next FF, resets on acquisition (runtime-verified)
+0x0E..+0x0F	2	uint16_t	bells_per_turn	
+0x10..+0x11	2	uint16_t	crosses_per_turn	
+0x12..+0x13	2	–	–	unmapped(2 bytes)
+0x14..+0x15	2	uint16_t	ff_count	
+0x16..+0x1D	8	–	–	unmapped(8 bytes)
+0x1E..+0x1F	2	uint16_t	artillery_bought	Europe artillery escalation counter (read x100 @0x035124; inc @0x035282; zeroed @0x03662F)
+0x20..+0x21	2	uint16_t	boycott_bits	bit i = good i boycotted (runtime-verified)
+0x22..+0x25	4	int32_t	royal_money	King's REF budget (runtime-verified: +18/turn at Discoverer)
+0x26..+0x29	4	int32_t	unknown_26	
+0x2A..+0x2D	4	uint32_t	gold	treasury (write-back updates UI)
+0x2E..+0x31	4	–	–	unmapped(4 bytes)
+0x32	1	uint8_t	home_x	spawn/relocation x (REF growth chain)
+0x33	1	uint8_t	home_y	
+0x34..+0x37	4	uint8_t	relations[4]	4x4 relation matrix row (DG 0x883C, row stride 0x13C; get func_007F34, symmetric set func_007F96):
	+0x34 England +0x35 France +0x36 Spain +0x37 Netherlands			
+0x38..+0x3F	8	–	–	unmapped(8 bytes)
+0x40	1	uint8_t	treaty_respect	counter, seed 2*(6-difficulty), halved w/ Franklin (@0x059B00; AI attack-abort @0x03F163; decrement site unlocated)
+0x41..+0x43	3	–	–	unmapped(3 bytes)
+0x44..+0x46	3	uint8_t	ref_bytes[3]	disputed: one runtime dump write-verified as REF dragoons/regulars/artillery, another found it stale; the King's code reads the globals 0x53DA..0x53E1 instead
+0x47..+0x4B	5	–	–	unmapped(5 bytes)
+0x4C..+0x5B	16	uint8_t	mkt_sensitivity[16]	per good (measured; not byte-cited)
+0x5C..+0x7B	32	int16_t	mkt_pool[16]	(measured; not byte-cited)
+0x7C..+0xBB	64	int32_t	mkt_traded[16]	(measured; not byte-cited)
+0xBC..+0xFB	64	int32_t	mkt_eu_supply[16]	(measured; not byte-cited)
+0xFC..+0x13B	64	int32_t	mkt_base[16]	(measured; not byte-cited)

A.5 AIPersonality — 0x34 bytes, base DGROUP:0x540E, 4 entries

European powers only (tribes use TribeData instead). Leader/region name pointers used by the diplomacy text filler are `0x540E + p·0x34` and `0x5426 + p·0x34`.

AIPersonality stride 0x34



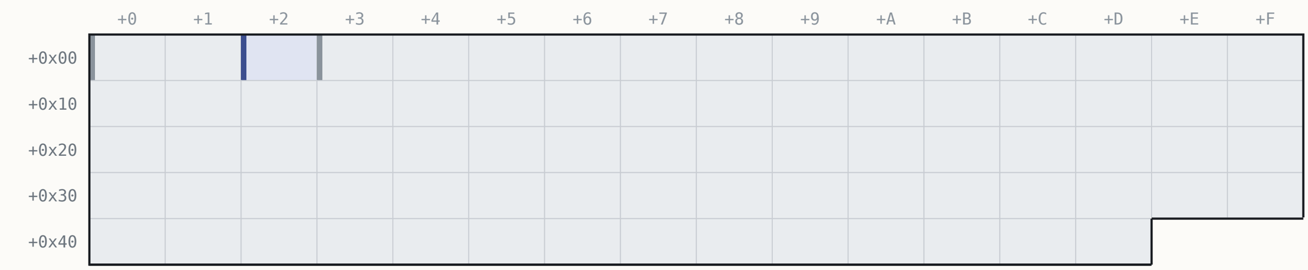
52 bytes (0x34) · mapped 52 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x17	24	char	leader_name[24]	e.g. "Walter Raleigh" (runtime-verified; NAMES @LEADERNAME)
+0x18..+0x2F	24	char	country_name[24]	e.g. "New England"
+0x30	1	uint8_t	unknown_30	(English = 0xC0; unlabelled)
+0x31	1	uint8_t	is_active	
+0x32	1	uint8_t	unknown_32	
+0x33	1	uint8_t	unknown_33	

A.6 TribeData — 0x4E bytes, base DGROUP:0x5AD6, 8 entries

Selected by `set_active_tribe` (`func_0081C6`): `[0x8D4E] = 0x5AD6 + tribe_idx·0x4E`.

TribeData stride 0x4E



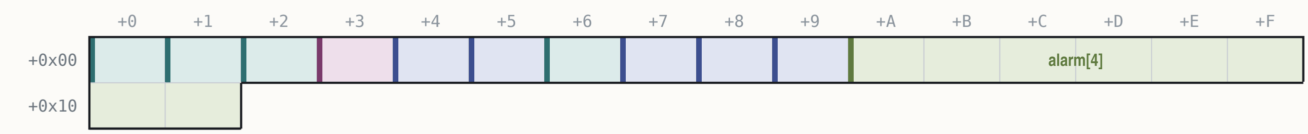
78 bytes (0x4E) · mapped 1 · unmapped 77

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	–	–	unmapped(2 bytes)
+0x02	1	uint8_t	`settlement_size_factor`	raze-formula multiplier (@0x04AB24 trace)
+0x03..+0x4D	75	–	–	unmapped(75 bytes)

A.7 NativeSettlement — 0x12 bytes, base DGROUP:0x54EC, ≥60 slots

The table is compacted on raze; walk from index 0 until an (0,0) coordinate pair.

NativeSettlement fully mapped



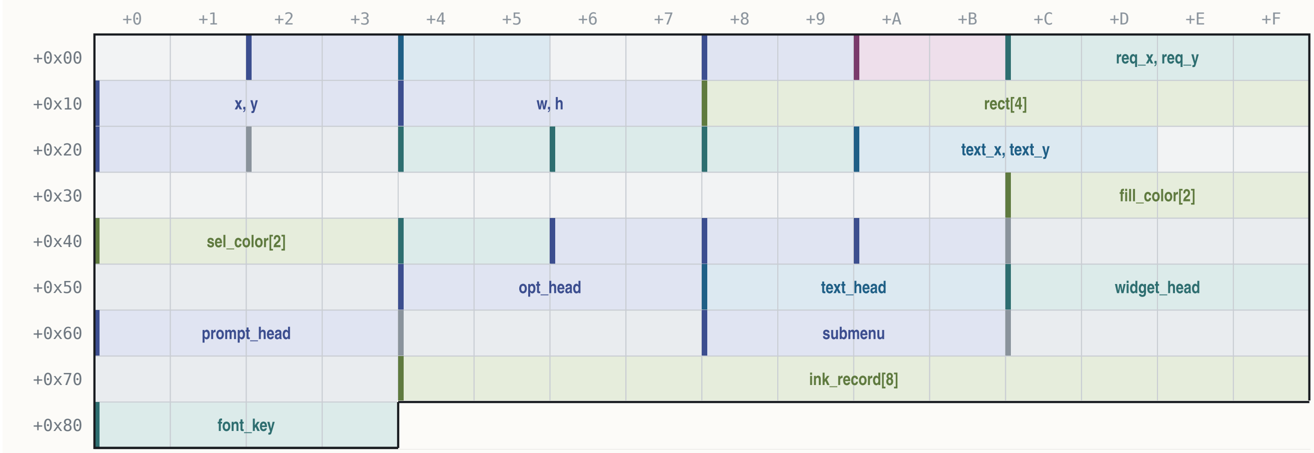
18 bytes (0x12) · mapped 18 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00	1	uint8_t	`map_x`	
+0x01	1	uint8_t	`map_y`	
+0x02	1	uint8_t	`owner_power`	4..11
+0x03	1	uint8_t	`flags`	0x02 taught, 0x04 mission/capital, 0x08 visited, 0x40 event
+0x04	1	uint8_t	`population`	CHIEFKILL size byte (user-verified raze payout)
+0x05	1	uint8_t	`mission`	0xFF none, else 0x10 | power 0..3 (user-verified)
+0x06	1	int8_t	`growth_counter`	+population per turn; spawns/grows at 20
+0x07	1	uint8_t	`sentinel`	always 0xFF
+0x08	1	uint8_t	`last_bought`	cargo idx of last good bought here
+0x09	1	uint8_t	`last_sold`	
+0x0A..+0x11	8	struct{uint8_t friction, attacks;}	`alarm[4]`	per European power
		+0x0A England +0x0C France +0x0E Spain +0x10 Netherlands		

A.8 VICEROY dialog struct (the @-directive dialog framework)

Allocated per dialog; accessed as a far pointer (`les bx,[bp+4]` in the finalizer `func_06D316`). Field cites are the construct/pump sites.

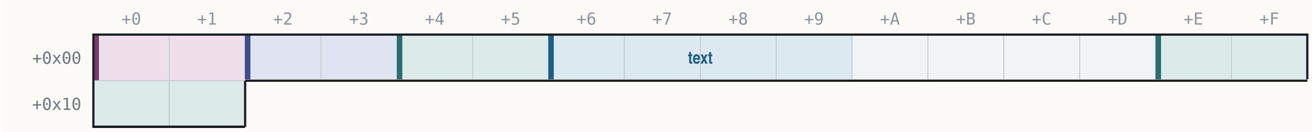
Dialog ~0x84+ bytes; trailing size unmapped



132 bytes (0x84) · mapped 98 · unmapped 34

OFFSET	B	TYPE	FIELD	NOTE
+0x02..+0x03	2	uint16_t	opt_count	option-row count (appender func_06C850 @0x06CA2B)
+0x04..+0x05	2	uint16_t	text_count	text-line count (appender func_06CA82 @0x06CB87)
+0x08..+0x09	2	uint16_t	third_count	third item-class count (@0x06CD57)
+0x0A..+0x0B	2	uint16_t	flags	0x10 borderless, 0x40 off-screen, 0x20 sibling-attach; checkbox sets |=5
+0x0C..+0x0F	4	int16_t	req_x, req_y	from @x/@y; -1 = centre sentinel (@0x06F2A6/@0x06F25E)
+0x10..+0x13	4	int16_t	x, y	final on-screen origin
+0x14..+0x17	4	uint16_t	w, h	box size
+0x18..+0x1F	8	uint16_t	rect[4]	final absolute rect (@0x06D5B9)
+0x20..+0x21	2	uint16_t	longest_line_px	(clamp @0x06D392)
+0x22..+0x23	2	uint16_t	pad	= 4, option-row x-indent component (@0x06C5AC)
+0x24..+0x25	2	uint16_t	content_x	(flags&0x10)?0:3; option rows at box_x+9 (@0x06D9D6)
+0x26..+0x27	2	uint16_t	row_y_seed	= inset'(3)+border(3), bumped past the text block (@0x06D440)
+0x28..+0x29	2	uint16_t	width_floor	init 0x50, @WIDTH override (@0x06CA7B)
+0x2A..+0x2D	4	uint16_t	text_x, text_y	text-line origin seeds (lines at box_x+5)
+0x3C..+0x3F	4	uint16_t	fill_color[2]	from [0x1F3C]/[0x1F3E] (= TEXTCOLR.SS sprite 1/2 pixel); value 7 = wood-tile fill sentinel
+0x40..+0x43	4	uint16_t	sel_color[2]	selection band from [0x1F40]/[0x1F42] (boot value 0x37)
+0x44..+0x45	2	uint16_t	ring2_color	from [0x1F44]
+0x46..+0x47	2	uint16_t	border	(flags&0x10)?0:3
+0x48..+0x49	2	uint16_t	inset	(flags&0x10)?0:2
+0x4A..+0x4B	2	uint16_t	content_h_cursor	summed as items append; H = 2*this + border
+0x4C..+0x53	8	–	–	unmapped(8 bytes)
+0x54..+0x57	4	void far *	opt_head	option-row list (painter func_06D9CC)
+0x58..+0x5B	4	void far *	text_head	text-line list (painter func_06CFE8)
+0x5C..+0x5F	4	void far *	widget_head	child/widget list (pump loop B @0x06E699)
+0x60..+0x63	4	void far *	prompt_head	prompt/third-class list (painter func_06DC64)
+0x64..+0x67	4	–	–	unmapped(4 bytes)
+0x68..+0x6B	4	void far *	submenu	attached submenu; on widget nodes: the sprite far-ptr blitted (@0x06D952)
+0x6C..+0x73	8	–	–	unmapped(8 bytes)
+0x74..+0x83	16	uint16_t	ink_record[8]	built by func_06C296: +2 normal<-[0x1F4A], +4 disabled<-[0x1F4C], +6 hilite<-[0x1F4E], +8/+A aux, +C/+E font ptr
+0x80..+0x83	4	void far *	font_key	identity/font latch (@SMALLFONT stores the FONTTINY latch here @0x06F211)

DialogRowNode one option row (appender @0x044DCE region)



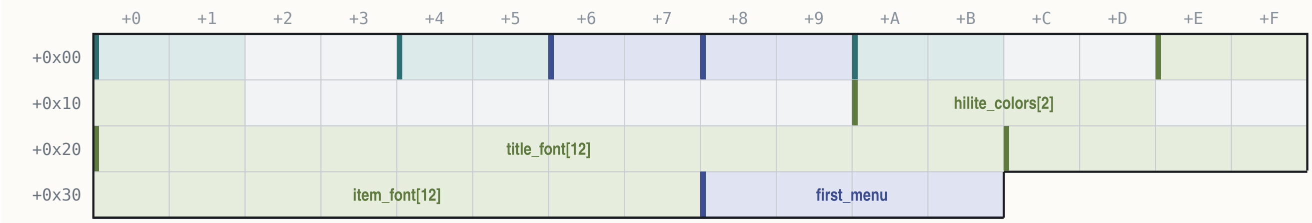
18 bytes (0x12) · mapped 14 · unmapped 4

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	uint16_t	flags	bit 0 = text empty -> pump skips
+0x02..+0x03	2	uint16_t	scalar	accelerator column or pixel width (callee untraced)
+0x04..+0x05	2	uint16_t	command_id	id the row fires
+0x06..+0x09	4	char far *	text	
+0x0E..+0x11	4	void far *	next	

A.9 Menu-bar structs (VICEROY page-0x0A module = MAPEDIT menu.obj)

The in-game menu bar object lives at `[0x896]` (built by `func_072090` @0x0720AC from MENU.TXT); MAPEDIT builds its bar from MAPMENU.TXT with the same module (`_menu_bar` DS:0x78). All offsets byte-cited in the VICEROY copy; the MAPEDIT node shapes match (result word +0, first menu +0x38, first item +0x1E).

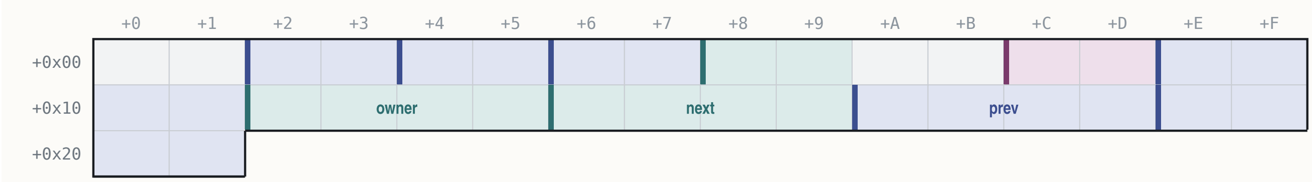
MenuBar menubar (creator func_044836)



60 bytes (0x3C) · mapped 46 · unmapped 14

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	uint16_t	result_id	selected command id (write @0x045895; 0 = none)
+0x04..+0x05	2	uint16_t	bar_y	= 1
+0x06..+0x07	2	uint16_t	title_gap	= 0x0C
+0x08..+0x09	2	uint16_t	item_leading	= 3
+0x0A..+0x0B	2	uint16_t	title_x_pad	= 1
+0x0E..+0x11	4	uint16_t	bar_colors[2]	from [0x149C]/[0x149E]
+0x1A..+0x1D	4	uint16_t	hilite_colors[2]	from [0x14A8]/[0x14AA]
+0x20..+0x2B	12	uint8_t	title_font[12]	font descriptor (far string ptr at +0x28)
+0x2C..+0x37	12	uint8_t	item_font[12]	(far string ptr at +0x34)
+0x38..+0x3B	4	void far *	first_menu	

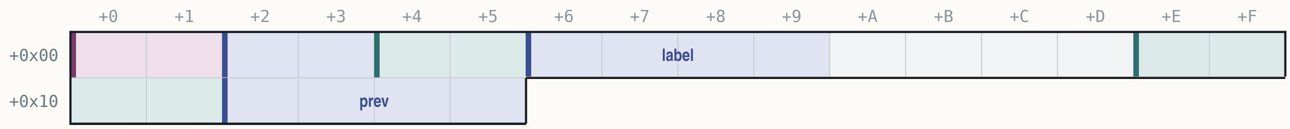
MenuNode menu node, 0x22 bytes (alloc @0x044BD9)



34 bytes (0x22) · mapped 30 · unmapped 4

OFFSET	B	TYPE	FIELD	NOTE
+0x02..+0x03	2	uint16_t	x	= prev.x + prev.width + gap (first title x = 0x0C)
+0x04..+0x05	2	uint16_t	title_w	
+0x06..+0x07	2	uint16_t	panel_inner_w	init 0x0A
+0x08..+0x09	2	uint16_t	hotkey	title hotkey char
+0x0C..+0x0D	2	uint16_t	flags	bit 0 = disabled
+0x0E..+0x11	4	char far *	title	
+0x12..+0x15	4	void far *	owner	owning menubar
+0x16..+0x19	4	void far *	next	
+0x1A..+0x1D	4	void far *	prev	
+0x1E..+0x21	4	void far *	first_item	

MenuItemNode item node



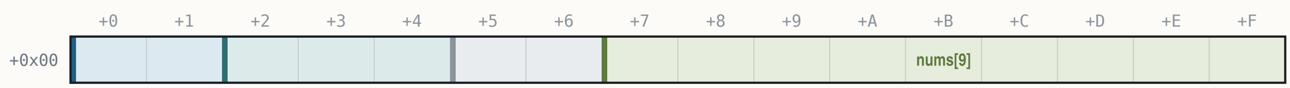
22 bytes (0x16) · mapped 18 · unmapped 4

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	uint16_t	flags	bit 0 disabled, bit 1 hidden
+0x02..+0x03	2	uint16_t	shortcut	
+0x04..+0x05	2	uint16_t	command_id	(returned into menubar +0)
+0x06..+0x09	4	char far *	label	empty first byte = separator
+0x0E..+0x11	4	void far *	next	
+0x12..+0x15	4	void far *	prev	

Dropdown layout (`func_044FA4`): panel x = menu.x; y = bar_y + title-height + 3; w = panel_inner_w + 2; h = visible·(item_font_h + leading) + leading + 2; clamps right ≤ 0x13D, bottom ≤ 0xC7 — instruction-identical in both programs (§28.8).

A.10 MAPEDIT terrain record and popup result

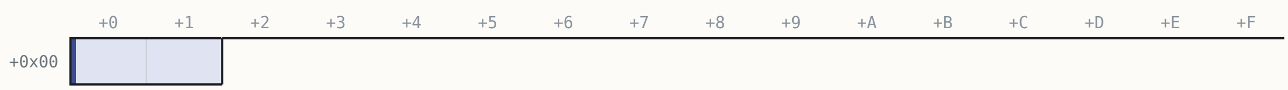
MapeditTerrainRec 16 bytes



16 bytes (0x10) · mapped 14 · unmapped 2

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	char *	name	near ptr into the NAMES text pool
+0x02..+0x04	3	uint8_t	num_a, num_b, num_c	NAMES numeric columns 1-3
+0x05..+0x06	2	–	–	unmapped(2 bytes)
+0x07..+0x0F	9	uint8_t	nums[9]	NAMES numeric columns 4-12 (column 13 never read)

MapeditPopup MAPEDIT popup object (popup.obj)



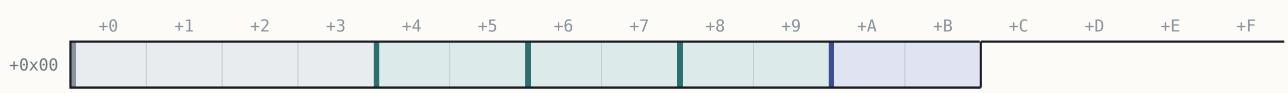
2 bytes (0x2) · mapped 2 · unmapped 0

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x01	2	uint16_t	result	selected item id 1..n; 0xFFFF = ESC (@popup_exec @0x6F5E)

A.11 Sprite sheet in memory (.SS handle)

A loaded sheet handle carries a header, then **12-byte frame records at +0x36**, indexed by the 1-based engine frame number: record = `handle + 0x36 + 12·(frame−1)` (VICEROY draw verb `func_00E76A`; the OPENING blit routine at its file 0x4520 uses the same layout).

SSFrameRec sheet dims at handle +0x4A/+0x4C



12 bytes (0xC) · mapped 8 · unmapped 4

OFFSET	B	TYPE	FIELD	NOTE
+0x00..+0x03	4	–	–	unmapped(4 bytes; pixel-data reference)
+0x04..+0x05	2	int16_t	anchor_x	= frame CENTRE x (screen x = anchor_x - w/2)
+0x06..+0x07	2	int16_t	anchor_y	= frame BOTTOM y (screen y = anchor_y - h + 1)
+0x08..+0x09	2	uint16_t	width	
+0x0A..+0x0B	2	uint16_t	height	(y-extent)

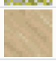
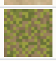
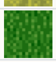
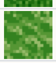
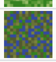
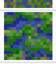
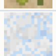
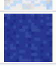
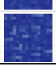
The (centre-x, bottom-y) anchor semantics were proven twice independently by pixel diff (the King-audience figure and throne-canopy banner land exactly where `ax−⌊w/2⌋, ay−h+1` predicts). Pixel value 0xFD in a decoded frame is transparent.

§B Appendix — sprite sheets and palette

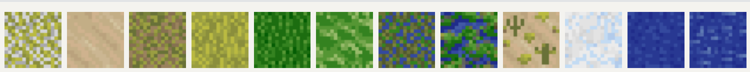
All numbers in this appendix were decoded directly from the shipped MADSPACK 2.0 containers (header `"MADSPACK 2.0"`, section directory, FAB-compressed sections; frames stored RLE with transparent index `0xFD`) — nothing is transcribed from secondary notes. The disc set contains **206 `.SS` sheets**. Frame numbering: **disk index = 0-based descriptor order; engine frame = disk + 1** (the convention proven in §29.1). Roles are stated only where the project established them; everything else is counted but left unlabelled.

B.1 TERRAIN.SS — 12 frames, all 16×16 (the base-ground sheet)

Loaded at boot and on map-enter; composited UNDER the PHYS0.SS overlays. Frame = ground id per the loaders in both programs (VICEROY fold at file `0x6204`; MAPEDIT `_tile_id` @`0x46CE`). MAPEDIT's mini-map colours sample pixel (8,8) of each frame.

DISK	ENGINE	TILE	SIZE	GROUND
0	1		16×16	Tundra
1	2		16×16	Desert
2	3		16×16	Plains
3	4		16×16	Prairie
4	5		16×16	Grassland
5	6		16×16	Savannah
6	7		16×16	Marsh
7	8		16×16	Swamp
8	9		16×16	Scrub floor (ground for ids 9 and `0x11` — forested Desert)
9	10		16×16	Arctic
10	11		16×16	Ocean
11	12		16×16	Sea Lane

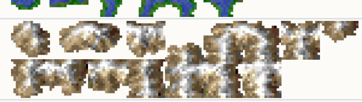
TERRAIN.SS — frame atlas 12 ground frames · 16×16


0 1 2 3 4 5 6 7 8 9 10 11

Disk-index numbering (engine frame = disk + 1, section 29.1); decoded from the MADSPACK container, transparent index `0xFD`.

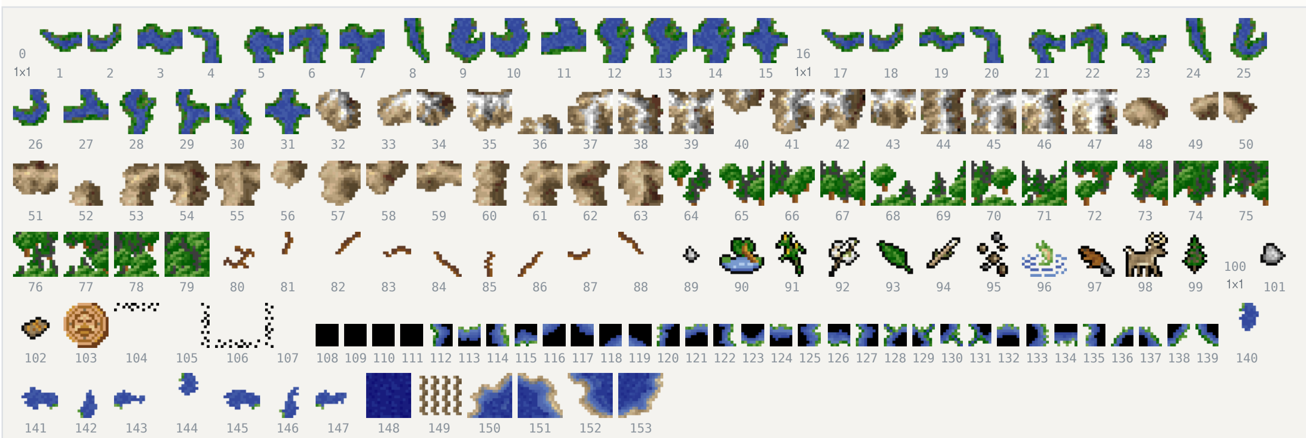
B.2 PHYS0.SS — 154 frames (terrain overlay sheet)

All frames 16×16 except the three 1×1 placeholders (disk 0, 16, 100 — never drawn: river mask 0 is remapped to the isolated form `0xF`) and the 8×8 beach-halo band (disk 108–139).

DISK BAND	ENGINE	SIZE	ROLE	
0	`0x01`	1×1	placeholder (major-river mask 0, unreachable)	
1–15	`0x02..0x10`	16×16	major rivers, 4-neighbour mask N8/S4/W2/E1; isolated = mask `0xF`	
16	`0x11`	1×1	placeholder (minor-river mask 0)	
17–31	`0x12..0x20`	16×16	minor rivers (same masks; majors/minors interconnect via terrain bit `0x40`)	
32–47	`0x21..0x30`	16×16	mountains, mask over neighbours with equal `byte&0xA0`	
48–63	`0x31..0x40`	16×16	hills (same adjacency; hills never connect to mountains)	

DISK BAND	ENGINE	SIZE	ROLE
64–79	0x41..0x50	16×16	forest, mask over neighbouring forests; desert scrub (id&7==1) never connects
80–88	0x51..0x59	16×16	roads: isolated = engine 0x51; else ONE frame per set 8-dir bit, engine 0x52+d
89–99, 101–102	0x5A..0x67	16×16	terrain-detail / prime-resource band (position hash + DTAB class; mountains draw ore/gold engine 0x66, hills rock engine 0x67)
100	0x65	1×1	placeholder inside the detail band
103	0x68	16×16	surf / lost-city-rumor circle (suppressed when the continent-plane owner nibble ≠ 0xF)
104–107	0x69..0x6C	16×16	dither-blend stencils N,E,S,W (class-boundary and fog-edge blends)
108–139	0x6D..0x8C	8×8	beach-halo quadrant sub-tiles, engine 0x6D+quad+4·code (code-0 frames disk 108–111 are all-zero punch-throughs)
140–147	0x8D..0x94	16×16	river mouths on water: base engine 0x8D (major) / 0x91 (minor) + cardinal direction
148	0x95	16×16	unexplored/fog tile
149	0x96	16×16	wave/hatch shore overlay (feature-layer bit 0x40; dormant in the standard game)
150–153	0x97..0x9A	16×16	the four straight-coast shorelines, by edge class

PHYS0.SS — frame atlas 154 overlay frames · rivers, mountains, hills, forest, roads, detail, halos, coasts



Disk-index numbering (engine frame = disk + 1, section 29.1); decoded from the MADSPACK container, transparent index 0xFD.

B.3 ICONS.SS — 131 frames (HUD, goods, units, markers)

Mixed sizes; established bands (disk numbering, engine gloss):

DISK	ENGINE	SIZE	ROLE
0–3	1..4	21×16	colony map markers (drawn with the pennant on the map and colony scene)
4	5	1×1	placeholder
5–7, 14–15, 127	6..8, 15..16, 0x80	13–14×16	ship unit icons (@UNIT icon column; e.g. Privateer eng 15, Frigate eng 16, Man-O-War eng 128 = disk 127)
18–21	0x13..0x16	16/8/4/2 px	map cursor set, engine 0x13+zoom (MAPEDIT blink cursor)
22–37	0x17..0x26	6–13×12	the 16 goods icons, @CARGO order (Europe market bar and colony stockpile bar, icon y=181)
67–69	0x44..0x46	14×13	button/flag plaques (colony flag panel = engine 0x44, frame = nation)
81–108	0x52..0x6D	6–14×16	unit figure band; foot-unit icons disk 100–105 + 109 (Colonist eng 101 = disk 100, Soldier eng 103 = disk 102)
118–121	0x77..0x7A	6×5	nation pennants, engine 0x77+power (colony markers)
122	0x7B	10×12	cargo crate (Europe dock slots at (147+12·slot,165); colony dock boxes)
124	0x7D	13×11	crown (colony Sons-of-Liberty/Tory band)
remainder	–	various	not yet role-assigned (incl. disk 38–66 second 12-px band, 70–80, 110–117, 123, 125–126, 128–130)

ICONS.SS — frame atlas

131 HUD frames · markers, ships, cursors, goods, units, pennants

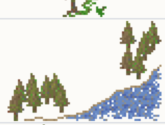
Disk-index numbering (engine frame = disk + 1, section 29.1); decoded from the MADSPACK container, transparent index `0xFD`.

B.4 BUILDING.SS — 48 frames (colony buildings)

Drawn frame = **def + 1** in engine numbering, i.e. **disk frame = def id**, named from NAMES.TXT `@BUILDING` (42 defs). Special cases at the colony composer: def 0 with build-query 0 → engine `0x11` (disk 16); defs `0x0F`/`0x11` with garrison → engine `0x2F`/`0x30` (disk 46/47); empty plots draw a per-category frame from the DS:`0x260` table minus one. 1×1/2×2 entries are placeholder descriptors.

DISK (=DEF)	SIZE		BUILDING
0	73×18		Stockade
1	73×18		Fort
2	73×18		Fortress
3	44×22		Armory
4	44×22		Magazine
5	44×22		Arsenal
6	75×48		Docks
7	75×48		Drydock
8	75×48		Shipyard
9	53×37		Town Hall
10	1×1	—	Town Hall (level 2 — placeholder art)
11	1×1	—	Town Hall (level 3 — placeholder art)
12	44×22		Schoolhouse
13	44×22		College
14	44×22		University
15	44×22		Warehouse
16	73×18		Warehouse Expansion (also the def-0 forced-stockade frame, engine `0x11`)
17	1×1	—	Stable (placeholder art)
18	23×27		Custom House
19	23×27		Printing Press
20	23×27		Newspaper

DISK (=DEF)	SIZE		BUILDING
21	23×27		Weaver's House
22	23×27		Weaver's Shop

DISK (=DEF)	SIZE		BUILDING
23	23×27		Textile Mill
24	23×27		Tobacconist's House
25	23×27		Tobacconist's Shop
26	23×27		Cigar Factory
27	23×27		Rum Distiller's House
28	23×27		Rum Distillery
29	23×27		Rum Factory
30	1×1	—	Capitol (placeholder art)
31	2×2	—	Capitol Expansion (placeholder art)
32	23×27		Fur Trader's House
33	23×27		Fur Trading Post
34	23×27		Fur Factory
35	44×22		Carpenter's Shop
36	44×22		Lumber Mill
37	53×37		Church
38	53×37		Cathedral
39	23×27		Blacksmith's House
40	23×27		Blacksmith's Shop
41	23×27		Iron Works
42	53×37		(extra frame; empty-plot/category art)
43	44×22		(extra frame)
44	23×27		(extra frame)
45	75×48		(extra frame)
46	44×22		garrison variant (engine 0x2F)
47	44×22		garrison variant (engine 0x30)

BUILDING.SS — frame atlas 48 colony-building frames · disk = def id

Disk-index numbering (engine frame = disk + 1, section 29.1); decoded from the MADSPACK container, transparent index 0xFD.

B.5 Small chrome sheets

SHEET	FRAMES	SIZES	ROLE
WOODTILE.SS	1	32×24	the wood background tile — menu bars, dropdowns, popup interiors, colony composer fill (fill-colour-7 sentinel selects it)
WOODFRAM.SS	1	274×170	woodcut-screen frame, centred from its sheet header
NAMEPLAT.SS	3	18×14, 16×14, 18×14	woodcut caption strip at y=162: left cap + repeated mid tile + right cap, centred on x=160
CURSOR.SS	2	17×17 both	mouse pointer (2 frames)
OPENTILE.SS	1	32×24	boot-menu plaque fill tile (tiled, phase-anchored at the box origin)
PARCH.SS	1	32×24	parchment fill tile

B.6 Inventory of the remaining sheets (counts from decode)

SHEET(S)	FRAMES EACH	FRAME SIZES	ROLE WHERE ESTABLISHED
BDARK.SS	46	2×2 – 75×48	**orphan — no load path in either EXE; never loaded**
CC-00 .. CC-24 (25 sheets)	1	31×86 – 115×114	the 25 Founding Father portraits (Continental Congress / FF pick)
CLOS-BEL / -FWK / -HAT / -LDY / -MAN / -MIL / -ROC	21 / 66 / 22 / 21 / 14 / 20 / 22	up to 204 px wide / 89 tall (CLOS-FWK)	closing-cinematic elements (CLOSING.EXE)
DEC-LOWA .. DEC-LOWZ (26)	8	5–11 × 22	Declaration of Independence lettering, lower case
DEC-UPPA .. DEC-UPPZ (26)	11	8–19 × 22	upper case
DEC-SQIG	11	28×22	lettering flourish
DUTCH1 / ENGLND1 / FRANCE1 / SPAIN1	1	172–178 × 120–122	King-audience nation banner, variant 1 (ENGLND1 = throne-canopy banner drawn at (32,0))
DUTCH2 / ENGLND2 / FRANCE2 / SPAIN2	1	170–176 × 128–133	nation banner, variant 2
IND0A0 .. IND7A3 (32 sheets)	1 (IND2A0: 2)	50×141 – 153×182	native chief speaker portraits (tribe, pose; loader name-patches "IND0A0")
KING.SS	1	79×161	King speaker portrait
KING1.SS	1	189×187	King-audience foreground figure (king + dog), drawn at (0,12) over KINGLSS1.PIK
KING2.SS	8	79×161	King speaker frames
KINGLOSE.SS	1	149×179	king crying — player wins the war
KINGWIN.SS	1	214×198	king triumphant — player loses
MPSLOGO.SS / MPSNAME.SS	16 / 29	155×119 / up to 302×26	MicroProse logo + name (opening)
MSS0 .. MSS5 (6)	1	60×68 – 149×95	advisor portraits (speaker channel `[0x1F5E]` 0..5)
MYR0 .. MYR3 (4)	1	67×68 – 96×93	European rival leader portraits (conversation channel 3)
OPENLOGO	1	276×50	opening title logo
OPENBONK / OPENCRD1-3 / OPENFISH / OPENGUY / OPENMON1-3 / OPENSHIP / OPENSUN / OPENWND1-2	18 / 7,7,5 / 13 / 54 / 15,32,21 / 8 / 7 / 10,11	various	opening-cinematic elements (OPENING.EXE anim table)
SCORE01 .. SCORE24 (24)	1	140–142 × 97–99	score-screen panels ("SCORE"+NN filename build)
WDCUT01 .. WDCUT13 (13)	1	192 × 113/115	woodcut event plates (no 00/14–16 files; captions 0/14–16 unshowable)

SHEET(S)	FRAMES EACH	FRAME SIZES	ROLE WHERE ESTABLISHED
WIN.SS / WIN-FWRK.SS	1 / 46	320×200 / up to 200×91	victory backdrop + fireworks

B.7 VICEROY.PAL — the master palette

The file is 1,024 bytes: 768 bytes of 6-bit VGA RGB (256 × 3) plus 256 trailing unused bytes. Values below are 8-bit `RRGGBB` after the standard expansion `(v<<2)|(v>>4)`. Screens whose `.PIK` backgrounds carry an embedded palette (the frontend/cinematic plates) replace this palette while shown; COLONY.PIK, for example, has none and renders on VICEROY.PAL.

BASE	+0 .. +15
0	000000 0000AA 00AA00 00AAAA AA0000 AA4900 AA5500 AAAAAA 555555 5555FF 55FF55 55FFFF FF0000 FF7100 FFFF55 FFFFFF
16	FBFBFB F3F3F3 EBEBEB E3E3E3 DBDBDB D3D3D3 CBCBCB C3C3C3 BEBEBE B6B6B6 AEAEAE A6A6A6 9E9E9E 969696 8E8E8E 868686
32	828282 797979 717171 696969 616161 595959 515151 494949 454545 3C3C3C 343434 2C2C2C 242424 1C1C1C 141414 0C0C0C
48	DBEFFF C3DBF3 B2CBEB 9EBADF 8EAAD7 799ACF 698AC3 5D79BA 4D65AE 4159A6 34499E 283892 202C8A 181C7D 101075 08086D
64	D7E3AA B6CF86 96BA69 75A64D 559634 348220 1C6D10 045D04 BABA41 A6AA41 9A9E41 8A8E3C 79823C 697138 5D6534 515930
80	CF9634 BE8630 B2792C A26928 965D20 86511C 79451C 6D3C18 CFB28E BAA27D AA926D 9A825D 867151 756145 655134 55452C
96	FFFFDB F7F3C7 F3E7B6 EBDBA2 E7CB92 DFB682 DBA675 D79265 FFFBEB F3EFDB EBE3CB E3DBBA DBCFAE D3C39E CBB692 C3AE86
112	F30000 E30000 D30000 C30000 B20000 A20000 920000 860000 4D65AE 5169B2 4961A6 4159A2 384D9E 30459A 2C3C96 283892
128	794934 75492C 694530 713C1C 613C28 65381C 593424 5D3018 512C20 49281C 3C2018 FF55FF FF55FF FF55FF FF55FF FF55FF
144	FFFFBE FFFF8E FFF35D FFE32C E3C328 C7A220 A67D1C 8A5D14 000000 000000 000000 000000 000000 000000 000000 000000
160	000000 ×16
176	000000 ×16
192	000000 ×14, 0C0C0C (206), 000000
208	000000 ×16
224	000000 ×16
240	000000 ×12, FF55FF FF55FF FF55FF (252–254), CFCFCF (255)

Cycling bands (VGA palette rotation; pixel indices never change):

- **54–60** — the harbour-water shimmer band inside the 48–63 blue ramp (pixel-verified pure palette animation on the Europe screen);
- **120–127** — the sea-lane sparkle band (rotated with an 8-step phase; fitted per capture in every map diff).

Index `0xFD` (253) doubles as the RLE transparent sentinel inside `.SS` frames; the tribe map-marker colours cited by the raze popup data are palette entries here (Aztec 149 = C7A220, Inca 97 = F7F3C7).